STEM NEXUS

MASTER QUESTION BANK

1,440 Questions | Elementary Grades 1–6 and Junior High Grades 7–9

GRADE LEVELS	SUBJECTS	THEMES / SUBJECT	QUESTIONS / THEME	TOTAL
9	4	4	10	1,440

Structure for every grade

Mathematics: 4 themes × 10 questions = 40. Science: 40. Engineering: 40. ICT (Technology): 40. Total per grade: 160 questions. Total across 9 grades: 1,440 questions.

Question format

Unique question ID • grade level • subject • theme • question • four answer options (A–D) • correct answer • explanation.

Review note

This is a structured starter question bank. Some question patterns vary by context or numerical values. Please review curriculum alignment, answer validity, and duplicate content with educators before using it for formal assessment.

Elementary School Grade 1

160 questions • 4 subjects • 4 themes per subject • 10 questions per theme

Mathematics

Theme 1: Numbers 1–20 — 10 questions

STEM-G1-MAT-T01-Q01 | Numbers 1–20 | Difficulty: Foundational

You have 9 objects and give away 3. How many objects are left?

A. 8

B. 7

C. 6

D. 5

Answer: C. Subtract the amount given away: 9 – 3 = 6.

STEM-G1-MAT-T01-Q02 | Numbers 1–20 | Difficulty: Foundational

There are 2 groups with 3 objects in each group. How many objects are there in total?

A. 5

B. 7

C. 8

D. 6

Answer: D. Multiply the number of groups by the objects in each group: 2 × 3 = 6.

STEM-G1-MAT-T01-Q03 | Numbers 1–20 | Difficulty: Foundational

18 objects are shared equally among 3 groups. How many objects are in each group?

A. 8

B. 7

C. 6

D. 5

Answer: C. Divide equally: 18 ÷ 3 = 6.

STEM-G1-MAT-T01-Q04 | Numbers 1–20 | Difficulty: Foundational

There are 10 objects. Then 2 more objects are added. How many objects are there altogether?

A. 13

B. 14

C. 11

D. 12

Answer: D. Add the two groups: 10 + 2 = 12.

STEM-G1-MAT-T01-Q05 | Numbers 1–20 | Difficulty: Foundational

You have 13 objects and give away 1. How many objects are left?

A. 14

B. 11

C. 13

D. 12

Answer: D. Subtract the amount given away: 13 – 1 = 12.

STEM-G1-MAT-T01-Q06 | Numbers 1–20 | Difficulty: Foundational

There are 2 groups with 4 objects in each group. How many objects are there in total?

A. 9

B. 8

C. 7

D. 10

Answer: B. Multiply the number of groups by the objects in each group: 2 × 4 = 8.

STEM-G1-MAT-T01-Q07 | Numbers 1–20 | Difficulty: Foundational

45 objects are shared equally among 5 groups. How many objects are in each group?

A. 10

B. 9

C. 8

D. 11

Answer: B. Divide equally: 45 ÷ 5 = 9.

STEM-G1-MAT-T01-Q08 | Numbers 1–20 | Difficulty: Foundational

There are 13 objects. Then 6 more objects are added. How many objects are there altogether?

A. 21

B. 18

C. 19

D. 20

Answer: C. Add the two groups: 13 + 6 = 19.

STEM-G1-MAT-T01-Q09 | Numbers 1–20 | Difficulty: Foundational

You have 8 objects and give away 3. How many objects are left?

A. 5

B. 4

C. 7

D. 6

Answer: A. Subtract the amount given away: 8 – 3 = 5.

STEM-G1-MAT-T01-Q10 | Numbers 1–20 | Difficulty: Foundational

There are 3 groups with 5 objects in each group. How many objects are there in total?

A. 16

B. 17

C. 15

D. 14

Answer: C. Multiply the number of groups by the objects in each group: 3 × 5 = 15.

Theme 2: Simple Addition and Subtraction — 10 questions

STEM-G1-MAT-T02-Q01 | Simple Addition and Subtraction | Difficulty: Foundational

You have 11 objects and give away 3. How many objects are left?

A. 7

B. 8

C. 9

D. 10

Answer: B. Subtract the amount given away: 11 – 3 = 8.

STEM-G1-MAT-T02-Q02 | Simple Addition and Subtraction | Difficulty: Foundational

There are 2 groups with 5 objects in each group. How many objects are there in total?

A. 11

B. 10

C. 9

D. 12

Answer: B. Multiply the number of groups by the objects in each group: 2 × 5 = 10.

STEM-G1-MAT-T02-Q03 | Simple Addition and Subtraction | Difficulty: Foundational

27 objects are shared equally among 3 groups. How many objects are in each group?

A. 9

B. 8

C. 11

D. 10

Answer: A. Divide equally: 27 ÷ 3 = 9.

STEM-G1-MAT-T02-Q04 | Simple Addition and Subtraction | Difficulty: Foundational

There are 9 objects. Then 4 more objects are added. How many objects are there altogether?

A. 14

B. 13

C. 15

D. 12

Answer: B. Add the two groups: 9 + 4 = 13.

STEM-G1-MAT-T02-Q05 | Simple Addition and Subtraction | Difficulty: Foundational

You have 7 objects and give away 7. How many objects are left?

A. 0

B. 2

C. 5

D. 1

Answer: A. Subtract the amount given away: 7 – 7 = 0.

STEM-G1-MAT-T02-Q06 | Simple Addition and Subtraction | Difficulty: Foundational

There are 3 groups with 3 objects in each group. How many objects are there in total?

A. 8

B. 10

C. 9

D. 11

Answer: C. Multiply the number of groups by the objects in each group: 3 × 3 = 9.

STEM-G1-MAT-T02-Q07 | Simple Addition and Subtraction | Difficulty: Foundational

24 objects are shared equally among 4 groups. How many objects are in each group?

A. 5

B. 6

C. 7

D. 8

Answer: B. Divide equally: 24 ÷ 4 = 6.

STEM-G1-MAT-T02-Q08 | Simple Addition and Subtraction | Difficulty: Foundational

There are 3 objects. Then 6 more objects are added. How many objects are there altogether?

A. 8

B. 11

C. 10

D. 9

Answer: D. Add the two groups: 3 + 6 = 9.

STEM-G1-MAT-T02-Q09 | Simple Addition and Subtraction | Difficulty: Foundational

You have 7 objects and give away 7. How many objects are left?

A. 1

B. 5

C. 0

D. 2

Answer: C. Subtract the amount given away: 7 – 7 = 0.

STEM-G1-MAT-T02-Q10 | Simple Addition and Subtraction | Difficulty: Foundational

There are 4 groups with 2 objects in each group. How many objects are there in total?

A. 9

B. 8

C. 7

D. 10

Answer: B. Multiply the number of groups by the objects in each group: 4 × 2 = 8.

Theme 3: Shapes and Patterns — 10 questions

STEM-G1-MAT-T03-Q01 | Shapes and Patterns | Difficulty: Foundational

You have 15 objects and give away 4. How many objects are left?

A. 11

B. 13

C. 12

D. 10

Answer: A. Subtract the amount given away: 15 – 4 = 11.

STEM-G1-MAT-T03-Q02 | Shapes and Patterns | Difficulty: Foundational

There are 5 groups with 3 objects in each group. How many objects are there in total?

A. 15

B. 17

C. 16

D. 14

Answer: A. Multiply the number of groups by the objects in each group: 5 × 3 = 15.

STEM-G1-MAT-T03-Q03 | Shapes and Patterns | Difficulty: Foundational

30 objects are shared equally among 3 groups. How many objects are in each group?

A. 9

B. 12

C. 10

D. 11

Answer: C. Divide equally: 30 ÷ 3 = 10.

STEM-G1-MAT-T03-Q04 | Shapes and Patterns | Difficulty: Foundational

There are 2 objects. Then 3 more objects are added. How many objects are there altogether?

A. 6

B. 4

C. 7

D. 5

Answer: D. Add the two groups: 2 + 3 = 5.

STEM-G1-MAT-T03-Q05 | Shapes and Patterns | Difficulty: Foundational

You have 14 objects and give away 7. How many objects are left?

A. 8

B. 7

C. 6

D. 9

Answer: B. Subtract the amount given away: 14 – 7 = 7.

STEM-G1-MAT-T03-Q06 | Shapes and Patterns | Difficulty: Foundational

There are 3 groups with 5 objects in each group. How many objects are there in total?

A. 17

B. 14

C. 16

D. 15

Answer: D. Multiply the number of groups by the objects in each group: 3 × 5 = 15.

STEM-G1-MAT-T03-Q07 | Shapes and Patterns | Difficulty: Foundational

6 objects are shared equally among 3 groups. How many objects are in each group?

A. 3

B. 4

C. 1

D. 2

Answer: D. Divide equally: 6 ÷ 3 = 2.

STEM-G1-MAT-T03-Q08 | Shapes and Patterns | Difficulty: Foundational

There are 11 objects. Then 2 more objects are added. How many objects are there altogether?

A. 15

B. 14

C. 12

D. 13

Answer: D. Add the two groups: 11 + 2 = 13.

STEM-G1-MAT-T03-Q09 | Shapes and Patterns | Difficulty: Foundational

You have 11 objects and give away 4. How many objects are left?

A. 9

B. 7

C. 6

D. 8

Answer: B. Subtract the amount given away: 11 – 4 = 7.

STEM-G1-MAT-T03-Q10 | Shapes and Patterns | Difficulty: Foundational

There are 3 groups with 4 objects in each group. How many objects are there in total?

A. 12

B. 11

C. 14

D. 13

Answer: A. Multiply the number of groups by the objects in each group: 3 × 4 = 12.

Theme 4: Length, Time, and Simple Comparisons — 10 questions

STEM-G1-MAT-T04-Q01 | Length, Time, and Simple Comparisons | Difficulty: Foundational

You have 9 objects and give away 3. How many objects are left?

A. 7

B. 8

C. 6

D. 5

Answer: C. Subtract the amount given away: 9 – 3 = 6.

STEM-G1-MAT-T04-Q02 | Length, Time, and Simple Comparisons | Difficulty: Foundational

There are 2 groups with 5 objects in each group. How many objects are there in total?

A. 10

B. 11

C. 12

D. 9

Answer: A. Multiply the number of groups by the objects in each group: 2 × 5 = 10.

STEM-G1-MAT-T04-Q03 | Length, Time, and Simple Comparisons | Difficulty: Foundational

20 objects are shared equally among 4 groups. How many objects are in each group?

A. 6

B. 4

C. 7

D. 5

Answer: D. Divide equally: 20 ÷ 4 = 5.

STEM-G1-MAT-T04-Q04 | Length, Time, and Simple Comparisons | Difficulty: Foundational

There are 6 objects. Then 5 more objects are added. How many objects are there altogether?

A. 12

B. 11

C. 13

D. 10

Answer: B. Add the two groups: 6 + 5 = 11.

STEM-G1-MAT-T04-Q05 | Length, Time, and Simple Comparisons | Difficulty: Foundational

You have 7 objects and give away 5. How many objects are left?

A. 4

B. 3

C. 1

D. 2

Answer: D. Subtract the amount given away: 7 – 5 = 2.

STEM-G1-MAT-T04-Q06 | Length, Time, and Simple Comparisons | Difficulty: Foundational

There are 5 groups with 3 objects in each group. How many objects are there in total?

A. 14

B. 17

C. 15

D. 16

Answer: C. Multiply the number of groups by the objects in each group: 5 × 3 = 15.

STEM-G1-MAT-T04-Q07 | Length, Time, and Simple Comparisons | Difficulty: Foundational

15 objects are shared equally among 3 groups. How many objects are in each group?

A. 4

B. 6

C. 5

D. 7

Answer: C. Divide equally: 15 ÷ 3 = 5.

STEM-G1-MAT-T04-Q08 | Length, Time, and Simple Comparisons | Difficulty: Foundational

There are 3 objects. Then 5 more objects are added. How many objects are there altogether?

A. 10

B. 9

C. 8

D. 7

Answer: C. Add the two groups: 3 + 5 = 8.

STEM-G1-MAT-T04-Q09 | Length, Time, and Simple Comparisons | Difficulty: Foundational

You have 14 objects and give away 1. How many objects are left?

A. 15

B. 12

C. 13

D. 14

Answer: C. Subtract the amount given away: 14 – 1 = 13.

STEM-G1-MAT-T04-Q10 | Length, Time, and Simple Comparisons | Difficulty: Foundational

There are 4 groups with 3 objects in each group. How many objects are there in total?

A. 11

B. 13

C. 12

D. 14

Answer: C. Multiply the number of groups by the objects in each group: 4 × 3 = 12.

Science

Theme 1: Body Parts and the Five Senses — 10 questions

STEM-G1-SCI-T01-Q01 | Body Parts and the Five Senses | Difficulty: Foundational

Which body part helps us hear sounds?

A. The ears

B. The nose

C. The eyes

D. The tongue

Answer: A. The ears detect sound vibrations.

STEM-G1-SCI-T01-Q02 | Body Parts and the Five Senses | Difficulty: Foundational

Which body part helps us hear sounds?

A. The nose

B. The eyes

C. The tongue

D. The ears

Answer: D. The ears detect sound vibrations.

STEM-G1-SCI-T01-Q03 | Body Parts and the Five Senses | Difficulty: Foundational

Which body part helps us hear sounds?

A. The nose

B. The tongue

C. The ears

D. The eyes

Answer: C. The ears detect sound vibrations.

STEM-G1-SCI-T01-Q04 | Body Parts and the Five Senses | Difficulty: Foundational

Which body part helps us hear sounds?

A. The nose

B. The ears

C. The tongue

D. The eyes

Answer: B. The ears detect sound vibrations.

STEM-G1-SCI-T01-Q05 | Body Parts and the Five Senses | Difficulty: Foundational

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STEM-G1-SCI-T01-Q06 | Body Parts and the Five Senses | Difficulty: Foundational

Which body part helps us hear sounds?

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B. The tongue

C. The ears

D. The eyes

Answer: C. The ears detect sound vibrations.

STEM-G1-SCI-T01-Q07 | Body Parts and the Five Senses | Difficulty: Foundational

Which body part helps us hear sounds?

A. The ears

B. The nose

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D. The tongue

Answer: A. The ears detect sound vibrations.

STEM-G1-SCI-T01-Q08 | Body Parts and the Five Senses | Difficulty: Foundational

Which body part helps us hear sounds?

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B. The ears

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D. The nose

Answer: B. The ears detect sound vibrations.

STEM-G1-SCI-T01-Q09 | Body Parts and the Five Senses | Difficulty: Foundational

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D. The eyes

Answer: A. The ears detect sound vibrations.

STEM-G1-SCI-T01-Q10 | Body Parts and the Five Senses | Difficulty: Foundational

Which body part helps us hear sounds?

A. The eyes

B. The tongue

C. The nose

D. The ears

Answer: D. The ears detect sound vibrations.

Theme 2: Animals and Plants — 10 questions

STEM-G1-SCI-T02-Q01 | Animals and Plants | Difficulty: Foundational

Which of these is a living thing?

A. A table

B. A pencil

C. A cat

D. A rock

Answer: C. Living things grow and need energy.

STEM-G1-SCI-T02-Q02 | Animals and Plants | Difficulty: Foundational

Which of these is a living thing?

A. A cat

B. A rock

C. A table

D. A pencil

Answer: A. Living things grow and need energy.

STEM-G1-SCI-T02-Q03 | Animals and Plants | Difficulty: Foundational

Which of these is a living thing?

A. A pencil

B. A cat

C. A rock

D. A table

Answer: B. Living things grow and need energy.

STEM-G1-SCI-T02-Q04 | Animals and Plants | Difficulty: Foundational

Which of these is a living thing?

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B. A table

C. A pencil

D. A rock

Answer: A. Living things grow and need energy.

STEM-G1-SCI-T02-Q05 | Animals and Plants | Difficulty: Foundational

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Answer: A. Living things grow and need energy.

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STEM-G1-SCI-T02-Q08 | Animals and Plants | Difficulty: Foundational

Which of these is a living thing?

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B. A table

C. A rock

D. A pencil

Answer: A. Living things grow and need energy.

STEM-G1-SCI-T02-Q09 | Animals and Plants | Difficulty: Foundational

Which of these is a living thing?

A. A table

B. A rock

C. A pencil

D. A cat

Answer: D. Living things grow and need energy.

STEM-G1-SCI-T02-Q10 | Animals and Plants | Difficulty: Foundational

Which of these is a living thing?

A. A rock

B. A cat

C. A pencil

D. A table

Answer: B. Living things grow and need energy.

Theme 3: Weather and the Environment — 10 questions

STEM-G1-SCI-T03-Q01 | Weather and the Environment | Difficulty: Foundational

Dark, thick clouds may indicate a chance of ...

A. A solar eclipse

B. An earthquake

C. A rainbow all day

D. Rain

Answer: D. Thick clouds can contain water droplets that may fall as rain.

STEM-G1-SCI-T03-Q02 | Weather and the Environment | Difficulty: Foundational

Dark, thick clouds may indicate a chance of ...

A. A solar eclipse

B. An earthquake

C. Rain

D. A rainbow all day

Answer: C. Thick clouds can contain water droplets that may fall as rain.

STEM-G1-SCI-T03-Q03 | Weather and the Environment | Difficulty: Foundational

Dark, thick clouds may indicate a chance of ...

A. An earthquake

B. Rain

C. A rainbow all day

D. A solar eclipse

Answer: B. Thick clouds can contain water droplets that may fall as rain.

STEM-G1-SCI-T03-Q04 | Weather and the Environment | Difficulty: Foundational

Dark, thick clouds may indicate a chance of ...

A. An earthquake

B. Rain

C. A solar eclipse

D. A rainbow all day

Answer: B. Thick clouds can contain water droplets that may fall as rain.

STEM-G1-SCI-T03-Q05 | Weather and the Environment | Difficulty: Foundational

Dark, thick clouds may indicate a chance of ...

A. A solar eclipse

B. An earthquake

C. Rain

D. A rainbow all day

Answer: C. Thick clouds can contain water droplets that may fall as rain.

STEM-G1-SCI-T03-Q06 | Weather and the Environment | Difficulty: Foundational

Dark, thick clouds may indicate a chance of ...

A. Rain

B. An earthquake

C. A solar eclipse

D. A rainbow all day

Answer: A. Thick clouds can contain water droplets that may fall as rain.

STEM-G1-SCI-T03-Q07 | Weather and the Environment | Difficulty: Foundational

Dark, thick clouds may indicate a chance of ...

A. A rainbow all day

B. A solar eclipse

C. An earthquake

D. Rain

Answer: D. Thick clouds can contain water droplets that may fall as rain.

STEM-G1-SCI-T03-Q08 | Weather and the Environment | Difficulty: Foundational

Dark, thick clouds may indicate a chance of ...

A. A solar eclipse

B. Rain

C. An earthquake

D. A rainbow all day

Answer: B. Thick clouds can contain water droplets that may fall as rain.

STEM-G1-SCI-T03-Q09 | Weather and the Environment | Difficulty: Foundational

Dark, thick clouds may indicate a chance of ...

A. A rainbow all day

B. An earthquake

C. Rain

D. A solar eclipse

Answer: C. Thick clouds can contain water droplets that may fall as rain.

STEM-G1-SCI-T03-Q10 | Weather and the Environment | Difficulty: Foundational

Dark, thick clouds may indicate a chance of ...

A. A solar eclipse

B. A rainbow all day

C. An earthquake

D. Rain

Answer: D. Thick clouds can contain water droplets that may fall as rain.

Theme 4: Objects and Their Properties — 10 questions

STEM-G1-SCI-T04-Q01 | Objects and Their Properties | Difficulty: Foundational

Which state of matter takes the shape of its container but has a nearly fixed volume?

A. Solid

B. Gas

C. Light

D. Liquid

Answer: D. A liquid flows and takes the shape of its container.

STEM-G1-SCI-T04-Q02 | Objects and Their Properties | Difficulty: Foundational

Which state of matter takes the shape of its container but has a nearly fixed volume?

A. Solid

B. Gas

C. Light

D. Liquid

Answer: D. A liquid flows and takes the shape of its container.

STEM-G1-SCI-T04-Q03 | Objects and Their Properties | Difficulty: Foundational

Which state of matter takes the shape of its container but has a nearly fixed volume?

A. Liquid

B. Gas

C. Light

D. Solid

Answer: A. A liquid flows and takes the shape of its container.

STEM-G1-SCI-T04-Q04 | Objects and Their Properties | Difficulty: Foundational

Which state of matter takes the shape of its container but has a nearly fixed volume?

A. Gas

B. Solid

C. Liquid

D. Light

Answer: C. A liquid flows and takes the shape of its container.

STEM-G1-SCI-T04-Q05 | Objects and Their Properties | Difficulty: Foundational

Which state of matter takes the shape of its container but has a nearly fixed volume?

A. Solid

B. Light

C. Gas

D. Liquid

Answer: D. A liquid flows and takes the shape of its container.

STEM-G1-SCI-T04-Q06 | Objects and Their Properties | Difficulty: Foundational

Which state of matter takes the shape of its container but has a nearly fixed volume?

A. Gas

B. Liquid

C. Solid

D. Light

Answer: B. A liquid flows and takes the shape of its container.

STEM-G1-SCI-T04-Q07 | Objects and Their Properties | Difficulty: Foundational

Which state of matter takes the shape of its container but has a nearly fixed volume?

A. Gas

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Answer: C. A liquid flows and takes the shape of its container.

STEM-G1-SCI-T04-Q08 | Objects and Their Properties | Difficulty: Foundational

Which state of matter takes the shape of its container but has a nearly fixed volume?

A. Light

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STEM-G1-SCI-T04-Q09 | Objects and Their Properties | Difficulty: Foundational

Which state of matter takes the shape of its container but has a nearly fixed volume?

A. Gas

B. Light

C. Liquid

D. Solid

Answer: C. A liquid flows and takes the shape of its container.

STEM-G1-SCI-T04-Q10 | Objects and Their Properties | Difficulty: Foundational

Which state of matter takes the shape of its container but has a nearly fixed volume?

A. Gas

B. Light

C. Liquid

D. Solid

Answer: C. A liquid flows and takes the shape of its container.

Engineering

Theme 1: Materials and Their Uses — 10 questions

STEM-G1-ENG-T01-Q01 | Materials and Their Uses | Difficulty: Foundational

What is the purpose of making a sketch before building an object?

A. To replace all testing

B. To plan its shape, dimensions, and parts

C. To avoid measuring

D. To make the object build itself

Answer: B. A sketch communicates a design before construction.

STEM-G1-ENG-T01-Q02 | Materials and Their Uses | Difficulty: Foundational

For a fair test, a team should ...

A. Avoid recording results

B. Change one factor while keeping other factors the same

C. Change every factor at once

D. Choose only the preferred result

Answer: B. Controlling variables helps identify what caused a result.

STEM-G1-ENG-T01-Q03 | Materials and Their Uses | Difficulty: Foundational

When selecting a material for a load-bearing part, what should be considered?

A. The designer's name

B. Its strength and suitability for the task

C. The packaging

D. Only its favorite color

Answer: B. Material choice depends on the load and operating conditions.

STEM-G1-ENG-T01-Q04 | Materials and Their Uses | Difficulty: Foundational

Why should joints in a model be inspected?

A. Joints remove gravity

B. Joints make mass zero

C. Joints are only decoration

D. Joints transfer forces between parts

Answer: D. Weak joints can become failure points in a structure.

STEM-G1-ENG-T01-Q05 | Materials and Their Uses | Difficulty: Foundational

After a test reveals a weakness, what should engineers do next?

A. Improve the design and test it again

B. Hide the result

C. Delete the notes

D. Ignore safety

Answer: A. Engineering is iterative and uses test results to guide improvements.

STEM-G1-ENG-T01-Q06 | Materials and Their Uses | Difficulty: Foundational

Which measurement is useful when comparing two bridge designs?

A. The maximum load each bridge can support

B. The bridge color

C. The model's name

D. The number of stickers

Answer: A. Load capacity is a relevant measure of bridge performance.

STEM-G1-ENG-T01-Q07 | Materials and Their Uses | Difficulty: Foundational

Why should an engineering prototype be tested before use?

A. To find out whether the design works and is safe

B. To make it larger every time

C. To avoid making improvements

D. To make it look more expensive

Answer: A. Testing reveals weaknesses and helps improve a design.

STEM-G1-ENG-T01-Q08 | Materials and Their Uses | Difficulty: Foundational

If a model structure tips over easily, what is a sensible first improvement?

A. Add weight to the top

B. Widen the base or strengthen the joints

C. Remove the base

D. Remove all supports

Answer: B. A wider base and strong joints can improve stability.

STEM-G1-ENG-T01-Q09 | Materials and Their Uses | Difficulty: Foundational

What is the purpose of making a sketch before building an object?

A. To plan its shape, dimensions, and parts

B. To make the object build itself

C. To replace all testing

D. To avoid measuring

Answer: A. A sketch communicates a design before construction.

STEM-G1-ENG-T01-Q10 | Materials and Their Uses | Difficulty: Foundational

For a fair test, a team should ...

A. Change every factor at once

B. Avoid recording results

C. Choose only the preferred result

D. Change one factor while keeping other factors the same

Answer: D. Controlling variables helps identify what caused a result.

Theme 2: Building Simple Structures — 10 questions

STEM-G1-ENG-T02-Q01 | Building Simple Structures | Difficulty: Foundational

What is the purpose of making a sketch before building an object?

A. To avoid measuring

B. To replace all testing

C. To make the object build itself

D. To plan its shape, dimensions, and parts

Answer: D. A sketch communicates a design before construction.

STEM-G1-ENG-T02-Q02 | Building Simple Structures | Difficulty: Foundational

For a fair test, a team should ...

A. Change one factor while keeping other factors the same

B. Choose only the preferred result

C. Avoid recording results

D. Change every factor at once

Answer: A. Controlling variables helps identify what caused a result.

STEM-G1-ENG-T02-Q03 | Building Simple Structures | Difficulty: Foundational

When selecting a material for a load-bearing part, what should be considered?

A. Its strength and suitability for the task

B. Only its favorite color

C. The packaging

D. The designer's name

Answer: A. Material choice depends on the load and operating conditions.

STEM-G1-ENG-T02-Q04 | Building Simple Structures | Difficulty: Foundational

Why should joints in a model be inspected?

A. Joints make mass zero

B. Joints remove gravity

C. Joints are only decoration

D. Joints transfer forces between parts

Answer: D. Weak joints can become failure points in a structure.

STEM-G1-ENG-T02-Q05 | Building Simple Structures | Difficulty: Foundational

After a test reveals a weakness, what should engineers do next?

A. Delete the notes

B. Improve the design and test it again

C. Hide the result

D. Ignore safety

Answer: B. Engineering is iterative and uses test results to guide improvements.

STEM-G1-ENG-T02-Q06 | Building Simple Structures | Difficulty: Foundational

Which measurement is useful when comparing two bridge designs?

A. The maximum load each bridge can support

B. The bridge color

C. The number of stickers

D. The model's name

Answer: A. Load capacity is a relevant measure of bridge performance.

STEM-G1-ENG-T02-Q07 | Building Simple Structures | Difficulty: Foundational

Why should an engineering prototype be tested before use?

A. To find out whether the design works and is safe

B. To avoid making improvements

C. To make it larger every time

D. To make it look more expensive

Answer: A. Testing reveals weaknesses and helps improve a design.

STEM-G1-ENG-T02-Q08 | Building Simple Structures | Difficulty: Foundational

If a model structure tips over easily, what is a sensible first improvement?

A. Remove all supports

B. Remove the base

C. Add weight to the top

D. Widen the base or strengthen the joints

Answer: D. A wider base and strong joints can improve stability.

STEM-G1-ENG-T02-Q09 | Building Simple Structures | Difficulty: Foundational

What is the purpose of making a sketch before building an object?

A. To replace all testing

B. To make the object build itself

C. To avoid measuring

D. To plan its shape, dimensions, and parts

Answer: D. A sketch communicates a design before construction.

STEM-G1-ENG-T02-Q10 | Building Simple Structures | Difficulty: Foundational

For a fair test, a team should ...

A. Avoid recording results

B. Change every factor at once

C. Change one factor while keeping other factors the same

D. Choose only the preferred result

Answer: C. Controlling variables helps identify what caused a result.

Theme 3: Wheels and Motion — 10 questions

STEM-G1-ENG-T03-Q01 | Wheels and Motion | Difficulty: Foundational

What is the purpose of making a sketch before building an object?

A. To replace all testing

B. To plan its shape, dimensions, and parts

C. To avoid measuring

D. To make the object build itself

Answer: B. A sketch communicates a design before construction.

STEM-G1-ENG-T03-Q02 | Wheels and Motion | Difficulty: Foundational

For a fair test, a team should ...

A. Change every factor at once

B. Change one factor while keeping other factors the same

C. Avoid recording results

D. Choose only the preferred result

Answer: B. Controlling variables helps identify what caused a result.

STEM-G1-ENG-T03-Q03 | Wheels and Motion | Difficulty: Foundational

When selecting a material for a load-bearing part, what should be considered?

A. The designer's name

B. Its strength and suitability for the task

C. Only its favorite color

D. The packaging

Answer: B. Material choice depends on the load and operating conditions.

STEM-G1-ENG-T03-Q04 | Wheels and Motion | Difficulty: Foundational

Why should joints in a model be inspected?

A. Joints make mass zero

B. Joints are only decoration

C. Joints transfer forces between parts

D. Joints remove gravity

Answer: C. Weak joints can become failure points in a structure.

STEM-G1-ENG-T03-Q05 | Wheels and Motion | Difficulty: Foundational

After a test reveals a weakness, what should engineers do next?

A. Hide the result

B. Improve the design and test it again

C. Delete the notes

D. Ignore safety

Answer: B. Engineering is iterative and uses test results to guide improvements.

STEM-G1-ENG-T03-Q06 | Wheels and Motion | Difficulty: Foundational

Which measurement is useful when comparing two bridge designs?

A. The number of stickers

B. The model's name

C. The bridge color

D. The maximum load each bridge can support

Answer: D. Load capacity is a relevant measure of bridge performance.

STEM-G1-ENG-T03-Q07 | Wheels and Motion | Difficulty: Foundational

Why should an engineering prototype be tested before use?

A. To make it look more expensive

B. To avoid making improvements

C. To make it larger every time

D. To find out whether the design works and is safe

Answer: D. Testing reveals weaknesses and helps improve a design.

STEM-G1-ENG-T03-Q08 | Wheels and Motion | Difficulty: Foundational

If a model structure tips over easily, what is a sensible first improvement?

A. Widen the base or strengthen the joints

B. Remove the base

C. Remove all supports

D. Add weight to the top

Answer: A. A wider base and strong joints can improve stability.

STEM-G1-ENG-T03-Q09 | Wheels and Motion | Difficulty: Foundational

What is the purpose of making a sketch before building an object?

A. To replace all testing

B. To plan its shape, dimensions, and parts

C. To avoid measuring

D. To make the object build itself

Answer: B. A sketch communicates a design before construction.

STEM-G1-ENG-T03-Q10 | Wheels and Motion | Difficulty: Foundational

For a fair test, a team should ...

A. Avoid recording results

B. Change every factor at once

C. Change one factor while keeping other factors the same

D. Choose only the preferred result

Answer: C. Controlling variables helps identify what caused a result.

Theme 4: Designing and Testing Objects — 10 questions

STEM-G1-ENG-T04-Q01 | Designing and Testing Objects | Difficulty: Foundational

What is the purpose of making a sketch before building an object?

A. To avoid measuring

B. To make the object build itself

C. To replace all testing

D. To plan its shape, dimensions, and parts

Answer: D. A sketch communicates a design before construction.

STEM-G1-ENG-T04-Q02 | Designing and Testing Objects | Difficulty: Foundational

For a fair test, a team should ...

A. Change every factor at once

B. Avoid recording results

C. Choose only the preferred result

D. Change one factor while keeping other factors the same

Answer: D. Controlling variables helps identify what caused a result.

STEM-G1-ENG-T04-Q03 | Designing and Testing Objects | Difficulty: Foundational

When selecting a material for a load-bearing part, what should be considered?

A. The designer's name

B. The packaging

C. Only its favorite color

D. Its strength and suitability for the task

Answer: D. Material choice depends on the load and operating conditions.

STEM-G1-ENG-T04-Q04 | Designing and Testing Objects | Difficulty: Foundational

Why should joints in a model be inspected?

A. Joints make mass zero

B. Joints remove gravity

C. Joints are only decoration

D. Joints transfer forces between parts

Answer: D. Weak joints can become failure points in a structure.

STEM-G1-ENG-T04-Q05 | Designing and Testing Objects | Difficulty: Foundational

After a test reveals a weakness, what should engineers do next?

A. Improve the design and test it again

B. Delete the notes

C. Ignore safety

D. Hide the result

Answer: A. Engineering is iterative and uses test results to guide improvements.

STEM-G1-ENG-T04-Q06 | Designing and Testing Objects | Difficulty: Foundational

Which measurement is useful when comparing two bridge designs?

A. The maximum load each bridge can support

B. The number of stickers

C. The bridge color

D. The model's name

Answer: A. Load capacity is a relevant measure of bridge performance.

STEM-G1-ENG-T04-Q07 | Designing and Testing Objects | Difficulty: Foundational

Why should an engineering prototype be tested before use?

A. To find out whether the design works and is safe

B. To make it look more expensive

C. To make it larger every time

D. To avoid making improvements

Answer: A. Testing reveals weaknesses and helps improve a design.

STEM-G1-ENG-T04-Q08 | Designing and Testing Objects | Difficulty: Foundational

If a model structure tips over easily, what is a sensible first improvement?

A. Remove all supports

B. Add weight to the top

C. Widen the base or strengthen the joints

D. Remove the base

Answer: C. A wider base and strong joints can improve stability.

STEM-G1-ENG-T04-Q09 | Designing and Testing Objects | Difficulty: Foundational

What is the purpose of making a sketch before building an object?

A. To plan its shape, dimensions, and parts

B. To make the object build itself

C. To avoid measuring

D. To replace all testing

Answer: A. A sketch communicates a design before construction.

STEM-G1-ENG-T04-Q10 | Designing and Testing Objects | Difficulty: Foundational

For a fair test, a team should ...

A. Avoid recording results

B. Change one factor while keeping other factors the same

C. Choose only the preferred result

D. Change every factor at once

Answer: B. Controlling variables helps identify what caused a result.

ICT (Technology)

Theme 1: Introduction to Computer Devices — 10 questions

STEM-G1-ICT-T01-Q01 | Introduction to Computer Devices | Difficulty: Foundational

Why should you keep your password private?

A. To make the internet faster

B. To make the screen brighter

C. To protect your account and personal information

D. To extend battery life

Answer: C. A password provides access to an account and should be protected.

STEM-G1-ICT-T01-Q02 | Introduction to Computer Devices | Difficulty: Foundational

A clear, ordered set of steps for solving a problem is called an ...

A. Wallpaper

B. Algorithm

C. Speaker

D. Resolution

Answer: B. An algorithm is an ordered set of instructions.

STEM-G1-ICT-T01-Q03 | Introduction to Computer Devices | Difficulty: Foundational

In a spreadsheet, a cell is ...

A. A type of cable

B. The intersection of a row and a column

C. An application icon

D. A webpage

Answer: B. A cell is a box at the intersection of a row and a column.

STEM-G1-ICT-T01-Q04 | Introduction to Computer Devices | Difficulty: Foundational

How should information found online be checked?

A. Compare sources and supporting evidence

B. Count the emojis

C. Choose the most dramatic headline

D. Share it before reading

Answer: A. Checking sources helps assess accuracy and credibility.

STEM-G1-ICT-T01-Q05 | Introduction to Computer Devices | Difficulty: Foundational

Finding and fixing errors in a program is called ...

A. Formatting

B. Streaming

C. Browsing

D. Debugging

Answer: D. Debugging is the process of locating and correcting errors.

STEM-G1-ICT-T01-Q06 | Introduction to Computer Devices | Difficulty: Foundational

Which device is commonly used to enter text into a computer?

A. Keyboard

B. Speaker

C. Projector

D. Monitor

Answer: A. A keyboard is an input device used for typing.

STEM-G1-ICT-T01-Q07 | Introduction to Computer Devices | Difficulty: Foundational

Software designed to help a user perform a task is commonly called an ...

A. Fan

B. Desk

C. Power cable

D. Application

Answer: D. An application is software that helps users perform specific tasks.

STEM-G1-ICT-T01-Q08 | Introduction to Computer Devices | Difficulty: Foundational

What should you do if an unknown person sends you a suspicious link?

A. Forward it to every contact

B. Download every attachment

C. Do not open it; check with a trusted adult or person

D. Enter your password

Answer: C. Suspicious links may lead to scams or harmful software.

STEM-G1-ICT-T01-Q09 | Introduction to Computer Devices | Difficulty: Foundational

Why should you keep your password private?

A. To extend battery life

B. To protect your account and personal information

C. To make the internet faster

D. To make the screen brighter

Answer: B. A password provides access to an account and should be protected.

STEM-G1-ICT-T01-Q10 | Introduction to Computer Devices | Difficulty: Foundational

A clear, ordered set of steps for solving a problem is called an ...

A. Wallpaper

B. Speaker

C. Resolution

D. Algorithm

Answer: D. An algorithm is an ordered set of instructions.

Theme 2: Input and Output Devices — 10 questions

STEM-G1-ICT-T02-Q01 | Input and Output Devices | Difficulty: Foundational

Why should you keep your password private?

A. To protect your account and personal information

B. To make the internet faster

C. To extend battery life

D. To make the screen brighter

Answer: A. A password provides access to an account and should be protected.

STEM-G1-ICT-T02-Q02 | Input and Output Devices | Difficulty: Foundational

A clear, ordered set of steps for solving a problem is called an ...

A. Resolution

B. Algorithm

C. Speaker

D. Wallpaper

Answer: B. An algorithm is an ordered set of instructions.

STEM-G1-ICT-T02-Q03 | Input and Output Devices | Difficulty: Foundational

In a spreadsheet, a cell is ...

A. A type of cable

B. A webpage

C. An application icon

D. The intersection of a row and a column

Answer: D. A cell is a box at the intersection of a row and a column.

STEM-G1-ICT-T02-Q04 | Input and Output Devices | Difficulty: Foundational

How should information found online be checked?

A. Compare sources and supporting evidence

B. Share it before reading

C. Choose the most dramatic headline

D. Count the emojis

Answer: A. Checking sources helps assess accuracy and credibility.

STEM-G1-ICT-T02-Q05 | Input and Output Devices | Difficulty: Foundational

Finding and fixing errors in a program is called ...

A. Debugging

B. Browsing

C. Streaming

D. Formatting

Answer: A. Debugging is the process of locating and correcting errors.

STEM-G1-ICT-T02-Q06 | Input and Output Devices | Difficulty: Foundational

Which device is commonly used to enter text into a computer?

A. Monitor

B. Speaker

C. Projector

D. Keyboard

Answer: D. A keyboard is an input device used for typing.

STEM-G1-ICT-T02-Q07 | Input and Output Devices | Difficulty: Foundational

Software designed to help a user perform a task is commonly called an ...

A. Desk

B. Application

C. Power cable

D. Fan

Answer: B. An application is software that helps users perform specific tasks.

STEM-G1-ICT-T02-Q08 | Input and Output Devices | Difficulty: Foundational

What should you do if an unknown person sends you a suspicious link?

A. Forward it to every contact

B. Download every attachment

C. Enter your password

D. Do not open it; check with a trusted adult or person

Answer: D. Suspicious links may lead to scams or harmful software.

STEM-G1-ICT-T02-Q09 | Input and Output Devices | Difficulty: Foundational

Why should you keep your password private?

A. To make the screen brighter

B. To protect your account and personal information

C. To make the internet faster

D. To extend battery life

Answer: B. A password provides access to an account and should be protected.

STEM-G1-ICT-T02-Q10 | Input and Output Devices | Difficulty: Foundational

A clear, ordered set of steps for solving a problem is called an ...

A. Speaker

B. Resolution

C. Algorithm

D. Wallpaper

Answer: C. An algorithm is an ordered set of instructions.

Theme 3: Using Devices Safely — 10 questions

STEM-G1-ICT-T03-Q01 | Using Devices Safely | Difficulty: Foundational

Why should you keep your password private?

A. To make the internet faster

B. To extend battery life

C. To protect your account and personal information

D. To make the screen brighter

Answer: C. A password provides access to an account and should be protected.

STEM-G1-ICT-T03-Q02 | Using Devices Safely | Difficulty: Foundational

A clear, ordered set of steps for solving a problem is called an ...

A. Resolution

B. Algorithm

C. Speaker

D. Wallpaper

Answer: B. An algorithm is an ordered set of instructions.

STEM-G1-ICT-T03-Q03 | Using Devices Safely | Difficulty: Foundational

In a spreadsheet, a cell is ...

A. A webpage

B. The intersection of a row and a column

C. An application icon

D. A type of cable

Answer: B. A cell is a box at the intersection of a row and a column.

STEM-G1-ICT-T03-Q04 | Using Devices Safely | Difficulty: Foundational

How should information found online be checked?

A. Count the emojis

B. Share it before reading

C. Choose the most dramatic headline

D. Compare sources and supporting evidence

Answer: D. Checking sources helps assess accuracy and credibility.

STEM-G1-ICT-T03-Q05 | Using Devices Safely | Difficulty: Foundational

Finding and fixing errors in a program is called ...

A. Formatting

B. Debugging

C. Streaming

D. Browsing

Answer: B. Debugging is the process of locating and correcting errors.

STEM-G1-ICT-T03-Q06 | Using Devices Safely | Difficulty: Foundational

Which device is commonly used to enter text into a computer?

A. Keyboard

B. Projector

C. Monitor

D. Speaker

Answer: A. A keyboard is an input device used for typing.

STEM-G1-ICT-T03-Q07 | Using Devices Safely | Difficulty: Foundational

Software designed to help a user perform a task is commonly called an ...

A. Power cable

B. Fan

C. Desk

D. Application

Answer: D. An application is software that helps users perform specific tasks.

STEM-G1-ICT-T03-Q08 | Using Devices Safely | Difficulty: Foundational

What should you do if an unknown person sends you a suspicious link?

A. Forward it to every contact

B. Download every attachment

C. Enter your password

D. Do not open it; check with a trusted adult or person

Answer: D. Suspicious links may lead to scams or harmful software.

STEM-G1-ICT-T03-Q09 | Using Devices Safely | Difficulty: Foundational

Why should you keep your password private?

A. To make the screen brighter

B. To extend battery life

C. To protect your account and personal information

D. To make the internet faster

Answer: C. A password provides access to an account and should be protected.

STEM-G1-ICT-T03-Q10 | Using Devices Safely | Difficulty: Foundational

A clear, ordered set of steps for solving a problem is called an ...

A. Speaker

B. Resolution

C. Algorithm

D. Wallpaper

Answer: C. An algorithm is an ordered set of instructions.

Theme 4: Simple Sequences of Instructions — 10 questions

STEM-G1-ICT-T04-Q01 | Simple Sequences of Instructions | Difficulty: Foundational

Why should you keep your password private?

A. To make the screen brighter

B. To protect your account and personal information

C. To make the internet faster

D. To extend battery life

Answer: B. A password provides access to an account and should be protected.

STEM-G1-ICT-T04-Q02 | Simple Sequences of Instructions | Difficulty: Foundational

A clear, ordered set of steps for solving a problem is called an ...

A. Speaker

B. Resolution

C. Algorithm

D. Wallpaper

Answer: C. An algorithm is an ordered set of instructions.

STEM-G1-ICT-T04-Q03 | Simple Sequences of Instructions | Difficulty: Foundational

In a spreadsheet, a cell is ...

A. The intersection of a row and a column

B. A type of cable

C. A webpage

D. An application icon

Answer: A. A cell is a box at the intersection of a row and a column.

STEM-G1-ICT-T04-Q04 | Simple Sequences of Instructions | Difficulty: Foundational

How should information found online be checked?

A. Compare sources and supporting evidence

B. Share it before reading

C. Count the emojis

D. Choose the most dramatic headline

Answer: A. Checking sources helps assess accuracy and credibility.

STEM-G1-ICT-T04-Q05 | Simple Sequences of Instructions | Difficulty: Foundational

Finding and fixing errors in a program is called ...

A. Browsing

B. Debugging

C. Formatting

D. Streaming

Answer: B. Debugging is the process of locating and correcting errors.

STEM-G1-ICT-T04-Q06 | Simple Sequences of Instructions | Difficulty: Foundational

Which device is commonly used to enter text into a computer?

A. Projector

B. Monitor

C. Speaker

D. Keyboard

Answer: D. A keyboard is an input device used for typing.

STEM-G1-ICT-T04-Q07 | Simple Sequences of Instructions | Difficulty: Foundational

Software designed to help a user perform a task is commonly called an ...

A. Fan

B. Application

C. Power cable

D. Desk

Answer: B. An application is software that helps users perform specific tasks.

STEM-G1-ICT-T04-Q08 | Simple Sequences of Instructions | Difficulty: Foundational

What should you do if an unknown person sends you a suspicious link?

A. Forward it to every contact

B. Download every attachment

C. Do not open it; check with a trusted adult or person

D. Enter your password

Answer: C. Suspicious links may lead to scams or harmful software.

STEM-G1-ICT-T04-Q09 | Simple Sequences of Instructions | Difficulty: Foundational

Why should you keep your password private?

A. To make the screen brighter

B. To make the internet faster

C. To extend battery life

D. To protect your account and personal information

Answer: D. A password provides access to an account and should be protected.

STEM-G1-ICT-T04-Q10 | Simple Sequences of Instructions | Difficulty: Foundational

A clear, ordered set of steps for solving a problem is called an ...

A. Resolution

B. Speaker

C. Algorithm

D. Wallpaper

Answer: C. An algorithm is an ordered set of instructions.

Elementary School Grade 2

160 questions • 4 subjects • 4 themes per subject • 10 questions per theme

Mathematics

Theme 1: Numbers up to 100 — 10 questions

STEM-G2-MAT-T01-Q01 | Numbers up to 100 | Difficulty: Foundational

You have 10 objects and give away 2. How many objects are left?

A. 10

B. 9

C. 8

D. 7

Answer: C. Subtract the amount given away: 10 – 2 = 8.

STEM-G2-MAT-T01-Q02 | Numbers up to 100 | Difficulty: Foundational

There are 4 groups with 4 objects in each group. How many objects are there in total?

A. 17

B. 15

C. 16

D. 18

Answer: C. Multiply the number of groups by the objects in each group: 4 × 4 = 16.

STEM-G2-MAT-T01-Q03 | Numbers up to 100 | Difficulty: Foundational

30 objects are shared equally among 3 groups. How many objects are in each group?

A. 11

B. 9

C. 12

D. 10

Answer: D. Divide equally: 30 ÷ 3 = 10.

STEM-G2-MAT-T01-Q04 | Numbers up to 100 | Difficulty: Foundational

There are 18 objects. Then 2 more objects are added. How many objects are there altogether?

A. 21

B. 20

C. 22

D. 19

Answer: B. Add the two groups: 18 + 2 = 20.

STEM-G2-MAT-T01-Q05 | Numbers up to 100 | Difficulty: Foundational

You have 7 objects and give away 7. How many objects are left?

A. 1

B. 5

C. 2

D. 0

Answer: D. Subtract the amount given away: 7 – 7 = 0.

STEM-G2-MAT-T01-Q06 | Numbers up to 100 | Difficulty: Foundational

There are 3 groups with 5 objects in each group. How many objects are there in total?

A. 16

B. 14

C. 15

D. 17

Answer: C. Multiply the number of groups by the objects in each group: 3 × 5 = 15.

STEM-G2-MAT-T01-Q07 | Numbers up to 100 | Difficulty: Foundational

32 objects are shared equally among 4 groups. How many objects are in each group?

A. 8

B. 9

C. 7

D. 10

Answer: A. Divide equally: 32 ÷ 4 = 8.

STEM-G2-MAT-T01-Q08 | Numbers up to 100 | Difficulty: Foundational

There are 2 objects. Then 4 more objects are added. How many objects are there altogether?

A. 5

B. 6

C. 7

D. 8

Answer: B. Add the two groups: 2 + 4 = 6.

STEM-G2-MAT-T01-Q09 | Numbers up to 100 | Difficulty: Foundational

You have 15 objects and give away 4. How many objects are left?

A. 12

B. 10

C. 11

D. 13

Answer: C. Subtract the amount given away: 15 – 4 = 11.

STEM-G2-MAT-T01-Q10 | Numbers up to 100 | Difficulty: Foundational

There are 3 groups with 4 objects in each group. How many objects are there in total?

A. 14

B. 12

C. 13

D. 11

Answer: B. Multiply the number of groups by the objects in each group: 3 × 4 = 12.

Theme 2: Two-Digit Addition and Subtraction — 10 questions

STEM-G2-MAT-T02-Q01 | Two-Digit Addition and Subtraction | Difficulty: Foundational

You have 11 objects and give away 2. How many objects are left?

A. 8

B. 9

C. 10

D. 11

Answer: B. Subtract the amount given away: 11 – 2 = 9.

STEM-G2-MAT-T02-Q02 | Two-Digit Addition and Subtraction | Difficulty: Foundational

There are 2 groups with 3 objects in each group. How many objects are there in total?

A. 5

B. 8

C. 7

D. 6

Answer: D. Multiply the number of groups by the objects in each group: 2 × 3 = 6.

STEM-G2-MAT-T02-Q03 | Two-Digit Addition and Subtraction | Difficulty: Foundational

25 objects are shared equally among 5 groups. How many objects are in each group?

A. 5

B. 6

C. 7

D. 4

Answer: A. Divide equally: 25 ÷ 5 = 5.

STEM-G2-MAT-T02-Q04 | Two-Digit Addition and Subtraction | Difficulty: Foundational

There are 12 objects. Then 2 more objects are added. How many objects are there altogether?

A. 14

B. 16

C. 13

D. 15

Answer: A. Add the two groups: 12 + 2 = 14.

STEM-G2-MAT-T02-Q05 | Two-Digit Addition and Subtraction | Difficulty: Foundational

You have 16 objects and give away 7. How many objects are left?

A. 11

B. 10

C. 9

D. 8

Answer: C. Subtract the amount given away: 16 – 7 = 9.

STEM-G2-MAT-T02-Q06 | Two-Digit Addition and Subtraction | Difficulty: Foundational

There are 5 groups with 2 objects in each group. How many objects are there in total?

A. 9

B. 10

C. 12

D. 11

Answer: B. Multiply the number of groups by the objects in each group: 5 × 2 = 10.

STEM-G2-MAT-T02-Q07 | Two-Digit Addition and Subtraction | Difficulty: Foundational

16 objects are shared equally among 2 groups. How many objects are in each group?

A. 9

B. 10

C. 8

D. 7

Answer: C. Divide equally: 16 ÷ 2 = 8.

STEM-G2-MAT-T02-Q08 | Two-Digit Addition and Subtraction | Difficulty: Foundational

There are 4 objects. Then 7 more objects are added. How many objects are there altogether?

A. 11

B. 10

C. 12

D. 13

Answer: A. Add the two groups: 4 + 7 = 11.

STEM-G2-MAT-T02-Q09 | Two-Digit Addition and Subtraction | Difficulty: Foundational

You have 15 objects and give away 1. How many objects are left?

A. 15

B. 16

C. 14

D. 13

Answer: C. Subtract the amount given away: 15 – 1 = 14.

STEM-G2-MAT-T02-Q10 | Two-Digit Addition and Subtraction | Difficulty: Foundational

There are 2 groups with 5 objects in each group. How many objects are there in total?

A. 11

B. 10

C. 12

D. 9

Answer: B. Multiply the number of groups by the objects in each group: 2 × 5 = 10.

Theme 3: Multiplication as Repeated Addition — 10 questions

STEM-G2-MAT-T03-Q01 | Multiplication as Repeated Addition | Difficulty: Foundational

You have 12 objects and give away 7. How many objects are left?

A. 4

B. 5

C. 7

D. 6

Answer: B. Subtract the amount given away: 12 – 7 = 5.

STEM-G2-MAT-T03-Q02 | Multiplication as Repeated Addition | Difficulty: Foundational

There are 2 groups with 2 objects in each group. How many objects are there in total?

A. 3

B. 6

C. 4

D. 5

Answer: C. Multiply the number of groups by the objects in each group: 2 × 2 = 4.

STEM-G2-MAT-T03-Q03 | Multiplication as Repeated Addition | Difficulty: Foundational

18 objects are shared equally among 2 groups. How many objects are in each group?

A. 8

B. 9

C. 11

D. 10

Answer: B. Divide equally: 18 ÷ 2 = 9.

STEM-G2-MAT-T03-Q04 | Multiplication as Repeated Addition | Difficulty: Foundational

There are 3 objects. Then 6 more objects are added. How many objects are there altogether?

A. 11

B. 10

C. 8

D. 9

Answer: D. Add the two groups: 3 + 6 = 9.

STEM-G2-MAT-T03-Q05 | Multiplication as Repeated Addition | Difficulty: Foundational

You have 10 objects and give away 6. How many objects are left?

A. 4

B. 3

C. 5

D. 6

Answer: A. Subtract the amount given away: 10 – 6 = 4.

STEM-G2-MAT-T03-Q06 | Multiplication as Repeated Addition | Difficulty: Foundational

There are 2 groups with 2 objects in each group. How many objects are there in total?

A. 6

B. 5

C. 4

D. 3

Answer: C. Multiply the number of groups by the objects in each group: 2 × 2 = 4.

STEM-G2-MAT-T03-Q07 | Multiplication as Repeated Addition | Difficulty: Foundational

16 objects are shared equally among 2 groups. How many objects are in each group?

A. 7

B. 9

C. 10

D. 8

Answer: D. Divide equally: 16 ÷ 2 = 8.

STEM-G2-MAT-T03-Q08 | Multiplication as Repeated Addition | Difficulty: Foundational

There are 17 objects. Then 4 more objects are added. How many objects are there altogether?

A. 20

B. 23

C. 22

D. 21

Answer: D. Add the two groups: 17 + 4 = 21.

STEM-G2-MAT-T03-Q09 | Multiplication as Repeated Addition | Difficulty: Foundational

You have 8 objects and give away 5. How many objects are left?

A. 4

B. 5

C. 2

D. 3

Answer: D. Subtract the amount given away: 8 – 5 = 3.

STEM-G2-MAT-T03-Q10 | Multiplication as Repeated Addition | Difficulty: Foundational

There are 3 groups with 2 objects in each group. How many objects are there in total?

A. 6

B. 8

C. 7

D. 5

Answer: A. Multiply the number of groups by the objects in each group: 3 × 2 = 6.

Theme 4: Measuring Length and Time — 10 questions

STEM-G2-MAT-T04-Q01 | Measuring Length and Time | Difficulty: Foundational

You have 15 objects and give away 8. How many objects are left?

A. 8

B. 7

C. 9

D. 6

Answer: B. Subtract the amount given away: 15 – 8 = 7.

STEM-G2-MAT-T04-Q02 | Measuring Length and Time | Difficulty: Foundational

There are 5 groups with 3 objects in each group. How many objects are there in total?

A. 15

B. 14

C. 16

D. 17

Answer: A. Multiply the number of groups by the objects in each group: 5 × 3 = 15.

STEM-G2-MAT-T04-Q03 | Measuring Length and Time | Difficulty: Foundational

16 objects are shared equally among 4 groups. How many objects are in each group?

A. 3

B. 4

C. 6

D. 5

Answer: B. Divide equally: 16 ÷ 4 = 4.

STEM-G2-MAT-T04-Q04 | Measuring Length and Time | Difficulty: Foundational

There are 17 objects. Then 3 more objects are added. How many objects are there altogether?

A. 20

B. 21

C. 19

D. 22

Answer: A. Add the two groups: 17 + 3 = 20.

STEM-G2-MAT-T04-Q05 | Measuring Length and Time | Difficulty: Foundational

You have 11 objects and give away 7. How many objects are left?

A. 3

B. 5

C. 6

D. 4

Answer: D. Subtract the amount given away: 11 – 7 = 4.

STEM-G2-MAT-T04-Q06 | Measuring Length and Time | Difficulty: Foundational

There are 3 groups with 3 objects in each group. How many objects are there in total?

A. 10

B. 11

C. 8

D. 9

Answer: D. Multiply the number of groups by the objects in each group: 3 × 3 = 9.

STEM-G2-MAT-T04-Q07 | Measuring Length and Time | Difficulty: Foundational

15 objects are shared equally among 3 groups. How many objects are in each group?

A. 5

B. 7

C. 4

D. 6

Answer: A. Divide equally: 15 ÷ 3 = 5.

STEM-G2-MAT-T04-Q08 | Measuring Length and Time | Difficulty: Foundational

There are 10 objects. Then 2 more objects are added. How many objects are there altogether?

A. 13

B. 14

C. 12

D. 11

Answer: C. Add the two groups: 10 + 2 = 12.

STEM-G2-MAT-T04-Q09 | Measuring Length and Time | Difficulty: Foundational

You have 15 objects and give away 2. How many objects are left?

A. 12

B. 13

C. 14

D. 15

Answer: B. Subtract the amount given away: 15 – 2 = 13.

STEM-G2-MAT-T04-Q10 | Measuring Length and Time | Difficulty: Foundational

There are 2 groups with 3 objects in each group. How many objects are there in total?

A. 8

B. 6

C. 7

D. 5

Answer: B. Multiply the number of groups by the objects in each group: 2 × 3 = 6.

Science

Theme 1: Needs of Living Things — 10 questions

STEM-G2-SCI-T01-Q01 | Needs of Living Things | Difficulty: Foundational

Green plants use light energy to carry out ...

A. Metal evaporation

B. Digestion

C. Photosynthesis

D. Freezing

Answer: C. Photosynthesis uses light energy to make sugars.

STEM-G2-SCI-T01-Q02 | Needs of Living Things | Difficulty: Foundational

Green plants use light energy to carry out ...

A. Digestion

B. Photosynthesis

C. Freezing

D. Metal evaporation

Answer: B. Photosynthesis uses light energy to make sugars.

STEM-G2-SCI-T01-Q03 | Needs of Living Things | Difficulty: Foundational

Green plants use light energy to carry out ...

A. Digestion

B. Metal evaporation

C. Freezing

D. Photosynthesis

Answer: D. Photosynthesis uses light energy to make sugars.

STEM-G2-SCI-T01-Q04 | Needs of Living Things | Difficulty: Foundational

Green plants use light energy to carry out ...

A. Freezing

B. Digestion

C. Metal evaporation

D. Photosynthesis

Answer: D. Photosynthesis uses light energy to make sugars.

STEM-G2-SCI-T01-Q05 | Needs of Living Things | Difficulty: Foundational

Green plants use light energy to carry out ...

A. Metal evaporation

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D. Photosynthesis

Answer: D. Photosynthesis uses light energy to make sugars.

STEM-G2-SCI-T01-Q06 | Needs of Living Things | Difficulty: Foundational

Green plants use light energy to carry out ...

A. Metal evaporation

B. Digestion

C. Freezing

D. Photosynthesis

Answer: D. Photosynthesis uses light energy to make sugars.

STEM-G2-SCI-T01-Q07 | Needs of Living Things | Difficulty: Foundational

Green plants use light energy to carry out ...

A. Freezing

B. Digestion

C. Metal evaporation

D. Photosynthesis

Answer: D. Photosynthesis uses light energy to make sugars.

STEM-G2-SCI-T01-Q08 | Needs of Living Things | Difficulty: Foundational

Green plants use light energy to carry out ...

A. Freezing

B. Digestion

C. Metal evaporation

D. Photosynthesis

Answer: D. Photosynthesis uses light energy to make sugars.

STEM-G2-SCI-T01-Q09 | Needs of Living Things | Difficulty: Foundational

Green plants use light energy to carry out ...

A. Freezing

B. Photosynthesis

C. Metal evaporation

D. Digestion

Answer: B. Photosynthesis uses light energy to make sugars.

STEM-G2-SCI-T01-Q10 | Needs of Living Things | Difficulty: Foundational

Green plants use light energy to carry out ...

A. Digestion

B. Photosynthesis

C. Freezing

D. Metal evaporation

Answer: B. Photosynthesis uses light energy to make sugars.

Theme 2: Animal Life Cycles — 10 questions

STEM-G2-SCI-T02-Q01 | Animal Life Cycles | Difficulty: Foundational

In a butterfly's life cycle, which stage comes after the caterpillar?

A. Chick

B. Pupa (chrysalis)

C. Egg

D. Tadpole

Answer: B. The caterpillar changes into a pupa, often called a chrysalis.

STEM-G2-SCI-T02-Q02 | Animal Life Cycles | Difficulty: Foundational

In a butterfly's life cycle, which stage comes after the caterpillar?

A. Egg

B. Tadpole

C. Chick

D. Pupa (chrysalis)

Answer: D. The caterpillar changes into a pupa, often called a chrysalis.

STEM-G2-SCI-T02-Q03 | Animal Life Cycles | Difficulty: Foundational

In a butterfly's life cycle, which stage comes after the caterpillar?

A. Egg

B. Tadpole

C. Pupa (chrysalis)

D. Chick

Answer: C. The caterpillar changes into a pupa, often called a chrysalis.

STEM-G2-SCI-T02-Q04 | Animal Life Cycles | Difficulty: Foundational

In a butterfly's life cycle, which stage comes after the caterpillar?

A. Pupa (chrysalis)

B. Tadpole

C. Chick

D. Egg

Answer: A. The caterpillar changes into a pupa, often called a chrysalis.

STEM-G2-SCI-T02-Q05 | Animal Life Cycles | Difficulty: Foundational

In a butterfly's life cycle, which stage comes after the caterpillar?

A. Tadpole

B. Chick

C. Egg

D. Pupa (chrysalis)

Answer: D. The caterpillar changes into a pupa, often called a chrysalis.

STEM-G2-SCI-T02-Q06 | Animal Life Cycles | Difficulty: Foundational

In a butterfly's life cycle, which stage comes after the caterpillar?

A. Chick

B. Egg

C. Pupa (chrysalis)

D. Tadpole

Answer: C. The caterpillar changes into a pupa, often called a chrysalis.

STEM-G2-SCI-T02-Q07 | Animal Life Cycles | Difficulty: Foundational

In a butterfly's life cycle, which stage comes after the caterpillar?

A. Tadpole

B. Egg

C. Pupa (chrysalis)

D. Chick

Answer: C. The caterpillar changes into a pupa, often called a chrysalis.

STEM-G2-SCI-T02-Q08 | Animal Life Cycles | Difficulty: Foundational

In a butterfly's life cycle, which stage comes after the caterpillar?

A. Chick

B. Egg

C. Tadpole

D. Pupa (chrysalis)

Answer: D. The caterpillar changes into a pupa, often called a chrysalis.

STEM-G2-SCI-T02-Q09 | Animal Life Cycles | Difficulty: Foundational

In a butterfly's life cycle, which stage comes after the caterpillar?

A. Chick

B. Egg

C. Pupa (chrysalis)

D. Tadpole

Answer: C. The caterpillar changes into a pupa, often called a chrysalis.

STEM-G2-SCI-T02-Q10 | Animal Life Cycles | Difficulty: Foundational

In a butterfly's life cycle, which stage comes after the caterpillar?

A. Chick

B. Tadpole

C. Egg

D. Pupa (chrysalis)

Answer: D. The caterpillar changes into a pupa, often called a chrysalis.

Theme 3: States and Changes of Matter — 10 questions

STEM-G2-SCI-T03-Q01 | States and Changes of Matter | Difficulty: Foundational

The change from liquid water to water vapor is called ...

A. Freezing

B. Evaporation

C. Melting

D. Condensation

Answer: B. Evaporation changes a liquid into a gas.

STEM-G2-SCI-T03-Q02 | States and Changes of Matter | Difficulty: Foundational

The change from liquid water to water vapor is called ...

A. Evaporation

B. Melting

C. Condensation

D. Freezing

Answer: A. Evaporation changes a liquid into a gas.

STEM-G2-SCI-T03-Q03 | States and Changes of Matter | Difficulty: Foundational

The change from liquid water to water vapor is called ...

A. Freezing

B. Condensation

C. Melting

D. Evaporation

Answer: D. Evaporation changes a liquid into a gas.

STEM-G2-SCI-T03-Q04 | States and Changes of Matter | Difficulty: Foundational

The change from liquid water to water vapor is called ...

A. Melting

B. Evaporation

C. Freezing

D. Condensation

Answer: B. Evaporation changes a liquid into a gas.

STEM-G2-SCI-T03-Q05 | States and Changes of Matter | Difficulty: Foundational

The change from liquid water to water vapor is called ...

A. Evaporation

B. Melting

C. Freezing

D. Condensation

Answer: A. Evaporation changes a liquid into a gas.

STEM-G2-SCI-T03-Q06 | States and Changes of Matter | Difficulty: Foundational

The change from liquid water to water vapor is called ...

A. Condensation

B. Melting

C. Evaporation

D. Freezing

Answer: C. Evaporation changes a liquid into a gas.

STEM-G2-SCI-T03-Q07 | States and Changes of Matter | Difficulty: Foundational

The change from liquid water to water vapor is called ...

A. Condensation

B. Melting

C. Freezing

D. Evaporation

Answer: D. Evaporation changes a liquid into a gas.

STEM-G2-SCI-T03-Q08 | States and Changes of Matter | Difficulty: Foundational

The change from liquid water to water vapor is called ...

A. Evaporation

B. Condensation

C. Melting

D. Freezing

Answer: A. Evaporation changes a liquid into a gas.

STEM-G2-SCI-T03-Q09 | States and Changes of Matter | Difficulty: Foundational

The change from liquid water to water vapor is called ...

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B. Melting

C. Freezing

D. Evaporation

Answer: D. Evaporation changes a liquid into a gas.

STEM-G2-SCI-T03-Q10 | States and Changes of Matter | Difficulty: Foundational

The change from liquid water to water vapor is called ...

A. Condensation

B. Freezing

C. Melting

D. Evaporation

Answer: D. Evaporation changes a liquid into a gas.

Theme 4: Earth, the Sun, and Weather — 10 questions

STEM-G2-SCI-T04-Q01 | Earth, the Sun, and Weather | Difficulty: Foundational

What is Earth's main natural source of light and heat?

A. A shooting star

B. The Sun

C. The Moon

D. Mars

Answer: B. The Sun supplies most of the energy driving Earth's systems.

STEM-G2-SCI-T04-Q02 | Earth, the Sun, and Weather | Difficulty: Foundational

What is Earth's main natural source of light and heat?

A. The Moon

B. A shooting star

C. Mars

D. The Sun

Answer: D. The Sun supplies most of the energy driving Earth's systems.

STEM-G2-SCI-T04-Q03 | Earth, the Sun, and Weather | Difficulty: Foundational

What is Earth's main natural source of light and heat?

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STEM-G2-SCI-T04-Q04 | Earth, the Sun, and Weather | Difficulty: Foundational

What is Earth's main natural source of light and heat?

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STEM-G2-SCI-T04-Q05 | Earth, the Sun, and Weather | Difficulty: Foundational

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D. The Moon

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STEM-G2-SCI-T04-Q06 | Earth, the Sun, and Weather | Difficulty: Foundational

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STEM-G2-SCI-T04-Q07 | Earth, the Sun, and Weather | Difficulty: Foundational

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STEM-G2-SCI-T04-Q09 | Earth, the Sun, and Weather | Difficulty: Foundational

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D. Mars

Answer: A. The Sun supplies most of the energy driving Earth's systems.

STEM-G2-SCI-T04-Q10 | Earth, the Sun, and Weather | Difficulty: Foundational

What is Earth's main natural source of light and heat?

A. Mars

B. The Moon

C. A shooting star

D. The Sun

Answer: D. The Sun supplies most of the energy driving Earth's systems.

Engineering

Theme 1: Choosing Suitable Materials — 10 questions

STEM-G2-ENG-T01-Q01 | Choosing Suitable Materials | Difficulty: Foundational

For a fair test, a team should ...

A. Avoid recording results

B. Change every factor at once

C. Choose only the preferred result

D. Change one factor while keeping other factors the same

Answer: D. Controlling variables helps identify what caused a result.

STEM-G2-ENG-T01-Q02 | Choosing Suitable Materials | Difficulty: Foundational

When selecting a material for a load-bearing part, what should be considered?

A. Its strength and suitability for the task

B. The packaging

C. Only its favorite color

D. The designer's name

Answer: A. Material choice depends on the load and operating conditions.

STEM-G2-ENG-T01-Q03 | Choosing Suitable Materials | Difficulty: Foundational

Why should joints in a model be inspected?

A. Joints are only decoration

B. Joints transfer forces between parts

C. Joints remove gravity

D. Joints make mass zero

Answer: B. Weak joints can become failure points in a structure.

STEM-G2-ENG-T01-Q04 | Choosing Suitable Materials | Difficulty: Foundational

After a test reveals a weakness, what should engineers do next?

A. Improve the design and test it again

B. Ignore safety

C. Hide the result

D. Delete the notes

Answer: A. Engineering is iterative and uses test results to guide improvements.

STEM-G2-ENG-T01-Q05 | Choosing Suitable Materials | Difficulty: Foundational

Which measurement is useful when comparing two bridge designs?

A. The model's name

B. The maximum load each bridge can support

C. The number of stickers

D. The bridge color

Answer: B. Load capacity is a relevant measure of bridge performance.

STEM-G2-ENG-T01-Q06 | Choosing Suitable Materials | Difficulty: Foundational

Why should an engineering prototype be tested before use?

A. To make it look more expensive

B. To avoid making improvements

C. To make it larger every time

D. To find out whether the design works and is safe

Answer: D. Testing reveals weaknesses and helps improve a design.

STEM-G2-ENG-T01-Q07 | Choosing Suitable Materials | Difficulty: Foundational

If a model structure tips over easily, what is a sensible first improvement?

A. Remove all supports

B. Widen the base or strengthen the joints

C. Add weight to the top

D. Remove the base

Answer: B. A wider base and strong joints can improve stability.

STEM-G2-ENG-T01-Q08 | Choosing Suitable Materials | Difficulty: Foundational

What is the purpose of making a sketch before building an object?

A. To replace all testing

B. To make the object build itself

C. To avoid measuring

D. To plan its shape, dimensions, and parts

Answer: D. A sketch communicates a design before construction.

STEM-G2-ENG-T01-Q09 | Choosing Suitable Materials | Difficulty: Foundational

For a fair test, a team should ...

A. Change one factor while keeping other factors the same

B. Change every factor at once

C. Avoid recording results

D. Choose only the preferred result

Answer: A. Controlling variables helps identify what caused a result.

STEM-G2-ENG-T01-Q10 | Choosing Suitable Materials | Difficulty: Foundational

When selecting a material for a load-bearing part, what should be considered?

A. Its strength and suitability for the task

B. Only its favorite color

C. The packaging

D. The designer's name

Answer: A. Material choice depends on the load and operating conditions.

Theme 2: Mini Bridges and Buildings — 10 questions

STEM-G2-ENG-T02-Q01 | Mini Bridges and Buildings | Difficulty: Foundational

For a fair test, a team should ...

A. Avoid recording results

B. Change one factor while keeping other factors the same

C. Change every factor at once

D. Choose only the preferred result

Answer: B. Controlling variables helps identify what caused a result.

STEM-G2-ENG-T02-Q02 | Mini Bridges and Buildings | Difficulty: Foundational

When selecting a material for a load-bearing part, what should be considered?

A. The designer's name

B. Only its favorite color

C. Its strength and suitability for the task

D. The packaging

Answer: C. Material choice depends on the load and operating conditions.

STEM-G2-ENG-T02-Q03 | Mini Bridges and Buildings | Difficulty: Foundational

Why should joints in a model be inspected?

A. Joints remove gravity

B. Joints transfer forces between parts

C. Joints are only decoration

D. Joints make mass zero

Answer: B. Weak joints can become failure points in a structure.

STEM-G2-ENG-T02-Q04 | Mini Bridges and Buildings | Difficulty: Foundational

After a test reveals a weakness, what should engineers do next?

A. Hide the result

B. Ignore safety

C. Delete the notes

D. Improve the design and test it again

Answer: D. Engineering is iterative and uses test results to guide improvements.

STEM-G2-ENG-T02-Q05 | Mini Bridges and Buildings | Difficulty: Foundational

Which measurement is useful when comparing two bridge designs?

A. The model's name

B. The bridge color

C. The number of stickers

D. The maximum load each bridge can support

Answer: D. Load capacity is a relevant measure of bridge performance.

STEM-G2-ENG-T02-Q06 | Mini Bridges and Buildings | Difficulty: Foundational

Why should an engineering prototype be tested before use?

A. To avoid making improvements

B. To find out whether the design works and is safe

C. To make it larger every time

D. To make it look more expensive

Answer: B. Testing reveals weaknesses and helps improve a design.

STEM-G2-ENG-T02-Q07 | Mini Bridges and Buildings | Difficulty: Foundational

If a model structure tips over easily, what is a sensible first improvement?

A. Widen the base or strengthen the joints

B. Add weight to the top

C. Remove the base

D. Remove all supports

Answer: A. A wider base and strong joints can improve stability.

STEM-G2-ENG-T02-Q08 | Mini Bridges and Buildings | Difficulty: Foundational

What is the purpose of making a sketch before building an object?

A. To make the object build itself

B. To plan its shape, dimensions, and parts

C. To replace all testing

D. To avoid measuring

Answer: B. A sketch communicates a design before construction.

STEM-G2-ENG-T02-Q09 | Mini Bridges and Buildings | Difficulty: Foundational

For a fair test, a team should ...

A. Choose only the preferred result

B. Change one factor while keeping other factors the same

C. Change every factor at once

D. Avoid recording results

Answer: B. Controlling variables helps identify what caused a result.

STEM-G2-ENG-T02-Q10 | Mini Bridges and Buildings | Difficulty: Foundational

When selecting a material for a load-bearing part, what should be considered?

A. Its strength and suitability for the task

B. Only its favorite color

C. The packaging

D. The designer's name

Answer: A. Material choice depends on the load and operating conditions.

Theme 3: Wheels, Axles, and Carts — 10 questions

STEM-G2-ENG-T03-Q01 | Wheels, Axles, and Carts | Difficulty: Foundational

For a fair test, a team should ...

A. Avoid recording results

B. Change every factor at once

C. Change one factor while keeping other factors the same

D. Choose only the preferred result

Answer: C. Controlling variables helps identify what caused a result.

STEM-G2-ENG-T03-Q02 | Wheels, Axles, and Carts | Difficulty: Foundational

When selecting a material for a load-bearing part, what should be considered?

A. The packaging

B. Only its favorite color

C. The designer's name

D. Its strength and suitability for the task

Answer: D. Material choice depends on the load and operating conditions.

STEM-G2-ENG-T03-Q03 | Wheels, Axles, and Carts | Difficulty: Foundational

Why should joints in a model be inspected?

A. Joints remove gravity

B. Joints make mass zero

C. Joints are only decoration

D. Joints transfer forces between parts

Answer: D. Weak joints can become failure points in a structure.

STEM-G2-ENG-T03-Q04 | Wheels, Axles, and Carts | Difficulty: Foundational

After a test reveals a weakness, what should engineers do next?

A. Delete the notes

B. Improve the design and test it again

C. Ignore safety

D. Hide the result

Answer: B. Engineering is iterative and uses test results to guide improvements.

STEM-G2-ENG-T03-Q05 | Wheels, Axles, and Carts | Difficulty: Foundational

Which measurement is useful when comparing two bridge designs?

A. The bridge color

B. The maximum load each bridge can support

C. The model's name

D. The number of stickers

Answer: B. Load capacity is a relevant measure of bridge performance.

STEM-G2-ENG-T03-Q06 | Wheels, Axles, and Carts | Difficulty: Foundational

Why should an engineering prototype be tested before use?

A. To avoid making improvements

B. To make it look more expensive

C. To make it larger every time

D. To find out whether the design works and is safe

Answer: D. Testing reveals weaknesses and helps improve a design.

STEM-G2-ENG-T03-Q07 | Wheels, Axles, and Carts | Difficulty: Foundational

If a model structure tips over easily, what is a sensible first improvement?

A. Add weight to the top

B. Widen the base or strengthen the joints

C. Remove the base

D. Remove all supports

Answer: B. A wider base and strong joints can improve stability.

STEM-G2-ENG-T03-Q08 | Wheels, Axles, and Carts | Difficulty: Foundational

What is the purpose of making a sketch before building an object?

A. To make the object build itself

B. To replace all testing

C. To avoid measuring

D. To plan its shape, dimensions, and parts

Answer: D. A sketch communicates a design before construction.

STEM-G2-ENG-T03-Q09 | Wheels, Axles, and Carts | Difficulty: Foundational

For a fair test, a team should ...

A. Avoid recording results

B. Choose only the preferred result

C. Change every factor at once

D. Change one factor while keeping other factors the same

Answer: D. Controlling variables helps identify what caused a result.

STEM-G2-ENG-T03-Q10 | Wheels, Axles, and Carts | Difficulty: Foundational

When selecting a material for a load-bearing part, what should be considered?

A. The designer's name

B. Only its favorite color

C. The packaging

D. Its strength and suitability for the task

Answer: D. Material choice depends on the load and operating conditions.

Theme 4: Solving Simple Problems — 10 questions

STEM-G2-ENG-T04-Q01 | Solving Simple Problems | Difficulty: Foundational

For a fair test, a team should ...

A. Avoid recording results

B. Change every factor at once

C. Change one factor while keeping other factors the same

D. Choose only the preferred result

Answer: C. Controlling variables helps identify what caused a result.

STEM-G2-ENG-T04-Q02 | Solving Simple Problems | Difficulty: Foundational

When selecting a material for a load-bearing part, what should be considered?

A. Its strength and suitability for the task

B. The designer's name

C. The packaging

D. Only its favorite color

Answer: A. Material choice depends on the load and operating conditions.

STEM-G2-ENG-T04-Q03 | Solving Simple Problems | Difficulty: Foundational

Why should joints in a model be inspected?

A. Joints remove gravity

B. Joints are only decoration

C. Joints make mass zero

D. Joints transfer forces between parts

Answer: D. Weak joints can become failure points in a structure.

STEM-G2-ENG-T04-Q04 | Solving Simple Problems | Difficulty: Foundational

After a test reveals a weakness, what should engineers do next?

A. Delete the notes

B. Hide the result

C. Ignore safety

D. Improve the design and test it again

Answer: D. Engineering is iterative and uses test results to guide improvements.

STEM-G2-ENG-T04-Q05 | Solving Simple Problems | Difficulty: Foundational

Which measurement is useful when comparing two bridge designs?

A. The model's name

B. The bridge color

C. The number of stickers

D. The maximum load each bridge can support

Answer: D. Load capacity is a relevant measure of bridge performance.

STEM-G2-ENG-T04-Q06 | Solving Simple Problems | Difficulty: Foundational

Why should an engineering prototype be tested before use?

A. To make it look more expensive

B. To find out whether the design works and is safe

C. To avoid making improvements

D. To make it larger every time

Answer: B. Testing reveals weaknesses and helps improve a design.

STEM-G2-ENG-T04-Q07 | Solving Simple Problems | Difficulty: Foundational

If a model structure tips over easily, what is a sensible first improvement?

A. Remove all supports

B. Add weight to the top

C. Widen the base or strengthen the joints

D. Remove the base

Answer: C. A wider base and strong joints can improve stability.

STEM-G2-ENG-T04-Q08 | Solving Simple Problems | Difficulty: Foundational

What is the purpose of making a sketch before building an object?

A. To plan its shape, dimensions, and parts

B. To replace all testing

C. To make the object build itself

D. To avoid measuring

Answer: A. A sketch communicates a design before construction.

STEM-G2-ENG-T04-Q09 | Solving Simple Problems | Difficulty: Foundational

For a fair test, a team should ...

A. Choose only the preferred result

B. Change every factor at once

C. Change one factor while keeping other factors the same

D. Avoid recording results

Answer: C. Controlling variables helps identify what caused a result.

STEM-G2-ENG-T04-Q10 | Solving Simple Problems | Difficulty: Foundational

When selecting a material for a load-bearing part, what should be considered?

A. The packaging

B. Only its favorite color

C. The designer's name

D. Its strength and suitability for the task

Answer: D. Material choice depends on the load and operating conditions.

ICT (Technology)

Theme 1: Computer Parts and Their Functions — 10 questions

STEM-G2-ICT-T01-Q01 | Computer Parts and Their Functions | Difficulty: Foundational

In a spreadsheet, a cell is ...

A. An application icon

B. The intersection of a row and a column

C. A type of cable

D. A webpage

Answer: B. A cell is a box at the intersection of a row and a column.

STEM-G2-ICT-T01-Q02 | Computer Parts and Their Functions | Difficulty: Foundational

How should information found online be checked?

A. Count the emojis

B. Compare sources and supporting evidence

C. Share it before reading

D. Choose the most dramatic headline

Answer: B. Checking sources helps assess accuracy and credibility.

STEM-G2-ICT-T01-Q03 | Computer Parts and Their Functions | Difficulty: Foundational

Finding and fixing errors in a program is called ...

A. Streaming

B. Browsing

C. Formatting

D. Debugging

Answer: D. Debugging is the process of locating and correcting errors.

STEM-G2-ICT-T01-Q04 | Computer Parts and Their Functions | Difficulty: Foundational

Which device is commonly used to enter text into a computer?

A. Monitor

B. Projector

C. Speaker

D. Keyboard

Answer: D. A keyboard is an input device used for typing.

STEM-G2-ICT-T01-Q05 | Computer Parts and Their Functions | Difficulty: Foundational

Software designed to help a user perform a task is commonly called an ...

A. Power cable

B. Fan

C. Application

D. Desk

Answer: C. An application is software that helps users perform specific tasks.

STEM-G2-ICT-T01-Q06 | Computer Parts and Their Functions | Difficulty: Foundational

What should you do if an unknown person sends you a suspicious link?

A. Enter your password

B. Download every attachment

C. Forward it to every contact

D. Do not open it; check with a trusted adult or person

Answer: D. Suspicious links may lead to scams or harmful software.

STEM-G2-ICT-T01-Q07 | Computer Parts and Their Functions | Difficulty: Foundational

Why should you keep your password private?

A. To protect your account and personal information

B. To make the screen brighter

C. To make the internet faster

D. To extend battery life

Answer: A. A password provides access to an account and should be protected.

STEM-G2-ICT-T01-Q08 | Computer Parts and Their Functions | Difficulty: Foundational

A clear, ordered set of steps for solving a problem is called an ...

A. Algorithm

B. Wallpaper

C. Speaker

D. Resolution

Answer: A. An algorithm is an ordered set of instructions.

STEM-G2-ICT-T01-Q09 | Computer Parts and Their Functions | Difficulty: Foundational

In a spreadsheet, a cell is ...

A. A type of cable

B. The intersection of a row and a column

C. An application icon

D. A webpage

Answer: B. A cell is a box at the intersection of a row and a column.

STEM-G2-ICT-T01-Q10 | Computer Parts and Their Functions | Difficulty: Foundational

How should information found online be checked?

A. Count the emojis

B. Choose the most dramatic headline

C. Share it before reading

D. Compare sources and supporting evidence

Answer: D. Checking sources helps assess accuracy and credibility.

Theme 2: Basic File Management — 10 questions

STEM-G2-ICT-T02-Q01 | Basic File Management | Difficulty: Foundational

In a spreadsheet, a cell is ...

A. An application icon

B. A type of cable

C. A webpage

D. The intersection of a row and a column

Answer: D. A cell is a box at the intersection of a row and a column.

STEM-G2-ICT-T02-Q02 | Basic File Management | Difficulty: Foundational

How should information found online be checked?

A. Choose the most dramatic headline

B. Share it before reading

C. Compare sources and supporting evidence

D. Count the emojis

Answer: C. Checking sources helps assess accuracy and credibility.

STEM-G2-ICT-T02-Q03 | Basic File Management | Difficulty: Foundational

Finding and fixing errors in a program is called ...

A. Formatting

B. Debugging

C. Browsing

D. Streaming

Answer: B. Debugging is the process of locating and correcting errors.

STEM-G2-ICT-T02-Q04 | Basic File Management | Difficulty: Foundational

Which device is commonly used to enter text into a computer?

A. Projector

B. Keyboard

C. Speaker

D. Monitor

Answer: B. A keyboard is an input device used for typing.

STEM-G2-ICT-T02-Q05 | Basic File Management | Difficulty: Foundational

Software designed to help a user perform a task is commonly called an ...

A. Desk

B. Application

C. Fan

D. Power cable

Answer: B. An application is software that helps users perform specific tasks.

STEM-G2-ICT-T02-Q06 | Basic File Management | Difficulty: Foundational

What should you do if an unknown person sends you a suspicious link?

A. Do not open it; check with a trusted adult or person

B. Download every attachment

C. Forward it to every contact

D. Enter your password

Answer: A. Suspicious links may lead to scams or harmful software.

STEM-G2-ICT-T02-Q07 | Basic File Management | Difficulty: Foundational

Why should you keep your password private?

A. To make the screen brighter

B. To extend battery life

C. To make the internet faster

D. To protect your account and personal information

Answer: D. A password provides access to an account and should be protected.

STEM-G2-ICT-T02-Q08 | Basic File Management | Difficulty: Foundational

A clear, ordered set of steps for solving a problem is called an ...

A. Speaker

B. Algorithm

C. Resolution

D. Wallpaper

Answer: B. An algorithm is an ordered set of instructions.

STEM-G2-ICT-T02-Q09 | Basic File Management | Difficulty: Foundational

In a spreadsheet, a cell is ...

A. A webpage

B. The intersection of a row and a column

C. An application icon

D. A type of cable

Answer: B. A cell is a box at the intersection of a row and a column.

STEM-G2-ICT-T02-Q10 | Basic File Management | Difficulty: Foundational

How should information found online be checked?

A. Count the emojis

B. Choose the most dramatic headline

C. Compare sources and supporting evidence

D. Share it before reading

Answer: C. Checking sources helps assess accuracy and credibility.

Theme 3: Digital Ethics and Safety — 10 questions

STEM-G2-ICT-T03-Q01 | Digital Ethics and Safety | Difficulty: Foundational

In a spreadsheet, a cell is ...

A. A webpage

B. A type of cable

C. The intersection of a row and a column

D. An application icon

Answer: C. A cell is a box at the intersection of a row and a column.

STEM-G2-ICT-T03-Q02 | Digital Ethics and Safety | Difficulty: Foundational

How should information found online be checked?

A. Count the emojis

B. Share it before reading

C. Compare sources and supporting evidence

D. Choose the most dramatic headline

Answer: C. Checking sources helps assess accuracy and credibility.

STEM-G2-ICT-T03-Q03 | Digital Ethics and Safety | Difficulty: Foundational

Finding and fixing errors in a program is called ...

A. Debugging

B. Browsing

C. Formatting

D. Streaming

Answer: A. Debugging is the process of locating and correcting errors.

STEM-G2-ICT-T03-Q04 | Digital Ethics and Safety | Difficulty: Foundational

Which device is commonly used to enter text into a computer?

A. Monitor

B. Speaker

C. Keyboard

D. Projector

Answer: C. A keyboard is an input device used for typing.

STEM-G2-ICT-T03-Q05 | Digital Ethics and Safety | Difficulty: Foundational

Software designed to help a user perform a task is commonly called an ...

A. Desk

B. Fan

C. Application

D. Power cable

Answer: C. An application is software that helps users perform specific tasks.

STEM-G2-ICT-T03-Q06 | Digital Ethics and Safety | Difficulty: Foundational

What should you do if an unknown person sends you a suspicious link?

A. Do not open it; check with a trusted adult or person

B. Download every attachment

C. Forward it to every contact

D. Enter your password

Answer: A. Suspicious links may lead to scams or harmful software.

STEM-G2-ICT-T03-Q07 | Digital Ethics and Safety | Difficulty: Foundational

Why should you keep your password private?

A. To protect your account and personal information

B. To extend battery life

C. To make the internet faster

D. To make the screen brighter

Answer: A. A password provides access to an account and should be protected.

STEM-G2-ICT-T03-Q08 | Digital Ethics and Safety | Difficulty: Foundational

A clear, ordered set of steps for solving a problem is called an ...

A. Wallpaper

B. Resolution

C. Algorithm

D. Speaker

Answer: C. An algorithm is an ordered set of instructions.

STEM-G2-ICT-T03-Q09 | Digital Ethics and Safety | Difficulty: Foundational

In a spreadsheet, a cell is ...

A. The intersection of a row and a column

B. A webpage

C. An application icon

D. A type of cable

Answer: A. A cell is a box at the intersection of a row and a column.

STEM-G2-ICT-T03-Q10 | Digital Ethics and Safety | Difficulty: Foundational

How should information found online be checked?

A. Share it before reading

B. Choose the most dramatic headline

C. Count the emojis

D. Compare sources and supporting evidence

Answer: D. Checking sources helps assess accuracy and credibility.

Theme 4: Patterns and Logic — 10 questions

STEM-G2-ICT-T04-Q01 | Patterns and Logic | Difficulty: Foundational

In a spreadsheet, a cell is ...

A. A webpage

B. A type of cable

C. An application icon

D. The intersection of a row and a column

Answer: D. A cell is a box at the intersection of a row and a column.

STEM-G2-ICT-T04-Q02 | Patterns and Logic | Difficulty: Foundational

How should information found online be checked?

A. Share it before reading

B. Compare sources and supporting evidence

C. Choose the most dramatic headline

D. Count the emojis

Answer: B. Checking sources helps assess accuracy and credibility.

STEM-G2-ICT-T04-Q03 | Patterns and Logic | Difficulty: Foundational

Finding and fixing errors in a program is called ...

A. Browsing

B. Formatting

C. Debugging

D. Streaming

Answer: C. Debugging is the process of locating and correcting errors.

STEM-G2-ICT-T04-Q04 | Patterns and Logic | Difficulty: Foundational

Which device is commonly used to enter text into a computer?

A. Monitor

B. Projector

C. Keyboard

D. Speaker

Answer: C. A keyboard is an input device used for typing.

STEM-G2-ICT-T04-Q05 | Patterns and Logic | Difficulty: Foundational

Software designed to help a user perform a task is commonly called an ...

A. Fan

B. Desk

C. Application

D. Power cable

Answer: C. An application is software that helps users perform specific tasks.

STEM-G2-ICT-T04-Q06 | Patterns and Logic | Difficulty: Foundational

What should you do if an unknown person sends you a suspicious link?

A. Do not open it; check with a trusted adult or person

B. Download every attachment

C. Enter your password

D. Forward it to every contact

Answer: A. Suspicious links may lead to scams or harmful software.

STEM-G2-ICT-T04-Q07 | Patterns and Logic | Difficulty: Foundational

Why should you keep your password private?

A. To make the internet faster

B. To make the screen brighter

C. To protect your account and personal information

D. To extend battery life

Answer: C. A password provides access to an account and should be protected.

STEM-G2-ICT-T04-Q08 | Patterns and Logic | Difficulty: Foundational

A clear, ordered set of steps for solving a problem is called an ...

A. Resolution

B. Wallpaper

C. Speaker

D. Algorithm

Answer: D. An algorithm is an ordered set of instructions.

STEM-G2-ICT-T04-Q09 | Patterns and Logic | Difficulty: Foundational

In a spreadsheet, a cell is ...

A. An application icon

B. A webpage

C. The intersection of a row and a column

D. A type of cable

Answer: C. A cell is a box at the intersection of a row and a column.

STEM-G2-ICT-T04-Q10 | Patterns and Logic | Difficulty: Foundational

How should information found online be checked?

A. Choose the most dramatic headline

B. Count the emojis

C. Compare sources and supporting evidence

D. Share it before reading

Answer: C. Checking sources helps assess accuracy and credibility.

Elementary School Grade 3

160 questions • 4 subjects • 4 themes per subject • 10 questions per theme

Mathematics

Theme 1: Multiplication and Division — 10 questions

STEM-G3-MAT-T01-Q01 | Multiplication and Division | Difficulty: Foundational

What is 58 – 13?

A. 43

B. 47

C. 45

D. 50

Answer: C. 58 – 13 = 45.

STEM-G3-MAT-T01-Q02 | Multiplication and Division | Difficulty: Foundational

What is 10 × 5?

A. 52

B. 55

C. 50

D. 48

Answer: C. 10 × 5 = 50.

STEM-G3-MAT-T01-Q03 | Multiplication and Division | Difficulty: Foundational

What is 108 ÷ 9?

A. 12

B. 10

C. 17

D. 14

Answer: A. 108 ÷ 9 = 12.

STEM-G3-MAT-T01-Q04 | Multiplication and Division | Difficulty: Foundational

What is 21 + 16?

A. 35

B. 42

C. 37

D. 39

Answer: C. 21 + 16 = 37.

STEM-G3-MAT-T01-Q05 | Multiplication and Division | Difficulty: Foundational

What is 34 – 13?

A. 26

B. 23

C. 21

D. 19

Answer: C. 34 – 13 = 21.

STEM-G3-MAT-T01-Q06 | Multiplication and Division | Difficulty: Foundational

What is 11 × 9?

A. 99

B. 101

C. 104

D. 97

Answer: A. 11 × 9 = 99.

STEM-G3-MAT-T01-Q07 | Multiplication and Division | Difficulty: Foundational

What is 27 ÷ 9?

A. 1

B. 3

C. 5

D. 8

Answer: B. 27 ÷ 9 = 3.

STEM-G3-MAT-T01-Q08 | Multiplication and Division | Difficulty: Foundational

What is 57 + 6?

A. 68

B. 65

C. 61

D. 63

Answer: D. 57 + 6 = 63.

STEM-G3-MAT-T01-Q09 | Multiplication and Division | Difficulty: Foundational

What is 34 – 17?

A. 15

B. 22

C. 17

D. 19

Answer: C. 34 – 17 = 17.

STEM-G3-MAT-T01-Q10 | Multiplication and Division | Difficulty: Foundational

What is 10 × 3?

A. 30

B. 28

C. 32

D. 35

Answer: A. 10 × 3 = 30.

Theme 2: Simple Fractions — 10 questions

STEM-G3-MAT-T02-Q01 | Simple Fractions | Difficulty: Foundational

What is 103 – 20?

A. 88

B. 81

C. 85

D. 83

Answer: D. 103 – 20 = 83.

STEM-G3-MAT-T02-Q02 | Simple Fractions | Difficulty: Foundational

What is 7 × 3?

A. 19

B. 21

C. 26

D. 23

Answer: B. 7 × 3 = 21.

STEM-G3-MAT-T02-Q03 | Simple Fractions | Difficulty: Foundational

What is 64 ÷ 8?

A. 10

B. 13

C. 8

D. 6

Answer: C. 64 ÷ 8 = 8.

STEM-G3-MAT-T02-Q04 | Simple Fractions | Difficulty: Foundational

What is 20 + 19?

A. 41

B. 44

C. 39

D. 37

Answer: C. 20 + 19 = 39.

STEM-G3-MAT-T02-Q05 | Simple Fractions | Difficulty: Foundational

What is 78 – 16?

A. 67

B. 64

C. 62

D. 60

Answer: C. 78 – 16 = 62.

STEM-G3-MAT-T02-Q06 | Simple Fractions | Difficulty: Foundational

What is 4 × 4?

A. 14

B. 18

C. 16

D. 21

Answer: C. 4 × 4 = 16.

STEM-G3-MAT-T02-Q07 | Simple Fractions | Difficulty: Foundational

What is 24 ÷ 2?

A. 17

B. 14

C. 10

D. 12

Answer: D. 24 ÷ 2 = 12.

STEM-G3-MAT-T02-Q08 | Simple Fractions | Difficulty: Foundational

What is 51 + 3?

A. 54

B. 52

C. 59

D. 56

Answer: A. 51 + 3 = 54.

STEM-G3-MAT-T02-Q09 | Simple Fractions | Difficulty: Foundational

What is 77 – 3?

A. 74

B. 76

C. 79

D. 72

Answer: A. 77 – 3 = 74.

STEM-G3-MAT-T02-Q10 | Simple Fractions | Difficulty: Foundational

What is 7 × 3?

A. 21

B. 23

C. 26

D. 19

Answer: A. 7 × 3 = 21.

Theme 3: Perimeter of Plane Shapes — 10 questions

STEM-G3-MAT-T03-Q01 | Perimeter of Plane Shapes | Difficulty: Foundational

What is 64 – 9?

A. 60

B. 53

C. 55

D. 57

Answer: C. 64 – 9 = 55.

STEM-G3-MAT-T03-Q02 | Perimeter of Plane Shapes | Difficulty: Foundational

What is 6 × 5?

A. 32

B. 35

C. 30

D. 28

Answer: C. 6 × 5 = 30.

STEM-G3-MAT-T03-Q03 | Perimeter of Plane Shapes | Difficulty: Foundational

What is 12 ÷ 3?

A. 2

B. 6

C. 9

D. 4

Answer: D. 12 ÷ 3 = 4.

STEM-G3-MAT-T03-Q04 | Perimeter of Plane Shapes | Difficulty: Foundational

What is 51 + 4?

A. 53

B. 55

C. 57

D. 60

Answer: B. 51 + 4 = 55.

STEM-G3-MAT-T03-Q05 | Perimeter of Plane Shapes | Difficulty: Foundational

What is 98 – 17?

A. 86

B. 81

C. 83

D. 79

Answer: B. 98 – 17 = 81.

STEM-G3-MAT-T03-Q06 | Perimeter of Plane Shapes | Difficulty: Foundational

What is 11 × 6?

A. 68

B. 71

C. 66

D. 64

Answer: C. 11 × 6 = 66.

STEM-G3-MAT-T03-Q07 | Perimeter of Plane Shapes | Difficulty: Foundational

What is 80 ÷ 8?

A. 10

B. 8

C. 12

D. 15

Answer: A. 80 ÷ 8 = 10.

STEM-G3-MAT-T03-Q08 | Perimeter of Plane Shapes | Difficulty: Foundational

What is 66 + 4?

A. 75

B. 68

C. 70

D. 72

Answer: C. 66 + 4 = 70.

STEM-G3-MAT-T03-Q09 | Perimeter of Plane Shapes | Difficulty: Foundational

What is 48 – 19?

A. 34

B. 29

C. 27

D. 31

Answer: B. 48 – 19 = 29.

STEM-G3-MAT-T03-Q10 | Perimeter of Plane Shapes | Difficulty: Foundational

What is 9 × 9?

A. 83

B. 79

C. 81

D. 86

Answer: C. 9 × 9 = 81.

Theme 4: Basic Measurement and Data — 10 questions

STEM-G3-MAT-T04-Q01 | Basic Measurement and Data | Difficulty: Foundational

What is 22 – 9?

A. 13

B. 18

C. 11

D. 15

Answer: A. 22 – 9 = 13.

STEM-G3-MAT-T04-Q02 | Basic Measurement and Data | Difficulty: Foundational

What is 12 × 9?

A. 106

B. 110

C. 113

D. 108

Answer: D. 12 × 9 = 108.

STEM-G3-MAT-T04-Q03 | Basic Measurement and Data | Difficulty: Foundational

What is 16 ÷ 8?

A. 4

B. 7

C. 0

D. 2

Answer: D. 16 ÷ 8 = 2.

STEM-G3-MAT-T04-Q04 | Basic Measurement and Data | Difficulty: Foundational

What is 22 + 10?

A. 32

B. 37

C. 34

D. 30

Answer: A. 22 + 10 = 32.

STEM-G3-MAT-T04-Q05 | Basic Measurement and Data | Difficulty: Foundational

What is 66 – 12?

A. 54

B. 52

C. 56

D. 59

Answer: A. 66 – 12 = 54.

STEM-G3-MAT-T04-Q06 | Basic Measurement and Data | Difficulty: Foundational

What is 9 × 8?

A. 70

B. 72

C. 77

D. 74

Answer: B. 9 × 8 = 72.

STEM-G3-MAT-T04-Q07 | Basic Measurement and Data | Difficulty: Foundational

What is 72 ÷ 9?

A. 13

B. 6

C. 8

D. 10

Answer: C. 72 ÷ 9 = 8.

STEM-G3-MAT-T04-Q08 | Basic Measurement and Data | Difficulty: Foundational

What is 19 + 13?

A. 37

B. 34

C. 30

D. 32

Answer: D. 19 + 13 = 32.

STEM-G3-MAT-T04-Q09 | Basic Measurement and Data | Difficulty: Foundational

What is 15 – 3?

A. 17

B. 14

C. 10

D. 12

Answer: D. 15 – 3 = 12.

STEM-G3-MAT-T04-Q10 | Basic Measurement and Data | Difficulty: Foundational

What is 6 × 9?

A. 56

B. 54

C. 52

D. 59

Answer: B. 6 × 9 = 54.

Science

Theme 1: Plant Parts and Their Functions — 10 questions

STEM-G3-SCI-T01-Q01 | Plant Parts and Their Functions | Difficulty: Foundational

Which plant part usually absorbs water from the soil?

A. Roots

B. Fruits

C. Flowers

D. Leaves

Answer: A. Roots absorb water and minerals from the soil.

STEM-G3-SCI-T01-Q02 | Plant Parts and Their Functions | Difficulty: Foundational

Which plant part usually absorbs water from the soil?

A. Leaves

B. Flowers

C. Roots

D. Fruits

Answer: C. Roots absorb water and minerals from the soil.

STEM-G3-SCI-T01-Q03 | Plant Parts and Their Functions | Difficulty: Foundational

Which plant part usually absorbs water from the soil?

A. Fruits

B. Leaves

C. Flowers

D. Roots

Answer: D. Roots absorb water and minerals from the soil.

STEM-G3-SCI-T01-Q04 | Plant Parts and Their Functions | Difficulty: Foundational

Which plant part usually absorbs water from the soil?

A. Roots

B. Leaves

C. Flowers

D. Fruits

Answer: A. Roots absorb water and minerals from the soil.

STEM-G3-SCI-T01-Q05 | Plant Parts and Their Functions | Difficulty: Foundational

Which plant part usually absorbs water from the soil?

A. Flowers

B. Roots

C. Leaves

D. Fruits

Answer: B. Roots absorb water and minerals from the soil.

STEM-G3-SCI-T01-Q06 | Plant Parts and Their Functions | Difficulty: Foundational

Which plant part usually absorbs water from the soil?

A. Flowers

B. Leaves

C. Roots

D. Fruits

Answer: C. Roots absorb water and minerals from the soil.

STEM-G3-SCI-T01-Q07 | Plant Parts and Their Functions | Difficulty: Foundational

Which plant part usually absorbs water from the soil?

A. Leaves

B. Fruits

C. Flowers

D. Roots

Answer: D. Roots absorb water and minerals from the soil.

STEM-G3-SCI-T01-Q08 | Plant Parts and Their Functions | Difficulty: Foundational

Which plant part usually absorbs water from the soil?

A. Roots

B. Flowers

C. Leaves

D. Fruits

Answer: A. Roots absorb water and minerals from the soil.

STEM-G3-SCI-T01-Q09 | Plant Parts and Their Functions | Difficulty: Foundational

Which plant part usually absorbs water from the soil?

A. Fruits

B. Leaves

C. Flowers

D. Roots

Answer: D. Roots absorb water and minerals from the soil.

STEM-G3-SCI-T01-Q10 | Plant Parts and Their Functions | Difficulty: Foundational

Which plant part usually absorbs water from the soil?

A. Fruits

B. Flowers

C. Roots

D. Leaves

Answer: C. Roots absorb water and minerals from the soil.

Theme 2: Simple Food Chains — 10 questions

STEM-G3-SCI-T02-Q01 | Simple Food Chains | Difficulty: Foundational

In the food chain grass → grasshopper → frog, what does the grasshopper eat?

A. Grass

B. An eagle

C. The frog

D. A rock

Answer: A. The grasshopper is a consumer that eats the producer, grass.

STEM-G3-SCI-T02-Q02 | Simple Food Chains | Difficulty: Foundational

In the food chain grass → grasshopper → frog, what does the grasshopper eat?

A. A rock

B. An eagle

C. The frog

D. Grass

Answer: D. The grasshopper is a consumer that eats the producer, grass.

STEM-G3-SCI-T02-Q03 | Simple Food Chains | Difficulty: Foundational

In the food chain grass → grasshopper → frog, what does the grasshopper eat?

A. Grass

B. A rock

C. An eagle

D. The frog

Answer: A. The grasshopper is a consumer that eats the producer, grass.

STEM-G3-SCI-T02-Q04 | Simple Food Chains | Difficulty: Foundational

In the food chain grass → grasshopper → frog, what does the grasshopper eat?

A. Grass

B. The frog

C. A rock

D. An eagle

Answer: A. The grasshopper is a consumer that eats the producer, grass.

STEM-G3-SCI-T02-Q05 | Simple Food Chains | Difficulty: Foundational

In the food chain grass → grasshopper → frog, what does the grasshopper eat?

A. Grass

B. An eagle

C. The frog

D. A rock

Answer: A. The grasshopper is a consumer that eats the producer, grass.

STEM-G3-SCI-T02-Q06 | Simple Food Chains | Difficulty: Foundational

In the food chain grass → grasshopper → frog, what does the grasshopper eat?

A. Grass

B. An eagle

C. A rock

D. The frog

Answer: A. The grasshopper is a consumer that eats the producer, grass.

STEM-G3-SCI-T02-Q07 | Simple Food Chains | Difficulty: Foundational

In the food chain grass → grasshopper → frog, what does the grasshopper eat?

A. Grass

B. The frog

C. An eagle

D. A rock

Answer: A. The grasshopper is a consumer that eats the producer, grass.

STEM-G3-SCI-T02-Q08 | Simple Food Chains | Difficulty: Foundational

In the food chain grass → grasshopper → frog, what does the grasshopper eat?

A. Grass

B. A rock

C. The frog

D. An eagle

Answer: A. The grasshopper is a consumer that eats the producer, grass.

STEM-G3-SCI-T02-Q09 | Simple Food Chains | Difficulty: Foundational

In the food chain grass → grasshopper → frog, what does the grasshopper eat?

A. The frog

B. A rock

C. An eagle

D. Grass

Answer: D. The grasshopper is a consumer that eats the producer, grass.

STEM-G3-SCI-T02-Q10 | Simple Food Chains | Difficulty: Foundational

In the food chain grass → grasshopper → frog, what does the grasshopper eat?

A. An eagle

B. A rock

C. Grass

D. The frog

Answer: C. The grasshopper is a consumer that eats the producer, grass.

Theme 3: Forces and Motion — 10 questions

STEM-G3-SCI-T03-Q01 | Forces and Motion | Difficulty: Foundational

A force can change an object's ...

A. The Sun's natural color

B. The number of days in a week

C. Its mass to zero

D. Motion or shape

Answer: D. A force can change speed, direction, or shape.

STEM-G3-SCI-T03-Q02 | Forces and Motion | Difficulty: Foundational

A force can change an object's ...

A. The Sun's natural color

B. Its mass to zero

C. Motion or shape

D. The number of days in a week

Answer: C. A force can change speed, direction, or shape.

STEM-G3-SCI-T03-Q03 | Forces and Motion | Difficulty: Foundational

A force can change an object's ...

A. The Sun's natural color

B. Its mass to zero

C. The number of days in a week

D. Motion or shape

Answer: D. A force can change speed, direction, or shape.

STEM-G3-SCI-T03-Q04 | Forces and Motion | Difficulty: Foundational

A force can change an object's ...

A. The Sun's natural color

B. Its mass to zero

C. Motion or shape

D. The number of days in a week

Answer: C. A force can change speed, direction, or shape.

STEM-G3-SCI-T03-Q05 | Forces and Motion | Difficulty: Foundational

A force can change an object's ...

A. Its mass to zero

B. The number of days in a week

C. Motion or shape

D. The Sun's natural color

Answer: C. A force can change speed, direction, or shape.

STEM-G3-SCI-T03-Q06 | Forces and Motion | Difficulty: Foundational

A force can change an object's ...

A. The number of days in a week

B. Its mass to zero

C. Motion or shape

D. The Sun's natural color

Answer: C. A force can change speed, direction, or shape.

STEM-G3-SCI-T03-Q07 | Forces and Motion | Difficulty: Foundational

A force can change an object's ...

A. Its mass to zero

B. The Sun's natural color

C. The number of days in a week

D. Motion or shape

Answer: D. A force can change speed, direction, or shape.

STEM-G3-SCI-T03-Q08 | Forces and Motion | Difficulty: Foundational

A force can change an object's ...

A. The number of days in a week

B. The Sun's natural color

C. Motion or shape

D. Its mass to zero

Answer: C. A force can change speed, direction, or shape.

STEM-G3-SCI-T03-Q09 | Forces and Motion | Difficulty: Foundational

A force can change an object's ...

A. Motion or shape

B. Its mass to zero

C. The Sun's natural color

D. The number of days in a week

Answer: A. A force can change speed, direction, or shape.

STEM-G3-SCI-T03-Q10 | Forces and Motion | Difficulty: Foundational

A force can change an object's ...

A. Its mass to zero

B. Motion or shape

C. The number of days in a week

D. The Sun's natural color

Answer: B. A force can change speed, direction, or shape.

Theme 4: Water Cycle and the Environment — 10 questions

STEM-G3-SCI-T04-Q01 | Water Cycle and the Environment | Difficulty: Foundational

Water changes from the ocean surface into vapor through ...

A. Precipitation

B. Infiltration

C. Evaporation

D. Condensation

Answer: C. Evaporation is the process in which liquid water becomes vapor.

STEM-G3-SCI-T04-Q02 | Water Cycle and the Environment | Difficulty: Foundational

Water changes from the ocean surface into vapor through ...

A. Evaporation

B. Precipitation

C. Infiltration

D. Condensation

Answer: A. Evaporation is the process in which liquid water becomes vapor.

STEM-G3-SCI-T04-Q03 | Water Cycle and the Environment | Difficulty: Foundational

Water changes from the ocean surface into vapor through ...

A. Evaporation

B. Condensation

C. Precipitation

D. Infiltration

Answer: A. Evaporation is the process in which liquid water becomes vapor.

STEM-G3-SCI-T04-Q04 | Water Cycle and the Environment | Difficulty: Foundational

Water changes from the ocean surface into vapor through ...

A. Infiltration

B. Evaporation

C. Condensation

D. Precipitation

Answer: B. Evaporation is the process in which liquid water becomes vapor.

STEM-G3-SCI-T04-Q05 | Water Cycle and the Environment | Difficulty: Foundational

Water changes from the ocean surface into vapor through ...

A. Evaporation

B. Condensation

C. Infiltration

D. Precipitation

Answer: A. Evaporation is the process in which liquid water becomes vapor.

STEM-G3-SCI-T04-Q06 | Water Cycle and the Environment | Difficulty: Foundational

Water changes from the ocean surface into vapor through ...

A. Evaporation

B. Infiltration

C. Precipitation

D. Condensation

Answer: A. Evaporation is the process in which liquid water becomes vapor.

STEM-G3-SCI-T04-Q07 | Water Cycle and the Environment | Difficulty: Foundational

Water changes from the ocean surface into vapor through ...

A. Infiltration

B. Evaporation

C. Condensation

D. Precipitation

Answer: B. Evaporation is the process in which liquid water becomes vapor.

STEM-G3-SCI-T04-Q08 | Water Cycle and the Environment | Difficulty: Foundational

Water changes from the ocean surface into vapor through ...

A. Precipitation

B. Infiltration

C. Evaporation

D. Condensation

Answer: C. Evaporation is the process in which liquid water becomes vapor.

STEM-G3-SCI-T04-Q09 | Water Cycle and the Environment | Difficulty: Foundational

Water changes from the ocean surface into vapor through ...

A. Precipitation

B. Condensation

C. Evaporation

D. Infiltration

Answer: C. Evaporation is the process in which liquid water becomes vapor.

STEM-G3-SCI-T04-Q10 | Water Cycle and the Environment | Difficulty: Foundational

Water changes from the ocean surface into vapor through ...

A. Condensation

B. Infiltration

C. Evaporation

D. Precipitation

Answer: C. Evaporation is the process in which liquid water becomes vapor.

Engineering

Theme 1: Strength of Shapes and Structures — 10 questions

STEM-G3-ENG-T01-Q01 | Strength of Shapes and Structures | Difficulty: Foundational

When selecting a material for a load-bearing part, what should be considered?

A. Only its favorite color

B. The packaging

C. The designer's name

D. Its strength and suitability for the task

Answer: D. Material choice depends on the load and operating conditions.

STEM-G3-ENG-T01-Q02 | Strength of Shapes and Structures | Difficulty: Foundational

Why should joints in a model be inspected?

A. Joints remove gravity

B. Joints are only decoration

C. Joints transfer forces between parts

D. Joints make mass zero

Answer: C. Weak joints can become failure points in a structure.

STEM-G3-ENG-T01-Q03 | Strength of Shapes and Structures | Difficulty: Foundational

After a test reveals a weakness, what should engineers do next?

A. Ignore safety

B. Improve the design and test it again

C. Hide the result

D. Delete the notes

Answer: B. Engineering is iterative and uses test results to guide improvements.

STEM-G3-ENG-T01-Q04 | Strength of Shapes and Structures | Difficulty: Foundational

Which measurement is useful when comparing two bridge designs?

A. The bridge color

B. The model's name

C. The number of stickers

D. The maximum load each bridge can support

Answer: D. Load capacity is a relevant measure of bridge performance.

STEM-G3-ENG-T01-Q05 | Strength of Shapes and Structures | Difficulty: Foundational

Why should an engineering prototype be tested before use?

A. To make it larger every time

B. To find out whether the design works and is safe

C. To avoid making improvements

D. To make it look more expensive

Answer: B. Testing reveals weaknesses and helps improve a design.

STEM-G3-ENG-T01-Q06 | Strength of Shapes and Structures | Difficulty: Foundational

If a model structure tips over easily, what is a sensible first improvement?

A. Add weight to the top

B. Remove all supports

C. Widen the base or strengthen the joints

D. Remove the base

Answer: C. A wider base and strong joints can improve stability.

STEM-G3-ENG-T01-Q07 | Strength of Shapes and Structures | Difficulty: Foundational

What is the purpose of making a sketch before building an object?

A. To avoid measuring

B. To make the object build itself

C. To plan its shape, dimensions, and parts

D. To replace all testing

Answer: C. A sketch communicates a design before construction.

STEM-G3-ENG-T01-Q08 | Strength of Shapes and Structures | Difficulty: Foundational

For a fair test, a team should ...

A. Avoid recording results

B. Change every factor at once

C. Change one factor while keeping other factors the same

D. Choose only the preferred result

Answer: C. Controlling variables helps identify what caused a result.

STEM-G3-ENG-T01-Q09 | Strength of Shapes and Structures | Difficulty: Foundational

When selecting a material for a load-bearing part, what should be considered?

A. The packaging

B. The designer's name

C. Its strength and suitability for the task

D. Only its favorite color

Answer: C. Material choice depends on the load and operating conditions.

STEM-G3-ENG-T01-Q10 | Strength of Shapes and Structures | Difficulty: Foundational

Why should joints in a model be inspected?

A. Joints are only decoration

B. Joints remove gravity

C. Joints transfer forces between parts

D. Joints make mass zero

Answer: C. Weak joints can become failure points in a structure.

Theme 2: Simple Lever Mechanisms — 10 questions

STEM-G3-ENG-T02-Q01 | Simple Lever Mechanisms | Difficulty: Foundational

When selecting a material for a load-bearing part, what should be considered?

A. The packaging

B. The designer's name

C. Only its favorite color

D. Its strength and suitability for the task

Answer: D. Material choice depends on the load and operating conditions.

STEM-G3-ENG-T02-Q02 | Simple Lever Mechanisms | Difficulty: Foundational

Why should joints in a model be inspected?

A. Joints are only decoration

B. Joints transfer forces between parts

C. Joints make mass zero

D. Joints remove gravity

Answer: B. Weak joints can become failure points in a structure.

STEM-G3-ENG-T02-Q03 | Simple Lever Mechanisms | Difficulty: Foundational

After a test reveals a weakness, what should engineers do next?

A. Ignore safety

B. Improve the design and test it again

C. Delete the notes

D. Hide the result

Answer: B. Engineering is iterative and uses test results to guide improvements.

STEM-G3-ENG-T02-Q04 | Simple Lever Mechanisms | Difficulty: Foundational

Which measurement is useful when comparing two bridge designs?

A. The bridge color

B. The maximum load each bridge can support

C. The number of stickers

D. The model's name

Answer: B. Load capacity is a relevant measure of bridge performance.

STEM-G3-ENG-T02-Q05 | Simple Lever Mechanisms | Difficulty: Foundational

Why should an engineering prototype be tested before use?

A. To avoid making improvements

B. To find out whether the design works and is safe

C. To make it larger every time

D. To make it look more expensive

Answer: B. Testing reveals weaknesses and helps improve a design.

STEM-G3-ENG-T02-Q06 | Simple Lever Mechanisms | Difficulty: Foundational

If a model structure tips over easily, what is a sensible first improvement?

A. Widen the base or strengthen the joints

B. Add weight to the top

C. Remove all supports

D. Remove the base

Answer: A. A wider base and strong joints can improve stability.

STEM-G3-ENG-T02-Q07 | Simple Lever Mechanisms | Difficulty: Foundational

What is the purpose of making a sketch before building an object?

A. To make the object build itself

B. To avoid measuring

C. To replace all testing

D. To plan its shape, dimensions, and parts

Answer: D. A sketch communicates a design before construction.

STEM-G3-ENG-T02-Q08 | Simple Lever Mechanisms | Difficulty: Foundational

For a fair test, a team should ...

A. Avoid recording results

B. Choose only the preferred result

C. Change every factor at once

D. Change one factor while keeping other factors the same

Answer: D. Controlling variables helps identify what caused a result.

STEM-G3-ENG-T02-Q09 | Simple Lever Mechanisms | Difficulty: Foundational

When selecting a material for a load-bearing part, what should be considered?

A. The designer's name

B. The packaging

C. Its strength and suitability for the task

D. Only its favorite color

Answer: C. Material choice depends on the load and operating conditions.

STEM-G3-ENG-T02-Q10 | Simple Lever Mechanisms | Difficulty: Foundational

Why should joints in a model be inspected?

A. Joints are only decoration

B. Joints transfer forces between parts

C. Joints make mass zero

D. Joints remove gravity

Answer: B. Weak joints can become failure points in a structure.

Theme 3: Designing a Simple Filter — 10 questions

STEM-G3-ENG-T03-Q01 | Designing a Simple Filter | Difficulty: Foundational

When selecting a material for a load-bearing part, what should be considered?

A. The packaging

B. Only its favorite color

C. Its strength and suitability for the task

D. The designer's name

Answer: C. Material choice depends on the load and operating conditions.

STEM-G3-ENG-T03-Q02 | Designing a Simple Filter | Difficulty: Foundational

Why should joints in a model be inspected?

A. Joints are only decoration

B. Joints transfer forces between parts

C. Joints make mass zero

D. Joints remove gravity

Answer: B. Weak joints can become failure points in a structure.

STEM-G3-ENG-T03-Q03 | Designing a Simple Filter | Difficulty: Foundational

After a test reveals a weakness, what should engineers do next?

A. Hide the result

B. Ignore safety

C. Improve the design and test it again

D. Delete the notes

Answer: C. Engineering is iterative and uses test results to guide improvements.

STEM-G3-ENG-T03-Q04 | Designing a Simple Filter | Difficulty: Foundational

Which measurement is useful when comparing two bridge designs?

A. The bridge color

B. The model's name

C. The number of stickers

D. The maximum load each bridge can support

Answer: D. Load capacity is a relevant measure of bridge performance.

STEM-G3-ENG-T03-Q05 | Designing a Simple Filter | Difficulty: Foundational

Why should an engineering prototype be tested before use?

A. To find out whether the design works and is safe

B. To make it look more expensive

C. To make it larger every time

D. To avoid making improvements

Answer: A. Testing reveals weaknesses and helps improve a design.

STEM-G3-ENG-T03-Q06 | Designing a Simple Filter | Difficulty: Foundational

If a model structure tips over easily, what is a sensible first improvement?

A. Widen the base or strengthen the joints

B. Add weight to the top

C. Remove all supports

D. Remove the base

Answer: A. A wider base and strong joints can improve stability.

STEM-G3-ENG-T03-Q07 | Designing a Simple Filter | Difficulty: Foundational

What is the purpose of making a sketch before building an object?

A. To make the object build itself

B. To replace all testing

C. To avoid measuring

D. To plan its shape, dimensions, and parts

Answer: D. A sketch communicates a design before construction.

STEM-G3-ENG-T03-Q08 | Designing a Simple Filter | Difficulty: Foundational

For a fair test, a team should ...

A. Change one factor while keeping other factors the same

B. Avoid recording results

C. Choose only the preferred result

D. Change every factor at once

Answer: A. Controlling variables helps identify what caused a result.

STEM-G3-ENG-T03-Q09 | Designing a Simple Filter | Difficulty: Foundational

When selecting a material for a load-bearing part, what should be considered?

A. Only its favorite color

B. The designer's name

C. The packaging

D. Its strength and suitability for the task

Answer: D. Material choice depends on the load and operating conditions.

STEM-G3-ENG-T03-Q10 | Designing a Simple Filter | Difficulty: Foundational

Why should joints in a model be inspected?

A. Joints transfer forces between parts

B. Joints remove gravity

C. Joints are only decoration

D. Joints make mass zero

Answer: A. Weak joints can become failure points in a structure.

Theme 4: Testing and Improving Designs — 10 questions

STEM-G3-ENG-T04-Q01 | Testing and Improving Designs | Difficulty: Foundational

When selecting a material for a load-bearing part, what should be considered?

A. Only its favorite color

B. The packaging

C. The designer's name

D. Its strength and suitability for the task

Answer: D. Material choice depends on the load and operating conditions.

STEM-G3-ENG-T04-Q02 | Testing and Improving Designs | Difficulty: Foundational

Why should joints in a model be inspected?

A. Joints make mass zero

B. Joints remove gravity

C. Joints are only decoration

D. Joints transfer forces between parts

Answer: D. Weak joints can become failure points in a structure.

STEM-G3-ENG-T04-Q03 | Testing and Improving Designs | Difficulty: Foundational

After a test reveals a weakness, what should engineers do next?

A. Improve the design and test it again

B. Delete the notes

C. Ignore safety

D. Hide the result

Answer: A. Engineering is iterative and uses test results to guide improvements.

STEM-G3-ENG-T04-Q04 | Testing and Improving Designs | Difficulty: Foundational

Which measurement is useful when comparing two bridge designs?

A. The maximum load each bridge can support

B. The bridge color

C. The number of stickers

D. The model's name

Answer: A. Load capacity is a relevant measure of bridge performance.

STEM-G3-ENG-T04-Q05 | Testing and Improving Designs | Difficulty: Foundational

Why should an engineering prototype be tested before use?

A. To avoid making improvements

B. To make it larger every time

C. To find out whether the design works and is safe

D. To make it look more expensive

Answer: C. Testing reveals weaknesses and helps improve a design.

STEM-G3-ENG-T04-Q06 | Testing and Improving Designs | Difficulty: Foundational

If a model structure tips over easily, what is a sensible first improvement?

A. Widen the base or strengthen the joints

B. Remove the base

C. Remove all supports

D. Add weight to the top

Answer: A. A wider base and strong joints can improve stability.

STEM-G3-ENG-T04-Q07 | Testing and Improving Designs | Difficulty: Foundational

What is the purpose of making a sketch before building an object?

A. To replace all testing

B. To plan its shape, dimensions, and parts

C. To make the object build itself

D. To avoid measuring

Answer: B. A sketch communicates a design before construction.

STEM-G3-ENG-T04-Q08 | Testing and Improving Designs | Difficulty: Foundational

For a fair test, a team should ...

A. Choose only the preferred result

B. Change every factor at once

C. Avoid recording results

D. Change one factor while keeping other factors the same

Answer: D. Controlling variables helps identify what caused a result.

STEM-G3-ENG-T04-Q09 | Testing and Improving Designs | Difficulty: Foundational

When selecting a material for a load-bearing part, what should be considered?

A. The designer's name

B. Its strength and suitability for the task

C. The packaging

D. Only its favorite color

Answer: B. Material choice depends on the load and operating conditions.

STEM-G3-ENG-T04-Q10 | Testing and Improving Designs | Difficulty: Foundational

Why should joints in a model be inspected?

A. Joints remove gravity

B. Joints are only decoration

C. Joints transfer forces between parts

D. Joints make mass zero

Answer: C. Weak joints can become failure points in a structure.

ICT (Technology)

Theme 1: Hardware and Software — 10 questions

STEM-G3-ICT-T01-Q01 | Hardware and Software | Difficulty: Foundational

Finding and fixing errors in a program is called ...

A. Streaming

B. Debugging

C. Browsing

D. Formatting

Answer: B. Debugging is the process of locating and correcting errors.

STEM-G3-ICT-T01-Q02 | Hardware and Software | Difficulty: Foundational

Which device is commonly used to enter text into a computer?

A. Speaker

B. Keyboard

C. Monitor

D. Projector

Answer: B. A keyboard is an input device used for typing.

STEM-G3-ICT-T01-Q03 | Hardware and Software | Difficulty: Foundational

Software designed to help a user perform a task is commonly called an ...

A. Fan

B. Power cable

C. Desk

D. Application

Answer: D. An application is software that helps users perform specific tasks.

STEM-G3-ICT-T01-Q04 | Hardware and Software | Difficulty: Foundational

What should you do if an unknown person sends you a suspicious link?

A. Download every attachment

B. Forward it to every contact

C. Do not open it; check with a trusted adult or person

D. Enter your password

Answer: C. Suspicious links may lead to scams or harmful software.

STEM-G3-ICT-T01-Q05 | Hardware and Software | Difficulty: Foundational

Why should you keep your password private?

A. To make the screen brighter

B. To protect your account and personal information

C. To extend battery life

D. To make the internet faster

Answer: B. A password provides access to an account and should be protected.

STEM-G3-ICT-T01-Q06 | Hardware and Software | Difficulty: Foundational

A clear, ordered set of steps for solving a problem is called an ...

A. Speaker

B. Algorithm

C. Resolution

D. Wallpaper

Answer: B. An algorithm is an ordered set of instructions.

STEM-G3-ICT-T01-Q07 | Hardware and Software | Difficulty: Foundational

In a spreadsheet, a cell is ...

A. The intersection of a row and a column

B. A webpage

C. An application icon

D. A type of cable

Answer: A. A cell is a box at the intersection of a row and a column.

STEM-G3-ICT-T01-Q08 | Hardware and Software | Difficulty: Foundational

How should information found online be checked?

A. Compare sources and supporting evidence

B. Choose the most dramatic headline

C. Count the emojis

D. Share it before reading

Answer: A. Checking sources helps assess accuracy and credibility.

STEM-G3-ICT-T01-Q09 | Hardware and Software | Difficulty: Foundational

Finding and fixing errors in a program is called ...

A. Debugging

B. Streaming

C. Browsing

D. Formatting

Answer: A. Debugging is the process of locating and correcting errors.

STEM-G3-ICT-T01-Q10 | Hardware and Software | Difficulty: Foundational

Which device is commonly used to enter text into a computer?

A. Monitor

B. Keyboard

C. Projector

D. Speaker

Answer: B. A keyboard is an input device used for typing.

Theme 2: Typing and Creating Documents — 10 questions

STEM-G3-ICT-T02-Q01 | Typing and Creating Documents | Difficulty: Foundational

Finding and fixing errors in a program is called ...

A. Formatting

B. Streaming

C. Browsing

D. Debugging

Answer: D. Debugging is the process of locating and correcting errors.

STEM-G3-ICT-T02-Q02 | Typing and Creating Documents | Difficulty: Foundational

Which device is commonly used to enter text into a computer?

A. Projector

B. Speaker

C. Monitor

D. Keyboard

Answer: D. A keyboard is an input device used for typing.

STEM-G3-ICT-T02-Q03 | Typing and Creating Documents | Difficulty: Foundational

Software designed to help a user perform a task is commonly called an ...

A. Application

B. Desk

C. Fan

D. Power cable

Answer: A. An application is software that helps users perform specific tasks.

STEM-G3-ICT-T02-Q04 | Typing and Creating Documents | Difficulty: Foundational

What should you do if an unknown person sends you a suspicious link?

A. Do not open it; check with a trusted adult or person

B. Enter your password

C. Forward it to every contact

D. Download every attachment

Answer: A. Suspicious links may lead to scams or harmful software.

STEM-G3-ICT-T02-Q05 | Typing and Creating Documents | Difficulty: Foundational

Why should you keep your password private?

A. To make the screen brighter

B. To make the internet faster

C. To protect your account and personal information

D. To extend battery life

Answer: C. A password provides access to an account and should be protected.

STEM-G3-ICT-T02-Q06 | Typing and Creating Documents | Difficulty: Foundational

A clear, ordered set of steps for solving a problem is called an ...

A. Resolution

B. Wallpaper

C. Speaker

D. Algorithm

Answer: D. An algorithm is an ordered set of instructions.

STEM-G3-ICT-T02-Q07 | Typing and Creating Documents | Difficulty: Foundational

In a spreadsheet, a cell is ...

A. The intersection of a row and a column

B. A webpage

C. A type of cable

D. An application icon

Answer: A. A cell is a box at the intersection of a row and a column.

STEM-G3-ICT-T02-Q08 | Typing and Creating Documents | Difficulty: Foundational

How should information found online be checked?

A. Share it before reading

B. Count the emojis

C. Choose the most dramatic headline

D. Compare sources and supporting evidence

Answer: D. Checking sources helps assess accuracy and credibility.

STEM-G3-ICT-T02-Q09 | Typing and Creating Documents | Difficulty: Foundational

Finding and fixing errors in a program is called ...

A. Formatting

B. Debugging

C. Streaming

D. Browsing

Answer: B. Debugging is the process of locating and correcting errors.

STEM-G3-ICT-T02-Q10 | Typing and Creating Documents | Difficulty: Foundational

Which device is commonly used to enter text into a computer?

A. Monitor

B. Keyboard

C. Projector

D. Speaker

Answer: B. A keyboard is an input device used for typing.

Theme 3: Safe Internet Search — 10 questions

STEM-G3-ICT-T03-Q01 | Safe Internet Search | Difficulty: Foundational

Finding and fixing errors in a program is called ...

A. Debugging

B. Formatting

C. Browsing

D. Streaming

Answer: A. Debugging is the process of locating and correcting errors.

STEM-G3-ICT-T03-Q02 | Safe Internet Search | Difficulty: Foundational

Which device is commonly used to enter text into a computer?

A. Projector

B. Monitor

C. Keyboard

D. Speaker

Answer: C. A keyboard is an input device used for typing.

STEM-G3-ICT-T03-Q03 | Safe Internet Search | Difficulty: Foundational

Software designed to help a user perform a task is commonly called an ...

A. Power cable

B. Application

C. Fan

D. Desk

Answer: B. An application is software that helps users perform specific tasks.

STEM-G3-ICT-T03-Q04 | Safe Internet Search | Difficulty: Foundational

What should you do if an unknown person sends you a suspicious link?

A. Do not open it; check with a trusted adult or person

B. Download every attachment

C. Enter your password

D. Forward it to every contact

Answer: A. Suspicious links may lead to scams or harmful software.

STEM-G3-ICT-T03-Q05 | Safe Internet Search | Difficulty: Foundational

Why should you keep your password private?

A. To protect your account and personal information

B. To make the screen brighter

C. To extend battery life

D. To make the internet faster

Answer: A. A password provides access to an account and should be protected.

STEM-G3-ICT-T03-Q06 | Safe Internet Search | Difficulty: Foundational

A clear, ordered set of steps for solving a problem is called an ...

A. Wallpaper

B. Algorithm

C. Speaker

D. Resolution

Answer: B. An algorithm is an ordered set of instructions.

STEM-G3-ICT-T03-Q07 | Safe Internet Search | Difficulty: Foundational

In a spreadsheet, a cell is ...

A. An application icon

B. A type of cable

C. The intersection of a row and a column

D. A webpage

Answer: C. A cell is a box at the intersection of a row and a column.

STEM-G3-ICT-T03-Q08 | Safe Internet Search | Difficulty: Foundational

How should information found online be checked?

A. Count the emojis

B. Compare sources and supporting evidence

C. Choose the most dramatic headline

D. Share it before reading

Answer: B. Checking sources helps assess accuracy and credibility.

STEM-G3-ICT-T03-Q09 | Safe Internet Search | Difficulty: Foundational

Finding and fixing errors in a program is called ...

A. Debugging

B. Formatting

C. Browsing

D. Streaming

Answer: A. Debugging is the process of locating and correcting errors.

STEM-G3-ICT-T03-Q10 | Safe Internet Search | Difficulty: Foundational

Which device is commonly used to enter text into a computer?

A. Projector

B. Keyboard

C. Monitor

D. Speaker

Answer: B. A keyboard is an input device used for typing.

Theme 4: Algorithms and Patterns — 10 questions

STEM-G3-ICT-T04-Q01 | Algorithms and Patterns | Difficulty: Foundational

Finding and fixing errors in a program is called ...

A. Debugging

B. Streaming

C. Formatting

D. Browsing

Answer: A. Debugging is the process of locating and correcting errors.

STEM-G3-ICT-T04-Q02 | Algorithms and Patterns | Difficulty: Foundational

Which device is commonly used to enter text into a computer?

A. Keyboard

B. Projector

C. Monitor

D. Speaker

Answer: A. A keyboard is an input device used for typing.

STEM-G3-ICT-T04-Q03 | Algorithms and Patterns | Difficulty: Foundational

Software designed to help a user perform a task is commonly called an ...

A. Desk

B. Fan

C. Power cable

D. Application

Answer: D. An application is software that helps users perform specific tasks.

STEM-G3-ICT-T04-Q04 | Algorithms and Patterns | Difficulty: Foundational

What should you do if an unknown person sends you a suspicious link?

A. Enter your password

B. Do not open it; check with a trusted adult or person

C. Download every attachment

D. Forward it to every contact

Answer: B. Suspicious links may lead to scams or harmful software.

STEM-G3-ICT-T04-Q05 | Algorithms and Patterns | Difficulty: Foundational

Why should you keep your password private?

A. To extend battery life

B. To make the internet faster

C. To protect your account and personal information

D. To make the screen brighter

Answer: C. A password provides access to an account and should be protected.

STEM-G3-ICT-T04-Q06 | Algorithms and Patterns | Difficulty: Foundational

A clear, ordered set of steps for solving a problem is called an ...

A. Algorithm

B. Wallpaper

C. Speaker

D. Resolution

Answer: A. An algorithm is an ordered set of instructions.

STEM-G3-ICT-T04-Q07 | Algorithms and Patterns | Difficulty: Foundational

In a spreadsheet, a cell is ...

A. An application icon

B. A type of cable

C. A webpage

D. The intersection of a row and a column

Answer: D. A cell is a box at the intersection of a row and a column.

STEM-G3-ICT-T04-Q08 | Algorithms and Patterns | Difficulty: Foundational

How should information found online be checked?

A. Choose the most dramatic headline

B. Count the emojis

C. Share it before reading

D. Compare sources and supporting evidence

Answer: D. Checking sources helps assess accuracy and credibility.

STEM-G3-ICT-T04-Q09 | Algorithms and Patterns | Difficulty: Foundational

Finding and fixing errors in a program is called ...

A. Browsing

B. Formatting

C. Debugging

D. Streaming

Answer: C. Debugging is the process of locating and correcting errors.

STEM-G3-ICT-T04-Q10 | Algorithms and Patterns | Difficulty: Foundational

Which device is commonly used to enter text into a computer?

A. Projector

B. Speaker

C. Keyboard

D. Monitor

Answer: C. A keyboard is an input device used for typing.

Elementary School Grade 4

160 questions • 4 subjects • 4 themes per subject • 10 questions per theme

Mathematics

Theme 1: Mixed Arithmetic Operations — 10 questions

STEM-G4-MAT-T01-Q01 | Mixed Arithmetic Operations | Difficulty: Intermediate

What is 68 – 8?

A. 58

B. 60

C. 65

D. 62

Answer: B. 68 – 8 = 60.

STEM-G4-MAT-T01-Q02 | Mixed Arithmetic Operations | Difficulty: Intermediate

What is 7 × 7?

A. 54

B. 51

C. 49

D. 47

Answer: C. 7 × 7 = 49.

STEM-G4-MAT-T01-Q03 | Mixed Arithmetic Operations | Difficulty: Intermediate

What is 54 ÷ 6?

A. 14

B. 7

C. 11

D. 9

Answer: D. 54 ÷ 6 = 9.

STEM-G4-MAT-T01-Q04 | Mixed Arithmetic Operations | Difficulty: Intermediate

What is 54 + 3?

A. 57

B. 62

C. 55

D. 59

Answer: A. 54 + 3 = 57.

STEM-G4-MAT-T01-Q05 | Mixed Arithmetic Operations | Difficulty: Intermediate

What is 91 – 19?

A. 74

B. 77

C. 70

D. 72

Answer: D. 91 – 19 = 72.

STEM-G4-MAT-T01-Q06 | Mixed Arithmetic Operations | Difficulty: Intermediate

What is 12 × 9?

A. 106

B. 108

C. 113

D. 110

Answer: B. 12 × 9 = 108.

STEM-G4-MAT-T01-Q07 | Mixed Arithmetic Operations | Difficulty: Intermediate

What is 4 ÷ 2?

A. 2

B. 4

C. 7

D. 0

Answer: A. 4 ÷ 2 = 2.

STEM-G4-MAT-T01-Q08 | Mixed Arithmetic Operations | Difficulty: Intermediate

What is 17 + 6?

A. 28

B. 25

C. 21

D. 23

Answer: D. 17 + 6 = 23.

STEM-G4-MAT-T01-Q09 | Mixed Arithmetic Operations | Difficulty: Intermediate

What is 76 – 19?

A. 57

B. 55

C. 59

D. 62

Answer: A. 76 – 19 = 57.

STEM-G4-MAT-T01-Q10 | Mixed Arithmetic Operations | Difficulty: Intermediate

What is 7 × 11?

A. 77

B. 75

C. 82

D. 79

Answer: A. 7 × 11 = 77.

Theme 2: Factors and Multiples — 10 questions

STEM-G4-MAT-T02-Q01 | Factors and Multiples | Difficulty: Intermediate

What is 94 – 14?

A. 85

B. 80

C. 78

D. 82

Answer: B. 94 – 14 = 80.

STEM-G4-MAT-T02-Q02 | Factors and Multiples | Difficulty: Intermediate

What is 9 × 3?

A. 32

B. 29

C. 27

D. 25

Answer: C. 9 × 3 = 27.

STEM-G4-MAT-T02-Q03 | Factors and Multiples | Difficulty: Intermediate

What is 60 ÷ 12?

A. 10

B. 3

C. 7

D. 5

Answer: D. 60 ÷ 12 = 5.

STEM-G4-MAT-T02-Q04 | Factors and Multiples | Difficulty: Intermediate

What is 40 + 17?

A. 62

B. 57

C. 59

D. 55

Answer: B. 40 + 17 = 57.

STEM-G4-MAT-T02-Q05 | Factors and Multiples | Difficulty: Intermediate

What is 54 – 18?

A. 41

B. 34

C. 38

D. 36

Answer: D. 54 – 18 = 36.

STEM-G4-MAT-T02-Q06 | Factors and Multiples | Difficulty: Intermediate

What is 11 × 4?

A. 49

B. 42

C. 46

D. 44

Answer: D. 11 × 4 = 44.

STEM-G4-MAT-T02-Q07 | Factors and Multiples | Difficulty: Intermediate

What is 108 ÷ 12?

A. 7

B. 14

C. 11

D. 9

Answer: D. 108 ÷ 12 = 9.

STEM-G4-MAT-T02-Q08 | Factors and Multiples | Difficulty: Intermediate

What is 30 + 18?

A. 53

B. 48

C. 50

D. 46

Answer: B. 30 + 18 = 48.

STEM-G4-MAT-T02-Q09 | Factors and Multiples | Difficulty: Intermediate

What is 84 – 6?

A. 80

B. 83

C. 76

D. 78

Answer: D. 84 – 6 = 78.

STEM-G4-MAT-T02-Q10 | Factors and Multiples | Difficulty: Intermediate

What is 3 × 3?

A. 7

B. 14

C. 9

D. 11

Answer: C. 3 × 3 = 9.

Theme 3: Area and Perimeter — 10 questions

STEM-G4-MAT-T03-Q01 | Area and Perimeter | Difficulty: Intermediate

What is 17 – 5?

A. 14

B. 10

C. 17

D. 12

Answer: D. 17 – 5 = 12.

STEM-G4-MAT-T03-Q02 | Area and Perimeter | Difficulty: Intermediate

What is 4 × 10?

A. 45

B. 38

C. 40

D. 42

Answer: C. 4 × 10 = 40.

STEM-G4-MAT-T03-Q03 | Area and Perimeter | Difficulty: Intermediate

What is 9 ÷ 3?

A. 8

B. 3

C. 1

D. 5

Answer: B. 9 ÷ 3 = 3.

STEM-G4-MAT-T03-Q04 | Area and Perimeter | Difficulty: Intermediate

What is 52 + 19?

A. 73

B. 69

C. 71

D. 76

Answer: C. 52 + 19 = 71.

STEM-G4-MAT-T03-Q05 | Area and Perimeter | Difficulty: Intermediate

What is 38 – 7?

A. 29

B. 33

C. 31

D. 36

Answer: C. 38 – 7 = 31.

STEM-G4-MAT-T03-Q06 | Area and Perimeter | Difficulty: Intermediate

What is 4 × 8?

A. 37

B. 34

C. 32

D. 30

Answer: C. 4 × 8 = 32.

STEM-G4-MAT-T03-Q07 | Area and Perimeter | Difficulty: Intermediate

What is 24 ÷ 8?

A. 3

B. 8

C. 1

D. 5

Answer: A. 24 ÷ 8 = 3.

STEM-G4-MAT-T03-Q08 | Area and Perimeter | Difficulty: Intermediate

What is 34 + 13?

A. 47

B. 45

C. 52

D. 49

Answer: A. 34 + 13 = 47.

STEM-G4-MAT-T03-Q09 | Area and Perimeter | Difficulty: Intermediate

What is 62 – 6?

A. 56

B. 54

C. 58

D. 61

Answer: A. 62 – 6 = 56.

STEM-G4-MAT-T03-Q10 | Area and Perimeter | Difficulty: Intermediate

What is 4 × 3?

A. 14

B. 10

C. 12

D. 17

Answer: C. 4 × 3 = 12.

Theme 4: Data and Averages — 10 questions

STEM-G4-MAT-T04-Q01 | Data and Averages | Difficulty: Intermediate

What is 40 – 12?

A. 30

B. 28

C. 26

D. 33

Answer: B. 40 – 12 = 28.

STEM-G4-MAT-T04-Q02 | Data and Averages | Difficulty: Intermediate

What is 5 × 5?

A. 23

B. 30

C. 25

D. 27

Answer: C. 5 × 5 = 25.

STEM-G4-MAT-T04-Q03 | Data and Averages | Difficulty: Intermediate

What is 9 ÷ 3?

A. 3

B. 1

C. 8

D. 5

Answer: A. 9 ÷ 3 = 3.

STEM-G4-MAT-T04-Q04 | Data and Averages | Difficulty: Intermediate

What is 82 + 18?

A. 105

B. 102

C. 100

D. 98

Answer: C. 82 + 18 = 100.

STEM-G4-MAT-T04-Q05 | Data and Averages | Difficulty: Intermediate

What is 85 – 5?

A. 80

B. 85

C. 82

D. 78

Answer: A. 85 – 5 = 80.

STEM-G4-MAT-T04-Q06 | Data and Averages | Difficulty: Intermediate

What is 11 × 7?

A. 82

B. 77

C. 75

D. 79

Answer: B. 11 × 7 = 77.

STEM-G4-MAT-T04-Q07 | Data and Averages | Difficulty: Intermediate

What is 63 ÷ 7?

A. 11

B. 14

C. 9

D. 7

Answer: C. 63 ÷ 7 = 9.

STEM-G4-MAT-T04-Q08 | Data and Averages | Difficulty: Intermediate

What is 24 + 15?

A. 44

B. 41

C. 39

D. 37

Answer: C. 24 + 15 = 39.

STEM-G4-MAT-T04-Q09 | Data and Averages | Difficulty: Intermediate

What is 30 – 11?

A. 21

B. 19

C. 24

D. 17

Answer: B. 30 – 11 = 19.

STEM-G4-MAT-T04-Q10 | Data and Averages | Difficulty: Intermediate

What is 10 × 10?

A. 102

B. 105

C. 98

D. 100

Answer: D. 10 × 10 = 100.

Science

Theme 1: Energy Sources and Transformations — 10 questions

STEM-G4-SCI-T01-Q01 | Energy Sources and Transformations | Difficulty: Intermediate

In an electric fan, electrical energy is mainly converted into ...

A. Gravitational potential energy only

B. Motion

C. Nuclear energy

D. Food energy

Answer: B. The electric motor spins the fan blades.

STEM-G4-SCI-T01-Q02 | Energy Sources and Transformations | Difficulty: Intermediate

In an electric fan, electrical energy is mainly converted into ...

A. Food energy

B. Nuclear energy

C. Motion

D. Gravitational potential energy only

Answer: C. The electric motor spins the fan blades.

STEM-G4-SCI-T01-Q03 | Energy Sources and Transformations | Difficulty: Intermediate

In an electric fan, electrical energy is mainly converted into ...

A. Nuclear energy

B. Food energy

C. Motion

D. Gravitational potential energy only

Answer: C. The electric motor spins the fan blades.

STEM-G4-SCI-T01-Q04 | Energy Sources and Transformations | Difficulty: Intermediate

In an electric fan, electrical energy is mainly converted into ...

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D. Motion

Answer: D. The electric motor spins the fan blades.

STEM-G4-SCI-T01-Q05 | Energy Sources and Transformations | Difficulty: Intermediate

In an electric fan, electrical energy is mainly converted into ...

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Answer: C. The electric motor spins the fan blades.

STEM-G4-SCI-T01-Q06 | Energy Sources and Transformations | Difficulty: Intermediate

In an electric fan, electrical energy is mainly converted into ...

A. Food energy

B. Gravitational potential energy only

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Answer: D. The electric motor spins the fan blades.

STEM-G4-SCI-T01-Q07 | Energy Sources and Transformations | Difficulty: Intermediate

In an electric fan, electrical energy is mainly converted into ...

A. Food energy

B. Motion

C. Gravitational potential energy only

D. Nuclear energy

Answer: B. The electric motor spins the fan blades.

STEM-G4-SCI-T01-Q08 | Energy Sources and Transformations | Difficulty: Intermediate

In an electric fan, electrical energy is mainly converted into ...

A. Motion

B. Gravitational potential energy only

C. Food energy

D. Nuclear energy

Answer: A. The electric motor spins the fan blades.

STEM-G4-SCI-T01-Q09 | Energy Sources and Transformations | Difficulty: Intermediate

In an electric fan, electrical energy is mainly converted into ...

A. Nuclear energy

B. Motion

C. Gravitational potential energy only

D. Food energy

Answer: B. The electric motor spins the fan blades.

STEM-G4-SCI-T01-Q10 | Energy Sources and Transformations | Difficulty: Intermediate

In an electric fan, electrical energy is mainly converted into ...

A. Nuclear energy

B. Food energy

C. Gravitational potential energy only

D. Motion

Answer: D. The electric motor spins the fan blades.

Theme 2: Ecosystems and Adaptations — 10 questions

STEM-G4-SCI-T02-Q01 | Ecosystems and Adaptations | Difficulty: Intermediate

A feeding relationship among organisms is represented by a ...

A. Conduction

B. Food chain

C. Rotation

D. Condensation

Answer: B. A food chain shows how energy moves between organisms.

STEM-G4-SCI-T02-Q02 | Ecosystems and Adaptations | Difficulty: Intermediate

A feeding relationship among organisms is represented by a ...

A. Food chain

B. Conduction

C. Condensation

D. Rotation

Answer: A. A food chain shows how energy moves between organisms.

STEM-G4-SCI-T02-Q03 | Ecosystems and Adaptations | Difficulty: Intermediate

A feeding relationship among organisms is represented by a ...

A. Rotation

B. Condensation

C. Conduction

D. Food chain

Answer: D. A food chain shows how energy moves between organisms.

STEM-G4-SCI-T02-Q04 | Ecosystems and Adaptations | Difficulty: Intermediate

A feeding relationship among organisms is represented by a ...

A. Rotation

B. Condensation

C. Food chain

D. Conduction

Answer: C. A food chain shows how energy moves between organisms.

STEM-G4-SCI-T02-Q05 | Ecosystems and Adaptations | Difficulty: Intermediate

A feeding relationship among organisms is represented by a ...

A. Condensation

B. Food chain

C. Conduction

D. Rotation

Answer: B. A food chain shows how energy moves between organisms.

STEM-G4-SCI-T02-Q06 | Ecosystems and Adaptations | Difficulty: Intermediate

A feeding relationship among organisms is represented by a ...

A. Conduction

B. Food chain

C. Condensation

D. Rotation

Answer: B. A food chain shows how energy moves between organisms.

STEM-G4-SCI-T02-Q07 | Ecosystems and Adaptations | Difficulty: Intermediate

A feeding relationship among organisms is represented by a ...

A. Condensation

B. Conduction

C. Rotation

D. Food chain

Answer: D. A food chain shows how energy moves between organisms.

STEM-G4-SCI-T02-Q08 | Ecosystems and Adaptations | Difficulty: Intermediate

A feeding relationship among organisms is represented by a ...

A. Conduction

B. Condensation

C. Rotation

D. Food chain

Answer: D. A food chain shows how energy moves between organisms.

STEM-G4-SCI-T02-Q09 | Ecosystems and Adaptations | Difficulty: Intermediate

A feeding relationship among organisms is represented by a ...

A. Conduction

B. Food chain

C. Rotation

D. Condensation

Answer: B. A food chain shows how energy moves between organisms.

STEM-G4-SCI-T02-Q10 | Ecosystems and Adaptations | Difficulty: Intermediate

A feeding relationship among organisms is represented by a ...

A. Rotation

B. Conduction

C. Condensation

D. Food chain

Answer: D. A food chain shows how energy moves between organisms.

Theme 3: Light and Sound — 10 questions

STEM-G4-SCI-T03-Q01 | Light and Sound | Difficulty: Intermediate

Which statement about light is correct?

A. Light travels only through wood

B. Light travels only through soil

C. Light travels only through thick fabric

D. Light can travel through a vacuum

Answer: D. Light does not require a medium and can travel through a vacuum.

STEM-G4-SCI-T03-Q02 | Light and Sound | Difficulty: Intermediate

Which statement about light is correct?

A. Light travels only through thick fabric

B. Light travels only through wood

C. Light can travel through a vacuum

D. Light travels only through soil

Answer: C. Light does not require a medium and can travel through a vacuum.

STEM-G4-SCI-T03-Q03 | Light and Sound | Difficulty: Intermediate

Which statement about light is correct?

A. Light can travel through a vacuum

B. Light travels only through wood

C. Light travels only through thick fabric

D. Light travels only through soil

Answer: A. Light does not require a medium and can travel through a vacuum.

STEM-G4-SCI-T03-Q04 | Light and Sound | Difficulty: Intermediate

Which statement about light is correct?

A. Light travels only through wood

B. Light can travel through a vacuum

C. Light travels only through soil

D. Light travels only through thick fabric

Answer: B. Light does not require a medium and can travel through a vacuum.

STEM-G4-SCI-T03-Q05 | Light and Sound | Difficulty: Intermediate

Which statement about light is correct?

A. Light travels only through soil

B. Light can travel through a vacuum

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STEM-G4-SCI-T03-Q06 | Light and Sound | Difficulty: Intermediate

Which statement about light is correct?

A. Light travels only through wood

B. Light travels only through thick fabric

C. Light travels only through soil

D. Light can travel through a vacuum

Answer: D. Light does not require a medium and can travel through a vacuum.

STEM-G4-SCI-T03-Q07 | Light and Sound | Difficulty: Intermediate

Which statement about light is correct?

A. Light can travel through a vacuum

B. Light travels only through wood

C. Light travels only through soil

D. Light travels only through thick fabric

Answer: A. Light does not require a medium and can travel through a vacuum.

STEM-G4-SCI-T03-Q08 | Light and Sound | Difficulty: Intermediate

Which statement about light is correct?

A. Light travels only through thick fabric

B. Light travels only through soil

C. Light can travel through a vacuum

D. Light travels only through wood

Answer: C. Light does not require a medium and can travel through a vacuum.

STEM-G4-SCI-T03-Q09 | Light and Sound | Difficulty: Intermediate

Which statement about light is correct?

A. Light travels only through thick fabric

B. Light travels only through wood

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Answer: D. Light does not require a medium and can travel through a vacuum.

STEM-G4-SCI-T03-Q10 | Light and Sound | Difficulty: Intermediate

Which statement about light is correct?

A. Light can travel through a vacuum

B. Light travels only through soil

C. Light travels only through thick fabric

D. Light travels only through wood

Answer: A. Light does not require a medium and can travel through a vacuum.

Theme 4: States of Matter and Their Changes — 10 questions

STEM-G4-SCI-T04-Q01 | States of Matter and Their Changes | Difficulty: Intermediate

The change from gas to liquid is called ...

A. Melting

B. Condensation

C. Evaporation

D. Sublimation

Answer: B. Condensation occurs when a gas loses energy and becomes liquid.

STEM-G4-SCI-T04-Q02 | States of Matter and Their Changes | Difficulty: Intermediate

The change from gas to liquid is called ...

A. Condensation

B. Melting

C. Evaporation

D. Sublimation

Answer: A. Condensation occurs when a gas loses energy and becomes liquid.

STEM-G4-SCI-T04-Q03 | States of Matter and Their Changes | Difficulty: Intermediate

The change from gas to liquid is called ...

A. Sublimation

B. Condensation

C. Evaporation

D. Melting

Answer: B. Condensation occurs when a gas loses energy and becomes liquid.

STEM-G4-SCI-T04-Q04 | States of Matter and Their Changes | Difficulty: Intermediate

The change from gas to liquid is called ...

A. Sublimation

B. Melting

C. Evaporation

D. Condensation

Answer: D. Condensation occurs when a gas loses energy and becomes liquid.

STEM-G4-SCI-T04-Q05 | States of Matter and Their Changes | Difficulty: Intermediate

The change from gas to liquid is called ...

A. Condensation

B. Melting

C. Evaporation

D. Sublimation

Answer: A. Condensation occurs when a gas loses energy and becomes liquid.

STEM-G4-SCI-T04-Q06 | States of Matter and Their Changes | Difficulty: Intermediate

The change from gas to liquid is called ...

A. Evaporation

B. Condensation

C. Sublimation

D. Melting

Answer: B. Condensation occurs when a gas loses energy and becomes liquid.

STEM-G4-SCI-T04-Q07 | States of Matter and Their Changes | Difficulty: Intermediate

The change from gas to liquid is called ...

A. Evaporation

B. Condensation

C. Sublimation

D. Melting

Answer: B. Condensation occurs when a gas loses energy and becomes liquid.

STEM-G4-SCI-T04-Q08 | States of Matter and Their Changes | Difficulty: Intermediate

The change from gas to liquid is called ...

A. Condensation

B. Melting

C. Sublimation

D. Evaporation

Answer: A. Condensation occurs when a gas loses energy and becomes liquid.

STEM-G4-SCI-T04-Q09 | States of Matter and Their Changes | Difficulty: Intermediate

The change from gas to liquid is called ...

A. Melting

B. Sublimation

C. Condensation

D. Evaporation

Answer: C. Condensation occurs when a gas loses energy and becomes liquid.

STEM-G4-SCI-T04-Q10 | States of Matter and Their Changes | Difficulty: Intermediate

The change from gas to liquid is called ...

A. Melting

B. Condensation

C. Evaporation

D. Sublimation

Answer: B. Condensation occurs when a gas loses energy and becomes liquid.

Engineering

Theme 1: Forces in Design — 10 questions

STEM-G4-ENG-T01-Q01 | Forces in Design | Difficulty: Intermediate

Why should joints in a model be inspected?

A. Joints transfer forces between parts

B. Joints make mass zero

C. Joints are only decoration

D. Joints remove gravity

Answer: A. Weak joints can become failure points in a structure.

STEM-G4-ENG-T01-Q02 | Forces in Design | Difficulty: Intermediate

After a test reveals a weakness, what should engineers do next?

A. Hide the result

B. Improve the design and test it again

C. Ignore safety

D. Delete the notes

Answer: B. Engineering is iterative and uses test results to guide improvements.

STEM-G4-ENG-T01-Q03 | Forces in Design | Difficulty: Intermediate

Which measurement is useful when comparing two bridge designs?

A. The number of stickers

B. The model's name

C. The bridge color

D. The maximum load each bridge can support

Answer: D. Load capacity is a relevant measure of bridge performance.

STEM-G4-ENG-T01-Q04 | Forces in Design | Difficulty: Intermediate

Why should an engineering prototype be tested before use?

A. To avoid making improvements

B. To make it look more expensive

C. To find out whether the design works and is safe

D. To make it larger every time

Answer: C. Testing reveals weaknesses and helps improve a design.

STEM-G4-ENG-T01-Q05 | Forces in Design | Difficulty: Intermediate

If a model structure tips over easily, what is a sensible first improvement?

A. Remove the base

B. Remove all supports

C. Add weight to the top

D. Widen the base or strengthen the joints

Answer: D. A wider base and strong joints can improve stability.

STEM-G4-ENG-T01-Q06 | Forces in Design | Difficulty: Intermediate

What is the purpose of making a sketch before building an object?

A. To plan its shape, dimensions, and parts

B. To avoid measuring

C. To replace all testing

D. To make the object build itself

Answer: A. A sketch communicates a design before construction.

STEM-G4-ENG-T01-Q07 | Forces in Design | Difficulty: Intermediate

For a fair test, a team should ...

A. Change one factor while keeping other factors the same

B. Choose only the preferred result

C. Avoid recording results

D. Change every factor at once

Answer: A. Controlling variables helps identify what caused a result.

STEM-G4-ENG-T01-Q08 | Forces in Design | Difficulty: Intermediate

When selecting a material for a load-bearing part, what should be considered?

A. Its strength and suitability for the task

B. Only its favorite color

C. The packaging

D. The designer's name

Answer: A. Material choice depends on the load and operating conditions.

STEM-G4-ENG-T01-Q09 | Forces in Design | Difficulty: Intermediate

Why should joints in a model be inspected?

A. Joints remove gravity

B. Joints transfer forces between parts

C. Joints make mass zero

D. Joints are only decoration

Answer: B. Weak joints can become failure points in a structure.

STEM-G4-ENG-T01-Q10 | Forces in Design | Difficulty: Intermediate

After a test reveals a weakness, what should engineers do next?

A. Hide the result

B. Delete the notes

C. Improve the design and test it again

D. Ignore safety

Answer: C. Engineering is iterative and uses test results to guide improvements.

Theme 2: Frames and Structural Stability — 10 questions

STEM-G4-ENG-T02-Q01 | Frames and Structural Stability | Difficulty: Intermediate

Why should joints in a model be inspected?

A. Joints transfer forces between parts

B. Joints remove gravity

C. Joints make mass zero

D. Joints are only decoration

Answer: A. Weak joints can become failure points in a structure.

STEM-G4-ENG-T02-Q02 | Frames and Structural Stability | Difficulty: Intermediate

After a test reveals a weakness, what should engineers do next?

A. Delete the notes

B. Ignore safety

C. Hide the result

D. Improve the design and test it again

Answer: D. Engineering is iterative and uses test results to guide improvements.

STEM-G4-ENG-T02-Q03 | Frames and Structural Stability | Difficulty: Intermediate

Which measurement is useful when comparing two bridge designs?

A. The number of stickers

B. The bridge color

C. The model's name

D. The maximum load each bridge can support

Answer: D. Load capacity is a relevant measure of bridge performance.

STEM-G4-ENG-T02-Q04 | Frames and Structural Stability | Difficulty: Intermediate

Why should an engineering prototype be tested before use?

A. To make it look more expensive

B. To make it larger every time

C. To avoid making improvements

D. To find out whether the design works and is safe

Answer: D. Testing reveals weaknesses and helps improve a design.

STEM-G4-ENG-T02-Q05 | Frames and Structural Stability | Difficulty: Intermediate

If a model structure tips over easily, what is a sensible first improvement?

A. Add weight to the top

B. Widen the base or strengthen the joints

C. Remove all supports

D. Remove the base

Answer: B. A wider base and strong joints can improve stability.

STEM-G4-ENG-T02-Q06 | Frames and Structural Stability | Difficulty: Intermediate

What is the purpose of making a sketch before building an object?

A. To replace all testing

B. To plan its shape, dimensions, and parts

C. To avoid measuring

D. To make the object build itself

Answer: B. A sketch communicates a design before construction.

STEM-G4-ENG-T02-Q07 | Frames and Structural Stability | Difficulty: Intermediate

For a fair test, a team should ...

A. Choose only the preferred result

B. Change every factor at once

C. Change one factor while keeping other factors the same

D. Avoid recording results

Answer: C. Controlling variables helps identify what caused a result.

STEM-G4-ENG-T02-Q08 | Frames and Structural Stability | Difficulty: Intermediate

When selecting a material for a load-bearing part, what should be considered?

A. Only its favorite color

B. The packaging

C. Its strength and suitability for the task

D. The designer's name

Answer: C. Material choice depends on the load and operating conditions.

STEM-G4-ENG-T02-Q09 | Frames and Structural Stability | Difficulty: Intermediate

Why should joints in a model be inspected?

A. Joints are only decoration

B. Joints make mass zero

C. Joints remove gravity

D. Joints transfer forces between parts

Answer: D. Weak joints can become failure points in a structure.

STEM-G4-ENG-T02-Q10 | Frames and Structural Stability | Difficulty: Intermediate

After a test reveals a weakness, what should engineers do next?

A. Hide the result

B. Delete the notes

C. Improve the design and test it again

D. Ignore safety

Answer: C. Engineering is iterative and uses test results to guide improvements.

Theme 3: Simple Electrical Circuits — 10 questions

STEM-G4-ENG-T03-Q01 | Simple Electrical Circuits | Difficulty: Intermediate

Why should joints in a model be inspected?

A. Joints are only decoration

B. Joints transfer forces between parts

C. Joints remove gravity

D. Joints make mass zero

Answer: B. Weak joints can become failure points in a structure.

STEM-G4-ENG-T03-Q02 | Simple Electrical Circuits | Difficulty: Intermediate

After a test reveals a weakness, what should engineers do next?

A. Hide the result

B. Delete the notes

C. Ignore safety

D. Improve the design and test it again

Answer: D. Engineering is iterative and uses test results to guide improvements.

STEM-G4-ENG-T03-Q03 | Simple Electrical Circuits | Difficulty: Intermediate

Which measurement is useful when comparing two bridge designs?

A. The number of stickers

B. The model's name

C. The maximum load each bridge can support

D. The bridge color

Answer: C. Load capacity is a relevant measure of bridge performance.

STEM-G4-ENG-T03-Q04 | Simple Electrical Circuits | Difficulty: Intermediate

Why should an engineering prototype be tested before use?

A. To find out whether the design works and is safe

B. To make it larger every time

C. To make it look more expensive

D. To avoid making improvements

Answer: A. Testing reveals weaknesses and helps improve a design.

STEM-G4-ENG-T03-Q05 | Simple Electrical Circuits | Difficulty: Intermediate

If a model structure tips over easily, what is a sensible first improvement?

A. Remove the base

B. Remove all supports

C. Add weight to the top

D. Widen the base or strengthen the joints

Answer: D. A wider base and strong joints can improve stability.

STEM-G4-ENG-T03-Q06 | Simple Electrical Circuits | Difficulty: Intermediate

What is the purpose of making a sketch before building an object?

A. To replace all testing

B. To make the object build itself

C. To plan its shape, dimensions, and parts

D. To avoid measuring

Answer: C. A sketch communicates a design before construction.

STEM-G4-ENG-T03-Q07 | Simple Electrical Circuits | Difficulty: Intermediate

For a fair test, a team should ...

A. Change one factor while keeping other factors the same

B. Choose only the preferred result

C. Avoid recording results

D. Change every factor at once

Answer: A. Controlling variables helps identify what caused a result.

STEM-G4-ENG-T03-Q08 | Simple Electrical Circuits | Difficulty: Intermediate

When selecting a material for a load-bearing part, what should be considered?

A. Its strength and suitability for the task

B. The designer's name

C. Only its favorite color

D. The packaging

Answer: A. Material choice depends on the load and operating conditions.

STEM-G4-ENG-T03-Q09 | Simple Electrical Circuits | Difficulty: Intermediate

Why should joints in a model be inspected?

A. Joints transfer forces between parts

B. Joints are only decoration

C. Joints make mass zero

D. Joints remove gravity

Answer: A. Weak joints can become failure points in a structure.

STEM-G4-ENG-T03-Q10 | Simple Electrical Circuits | Difficulty: Intermediate

After a test reveals a weakness, what should engineers do next?

A. Improve the design and test it again

B. Delete the notes

C. Hide the result

D. Ignore safety

Answer: A. Engineering is iterative and uses test results to guide improvements.

Theme 4: The Engineering Design Process — 10 questions

STEM-G4-ENG-T04-Q01 | The Engineering Design Process | Difficulty: Intermediate

Why should joints in a model be inspected?

A. Joints make mass zero

B. Joints transfer forces between parts

C. Joints are only decoration

D. Joints remove gravity

Answer: B. Weak joints can become failure points in a structure.

STEM-G4-ENG-T04-Q02 | The Engineering Design Process | Difficulty: Intermediate

After a test reveals a weakness, what should engineers do next?

A. Delete the notes

B. Ignore safety

C. Improve the design and test it again

D. Hide the result

Answer: C. Engineering is iterative and uses test results to guide improvements.

STEM-G4-ENG-T04-Q03 | The Engineering Design Process | Difficulty: Intermediate

Which measurement is useful when comparing two bridge designs?

A. The number of stickers

B. The model's name

C. The maximum load each bridge can support

D. The bridge color

Answer: C. Load capacity is a relevant measure of bridge performance.

STEM-G4-ENG-T04-Q04 | The Engineering Design Process | Difficulty: Intermediate

Why should an engineering prototype be tested before use?

A. To avoid making improvements

B. To make it larger every time

C. To make it look more expensive

D. To find out whether the design works and is safe

Answer: D. Testing reveals weaknesses and helps improve a design.

STEM-G4-ENG-T04-Q05 | The Engineering Design Process | Difficulty: Intermediate

If a model structure tips over easily, what is a sensible first improvement?

A. Remove all supports

B. Widen the base or strengthen the joints

C. Remove the base

D. Add weight to the top

Answer: B. A wider base and strong joints can improve stability.

STEM-G4-ENG-T04-Q06 | The Engineering Design Process | Difficulty: Intermediate

What is the purpose of making a sketch before building an object?

A. To make the object build itself

B. To plan its shape, dimensions, and parts

C. To replace all testing

D. To avoid measuring

Answer: B. A sketch communicates a design before construction.

STEM-G4-ENG-T04-Q07 | The Engineering Design Process | Difficulty: Intermediate

For a fair test, a team should ...

A. Change one factor while keeping other factors the same

B. Avoid recording results

C. Choose only the preferred result

D. Change every factor at once

Answer: A. Controlling variables helps identify what caused a result.

STEM-G4-ENG-T04-Q08 | The Engineering Design Process | Difficulty: Intermediate

When selecting a material for a load-bearing part, what should be considered?

A. Its strength and suitability for the task

B. The packaging

C. Only its favorite color

D. The designer's name

Answer: A. Material choice depends on the load and operating conditions.

STEM-G4-ENG-T04-Q09 | The Engineering Design Process | Difficulty: Intermediate

Why should joints in a model be inspected?

A. Joints make mass zero

B. Joints transfer forces between parts

C. Joints are only decoration

D. Joints remove gravity

Answer: B. Weak joints can become failure points in a structure.

STEM-G4-ENG-T04-Q10 | The Engineering Design Process | Difficulty: Intermediate

After a test reveals a weakness, what should engineers do next?

A. Delete the notes

B. Ignore safety

C. Improve the design and test it again

D. Hide the result

Answer: C. Engineering is iterative and uses test results to guide improvements.

ICT (Technology)

Theme 1: Operating Systems and Applications — 10 questions

STEM-G4-ICT-T01-Q01 | Operating Systems and Applications | Difficulty: Intermediate

Software designed to help a user perform a task is commonly called an ...

A. Power cable

B. Desk

C. Application

D. Fan

Answer: C. An application is software that helps users perform specific tasks.

STEM-G4-ICT-T01-Q02 | Operating Systems and Applications | Difficulty: Intermediate

What should you do if an unknown person sends you a suspicious link?

A. Enter your password

B. Download every attachment

C. Do not open it; check with a trusted adult or person

D. Forward it to every contact

Answer: C. Suspicious links may lead to scams or harmful software.

STEM-G4-ICT-T01-Q03 | Operating Systems and Applications | Difficulty: Intermediate

Why should you keep your password private?

A. To extend battery life

B. To protect your account and personal information

C. To make the internet faster

D. To make the screen brighter

Answer: B. A password provides access to an account and should be protected.

STEM-G4-ICT-T01-Q04 | Operating Systems and Applications | Difficulty: Intermediate

A clear, ordered set of steps for solving a problem is called an ...

A. Resolution

B. Speaker

C. Algorithm

D. Wallpaper

Answer: C. An algorithm is an ordered set of instructions.

STEM-G4-ICT-T01-Q05 | Operating Systems and Applications | Difficulty: Intermediate

In a spreadsheet, a cell is ...

A. An application icon

B. The intersection of a row and a column

C. A webpage

D. A type of cable

Answer: B. A cell is a box at the intersection of a row and a column.

STEM-G4-ICT-T01-Q06 | Operating Systems and Applications | Difficulty: Intermediate

How should information found online be checked?

A. Count the emojis

B. Share it before reading

C. Compare sources and supporting evidence

D. Choose the most dramatic headline

Answer: C. Checking sources helps assess accuracy and credibility.

STEM-G4-ICT-T01-Q07 | Operating Systems and Applications | Difficulty: Intermediate

Finding and fixing errors in a program is called ...

A. Debugging

B. Streaming

C. Browsing

D. Formatting

Answer: A. Debugging is the process of locating and correcting errors.

STEM-G4-ICT-T01-Q08 | Operating Systems and Applications | Difficulty: Intermediate

Which device is commonly used to enter text into a computer?

A. Speaker

B. Projector

C. Keyboard

D. Monitor

Answer: C. A keyboard is an input device used for typing.

STEM-G4-ICT-T01-Q09 | Operating Systems and Applications | Difficulty: Intermediate

Software designed to help a user perform a task is commonly called an ...

A. Fan

B. Power cable

C. Desk

D. Application

Answer: D. An application is software that helps users perform specific tasks.

STEM-G4-ICT-T01-Q10 | Operating Systems and Applications | Difficulty: Intermediate

What should you do if an unknown person sends you a suspicious link?

A. Do not open it; check with a trusted adult or person

B. Enter your password

C. Forward it to every contact

D. Download every attachment

Answer: A. Suspicious links may lead to scams or harmful software.

Theme 2: Digital Documents and Presentations — 10 questions

STEM-G4-ICT-T02-Q01 | Digital Documents and Presentations | Difficulty: Intermediate

Software designed to help a user perform a task is commonly called an ...

A. Desk

B. Fan

C. Power cable

D. Application

Answer: D. An application is software that helps users perform specific tasks.

STEM-G4-ICT-T02-Q02 | Digital Documents and Presentations | Difficulty: Intermediate

What should you do if an unknown person sends you a suspicious link?

A. Download every attachment

B. Forward it to every contact

C. Do not open it; check with a trusted adult or person

D. Enter your password

Answer: C. Suspicious links may lead to scams or harmful software.

STEM-G4-ICT-T02-Q03 | Digital Documents and Presentations | Difficulty: Intermediate

Why should you keep your password private?

A. To protect your account and personal information

B. To make the internet faster

C. To extend battery life

D. To make the screen brighter

Answer: A. A password provides access to an account and should be protected.

STEM-G4-ICT-T02-Q04 | Digital Documents and Presentations | Difficulty: Intermediate

A clear, ordered set of steps for solving a problem is called an ...

A. Wallpaper

B. Algorithm

C. Resolution

D. Speaker

Answer: B. An algorithm is an ordered set of instructions.

STEM-G4-ICT-T02-Q05 | Digital Documents and Presentations | Difficulty: Intermediate

In a spreadsheet, a cell is ...

A. An application icon

B. A webpage

C. A type of cable

D. The intersection of a row and a column

Answer: D. A cell is a box at the intersection of a row and a column.

STEM-G4-ICT-T02-Q06 | Digital Documents and Presentations | Difficulty: Intermediate

How should information found online be checked?

A. Choose the most dramatic headline

B. Compare sources and supporting evidence

C. Share it before reading

D. Count the emojis

Answer: B. Checking sources helps assess accuracy and credibility.

STEM-G4-ICT-T02-Q07 | Digital Documents and Presentations | Difficulty: Intermediate

Finding and fixing errors in a program is called ...

A. Streaming

B. Formatting

C. Browsing

D. Debugging

Answer: D. Debugging is the process of locating and correcting errors.

STEM-G4-ICT-T02-Q08 | Digital Documents and Presentations | Difficulty: Intermediate

Which device is commonly used to enter text into a computer?

A. Monitor

B. Keyboard

C. Speaker

D. Projector

Answer: B. A keyboard is an input device used for typing.

STEM-G4-ICT-T02-Q09 | Digital Documents and Presentations | Difficulty: Intermediate

Software designed to help a user perform a task is commonly called an ...

A. Fan

B. Application

C. Desk

D. Power cable

Answer: B. An application is software that helps users perform specific tasks.

STEM-G4-ICT-T02-Q10 | Digital Documents and Presentations | Difficulty: Intermediate

What should you do if an unknown person sends you a suspicious link?

A. Enter your password

B. Do not open it; check with a trusted adult or person

C. Forward it to every contact

D. Download every attachment

Answer: B. Suspicious links may lead to scams or harmful software.

Theme 3: Reliable Information and Misinformation — 10 questions

STEM-G4-ICT-T03-Q01 | Reliable Information and Misinformation | Difficulty: Intermediate

Software designed to help a user perform a task is commonly called an ...

A. Fan

B. Power cable

C. Desk

D. Application

Answer: D. An application is software that helps users perform specific tasks.

STEM-G4-ICT-T03-Q02 | Reliable Information and Misinformation | Difficulty: Intermediate

What should you do if an unknown person sends you a suspicious link?

A. Do not open it; check with a trusted adult or person

B. Forward it to every contact

C. Enter your password

D. Download every attachment

Answer: A. Suspicious links may lead to scams or harmful software.

STEM-G4-ICT-T03-Q03 | Reliable Information and Misinformation | Difficulty: Intermediate

Why should you keep your password private?

A. To make the internet faster

B. To protect your account and personal information

C. To extend battery life

D. To make the screen brighter

Answer: B. A password provides access to an account and should be protected.

STEM-G4-ICT-T03-Q04 | Reliable Information and Misinformation | Difficulty: Intermediate

A clear, ordered set of steps for solving a problem is called an ...

A. Wallpaper

B. Algorithm

C. Resolution

D. Speaker

Answer: B. An algorithm is an ordered set of instructions.

STEM-G4-ICT-T03-Q05 | Reliable Information and Misinformation | Difficulty: Intermediate

In a spreadsheet, a cell is ...

A. An application icon

B. A type of cable

C. The intersection of a row and a column

D. A webpage

Answer: C. A cell is a box at the intersection of a row and a column.

STEM-G4-ICT-T03-Q06 | Reliable Information and Misinformation | Difficulty: Intermediate

How should information found online be checked?

A. Share it before reading

B. Choose the most dramatic headline

C. Compare sources and supporting evidence

D. Count the emojis

Answer: C. Checking sources helps assess accuracy and credibility.

STEM-G4-ICT-T03-Q07 | Reliable Information and Misinformation | Difficulty: Intermediate

Finding and fixing errors in a program is called ...

A. Debugging

B. Streaming

C. Formatting

D. Browsing

Answer: A. Debugging is the process of locating and correcting errors.

STEM-G4-ICT-T03-Q08 | Reliable Information and Misinformation | Difficulty: Intermediate

Which device is commonly used to enter text into a computer?

A. Speaker

B. Monitor

C. Projector

D. Keyboard

Answer: D. A keyboard is an input device used for typing.

STEM-G4-ICT-T03-Q09 | Reliable Information and Misinformation | Difficulty: Intermediate

Software designed to help a user perform a task is commonly called an ...

A. Application

B. Fan

C. Desk

D. Power cable

Answer: A. An application is software that helps users perform specific tasks.

STEM-G4-ICT-T03-Q10 | Reliable Information and Misinformation | Difficulty: Intermediate

What should you do if an unknown person sends you a suspicious link?

A. Forward it to every contact

B. Download every attachment

C. Enter your password

D. Do not open it; check with a trusted adult or person

Answer: D. Suspicious links may lead to scams or harmful software.

Theme 4: Computational Thinking and Flowcharts — 10 questions

STEM-G4-ICT-T04-Q01 | Computational Thinking and Flowcharts | Difficulty: Intermediate

Software designed to help a user perform a task is commonly called an ...

A. Application

B. Fan

C. Desk

D. Power cable

Answer: A. An application is software that helps users perform specific tasks.

STEM-G4-ICT-T04-Q02 | Computational Thinking and Flowcharts | Difficulty: Intermediate

What should you do if an unknown person sends you a suspicious link?

A. Enter your password

B. Forward it to every contact

C. Do not open it; check with a trusted adult or person

D. Download every attachment

Answer: C. Suspicious links may lead to scams or harmful software.

STEM-G4-ICT-T04-Q03 | Computational Thinking and Flowcharts | Difficulty: Intermediate

Why should you keep your password private?

A. To make the internet faster

B. To make the screen brighter

C. To protect your account and personal information

D. To extend battery life

Answer: C. A password provides access to an account and should be protected.

STEM-G4-ICT-T04-Q04 | Computational Thinking and Flowcharts | Difficulty: Intermediate

A clear, ordered set of steps for solving a problem is called an ...

A. Speaker

B. Algorithm

C. Resolution

D. Wallpaper

Answer: B. An algorithm is an ordered set of instructions.

STEM-G4-ICT-T04-Q05 | Computational Thinking and Flowcharts | Difficulty: Intermediate

In a spreadsheet, a cell is ...

A. A webpage

B. A type of cable

C. An application icon

D. The intersection of a row and a column

Answer: D. A cell is a box at the intersection of a row and a column.

STEM-G4-ICT-T04-Q06 | Computational Thinking and Flowcharts | Difficulty: Intermediate

How should information found online be checked?

A. Compare sources and supporting evidence

B. Count the emojis

C. Share it before reading

D. Choose the most dramatic headline

Answer: A. Checking sources helps assess accuracy and credibility.

STEM-G4-ICT-T04-Q07 | Computational Thinking and Flowcharts | Difficulty: Intermediate

Finding and fixing errors in a program is called ...

A. Debugging

B. Formatting

C. Browsing

D. Streaming

Answer: A. Debugging is the process of locating and correcting errors.

STEM-G4-ICT-T04-Q08 | Computational Thinking and Flowcharts | Difficulty: Intermediate

Which device is commonly used to enter text into a computer?

A. Monitor

B. Projector

C. Keyboard

D. Speaker

Answer: C. A keyboard is an input device used for typing.

STEM-G4-ICT-T04-Q09 | Computational Thinking and Flowcharts | Difficulty: Intermediate

Software designed to help a user perform a task is commonly called an ...

A. Application

B. Fan

C. Desk

D. Power cable

Answer: A. An application is software that helps users perform specific tasks.

STEM-G4-ICT-T04-Q10 | Computational Thinking and Flowcharts | Difficulty: Intermediate

What should you do if an unknown person sends you a suspicious link?

A. Download every attachment

B. Enter your password

C. Do not open it; check with a trusted adult or person

D. Forward it to every contact

Answer: C. Suspicious links may lead to scams or harmful software.

Elementary School Grade 5

160 questions • 4 subjects • 4 themes per subject • 10 questions per theme

Mathematics

Theme 1: Fractions and Decimals — 10 questions

STEM-G5-MAT-T01-Q01 | Fractions and Decimals | Difficulty: Intermediate

Calculate $9^2 + 7$.

A. 85

B. 91

C. 88

D. 98

Answer: C. $9^2 = 81$, then 81 + 7 = 88.

STEM-G5-MAT-T01-Q02 | Fractions and Decimals | Difficulty: Intermediate

If a = 18 and b = 12, calculate 2a + 3b.

A. 69

B. 72

C. 75

D. 82

Answer: B. 2(18) + 3(12) = 36 + 36 = 72.

STEM-G5-MAT-T01-Q03 | Fractions and Decimals | Difficulty: Intermediate

A rectangle is 16 cm long and 6 cm wide. What is its perimeter?

A. 54

B. 41

C. 44

D. 47

Answer: C. Perimeter = 2(length + width) = 2(16 + 6) = 44 cm.

STEM-G5-MAT-T01-Q04 | Fractions and Decimals | Difficulty: Intermediate

A rectangular area measures 14 cm by 5 cm. What is its area?

A. 70

B. 73

C. 67

D. 80

Answer: A. Area = 14 × 5 = 70 cm^2.

STEM-G5-MAT-T01-Q05 | Fractions and Decimals | Difficulty: Intermediate

If x = 7 and y = 4, what is x × y?

A. 25

B. 31

C. 28

D. 38

Answer: C. Substitute the values: 7 × 4 = 28.

STEM-G5-MAT-T01-Q06 | Fractions and Decimals | Difficulty: Intermediate

Calculate $8^2 + 6$.

A. 73

B. 80

C. 67

D. 70

Answer: D. $8^2 = 64$, then 64 + 6 = 70.

STEM-G5-MAT-T01-Q07 | Fractions and Decimals | Difficulty: Intermediate

If a = 3 and b = 7, calculate 2a + 3b.

A. 37

B. 30

C. 24

D. 27

Answer: D. 2(3) + 3(7) = 6 + 21 = 27.

STEM-G5-MAT-T01-Q08 | Fractions and Decimals | Difficulty: Intermediate

A rectangle is 5 cm long and 7 cm wide. What is its perimeter?

A. 27

B. 21

C. 34

D. 24

Answer: D. Perimeter = 2(length + width) = 2(5 + 7) = 24 cm.

STEM-G5-MAT-T01-Q09 | Fractions and Decimals | Difficulty: Intermediate

A rectangular area measures 8 cm by 12 cm. What is its area?

A. 99

B. 93

C. 96

D. 106

Answer: C. Area = 8 × 12 = 96 cm².

STEM-G5-MAT-T01-Q10 | Fractions and Decimals | Difficulty: Intermediate

If x = 7 and y = 9, what is x × y?

A. 66

B. 63

C. 60

D. 73

Answer: B. Substitute the values: 7 × 9 = 63.

Theme 2: Percentages and Ratios — 10 questions

STEM-G5-MAT-T02-Q01 | Percentages and Ratios | Difficulty: Intermediate

Calculate 16² + 6.

A. 259

B. 265

C. 262

D. 272

Answer: C. 16² = 256, then 256 + 6 = 262.

STEM-G5-MAT-T02-Q02 | Percentages and Ratios | Difficulty: Intermediate

If a = 18 and b = 3, calculate 2a + 3b.

A. 55

B. 42

C. 48

D. 45

Answer: D. 2(18) + 3(3) = 36 + 9 = 45.

STEM-G5-MAT-T02-Q03 | Percentages and Ratios | Difficulty: Intermediate

A rectangle is 12 cm long and 12 cm wide. What is its perimeter?

A. 51

B. 48

C. 45

D. 58

Answer: B. Perimeter = 2(length + width) = 2(12 + 12) = 48 cm.

STEM-G5-MAT-T02-Q04 | Percentages and Ratios | Difficulty: Intermediate

A rectangular area measures 15 cm by 6 cm. What is its area?

A. 100

B. 93

C. 90

D. 87

Answer: C. Area = 15 × 6 = 90 cm².

STEM-G5-MAT-T02-Q05 | Percentages and Ratios | Difficulty: Intermediate

If x = 8 and y = 4, what is x × y?

A. 32

B. 29

C. 42

D. 35

Answer: A. Substitute the values: 8 × 4 = 32.

STEM-G5-MAT-T02-Q06 | Percentages and Ratios | Difficulty: Intermediate

Calculate $8^2 + 11$.

A. 72

B. 78

C. 75

D. 85

Answer: C. $8^2 = 64$, then 64 + 11 = 75.

STEM-G5-MAT-T02-Q07 | Percentages and Ratios | Difficulty: Intermediate

If a = 7 and b = 2, calculate 2a + 3b.

A. 17

B. 23

C. 20

D. 30

Answer: C. 2(7) + 3(2) = 14 + 6 = 20.

STEM-G5-MAT-T02-Q08 | Percentages and Ratios | Difficulty: Intermediate

A rectangle is 15 cm long and 8 cm wide. What is its perimeter?

A. 43

B. 49

C. 56

D. 46

Answer: D. Perimeter = 2(length + width) = 2(15 + 8) = 46 cm.

STEM-G5-MAT-T02-Q09 | Percentages and Ratios | Difficulty: Intermediate

A rectangular area measures 4 cm by 7 cm. What is its area?

A. 31

B. 28

C. 38

D. 25

Answer: B. Area = 4 × 7 = 28 cm².

STEM-G5-MAT-T02-Q10 | Percentages and Ratios | Difficulty: Intermediate

If x = 6 and y = 7, what is x × y?

A. 52

B. 42

C. 39

D. 45

Answer: B. Substitute the values: 6 × 7 = 42.

Theme 3: Volume of Solid Shapes — 10 questions

STEM-G5-MAT-T03-Q01 | Volume of Solid Shapes | Difficulty: Intermediate

Calculate $18^2 + 6$.

A. 327

B. 340

C. 333

D. 330

Answer: D. $18^2 = 324$, then $324 + 6 = 330$.

STEM-G5-MAT-T03-Q02 | Volume of Solid Shapes | Difficulty: Intermediate

If a = 7 and b = 5, calculate 2a + 3b.

A. 32

B. 26

C. 29

D. 39

Answer: C. 2(7) + 3(5) = 14 + 15 = 29.

STEM-G5-MAT-T03-Q03 | Volume of Solid Shapes | Difficulty: Intermediate

A rectangle is 11 cm long and 8 cm wide. What is its perimeter?

A. 35

B. 41

C. 38

D. 48

Answer: C. Perimeter = 2(length + width) = 2(11 + 8) = 38 cm.

STEM-G5-MAT-T03-Q04 | Volume of Solid Shapes | Difficulty: Intermediate

A rectangular area measures 14 cm by 8 cm. What is its area?

A. 115

B. 109

C. 112

D. 122

Answer: C. Area = 14 × 8 = 112 cm^2.

STEM-G5-MAT-T03-Q05 | Volume of Solid Shapes | Difficulty: Intermediate

If x = 18 and y = 10, what is x × y?

A. 190

B. 180

C. 177

D. 183

Answer: B. Substitute the values: 18 × 10 = 180.

STEM-G5-MAT-T03-Q06 | Volume of Solid Shapes | Difficulty: Intermediate

Calculate $11^2 + 8$.

A. 129

B. 132

C. 139

D. 126

Answer: A. $11^2 = 121$, then $121 + 8 = 129$.

STEM-G5-MAT-T03-Q07 | Volume of Solid Shapes | Difficulty: Intermediate

If a = 6 and b = 3, calculate 2a + 3b.

A. 24

B. 31

C. 18

D. 21

Answer: D. 2(6) + 3(3) = 12 + 9 = 21.

STEM-G5-MAT-T03-Q08 | Volume of Solid Shapes | Difficulty: Intermediate

A rectangle is 17 cm long and 7 cm wide. What is its perimeter?

A. 58

B. 45

C. 51

D. 48

Answer: D. Perimeter = 2(length + width) = 2(17 + 7) = 48 cm.

STEM-G5-MAT-T03-Q09 | Volume of Solid Shapes | Difficulty: Intermediate

A rectangular area measures 4 cm by 3 cm. What is its area?

A. 12

B. 9

C. 22

D. 15

Answer: A. Area = 4 × 3 = 12 cm².

STEM-G5-MAT-T03-Q10 | Volume of Solid Shapes | Difficulty: Intermediate

If x = 16 and y = 3, what is x × y?

A. 51

B. 58

C. 45

D. 48

Answer: D. Substitute the values: 16 × 3 = 48.

Theme 4: Coordinates and Data Handling — 10 questions

STEM-G5-MAT-T04-Q01 | Coordinates and Data Handling | Difficulty: Intermediate

Calculate $7^2 + 2$.

A. 54

B. 48

C. 51

D. 61

Answer: C. $7^2 = 49$, then 49 + 2 = 51.

STEM-G5-MAT-T04-Q02 | Coordinates and Data Handling | Difficulty: Intermediate

If a = 6 and b = 2, calculate 2a + 3b.

A. 15

B. 28

C. 18

D. 21

Answer: C. 2(6) + 3(2) = 12 + 6 = 18.

STEM-G5-MAT-T04-Q03 | Coordinates and Data Handling | Difficulty: Intermediate

A rectangle is 11 cm long and 12 cm wide. What is its perimeter?

A. 46

B. 49

C. 43

D. 56

Answer: A. Perimeter = 2(length + width) = 2(11 + 12) = 46 cm.

STEM-G5-MAT-T04-Q04 | Coordinates and Data Handling | Difficulty: Intermediate

A rectangular area measures 18 cm by 7 cm. What is its area?

A. 129

B. 126

C. 123

D. 136

Answer: B. Area = 18 × 7 = 126 cm².

STEM-G5-MAT-T04-Q05 | Coordinates and Data Handling | Difficulty: Intermediate

If x = 11 and y = 11, what is x × y?

A. 118

B. 131

C. 124

D. 121

Answer: D. Substitute the values: 11 × 11 = 121.

STEM-G5-MAT-T04-Q06 | Coordinates and Data Handling | Difficulty: Intermediate

Calculate $6^2 + 5$.

A. 51

B. 41

C. 38

D. 44

Answer: B. $6^2 = 36$, then 36 + 5 = 41.

STEM-G5-MAT-T04-Q07 | Coordinates and Data Handling | Difficulty: Intermediate

If a = 16 and b = 2, calculate 2a + 3b.

A. 38

B. 35

C. 48

D. 41

Answer: A. 2(16) + 3(2) = 32 + 6 = 38.

STEM-G5-MAT-T04-Q08 | Coordinates and Data Handling | Difficulty: Intermediate

A rectangle is 3 cm long and 8 cm wide. What is its perimeter?

A. 19

B. 25

C. 32

D. 22

Answer: D. Perimeter = 2(length + width) = 2(3 + 8) = 22 cm.

STEM-G5-MAT-T04-Q09 | Coordinates and Data Handling | Difficulty: Intermediate

A rectangular area measures 15 cm by 8 cm. What is its area?

A. 130

B. 117

C. 120

D. 123

Answer: C. Area = 15 × 8 = 120 cm^2.

STEM-G5-MAT-T04-Q10 | Coordinates and Data Handling | Difficulty: Intermediate

If x = 17 and y = 3, what is x × y?

A. 51

B. 61

C. 54

D. 48

Answer: A. Substitute the values: 17 × 3 = 51.

Science

Theme 1: Human Body Systems — 10 questions

STEM-G5-SCI-T01-Q01 | Human Body Systems | Difficulty: Intermediate

Which organ pumps blood around the body?

A. The stomach

B. The lungs

C. The small intestine

D. The heart

Answer: D. The heart pumps blood through blood vessels.

STEM-G5-SCI-T01-Q02 | Human Body Systems | Difficulty: Intermediate

Which organ pumps blood around the body?

A. The stomach

B. The heart

C. The lungs

D. The small intestine

Answer: B. The heart pumps blood through blood vessels.

STEM-G5-SCI-T01-Q03 | Human Body Systems | Difficulty: Intermediate

Which organ pumps blood around the body?

A. The stomach

B. The small intestine

C. The heart

D. The lungs

Answer: C. The heart pumps blood through blood vessels.

STEM-G5-SCI-T01-Q04 | Human Body Systems | Difficulty: Intermediate

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Answer: D. The heart pumps blood through blood vessels.

STEM-G5-SCI-T01-Q05 | Human Body Systems | Difficulty: Intermediate

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STEM-G5-SCI-T01-Q06 | Human Body Systems | Difficulty: Intermediate

Which organ pumps blood around the body?

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B. The heart

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Answer: B. The heart pumps blood through blood vessels.

STEM-G5-SCI-T01-Q07 | Human Body Systems | Difficulty: Intermediate

Which organ pumps blood around the body?

A. The small intestine

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Answer: D. The heart pumps blood through blood vessels.

STEM-G5-SCI-T01-Q08 | Human Body Systems | Difficulty: Intermediate

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Answer: A. The heart pumps blood through blood vessels.

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Which organ pumps blood around the body?

A. The stomach

B. The small intestine

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D. The heart

Answer: D. The heart pumps blood through blood vessels.

Theme 2: The Solar System and Planets — 10 questions

STEM-G5-SCI-T02-Q01 | The Solar System and Planets | Difficulty: Intermediate

Which planet is especially well known for its prominent ring system?

A. Mars

B. Mercury

C. Earth

D. Saturn

Answer: D. Saturn has a large and prominent ring system.

STEM-G5-SCI-T02-Q02 | The Solar System and Planets | Difficulty: Intermediate

Which planet is especially well known for its prominent ring system?

A. Earth

B. Saturn

C. Mercury

D. Mars

Answer: B. Saturn has a large and prominent ring system.

STEM-G5-SCI-T02-Q03 | The Solar System and Planets | Difficulty: Intermediate

Which planet is especially well known for its prominent ring system?

A. Saturn

B. Mars

C. Mercury

D. Earth

Answer: A. Saturn has a large and prominent ring system.

STEM-G5-SCI-T02-Q04 | The Solar System and Planets | Difficulty: Intermediate

Which planet is especially well known for its prominent ring system?

A. Earth

B. Mars

C. Mercury

D. Saturn

Answer: D. Saturn has a large and prominent ring system.

STEM-G5-SCI-T02-Q05 | The Solar System and Planets | Difficulty: Intermediate

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C. Earth

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STEM-G5-SCI-T02-Q06 | The Solar System and Planets | Difficulty: Intermediate

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A. Mercury

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C. Saturn

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STEM-G5-SCI-T02-Q07 | The Solar System and Planets | Difficulty: Intermediate

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C. Saturn

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Answer: C. Saturn has a large and prominent ring system.

STEM-G5-SCI-T02-Q09 | The Solar System and Planets | Difficulty: Intermediate

Which planet is especially well known for its prominent ring system?

A. Saturn

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Answer: A. Saturn has a large and prominent ring system.

STEM-G5-SCI-T02-Q10 | The Solar System and Planets | Difficulty: Intermediate

Which planet is especially well known for its prominent ring system?

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Answer: D. Saturn has a large and prominent ring system.

Theme 3: Simple Machines — 10 questions

STEM-G5-SCI-T03-Q01 | Simple Machines | Difficulty: Intermediate

Which simple machine uses a fulcrum to help lift a load?

A. A lever

B. A compass

C. A thermometer

D. A barometer

Answer: A. A lever is a rigid bar that rotates around a fulcrum.

STEM-G5-SCI-T03-Q02 | Simple Machines | Difficulty: Intermediate

Which simple machine uses a fulcrum to help lift a load?

A. A barometer

B. A thermometer

C. A lever

D. A compass

Answer: C. A lever is a rigid bar that rotates around a fulcrum.

STEM-G5-SCI-T03-Q03 | Simple Machines | Difficulty: Intermediate

Which simple machine uses a fulcrum to help lift a load?

A. A barometer

B. A lever

C. A thermometer

D. A compass

Answer: B. A lever is a rigid bar that rotates around a fulcrum.

STEM-G5-SCI-T03-Q04 | Simple Machines | Difficulty: Intermediate

Which simple machine uses a fulcrum to help lift a load?

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STEM-G5-SCI-T03-Q05 | Simple Machines | Difficulty: Intermediate

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STEM-G5-SCI-T03-Q06 | Simple Machines | Difficulty: Intermediate

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Theme 4: Food Chains and Ecosystem Balance — 10 questions

STEM-G5-SCI-T04-Q01 | Food Chains and Ecosystem Balance | Difficulty: Intermediate

If the number of predators decreases, the prey population may ...

A. Never change

B. Turn into producers

C. Always become zero

D. Increase

Answer: D. Fewer predators can reduce predation pressure on prey.

STEM-G5-SCI-T04-Q02 | Food Chains and Ecosystem Balance | Difficulty: Intermediate

If the number of predators decreases, the prey population may ...

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STEM-G5-SCI-T04-Q03 | Food Chains and Ecosystem Balance | Difficulty: Intermediate

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STEM-G5-SCI-T04-Q05 | Food Chains and Ecosystem Balance | Difficulty: Intermediate

If the number of predators decreases, the prey population may ...

A. Increase

B. Always become zero

C. Never change

D. Turn into producers

Answer: A. Fewer predators can reduce predation pressure on prey.

STEM-G5-SCI-T04-Q06 | Food Chains and Ecosystem Balance | Difficulty: Intermediate

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STEM-G5-SCI-T04-Q07 | Food Chains and Ecosystem Balance | Difficulty: Intermediate

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STEM-G5-SCI-T04-Q08 | Food Chains and Ecosystem Balance | Difficulty: Intermediate

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STEM-G5-SCI-T04-Q09 | Food Chains and Ecosystem Balance | Difficulty: Intermediate

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STEM-G5-SCI-T04-Q10 | Food Chains and Ecosystem Balance | Difficulty: Intermediate

If the number of predators decreases, the prey population may ...

A. Never change

B. Always become zero

C. Increase

D. Turn into producers

Answer: C. Fewer predators can reduce predation pressure on prey.

Engineering

Theme 1: Simple Machines — 10 questions

STEM-G5-ENG-T01-Q01 | Simple Machines | Difficulty: Intermediate

After a test reveals a weakness, what should engineers do next?

A. Ignore safety

B. Hide the result

C. Improve the design and test it again

D. Delete the notes

Answer: C. Engineering is iterative and uses test results to guide improvements.

STEM-G5-ENG-T01-Q02 | Simple Machines | Difficulty: Intermediate

Which measurement is useful when comparing two bridge designs?

A. The number of stickers

B. The bridge color

C. The model's name

D. The maximum load each bridge can support

Answer: D. Load capacity is a relevant measure of bridge performance.

STEM-G5-ENG-T01-Q03 | Simple Machines | Difficulty: Intermediate

Why should an engineering prototype be tested before use?

A. To find out whether the design works and is safe

B. To avoid making improvements

C. To make it larger every time

D. To make it look more expensive

Answer: A. Testing reveals weaknesses and helps improve a design.

STEM-G5-ENG-T01-Q04 | Simple Machines | Difficulty: Intermediate

If a model structure tips over easily, what is a sensible first improvement?

A. Add weight to the top

B. Remove all supports

C. Remove the base

D. Widen the base or strengthen the joints

Answer: D. A wider base and strong joints can improve stability.

STEM-G5-ENG-T01-Q05 | Simple Machines | Difficulty: Intermediate

What is the purpose of making a sketch before building an object?

A. To replace all testing

B. To plan its shape, dimensions, and parts

C. To make the object build itself

D. To avoid measuring

Answer: B. A sketch communicates a design before construction.

STEM-G5-ENG-T01-Q06 | Simple Machines | Difficulty: Intermediate

For a fair test, a team should ...

A. Change every factor at once

B. Choose only the preferred result

C. Change one factor while keeping other factors the same

D. Avoid recording results

Answer: C. Controlling variables helps identify what caused a result.

STEM-G5-ENG-T01-Q07 | Simple Machines | Difficulty: Intermediate

When selecting a material for a load-bearing part, what should be considered?

A. Its strength and suitability for the task

B. Only its favorite color

C. The packaging

D. The designer's name

Answer: A. Material choice depends on the load and operating conditions.

STEM-G5-ENG-T01-Q08 | Simple Machines | Difficulty: Intermediate

Why should joints in a model be inspected?

A. Joints remove gravity

B. Joints transfer forces between parts

C. Joints are only decoration

D. Joints make mass zero

Answer: B. Weak joints can become failure points in a structure.

STEM-G5-ENG-T01-Q09 | Simple Machines | Difficulty: Intermediate

After a test reveals a weakness, what should engineers do next?

A. Ignore safety

B. Delete the notes

C. Improve the design and test it again

D. Hide the result

Answer: C. Engineering is iterative and uses test results to guide improvements.

STEM-G5-ENG-T01-Q10 | Simple Machines | Difficulty: Intermediate

Which measurement is useful when comparing two bridge designs?

A. The model's name

B. The number of stickers

C. The maximum load each bridge can support

D. The bridge color

Answer: C. Load capacity is a relevant measure of bridge performance.

Theme 2: Bridges and Load Distribution — 10 questions

STEM-G5-ENG-T02-Q01 | Bridges and Load Distribution | Difficulty: Intermediate

After a test reveals a weakness, what should engineers do next?

A. Improve the design and test it again

B. Ignore safety

C. Hide the result

D. Delete the notes

Answer: A. Engineering is iterative and uses test results to guide improvements.

STEM-G5-ENG-T02-Q02 | Bridges and Load Distribution | Difficulty: Intermediate

Which measurement is useful when comparing two bridge designs?

A. The bridge color

B. The number of stickers

C. The model's name

D. The maximum load each bridge can support

Answer: D. Load capacity is a relevant measure of bridge performance.

STEM-G5-ENG-T02-Q03 | Bridges and Load Distribution | Difficulty: Intermediate

Why should an engineering prototype be tested before use?

A. To avoid making improvements

B. To find out whether the design works and is safe

C. To make it look more expensive

D. To make it larger every time

Answer: B. Testing reveals weaknesses and helps improve a design.

STEM-G5-ENG-T02-Q04 | Bridges and Load Distribution | Difficulty: Intermediate

If a model structure tips over easily, what is a sensible first improvement?

A. Remove the base

B. Add weight to the top

C. Widen the base or strengthen the joints

D. Remove all supports

Answer: C. A wider base and strong joints can improve stability.

STEM-G5-ENG-T02-Q05 | Bridges and Load Distribution | Difficulty: Intermediate

What is the purpose of making a sketch before building an object?

A. To replace all testing

B. To make the object build itself

C. To plan its shape, dimensions, and parts

D. To avoid measuring

Answer: C. A sketch communicates a design before construction.

STEM-G5-ENG-T02-Q06 | Bridges and Load Distribution | Difficulty: Intermediate

For a fair test, a team should ...

A. Avoid recording results

B. Choose only the preferred result

C. Change one factor while keeping other factors the same

D. Change every factor at once

Answer: C. Controlling variables helps identify what caused a result.

STEM-G5-ENG-T02-Q07 | Bridges and Load Distribution | Difficulty: Intermediate

When selecting a material for a load-bearing part, what should be considered?

A. Its strength and suitability for the task

B. The designer's name

C. The packaging

D. Only its favorite color

Answer: A. Material choice depends on the load and operating conditions.

STEM-G5-ENG-T02-Q08 | Bridges and Load Distribution | Difficulty: Intermediate

Why should joints in a model be inspected?

A. Joints remove gravity

B. Joints make mass zero

C. Joints transfer forces between parts

D. Joints are only decoration

Answer: C. Weak joints can become failure points in a structure.

STEM-G5-ENG-T02-Q09 | Bridges and Load Distribution | Difficulty: Intermediate

After a test reveals a weakness, what should engineers do next?

A. Delete the notes

B. Ignore safety

C. Hide the result

D. Improve the design and test it again

Answer: D. Engineering is iterative and uses test results to guide improvements.

STEM-G5-ENG-T02-Q10 | Bridges and Load Distribution | Difficulty: Intermediate

Which measurement is useful when comparing two bridge designs?

A. The number of stickers

B. The bridge color

C. The model's name

D. The maximum load each bridge can support

Answer: D. Load capacity is a relevant measure of bridge performance.

Theme 3: Energy in Mechanical Systems — 10 questions

STEM-G5-ENG-T03-Q01 | Energy in Mechanical Systems | Difficulty: Intermediate

After a test reveals a weakness, what should engineers do next?

A. Delete the notes

B. Ignore safety

C. Improve the design and test it again

D. Hide the result

Answer: C. Engineering is iterative and uses test results to guide improvements.

STEM-G5-ENG-T03-Q02 | Energy in Mechanical Systems | Difficulty: Intermediate

Which measurement is useful when comparing two bridge designs?

A. The number of stickers

B. The model's name

C. The maximum load each bridge can support

D. The bridge color

Answer: C. Load capacity is a relevant measure of bridge performance.

STEM-G5-ENG-T03-Q03 | Energy in Mechanical Systems | Difficulty: Intermediate

Why should an engineering prototype be tested before use?

A. To make it look more expensive

B. To make it larger every time

C. To find out whether the design works and is safe

D. To avoid making improvements

Answer: C. Testing reveals weaknesses and helps improve a design.

STEM-G5-ENG-T03-Q04 | Energy in Mechanical Systems | Difficulty: Intermediate

If a model structure tips over easily, what is a sensible first improvement?

A. Remove the base

B. Widen the base or strengthen the joints

C. Add weight to the top

D. Remove all supports

Answer: B. A wider base and strong joints can improve stability.

STEM-G5-ENG-T03-Q05 | Energy in Mechanical Systems | Difficulty: Intermediate

What is the purpose of making a sketch before building an object?

A. To replace all testing

B. To make the object build itself

C. To plan its shape, dimensions, and parts

D. To avoid measuring

Answer: C. A sketch communicates a design before construction.

STEM-G5-ENG-T03-Q06 | Energy in Mechanical Systems | Difficulty: Intermediate

For a fair test, a team should ...

A. Choose only the preferred result

B. Avoid recording results

C. Change one factor while keeping other factors the same

D. Change every factor at once

Answer: C. Controlling variables helps identify what caused a result.

STEM-G5-ENG-T03-Q07 | Energy in Mechanical Systems | Difficulty: Intermediate

When selecting a material for a load-bearing part, what should be considered?

A. The packaging

B. Only its favorite color

C. Its strength and suitability for the task

D. The designer's name

Answer: C. Material choice depends on the load and operating conditions.

STEM-G5-ENG-T03-Q08 | Energy in Mechanical Systems | Difficulty: Intermediate

Why should joints in a model be inspected?

A. Joints remove gravity

B. Joints are only decoration

C. Joints make mass zero

D. Joints transfer forces between parts

Answer: D. Weak joints can become failure points in a structure.

STEM-G5-ENG-T03-Q09 | Energy in Mechanical Systems | Difficulty: Intermediate

After a test reveals a weakness, what should engineers do next?

A. Delete the notes

B. Improve the design and test it again

C. Ignore safety

D. Hide the result

Answer: B. Engineering is iterative and uses test results to guide improvements.

STEM-G5-ENG-T03-Q10 | Energy in Mechanical Systems | Difficulty: Intermediate

Which measurement is useful when comparing two bridge designs?

A. The bridge color

B. The maximum load each bridge can support

C. The model's name

D. The number of stickers

Answer: B. Load capacity is a relevant measure of bridge performance.

Theme 4: Prototyping and Testing — 10 questions

STEM-G5-ENG-T04-Q01 | Prototyping and Testing | Difficulty: Intermediate

After a test reveals a weakness, what should engineers do next?

A. Delete the notes

B. Hide the result

C. Ignore safety

D. Improve the design and test it again

Answer: D. Engineering is iterative and uses test results to guide improvements.

STEM-G5-ENG-T04-Q02 | Prototyping and Testing | Difficulty: Intermediate

Which measurement is useful when comparing two bridge designs?

A. The bridge color

B. The model's name

C. The maximum load each bridge can support

D. The number of stickers

Answer: C. Load capacity is a relevant measure of bridge performance.

STEM-G5-ENG-T04-Q03 | Prototyping and Testing | Difficulty: Intermediate

Why should an engineering prototype be tested before use?

A. To make it larger every time

B. To find out whether the design works and is safe

C. To make it look more expensive

D. To avoid making improvements

Answer: B. Testing reveals weaknesses and helps improve a design.

STEM-G5-ENG-T04-Q04 | Prototyping and Testing | Difficulty: Intermediate

If a model structure tips over easily, what is a sensible first improvement?

A. Remove all supports

B. Add weight to the top

C. Widen the base or strengthen the joints

D. Remove the base

Answer: C. A wider base and strong joints can improve stability.

STEM-G5-ENG-T04-Q05 | Prototyping and Testing | Difficulty: Intermediate

What is the purpose of making a sketch before building an object?

A. To plan its shape, dimensions, and parts

B. To avoid measuring

C. To replace all testing

D. To make the object build itself

Answer: A. A sketch communicates a design before construction.

STEM-G5-ENG-T04-Q06 | Prototyping and Testing | Difficulty: Intermediate

For a fair test, a team should ...

A. Choose only the preferred result

B. Change every factor at once

C. Avoid recording results

D. Change one factor while keeping other factors the same

Answer: D. Controlling variables helps identify what caused a result.

STEM-G5-ENG-T04-Q07 | Prototyping and Testing | Difficulty: Intermediate

When selecting a material for a load-bearing part, what should be considered?

A. The packaging

B. Only its favorite color

C. The designer's name

D. Its strength and suitability for the task

Answer: D. Material choice depends on the load and operating conditions.

STEM-G5-ENG-T04-Q08 | Prototyping and Testing | Difficulty: Intermediate

Why should joints in a model be inspected?

A. Joints make mass zero

B. Joints remove gravity

C. Joints are only decoration

D. Joints transfer forces between parts

Answer: D. Weak joints can become failure points in a structure.

STEM-G5-ENG-T04-Q09 | Prototyping and Testing | Difficulty: Intermediate

After a test reveals a weakness, what should engineers do next?

A. Hide the result

B. Ignore safety

C. Improve the design and test it again

D. Delete the notes

Answer: C. Engineering is iterative and uses test results to guide improvements.

STEM-G5-ENG-T04-Q10 | Prototyping and Testing | Difficulty: Intermediate

Which measurement is useful when comparing two bridge designs?

A. The bridge color

B. The maximum load each bridge can support

C. The model's name

D. The number of stickers

Answer: B. Load capacity is a relevant measure of bridge performance.

ICT (Technology)

Theme 1: Word Processing and Presentations — 10 questions

STEM-G5-ICT-T01-Q01 | Word Processing and Presentations | Difficulty: Intermediate

Why should you keep your password private?

A. To extend battery life

B. To make the internet faster

C. To make the screen brighter

D. To protect your account and personal information

Answer: D. A password provides access to an account and should be protected.

STEM-G5-ICT-T01-Q02 | Word Processing and Presentations | Difficulty: Intermediate

A clear, ordered set of steps for solving a problem is called an ...

A. Resolution

B. Speaker

C. Wallpaper

D. Algorithm

Answer: D. An algorithm is an ordered set of instructions.

STEM-G5-ICT-T01-Q03 | Word Processing and Presentations | Difficulty: Intermediate

In a spreadsheet, a cell is ...

A. A type of cable

B. The intersection of a row and a column

C. An application icon

D. A webpage

Answer: B. A cell is a box at the intersection of a row and a column.

STEM-G5-ICT-T01-Q04 | Word Processing and Presentations | Difficulty: Intermediate

How should information found online be checked?

A. Compare sources and supporting evidence

B. Choose the most dramatic headline

C. Count the emojis

D. Share it before reading

Answer: A. Checking sources helps assess accuracy and credibility.

STEM-G5-ICT-T01-Q05 | Word Processing and Presentations | Difficulty: Intermediate

Finding and fixing errors in a program is called ...

A. Streaming

B. Formatting

C. Debugging

D. Browsing

Answer: C. Debugging is the process of locating and correcting errors.

STEM-G5-ICT-T01-Q06 | Word Processing and Presentations | Difficulty: Intermediate

Which device is commonly used to enter text into a computer?

A. Keyboard

B. Speaker

C. Projector

D. Monitor

Answer: A. A keyboard is an input device used for typing.

STEM-G5-ICT-T01-Q07 | Word Processing and Presentations | Difficulty: Intermediate

Software designed to help a user perform a task is commonly called an ...

A. Power cable

B. Desk

C. Fan

D. Application

Answer: D. An application is software that helps users perform specific tasks.

STEM-G5-ICT-T01-Q08 | Word Processing and Presentations | Difficulty: Intermediate

What should you do if an unknown person sends you a suspicious link?

A. Do not open it; check with a trusted adult or person

B. Download every attachment

C. Forward it to every contact

D. Enter your password

Answer: A. Suspicious links may lead to scams or harmful software.

STEM-G5-ICT-T01-Q09 | Word Processing and Presentations | Difficulty: Intermediate

Why should you keep your password private?

A. To extend battery life

B. To make the screen brighter

C. To make the internet faster

D. To protect your account and personal information

Answer: D. A password provides access to an account and should be protected.

STEM-G5-ICT-T01-Q10 | Word Processing and Presentations | Difficulty: Intermediate

A clear, ordered set of steps for solving a problem is called an ...

A. Speaker

B. Algorithm

C. Resolution

D. Wallpaper

Answer: B. An algorithm is an ordered set of instructions.

Theme 2: Digital Data and Tables — 10 questions

STEM-G5-ICT-T02-Q01 | Digital Data and Tables | Difficulty: Intermediate

Why should you keep your password private?

A. To protect your account and personal information

B. To extend battery life

C. To make the screen brighter

D. To make the internet faster

Answer: A. A password provides access to an account and should be protected.

STEM-G5-ICT-T02-Q02 | Digital Data and Tables | Difficulty: Intermediate

A clear, ordered set of steps for solving a problem is called an ...

A. Algorithm

B. Speaker

C. Wallpaper

D. Resolution

Answer: A. An algorithm is an ordered set of instructions.

STEM-G5-ICT-T02-Q03 | Digital Data and Tables | Difficulty: Intermediate

In a spreadsheet, a cell is ...

A. A type of cable

B. A webpage

C. The intersection of a row and a column

D. An application icon

Answer: C. A cell is a box at the intersection of a row and a column.

STEM-G5-ICT-T02-Q04 | Digital Data and Tables | Difficulty: Intermediate

How should information found online be checked?

A. Choose the most dramatic headline

B. Count the emojis

C. Share it before reading

D. Compare sources and supporting evidence

Answer: D. Checking sources helps assess accuracy and credibility.

STEM-G5-ICT-T02-Q05 | Digital Data and Tables | Difficulty: Intermediate

Finding and fixing errors in a program is called ...

A. Browsing

B. Debugging

C. Formatting

D. Streaming

Answer: B. Debugging is the process of locating and correcting errors.

STEM-G5-ICT-T02-Q06 | Digital Data and Tables | Difficulty: Intermediate

Which device is commonly used to enter text into a computer?

A. Speaker

B. Monitor

C. Projector

D. Keyboard

Answer: D. A keyboard is an input device used for typing.

STEM-G5-ICT-T02-Q07 | Digital Data and Tables | Difficulty: Intermediate

Software designed to help a user perform a task is commonly called an ...

A. Desk

B. Application

C. Power cable

D. Fan

Answer: B. An application is software that helps users perform specific tasks.

STEM-G5-ICT-T02-Q08 | Digital Data and Tables | Difficulty: Intermediate

What should you do if an unknown person sends you a suspicious link?

A. Forward it to every contact

B. Enter your password

C. Do not open it; check with a trusted adult or person

D. Download every attachment

Answer: C. Suspicious links may lead to scams or harmful software.

STEM-G5-ICT-T02-Q09 | Digital Data and Tables | Difficulty: Intermediate

Why should you keep your password private?

A. To make the screen brighter

B. To protect your account and personal information

C. To extend battery life

D. To make the internet faster

Answer: B. A password provides access to an account and should be protected.

STEM-G5-ICT-T02-Q10 | Digital Data and Tables | Difficulty: Intermediate

A clear, ordered set of steps for solving a problem is called an ...

A. Wallpaper

B. Resolution

C. Algorithm

D. Speaker

Answer: C. An algorithm is an ordered set of instructions.

Theme 3: Privacy and Passwords — 10 questions

STEM-G5-ICT-T03-Q01 | Privacy and Passwords | Difficulty: Intermediate

Why should you keep your password private?

A. To make the internet faster

B. To make the screen brighter

C. To protect your account and personal information

D. To extend battery life

Answer: C. A password provides access to an account and should be protected.

STEM-G5-ICT-T03-Q02 | Privacy and Passwords | Difficulty: Intermediate

A clear, ordered set of steps for solving a problem is called an ...

A. Resolution

B. Wallpaper

C. Algorithm

D. Speaker

Answer: C. An algorithm is an ordered set of instructions.

STEM-G5-ICT-T03-Q03 | Privacy and Passwords | Difficulty: Intermediate

In a spreadsheet, a cell is ...

A. A type of cable

B. An application icon

C. A webpage

D. The intersection of a row and a column

Answer: D. A cell is a box at the intersection of a row and a column.

STEM-G5-ICT-T03-Q04 | Privacy and Passwords | Difficulty: Intermediate

How should information found online be checked?

A. Count the emojis

B. Compare sources and supporting evidence

C. Choose the most dramatic headline

D. Share it before reading

Answer: B. Checking sources helps assess accuracy and credibility.

STEM-G5-ICT-T03-Q05 | Privacy and Passwords | Difficulty: Intermediate

Finding and fixing errors in a program is called ...

A. Formatting

B. Debugging

C. Streaming

D. Browsing

Answer: B. Debugging is the process of locating and correcting errors.

STEM-G5-ICT-T03-Q06 | Privacy and Passwords | Difficulty: Intermediate

Which device is commonly used to enter text into a computer?

A. Projector

B. Speaker

C. Monitor

D. Keyboard

Answer: D. A keyboard is an input device used for typing.

STEM-G5-ICT-T03-Q07 | Privacy and Passwords | Difficulty: Intermediate

Software designed to help a user perform a task is commonly called an ...

A. Fan

B. Application

C. Desk

D. Power cable

Answer: B. An application is software that helps users perform specific tasks.

STEM-G5-ICT-T03-Q08 | Privacy and Passwords | Difficulty: Intermediate

What should you do if an unknown person sends you a suspicious link?

A. Forward it to every contact

B. Enter your password

C. Download every attachment

D. Do not open it; check with a trusted adult or person

Answer: D. Suspicious links may lead to scams or harmful software.

STEM-G5-ICT-T03-Q09 | Privacy and Passwords | Difficulty: Intermediate

Why should you keep your password private?

A. To protect your account and personal information

B. To make the screen brighter

C. To make the internet faster

D. To extend battery life

Answer: A. A password provides access to an account and should be protected.

STEM-G5-ICT-T03-Q10 | Privacy and Passwords | Difficulty: Intermediate

A clear, ordered set of steps for solving a problem is called an ...

A. Algorithm

B. Wallpaper

C. Resolution

D. Speaker

Answer: A. An algorithm is an ordered set of instructions.

Theme 4: Block-Based Coding and Loops — 10 questions

STEM-G5-ICT-T04-Q01 | Block-Based Coding and Loops | Difficulty: Intermediate

Why should you keep your password private?

A. To extend battery life

B. To protect your account and personal information

C. To make the screen brighter

D. To make the internet faster

Answer: B. A password provides access to an account and should be protected.

STEM-G5-ICT-T04-Q02 | Block-Based Coding and Loops | Difficulty: Intermediate

A clear, ordered set of steps for solving a problem is called an ...

A. Resolution

B. Algorithm

C. Speaker

D. Wallpaper

Answer: B. An algorithm is an ordered set of instructions.

STEM-G5-ICT-T04-Q03 | Block-Based Coding and Loops | Difficulty: Intermediate

In a spreadsheet, a cell is ...

A. An application icon

B. The intersection of a row and a column

C. A type of cable

D. A webpage

Answer: B. A cell is a box at the intersection of a row and a column.

STEM-G5-ICT-T04-Q04 | Block-Based Coding and Loops | Difficulty: Intermediate

How should information found online be checked?

A. Choose the most dramatic headline

B. Count the emojis

C. Compare sources and supporting evidence

D. Share it before reading

Answer: C. Checking sources helps assess accuracy and credibility.

STEM-G5-ICT-T04-Q05 | Block-Based Coding and Loops | Difficulty: Intermediate

Finding and fixing errors in a program is called ...

A. Formatting

B. Streaming

C. Debugging

D. Browsing

Answer: C. Debugging is the process of locating and correcting errors.

STEM-G5-ICT-T04-Q06 | Block-Based Coding and Loops | Difficulty: Intermediate

Which device is commonly used to enter text into a computer?

A. Keyboard

B. Projector

C. Speaker

D. Monitor

Answer: A. A keyboard is an input device used for typing.

STEM-G5-ICT-T04-Q07 | Block-Based Coding and Loops | Difficulty: Intermediate

Software designed to help a user perform a task is commonly called an ...

A. Power cable

B. Application

C. Desk

D. Fan

Answer: B. An application is software that helps users perform specific tasks.

STEM-G5-ICT-T04-Q08 | Block-Based Coding and Loops | Difficulty: Intermediate

What should you do if an unknown person sends you a suspicious link?

A. Forward it to every contact

B. Download every attachment

C. Do not open it; check with a trusted adult or person

D. Enter your password

Answer: C. Suspicious links may lead to scams or harmful software.

STEM-G5-ICT-T04-Q09 | Block-Based Coding and Loops | Difficulty: Intermediate

Why should you keep your password private?

A. To make the screen brighter

B. To extend battery life

C. To make the internet faster

D. To protect your account and personal information

Answer: D. A password provides access to an account and should be protected.

STEM-G5-ICT-T04-Q10 | Block-Based Coding and Loops | Difficulty: Intermediate

A clear, ordered set of steps for solving a problem is called an ...

A. Resolution

B. Algorithm

C. Speaker

D. Wallpaper

Answer: B. An algorithm is an ordered set of instructions.

Elementary School Grade 6

160 questions • 4 subjects • 4 themes per subject • 10 questions per theme

Mathematics

Theme 1: Ratios and Scales — 10 questions

STEM-G6-MAT-T01-Q01 | Ratios and Scales | Difficulty: Intermediate

Calculate $13^2 + 5$.

A. 177

B. 174

C. 184

D. 171

Answer: B. $13^2 = 169$, then $169 + 5 = 174$.

STEM-G6-MAT-T01-Q02 | Ratios and Scales | Difficulty: Intermediate

If a = 14 and b = 2, calculate 2a + 3b.

A. 37

B. 34

C. 31

D. 44

Answer: B. 2(14) + 3(2) = 28 + 6 = 34.

STEM-G6-MAT-T01-Q03 | Ratios and Scales | Difficulty: Intermediate

A rectangle is 10 cm long and 4 cm wide. What is its perimeter?

A. 38

B. 31

C. 25

D. 28

Answer: D. Perimeter = 2(length + width) = 2(10 + 4) = 28 cm.

STEM-G6-MAT-T01-Q04 | Ratios and Scales | Difficulty: Intermediate

A rectangular area measures 18 cm by 10 cm. What is its area?

A. 180

B. 190

C. 183

D. 177

Answer: A. Area = 18 × 10 = 180 cm^2.

STEM-G6-MAT-T01-Q05 | Ratios and Scales | Difficulty: Intermediate

If x = 11 and y = 11, what is x × y?

A. 121

B. 124

C. 118

D. 131

Answer: A. Substitute the values: 11 × 11 = 121.

STEM-G6-MAT-T01-Q06 | Ratios and Scales | Difficulty: Intermediate

Calculate $10^2 + 12$.

A. 115

B. 109

C. 122

D. 112

Answer: D. $10^2 = 100$, then 100 + 12 = 112.

STEM-G6-MAT-T01-Q07 | Ratios and Scales | Difficulty: Intermediate

If a = 18 and b = 12, calculate 2a + 3b.

A. 69

B. 82

C. 72

D. 75

Answer: C. 2(18) + 3(12) = 36 + 36 = 72.

STEM-G6-MAT-T01-Q08 | Ratios and Scales | Difficulty: Intermediate

A rectangle is 6 cm long and 9 cm wide. What is its perimeter?

A. 33

B. 27

C. 30

D. 40

Answer: C. Perimeter = 2(length + width) = 2(6 + 9) = 30 cm.

STEM-G6-MAT-T01-Q09 | Ratios and Scales | Difficulty: Intermediate

A rectangular area measures 10 cm by 7 cm. What is its area?

A. 67

B. 70

C. 73

D. 80

Answer: B. Area = 10 × 7 = 70 cm².

STEM-G6-MAT-T01-Q10 | Ratios and Scales | Difficulty: Intermediate

If x = 11 and y = 8, what is x × y?

A. 85

B. 91

C. 98

D. 88

Answer: D. Substitute the values: 11 × 8 = 88.

Theme 2: Percentages and Number Operations — 10 questions

STEM-G6-MAT-T02-Q01 | Percentages and Number Operations | Difficulty: Intermediate

Calculate 4² + 6.

A. 19

B. 32

C. 25

D. 22

Answer: D. 4² = 16, then 16 + 6 = 22.

STEM-G6-MAT-T02-Q02 | Percentages and Number Operations | Difficulty: Intermediate

If a = 8 and b = 10, calculate 2a + 3b.

A. 49

B. 56

C. 43

D. 46

Answer: D. 2(8) + 3(10) = 16 + 30 = 46.

STEM-G6-MAT-T02-Q03 | Percentages and Number Operations | Difficulty: Intermediate

A rectangle is 15 cm long and 3 cm wide. What is its perimeter?

A. 33

B. 39

C. 46

D. 36

Answer: D. Perimeter = 2(length + width) = 2(15 + 3) = 36 cm.

STEM-G6-MAT-T02-Q04 | Percentages and Number Operations | Difficulty: Intermediate

A rectangular area measures 7 cm by 8 cm. What is its area?

A. 66

B. 53

C. 59

D. 56

Answer: D. Area = 7 × 8 = 56 cm².

STEM-G6-MAT-T02-Q05 | Percentages and Number Operations | Difficulty: Intermediate

If x = 17 and y = 7, what is x × y?

A. 119

B. 129

C. 116

D. 122

Answer: A. Substitute the values: 17 × 7 = 119.

STEM-G6-MAT-T02-Q06 | Percentages and Number Operations | Difficulty: Intermediate

Calculate $9^2 + 8$.

A. 89

B. 86

C. 92

D. 99

Answer: A. $9^2 = 81$, then 81 + 8 = 89.

STEM-G6-MAT-T02-Q07 | Percentages and Number Operations | Difficulty: Intermediate

If a = 3 and b = 12, calculate 2a + 3b.

A. 42

B. 52

C. 45

D. 39

Answer: A. 2(3) + 3(12) = 6 + 36 = 42.

STEM-G6-MAT-T02-Q08 | Percentages and Number Operations | Difficulty: Intermediate

A rectangle is 14 cm long and 8 cm wide. What is its perimeter?

A. 41

B. 44

C. 54

D. 47

Answer: B. Perimeter = 2(length + width) = 2(14 + 8) = 44 cm.

STEM-G6-MAT-T02-Q09 | Percentages and Number Operations | Difficulty: Intermediate

A rectangular area measures 14 cm by 5 cm. What is its area?

A. 67

B. 80

C. 73

D. 70

Answer: D. Area = 14 × 5 = 70 cm².

STEM-G6-MAT-T02-Q10 | Percentages and Number Operations | Difficulty: Intermediate

If x = 5 and y = 7, what is x × y?

A. 38

B. 45

C. 32

D. 35

Answer: D. Substitute the values: 5 × 7 = 35.

Theme 3: Area and Volume — 10 questions

STEM-G6-MAT-T03-Q01 | Area and Volume | Difficulty: Intermediate

Calculate $3^2 + 2$.

A. 8

B. 14

C. 21

D. 11

Answer: D. $3^2 = 9$, then $9 + 2 = 11$.

STEM-G6-MAT-T03-Q02 | Area and Volume | Difficulty: Intermediate

If a = 7 and b = 4, calculate 2a + 3b.

A. 26

B. 36

C. 23

D. 29

Answer: A. 2(7) + 3(4) = 14 + 12 = 26.

STEM-G6-MAT-T03-Q03 | Area and Volume | Difficulty: Intermediate

A rectangle is 6 cm long and 6 cm wide. What is its perimeter?

A. 21

B. 34

C. 24

D. 27

Answer: C. Perimeter = 2(length + width) = 2(6 + 6) = 24 cm.

STEM-G6-MAT-T03-Q04 | Area and Volume | Difficulty: Intermediate

A rectangular area measures 13 cm by 9 cm. What is its area?

A. 114

B. 120

C. 127

D. 117

Answer: D. Area = 13 × 9 = 117 cm^2.

STEM-G6-MAT-T03-Q05 | Area and Volume | Difficulty: Intermediate

If x = 4 and y = 3, what is x × y?

A. 22

B. 15

C. 12

D. 9

Answer: C. Substitute the values: 4 × 3 = 12.

STEM-G6-MAT-T03-Q06 | Area and Volume | Difficulty: Intermediate

Calculate $15^2 + 4$.

A. 239

B. 226

C. 229

D. 232

Answer: C. $15^2 = 225$, then 225 + 4 = 229.

STEM-G6-MAT-T03-Q07 | Area and Volume | Difficulty: Intermediate

If a = 4 and b = 4, calculate 2a + 3b.

A. 30

B. 23

C. 20

D. 17

Answer: C. 2(4) + 3(4) = 8 + 12 = 20.

STEM-G6-MAT-T03-Q08 | Area and Volume | Difficulty: Intermediate

A rectangle is 17 cm long and 4 cm wide. What is its perimeter?

A. 39

B. 45

C. 52

D. 42

Answer: D. Perimeter = 2(length + width) = 2(17 + 4) = 42 cm.

STEM-G6-MAT-T03-Q09 | Area and Volume | Difficulty: Intermediate

A rectangular area measures 9 cm by 9 cm. What is its area?

A. 78

B. 84

C. 81

D. 91

Answer: C. Area = 9 × 9 = 81 cm².

STEM-G6-MAT-T03-Q10 | Area and Volume | Difficulty: Intermediate

If x = 5 and y = 2, what is x × y?

A. 7

B. 10

C. 20

D. 13

Answer: B. Substitute the values: 5 × 2 = 10.

Theme 4: Data Handling and Basic Probability — 10 questions

STEM-G6-MAT-T04-Q01 | Data Handling and Basic Probability | Difficulty: Intermediate

Calculate $17^2 + 12$.

A. 304

B. 301

C. 311

D. 298

Answer: B. $17^2 = 289$, then 289 + 12 = 301.

STEM-G6-MAT-T04-Q02 | Data Handling and Basic Probability | Difficulty: Intermediate

If a = 14 and b = 3, calculate 2a + 3b.

A. 37

B. 34

C. 40

D. 47

Answer: A. 2(14) + 3(3) = 28 + 9 = 37.

STEM-G6-MAT-T04-Q03 | Data Handling and Basic Probability | Difficulty: Intermediate

A rectangle is 3 cm long and 11 cm wide. What is its perimeter?

A. 38

B. 28

C. 31

D. 25

Answer: B. Perimeter = 2(length + width) = 2(3 + 11) = 28 cm.

STEM-G6-MAT-T04-Q04 | Data Handling and Basic Probability | Difficulty: Intermediate

A rectangular area measures 3 cm by 3 cm. What is its area?

A. 6

B. 9

C. 19

D. 12

Answer: B. Area = 3 × 3 = 9 cm².

STEM-G6-MAT-T04-Q05 | Data Handling and Basic Probability | Difficulty: Intermediate

If x = 11 and y = 10, what is x × y?

A. 110

B. 107

C. 113

D. 120

Answer: A. Substitute the values: 11 × 10 = 110.

STEM-G6-MAT-T04-Q06 | Data Handling and Basic Probability | Difficulty: Intermediate

Calculate $8^2 + 2$.

A. 63

B. 66

C. 69

D. 76

Answer: B. $8^2 = 64$, then 64 + 2 = 66.

STEM-G6-MAT-T04-Q07 | Data Handling and Basic Probability | Difficulty: Intermediate

If a = 13 and b = 12, calculate 2a + 3b.

A. 72

B. 59

C. 62

D. 65

Answer: C. 2(13) + 3(12) = 26 + 36 = 62.

STEM-G6-MAT-T04-Q08 | Data Handling and Basic Probability | Difficulty: Intermediate

A rectangle is 7 cm long and 6 cm wide. What is its perimeter?

A. 26

B. 36

C. 29

D. 23

Answer: A. Perimeter = 2(length + width) = 2(7 + 6) = 26 cm.

STEM-G6-MAT-T04-Q09 | Data Handling and Basic Probability | Difficulty: Intermediate

A rectangular area measures 10 cm by 5 cm. What is its area?

A. 50

B. 60

C. 53

D. 47

Answer: A. Area = 10 × 5 = 50 cm^2.

STEM-G6-MAT-T04-Q10 | Data Handling and Basic Probability | Difficulty: Intermediate

If x = 17 and y = 12, what is x × y?

A. 201

B. 204

C. 207

D. 214

Answer: B. Substitute the values: 17 × 12 = 204.

Science

Theme 1: Earth's Rotation and Revolution — 10 questions

STEM-G6-SCI-T01-Q01 | Earth's Rotation and Revolution | Difficulty: Intermediate

What causes day and night on Earth?

A. Earth changing size

B. Cloud movement

C. Earth's rotation

D. The Moon's revolution

Answer: C. Earth rotates on its axis, bringing different areas into daylight.

STEM-G6-SCI-T01-Q02 | Earth's Rotation and Revolution | Difficulty: Intermediate

What causes day and night on Earth?

A. Earth changing size

B. The Moon's revolution

C. Earth's rotation

D. Cloud movement

Answer: C. Earth rotates on its axis, bringing different areas into daylight.

STEM-G6-SCI-T01-Q03 | Earth's Rotation and Revolution | Difficulty: Intermediate

What causes day and night on Earth?

A. Earth's rotation

B. Cloud movement

C. The Moon's revolution

D. Earth changing size

Answer: A. Earth rotates on its axis, bringing different areas into daylight.

STEM-G6-SCI-T01-Q04 | Earth's Rotation and Revolution | Difficulty: Intermediate

What causes day and night on Earth?

A. Cloud movement

B. Earth's rotation

C. Earth changing size

D. The Moon's revolution

Answer: B. Earth rotates on its axis, bringing different areas into daylight.

STEM-G6-SCI-T01-Q05 | Earth's Rotation and Revolution | Difficulty: Intermediate

What causes day and night on Earth?

A. Earth's rotation

B. Cloud movement

C. The Moon's revolution

D. Earth changing size

Answer: A. Earth rotates on its axis, bringing different areas into daylight.

STEM-G6-SCI-T01-Q06 | Earth's Rotation and Revolution | Difficulty: Intermediate

What causes day and night on Earth?

A. Cloud movement

B. Earth changing size

C. Earth's rotation

D. The Moon's revolution

Answer: C. Earth rotates on its axis, bringing different areas into daylight.

STEM-G6-SCI-T01-Q07 | Earth's Rotation and Revolution | Difficulty: Intermediate

What causes day and night on Earth?

A. The Moon's revolution

B. Earth changing size

C. Cloud movement

D. Earth's rotation

Answer: D. Earth rotates on its axis, bringing different areas into daylight.

STEM-G6-SCI-T01-Q08 | Earth's Rotation and Revolution | Difficulty: Intermediate

What causes day and night on Earth?

A. Earth's rotation

B. The Moon's revolution

C. Cloud movement

D. Earth changing size

Answer: A. Earth rotates on its axis, bringing different areas into daylight.

STEM-G6-SCI-T01-Q09 | Earth's Rotation and Revolution | Difficulty: Intermediate

What causes day and night on Earth?

A. Earth changing size

B. Cloud movement

C. The Moon's revolution

D. Earth's rotation

Answer: D. Earth rotates on its axis, bringing different areas into daylight.

STEM-G6-SCI-T01-Q10 | Earth's Rotation and Revolution | Difficulty: Intermediate

What causes day and night on Earth?

A. Earth changing size

B. The Moon's revolution

C. Earth's rotation

D. Cloud movement

Answer: C. Earth rotates on its axis, bringing different areas into daylight.

Theme 2: Electricity and Magnetism — 10 questions

STEM-G6-SCI-T02-Q01 | Electricity and Magnetism | Difficulty: Intermediate

Which material is a good electrical conductor?

A. Dry wood

B. Rubber

C. Copper

D. Plastic

Answer: C. Copper allows electric charges to move easily.

STEM-G6-SCI-T02-Q02 | Electricity and Magnetism | Difficulty: Intermediate

Which material is a good electrical conductor?

A. Rubber

B. Copper

C. Dry wood

D. Plastic

Answer: B. Copper allows electric charges to move easily.

STEM-G6-SCI-T02-Q03 | Electricity and Magnetism | Difficulty: Intermediate

Which material is a good electrical conductor?

A. Plastic

B. Rubber

C. Dry wood

D. Copper

Answer: D. Copper allows electric charges to move easily.

STEM-G6-SCI-T02-Q04 | Electricity and Magnetism | Difficulty: Intermediate

Which material is a good electrical conductor?

A. Dry wood

B. Rubber

C. Copper

D. Plastic

Answer: C. Copper allows electric charges to move easily.

STEM-G6-SCI-T02-Q05 | Electricity and Magnetism | Difficulty: Intermediate

Which material is a good electrical conductor?

A. Plastic

B. Copper

C. Dry wood

D. Rubber

Answer: B. Copper allows electric charges to move easily.

STEM-G6-SCI-T02-Q06 | Electricity and Magnetism | Difficulty: Intermediate

Which material is a good electrical conductor?

A. Plastic

B. Copper

C. Dry wood

D. Rubber

Answer: B. Copper allows electric charges to move easily.

STEM-G6-SCI-T02-Q07 | Electricity and Magnetism | Difficulty: Intermediate

Which material is a good electrical conductor?

A. Copper

B. Dry wood

C. Plastic

D. Rubber

Answer: A. Copper allows electric charges to move easily.

STEM-G6-SCI-T02-Q08 | Electricity and Magnetism | Difficulty: Intermediate

Which material is a good electrical conductor?

A. Rubber

B. Plastic

C. Dry wood

D. Copper

Answer: D. Copper allows electric charges to move easily.

STEM-G6-SCI-T02-Q09 | Electricity and Magnetism | Difficulty: Intermediate

Which material is a good electrical conductor?

A. Dry wood

B. Copper

C. Rubber

D. Plastic

Answer: B. Copper allows electric charges to move easily.

STEM-G6-SCI-T02-Q10 | Electricity and Magnetism | Difficulty: Intermediate

Which material is a good electrical conductor?

A. Rubber

B. Dry wood

C. Copper

D. Plastic

Answer: C. Copper allows electric charges to move easily.

Theme 3: Reproduction in Living Things — 10 questions

STEM-G6-SCI-T03-Q01 | Reproduction in Living Things | Difficulty: Intermediate

Reproduction involving the formation of seeds is generally ...

A. Only artificial vegetative reproduction

B. Evaporation

C. Sexual reproduction

D. Metal fragmentation

Answer: C. Seeds form through sexual reproduction in flowering plants.

STEM-G6-SCI-T03-Q02 | Reproduction in Living Things | Difficulty: Intermediate

Reproduction involving the formation of seeds is generally ...

A. Metal fragmentation

B. Sexual reproduction

C. Only artificial vegetative reproduction

D. Evaporation

Answer: B. Seeds form through sexual reproduction in flowering plants.

STEM-G6-SCI-T03-Q03 | Reproduction in Living Things | Difficulty: Intermediate

Reproduction involving the formation of seeds is generally ...

A. Only artificial vegetative reproduction

B. Evaporation

C. Sexual reproduction

D. Metal fragmentation

Answer: C. Seeds form through sexual reproduction in flowering plants.

STEM-G6-SCI-T03-Q04 | Reproduction in Living Things | Difficulty: Intermediate

Reproduction involving the formation of seeds is generally ...

A. Evaporation

B. Sexual reproduction

C. Only artificial vegetative reproduction

D. Metal fragmentation

Answer: B. Seeds form through sexual reproduction in flowering plants.

STEM-G6-SCI-T03-Q05 | Reproduction in Living Things | Difficulty: Intermediate

Reproduction involving the formation of seeds is generally ...

A. Evaporation

B. Metal fragmentation

C. Only artificial vegetative reproduction

D. Sexual reproduction

Answer: D. Seeds form through sexual reproduction in flowering plants.

STEM-G6-SCI-T03-Q06 | Reproduction in Living Things | Difficulty: Intermediate

Reproduction involving the formation of seeds is generally ...

A. Only artificial vegetative reproduction

B. Evaporation

C. Sexual reproduction

D. Metal fragmentation

Answer: C. Seeds form through sexual reproduction in flowering plants.

STEM-G6-SCI-T03-Q07 | Reproduction in Living Things | Difficulty: Intermediate

Reproduction involving the formation of seeds is generally ...

A. Evaporation

B. Metal fragmentation

C. Only artificial vegetative reproduction

D. Sexual reproduction

Answer: D. Seeds form through sexual reproduction in flowering plants.

STEM-G6-SCI-T03-Q08 | Reproduction in Living Things | Difficulty: Intermediate

Reproduction involving the formation of seeds is generally ...

A. Evaporation

B. Sexual reproduction

C. Metal fragmentation

D. Only artificial vegetative reproduction

Answer: B. Seeds form through sexual reproduction in flowering plants.

STEM-G6-SCI-T03-Q09 | Reproduction in Living Things | Difficulty: Intermediate

Reproduction involving the formation of seeds is generally ...

A. Evaporation

B. Only artificial vegetative reproduction

C. Metal fragmentation

D. Sexual reproduction

Answer: D. Seeds form through sexual reproduction in flowering plants.

STEM-G6-SCI-T03-Q10 | Reproduction in Living Things | Difficulty: Intermediate

Reproduction involving the formation of seeds is generally ...

A. Metal fragmentation

B. Sexual reproduction

C. Only artificial vegetative reproduction

D. Evaporation

Answer: B. Seeds form through sexual reproduction in flowering plants.

Theme 4: Energy and Environmental Conservation — 10 questions

STEM-G6-SCI-T04-Q01 | Energy and Environmental Conservation | Difficulty: Intermediate

Which action helps conserve electrical energy?

A. Turning off lights when they are not needed

B. Keeping the refrigerator door open

C. Leaving every light on

D. Charging devices unnecessarily

Answer: A. Avoiding unnecessary electricity use saves energy.

STEM-G6-SCI-T04-Q02 | Energy and Environmental Conservation | Difficulty: Intermediate

Which action helps conserve electrical energy?

A. Leaving every light on

B. Turning off lights when they are not needed

C. Charging devices unnecessarily

D. Keeping the refrigerator door open

Answer: B. Avoiding unnecessary electricity use saves energy.

STEM-G6-SCI-T04-Q03 | Energy and Environmental Conservation | Difficulty: Intermediate

Which action helps conserve electrical energy?

A. Leaving every light on

B. Turning off lights when they are not needed

C. Charging devices unnecessarily

D. Keeping the refrigerator door open

Answer: B. Avoiding unnecessary electricity use saves energy.

STEM-G6-SCI-T04-Q04 | Energy and Environmental Conservation | Difficulty: Intermediate

Which action helps conserve electrical energy?

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C. Leaving every light on

D. Charging devices unnecessarily

Answer: B. Avoiding unnecessary electricity use saves energy.

STEM-G6-SCI-T04-Q05 | Energy and Environmental Conservation | Difficulty: Intermediate

Which action helps conserve electrical energy?

A. Leaving every light on

B. Charging devices unnecessarily

C. Turning off lights when they are not needed

D. Keeping the refrigerator door open

Answer: C. Avoiding unnecessary electricity use saves energy.

STEM-G6-SCI-T04-Q06 | Energy and Environmental Conservation | Difficulty: Intermediate

Which action helps conserve electrical energy?

A. Leaving every light on

B. Turning off lights when they are not needed

C. Charging devices unnecessarily

D. Keeping the refrigerator door open

Answer: B. Avoiding unnecessary electricity use saves energy.

STEM-G6-SCI-T04-Q07 | Energy and Environmental Conservation | Difficulty: Intermediate

Which action helps conserve electrical energy?

A. Keeping the refrigerator door open

B. Charging devices unnecessarily

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D. Turning off lights when they are not needed

Answer: D. Avoiding unnecessary electricity use saves energy.

STEM-G6-SCI-T04-Q08 | Energy and Environmental Conservation | Difficulty: Intermediate

Which action helps conserve electrical energy?

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STEM-G6-SCI-T04-Q09 | Energy and Environmental Conservation | Difficulty: Intermediate

Which action helps conserve electrical energy?

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D. Charging devices unnecessarily

Answer: C. Avoiding unnecessary electricity use saves energy.

STEM-G6-SCI-T04-Q10 | Energy and Environmental Conservation | Difficulty: Intermediate

Which action helps conserve electrical energy?

A. Turning off lights when they are not needed

B. Leaving every light on

C. Charging devices unnecessarily

D. Keeping the refrigerator door open

Answer: A. Avoiding unnecessary electricity use saves energy.

Engineering

Theme 1: Gears and Transmission — 10 questions

STEM-G6-ENG-T01-Q01 | Gears and Transmission | Difficulty: Intermediate

Which measurement is useful when comparing two bridge designs?

A. The number of stickers

B. The maximum load each bridge can support

C. The bridge color

D. The model's name

Answer: B. Load capacity is a relevant measure of bridge performance.

STEM-G6-ENG-T01-Q02 | Gears and Transmission | Difficulty: Intermediate

Why should an engineering prototype be tested before use?

A. To make it look more expensive

B. To find out whether the design works and is safe

C. To make it larger every time

D. To avoid making improvements

Answer: B. Testing reveals weaknesses and helps improve a design.

STEM-G6-ENG-T01-Q03 | Gears and Transmission | Difficulty: Intermediate

If a model structure tips over easily, what is a sensible first improvement?

A. Add weight to the top

B. Remove the base

C. Widen the base or strengthen the joints

D. Remove all supports

Answer: C. A wider base and strong joints can improve stability.

STEM-G6-ENG-T01-Q04 | Gears and Transmission | Difficulty: Intermediate

What is the purpose of making a sketch before building an object?

A. To plan its shape, dimensions, and parts

B. To avoid measuring

C. To make the object build itself

D. To replace all testing

Answer: A. A sketch communicates a design before construction.

STEM-G6-ENG-T01-Q05 | Gears and Transmission | Difficulty: Intermediate

For a fair test, a team should ...

A. Change one factor while keeping other factors the same

B. Choose only the preferred result

C. Avoid recording results

D. Change every factor at once

Answer: A. Controlling variables helps identify what caused a result.

STEM-G6-ENG-T01-Q06 | Gears and Transmission | Difficulty: Intermediate

When selecting a material for a load-bearing part, what should be considered?

A. The designer's name

B. Only its favorite color

C. Its strength and suitability for the task

D. The packaging

Answer: C. Material choice depends on the load and operating conditions.

STEM-G6-ENG-T01-Q07 | Gears and Transmission | Difficulty: Intermediate

Why should joints in a model be inspected?

A. Joints transfer forces between parts

B. Joints make mass zero

C. Joints are only decoration

D. Joints remove gravity

Answer: A. Weak joints can become failure points in a structure.

STEM-G6-ENG-T01-Q08 | Gears and Transmission | Difficulty: Intermediate

After a test reveals a weakness, what should engineers do next?

A. Improve the design and test it again

B. Hide the result

C. Ignore safety

D. Delete the notes

Answer: A. Engineering is iterative and uses test results to guide improvements.

STEM-G6-ENG-T01-Q09 | Gears and Transmission | Difficulty: Intermediate

Which measurement is useful when comparing two bridge designs?

A. The number of stickers

B. The bridge color

C. The model's name

D. The maximum load each bridge can support

Answer: D. Load capacity is a relevant measure of bridge performance.

STEM-G6-ENG-T01-Q10 | Gears and Transmission | Difficulty: Intermediate

Why should an engineering prototype be tested before use?

A. To make it larger every time

B. To make it look more expensive

C. To avoid making improvements

D. To find out whether the design works and is safe

Answer: D. Testing reveals weaknesses and helps improve a design.

Theme 2: Load-Bearing Structures — 10 questions

STEM-G6-ENG-T02-Q01 | Load-Bearing Structures | Difficulty: Intermediate

Which measurement is useful when comparing two bridge designs?

A. The model's name

B. The maximum load each bridge can support

C. The number of stickers

D. The bridge color

Answer: B. Load capacity is a relevant measure of bridge performance.

STEM-G6-ENG-T02-Q02 | Load-Bearing Structures | Difficulty: Intermediate

Why should an engineering prototype be tested before use?

A. To find out whether the design works and is safe

B. To make it larger every time

C. To avoid making improvements

D. To make it look more expensive

Answer: A. Testing reveals weaknesses and helps improve a design.

STEM-G6-ENG-T02-Q03 | Load-Bearing Structures | Difficulty: Intermediate

If a model structure tips over easily, what is a sensible first improvement?

A. Widen the base or strengthen the joints

B. Remove the base

C. Add weight to the top

D. Remove all supports

Answer: A. A wider base and strong joints can improve stability.

STEM-G6-ENG-T02-Q04 | Load-Bearing Structures | Difficulty: Intermediate

What is the purpose of making a sketch before building an object?

A. To plan its shape, dimensions, and parts

B. To avoid measuring

C. To make the object build itself

D. To replace all testing

Answer: A. A sketch communicates a design before construction.

STEM-G6-ENG-T02-Q05 | Load-Bearing Structures | Difficulty: Intermediate

For a fair test, a team should ...

A. Avoid recording results

B. Change every factor at once

C. Choose only the preferred result

D. Change one factor while keeping other factors the same

Answer: D. Controlling variables helps identify what caused a result.

STEM-G6-ENG-T02-Q06 | Load-Bearing Structures | Difficulty: Intermediate

When selecting a material for a load-bearing part, what should be considered?

A. The designer's name

B. Only its favorite color

C. The packaging

D. Its strength and suitability for the task

Answer: D. Material choice depends on the load and operating conditions.

STEM-G6-ENG-T02-Q07 | Load-Bearing Structures | Difficulty: Intermediate

Why should joints in a model be inspected?

A. Joints remove gravity

B. Joints are only decoration

C. Joints make mass zero

D. Joints transfer forces between parts

Answer: D. Weak joints can become failure points in a structure.

STEM-G6-ENG-T02-Q08 | Load-Bearing Structures | Difficulty: Intermediate

After a test reveals a weakness, what should engineers do next?

A. Delete the notes

B. Improve the design and test it again

C. Hide the result

D. Ignore safety

Answer: B. Engineering is iterative and uses test results to guide improvements.

STEM-G6-ENG-T02-Q09 | Load-Bearing Structures | Difficulty: Intermediate

Which measurement is useful when comparing two bridge designs?

A. The number of stickers

B. The model's name

C. The bridge color

D. The maximum load each bridge can support

Answer: D. Load capacity is a relevant measure of bridge performance.

STEM-G6-ENG-T02-Q10 | Load-Bearing Structures | Difficulty: Intermediate

Why should an engineering prototype be tested before use?

A. To make it larger every time

B. To avoid making improvements

C. To find out whether the design works and is safe

D. To make it look more expensive

Answer: C. Testing reveals weaknesses and helps improve a design.

Theme 3: Basic Renewable Energy — 10 questions

STEM-G6-ENG-T03-Q01 | Basic Renewable Energy | Difficulty: Intermediate

Which measurement is useful when comparing two bridge designs?

A. The number of stickers

B. The maximum load each bridge can support

C. The model's name

D. The bridge color

Answer: B. Load capacity is a relevant measure of bridge performance.

STEM-G6-ENG-T03-Q02 | Basic Renewable Energy | Difficulty: Intermediate

Why should an engineering prototype be tested before use?

A. To find out whether the design works and is safe

B. To make it larger every time

C. To make it look more expensive

D. To avoid making improvements

Answer: A. Testing reveals weaknesses and helps improve a design.

STEM-G6-ENG-T03-Q03 | Basic Renewable Energy | Difficulty: Intermediate

If a model structure tips over easily, what is a sensible first improvement?

A. Add weight to the top

B. Remove all supports

C. Widen the base or strengthen the joints

D. Remove the base

Answer: C. A wider base and strong joints can improve stability.

STEM-G6-ENG-T03-Q04 | Basic Renewable Energy | Difficulty: Intermediate

What is the purpose of making a sketch before building an object?

A. To make the object build itself

B. To avoid measuring

C. To plan its shape, dimensions, and parts

D. To replace all testing

Answer: C. A sketch communicates a design before construction.

STEM-G6-ENG-T03-Q05 | Basic Renewable Energy | Difficulty: Intermediate

For a fair test, a team should ...

A. Avoid recording results

B. Choose only the preferred result

C. Change one factor while keeping other factors the same

D. Change every factor at once

Answer: C. Controlling variables helps identify what caused a result.

STEM-G6-ENG-T03-Q06 | Basic Renewable Energy | Difficulty: Intermediate

When selecting a material for a load-bearing part, what should be considered?

A. Only its favorite color

B. The designer's name

C. The packaging

D. Its strength and suitability for the task

Answer: D. Material choice depends on the load and operating conditions.

STEM-G6-ENG-T03-Q07 | Basic Renewable Energy | Difficulty: Intermediate

Why should joints in a model be inspected?

A. Joints remove gravity

B. Joints are only decoration

C. Joints make mass zero

D. Joints transfer forces between parts

Answer: D. Weak joints can become failure points in a structure.

STEM-G6-ENG-T03-Q08 | Basic Renewable Energy | Difficulty: Intermediate

After a test reveals a weakness, what should engineers do next?

A. Ignore safety

B. Delete the notes

C. Hide the result

D. Improve the design and test it again

Answer: D. Engineering is iterative and uses test results to guide improvements.

STEM-G6-ENG-T03-Q09 | Basic Renewable Energy | Difficulty: Intermediate

Which measurement is useful when comparing two bridge designs?

A. The bridge color

B. The number of stickers

C. The maximum load each bridge can support

D. The model's name

Answer: C. Load capacity is a relevant measure of bridge performance.

STEM-G6-ENG-T03-Q10 | Basic Renewable Energy | Difficulty: Intermediate

Why should an engineering prototype be tested before use?

A. To find out whether the design works and is safe

B. To make it look more expensive

C. To avoid making improvements

D. To make it larger every time

Answer: A. Testing reveals weaknesses and helps improve a design.

Theme 4: Measurement and Design Optimization — 10 questions

STEM-G6-ENG-T04-Q01 | Measurement and Design Optimization | Difficulty: Intermediate

Which measurement is useful when comparing two bridge designs?

A. The maximum load each bridge can support

B. The bridge color

C. The model's name

D. The number of stickers

Answer: A. Load capacity is a relevant measure of bridge performance.

STEM-G6-ENG-T04-Q02 | Measurement and Design Optimization | Difficulty: Intermediate

Why should an engineering prototype be tested before use?

A. To avoid making improvements

B. To find out whether the design works and is safe

C. To make it larger every time

D. To make it look more expensive

Answer: B. Testing reveals weaknesses and helps improve a design.

STEM-G6-ENG-T04-Q03 | Measurement and Design Optimization | Difficulty: Intermediate

If a model structure tips over easily, what is a sensible first improvement?

A. Widen the base or strengthen the joints

B. Remove the base

C. Remove all supports

D. Add weight to the top

Answer: A. A wider base and strong joints can improve stability.

STEM-G6-ENG-T04-Q04 | Measurement and Design Optimization | Difficulty: Intermediate

What is the purpose of making a sketch before building an object?

A. To avoid measuring

B. To make the object build itself

C. To replace all testing

D. To plan its shape, dimensions, and parts

Answer: D. A sketch communicates a design before construction.

STEM-G6-ENG-T04-Q05 | Measurement and Design Optimization | Difficulty: Intermediate

For a fair test, a team should ...

A. Choose only the preferred result

B. Avoid recording results

C. Change every factor at once

D. Change one factor while keeping other factors the same

Answer: D. Controlling variables helps identify what caused a result.

STEM-G6-ENG-T04-Q06 | Measurement and Design Optimization | Difficulty: Intermediate

When selecting a material for a load-bearing part, what should be considered?

A. The designer's name

B. Only its favorite color

C. The packaging

D. Its strength and suitability for the task

Answer: D. Material choice depends on the load and operating conditions.

STEM-G6-ENG-T04-Q07 | Measurement and Design Optimization | Difficulty: Intermediate

Why should joints in a model be inspected?

A. Joints are only decoration

B. Joints make mass zero

C. Joints remove gravity

D. Joints transfer forces between parts

Answer: D. Weak joints can become failure points in a structure.

STEM-G6-ENG-T04-Q08 | Measurement and Design Optimization | Difficulty: Intermediate

After a test reveals a weakness, what should engineers do next?

A. Improve the design and test it again

B. Delete the notes

C. Hide the result

D. Ignore safety

Answer: A. Engineering is iterative and uses test results to guide improvements.

STEM-G6-ENG-T04-Q09 | Measurement and Design Optimization | Difficulty: Intermediate

Which measurement is useful when comparing two bridge designs?

A. The model's name

B. The number of stickers

C. The maximum load each bridge can support

D. The bridge color

Answer: C. Load capacity is a relevant measure of bridge performance.

STEM-G6-ENG-T04-Q10 | Measurement and Design Optimization | Difficulty: Intermediate

Why should an engineering prototype be tested before use?

A. To find out whether the design works and is safe

B. To make it larger every time

C. To make it look more expensive

D. To avoid making improvements

Answer: A. Testing reveals weaknesses and helps improve a design.

ICT (Technology)

Theme 1: Spreadsheets and Basic Formulas — 10 questions

STEM-G6-ICT-T01-Q01 | Spreadsheets and Basic Formulas | Difficulty: Intermediate

In a spreadsheet, a cell is ...

A. A webpage

B. The intersection of a row and a column

C. An application icon

D. A type of cable

Answer: B. A cell is a box at the intersection of a row and a column.

STEM-G6-ICT-T01-Q02 | Spreadsheets and Basic Formulas | Difficulty: Intermediate

How should information found online be checked?

A. Share it before reading

B. Choose the most dramatic headline

C. Count the emojis

D. Compare sources and supporting evidence

Answer: D. Checking sources helps assess accuracy and credibility.

STEM-G6-ICT-T01-Q03 | Spreadsheets and Basic Formulas | Difficulty: Intermediate

Finding and fixing errors in a program is called ...

A. Streaming

B. Formatting

C. Debugging

D. Browsing

Answer: C. Debugging is the process of locating and correcting errors.

STEM-G6-ICT-T01-Q04 | Spreadsheets and Basic Formulas | Difficulty: Intermediate

Which device is commonly used to enter text into a computer?

A. Keyboard

B. Projector

C. Speaker

D. Monitor

Answer: A. A keyboard is an input device used for typing.

STEM-G6-ICT-T01-Q05 | Spreadsheets and Basic Formulas | Difficulty: Intermediate

Software designed to help a user perform a task is commonly called an ...

A. Power cable

B. Desk

C. Fan

D. Application

Answer: D. An application is software that helps users perform specific tasks.

STEM-G6-ICT-T01-Q06 | Spreadsheets and Basic Formulas | Difficulty: Intermediate

What should you do if an unknown person sends you a suspicious link?

A. Enter your password

B. Forward it to every contact

C. Do not open it; check with a trusted adult or person

D. Download every attachment

Answer: C. Suspicious links may lead to scams or harmful software.

STEM-G6-ICT-T01-Q07 | Spreadsheets and Basic Formulas | Difficulty: Intermediate

Why should you keep your password private?

A. To make the internet faster

B. To make the screen brighter

C. To extend battery life

D. To protect your account and personal information

Answer: D. A password provides access to an account and should be protected.

STEM-G6-ICT-T01-Q08 | Spreadsheets and Basic Formulas | Difficulty: Intermediate

A clear, ordered set of steps for solving a problem is called an ...

A. Algorithm

B. Wallpaper

C. Speaker

D. Resolution

Answer: A. An algorithm is an ordered set of instructions.

STEM-G6-ICT-T01-Q09 | Spreadsheets and Basic Formulas | Difficulty: Intermediate

In a spreadsheet, a cell is ...

A. A type of cable

B. An application icon

C. The intersection of a row and a column

D. A webpage

Answer: C. A cell is a box at the intersection of a row and a column.

STEM-G6-ICT-T01-Q10 | Spreadsheets and Basic Formulas | Difficulty: Intermediate

How should information found online be checked?

A. Compare sources and supporting evidence

B. Count the emojis

C. Share it before reading

D. Choose the most dramatic headline

Answer: A. Checking sources helps assess accuracy and credibility.

Theme 2: Internet, Networks, and Cloud Services — 10 questions

STEM-G6-ICT-T02-Q01 | Internet, Networks, and Cloud Services | Difficulty: Intermediate

In a spreadsheet, a cell is ...

A. The intersection of a row and a column

B. A type of cable

C. A webpage

D. An application icon

Answer: A. A cell is a box at the intersection of a row and a column.

STEM-G6-ICT-T02-Q02 | Internet, Networks, and Cloud Services | Difficulty: Intermediate

How should information found online be checked?

A. Compare sources and supporting evidence

B. Choose the most dramatic headline

C. Share it before reading

D. Count the emojis

Answer: A. Checking sources helps assess accuracy and credibility.

STEM-G6-ICT-T02-Q03 | Internet, Networks, and Cloud Services | Difficulty: Intermediate

Finding and fixing errors in a program is called ...

A. Formatting

B. Browsing

C. Debugging

D. Streaming

Answer: C. Debugging is the process of locating and correcting errors.

STEM-G6-ICT-T02-Q04 | Internet, Networks, and Cloud Services | Difficulty: Intermediate

Which device is commonly used to enter text into a computer?

A. Projector

B. Speaker

C. Keyboard

D. Monitor

Answer: C. A keyboard is an input device used for typing.

STEM-G6-ICT-T02-Q05 | Internet, Networks, and Cloud Services | Difficulty: Intermediate

Software designed to help a user perform a task is commonly called an ...

A. Application

B. Power cable

C. Desk

D. Fan

Answer: A. An application is software that helps users perform specific tasks.

STEM-G6-ICT-T02-Q06 | Internet, Networks, and Cloud Services | Difficulty: Intermediate

What should you do if an unknown person sends you a suspicious link?

A. Download every attachment

B. Enter your password

C. Do not open it; check with a trusted adult or person

D. Forward it to every contact

Answer: C. Suspicious links may lead to scams or harmful software.

STEM-G6-ICT-T02-Q07 | Internet, Networks, and Cloud Services | Difficulty: Intermediate

Why should you keep your password private?

A. To protect your account and personal information

B. To make the internet faster

C. To make the screen brighter

D. To extend battery life

Answer: A. A password provides access to an account and should be protected.

STEM-G6-ICT-T02-Q08 | Internet, Networks, and Cloud Services | Difficulty: Intermediate

A clear, ordered set of steps for solving a problem is called an ...

A. Wallpaper

B. Algorithm

C. Speaker

D. Resolution

Answer: B. An algorithm is an ordered set of instructions.

STEM-G6-ICT-T02-Q09 | Internet, Networks, and Cloud Services | Difficulty: Intermediate

In a spreadsheet, a cell is ...

A. An application icon

B. The intersection of a row and a column

C. A webpage

D. A type of cable

Answer: B. A cell is a box at the intersection of a row and a column.

STEM-G6-ICT-T02-Q10 | Internet, Networks, and Cloud Services | Difficulty: Intermediate

How should information found online be checked?

A. Compare sources and supporting evidence

B. Choose the most dramatic headline

C. Share it before reading

D. Count the emojis

Answer: A. Checking sources helps assess accuracy and credibility.

Theme 3: Digital Footprints and Security — 10 questions

STEM-G6-ICT-T03-Q01 | Digital Footprints and Security | Difficulty: Intermediate

In a spreadsheet, a cell is ...

A. An application icon

B. A webpage

C. A type of cable

D. The intersection of a row and a column

Answer: D. A cell is a box at the intersection of a row and a column.

STEM-G6-ICT-T03-Q02 | Digital Footprints and Security | Difficulty: Intermediate

How should information found online be checked?

A. Choose the most dramatic headline

B. Share it before reading

C. Count the emojis

D. Compare sources and supporting evidence

Answer: D. Checking sources helps assess accuracy and credibility.

STEM-G6-ICT-T03-Q03 | Digital Footprints and Security | Difficulty: Intermediate

Finding and fixing errors in a program is called ...

A. Debugging

B. Formatting

C. Streaming

D. Browsing

Answer: A. Debugging is the process of locating and correcting errors.

STEM-G6-ICT-T03-Q04 | Digital Footprints and Security | Difficulty: Intermediate

Which device is commonly used to enter text into a computer?

A. Speaker

B. Projector

C. Monitor

D. Keyboard

Answer: D. A keyboard is an input device used for typing.

STEM-G6-ICT-T03-Q05 | Digital Footprints and Security | Difficulty: Intermediate

Software designed to help a user perform a task is commonly called an ...

A. Power cable

B. Fan

C. Application

D. Desk

Answer: C. An application is software that helps users perform specific tasks.

STEM-G6-ICT-T03-Q06 | Digital Footprints and Security | Difficulty: Intermediate

What should you do if an unknown person sends you a suspicious link?

A. Enter your password

B. Forward it to every contact

C. Do not open it; check with a trusted adult or person

D. Download every attachment

Answer: C. Suspicious links may lead to scams or harmful software.

STEM-G6-ICT-T03-Q07 | Digital Footprints and Security | Difficulty: Intermediate

Why should you keep your password private?

A. To make the screen brighter

B. To protect your account and personal information

C. To extend battery life

D. To make the internet faster

Answer: B. A password provides access to an account and should be protected.

STEM-G6-ICT-T03-Q08 | Digital Footprints and Security | Difficulty: Intermediate

A clear, ordered set of steps for solving a problem is called an ...

A. Speaker

B. Algorithm

C. Wallpaper

D. Resolution

Answer: B. An algorithm is an ordered set of instructions.

STEM-G6-ICT-T03-Q09 | Digital Footprints and Security | Difficulty: Intermediate

In a spreadsheet, a cell is ...

A. A webpage

B. The intersection of a row and a column

C. A type of cable

D. An application icon

Answer: B. A cell is a box at the intersection of a row and a column.

STEM-G6-ICT-T03-Q10 | Digital Footprints and Security | Difficulty: Intermediate

How should information found online be checked?

A. Choose the most dramatic headline

B. Share it before reading

C. Count the emojis

D. Compare sources and supporting evidence

Answer: D. Checking sources helps assess accuracy and credibility.

Theme 4: Algorithms, Conditions, and Debugging — 10 questions

STEM-G6-ICT-T04-Q01 | Algorithms, Conditions, and Debugging | Difficulty: Intermediate

In a spreadsheet, a cell is ...

A. An application icon

B. A webpage

C. The intersection of a row and a column

D. A type of cable

Answer: C. A cell is a box at the intersection of a row and a column.

STEM-G6-ICT-T04-Q02 | Algorithms, Conditions, and Debugging | Difficulty: Intermediate

How should information found online be checked?

A. Share it before reading

B. Choose the most dramatic headline

C. Compare sources and supporting evidence

D. Count the emojis

Answer: C. Checking sources helps assess accuracy and credibility.

STEM-G6-ICT-T04-Q03 | Algorithms, Conditions, and Debugging | Difficulty: Intermediate

Finding and fixing errors in a program is called ...

A. Streaming

B. Browsing

C. Debugging

D. Formatting

Answer: C. Debugging is the process of locating and correcting errors.

STEM-G6-ICT-T04-Q04 | Algorithms, Conditions, and Debugging | Difficulty: Intermediate

Which device is commonly used to enter text into a computer?

A. Monitor

B. Projector

C. Keyboard

D. Speaker

Answer: C. A keyboard is an input device used for typing.

STEM-G6-ICT-T04-Q05 | Algorithms, Conditions, and Debugging | Difficulty: Intermediate

Software designed to help a user perform a task is commonly called an ...

A. Desk

B. Power cable

C. Fan

D. Application

Answer: D. An application is software that helps users perform specific tasks.

STEM-G6-ICT-T04-Q06 | Algorithms, Conditions, and Debugging | Difficulty: Intermediate

What should you do if an unknown person sends you a suspicious link?

A. Do not open it; check with a trusted adult or person

B. Enter your password

C. Download every attachment

D. Forward it to every contact

Answer: A. Suspicious links may lead to scams or harmful software.

STEM-G6-ICT-T04-Q07 | Algorithms, Conditions, and Debugging | Difficulty: Intermediate

Why should you keep your password private?

A. To make the screen brighter

B. To protect your account and personal information

C. To make the internet faster

D. To extend battery life

Answer: B. A password provides access to an account and should be protected.

STEM-G6-ICT-T04-Q08 | Algorithms, Conditions, and Debugging | Difficulty: Intermediate

A clear, ordered set of steps for solving a problem is called an ...

A. Resolution

B. Algorithm

C. Wallpaper

D. Speaker

Answer: B. An algorithm is an ordered set of instructions.

STEM-G6-ICT-T04-Q09 | Algorithms, Conditions, and Debugging | Difficulty: Intermediate

In a spreadsheet, a cell is ...

A. A webpage

B. An application icon

C. A type of cable

D. The intersection of a row and a column

Answer: D. A cell is a box at the intersection of a row and a column.

STEM-G6-ICT-T04-Q10 | Algorithms, Conditions, and Debugging | Difficulty: Intermediate

How should information found online be checked?

A. Count the emojis

B. Choose the most dramatic headline

C. Compare sources and supporting evidence

D. Share it before reading

Answer: C. Checking sources helps assess accuracy and credibility.

Junior High School Grade 7

160 questions • 4 subjects • 4 themes per subject • 10 questions per theme

Mathematics

Theme 1: Integers and Fractions — 10 questions

STEM-G7-MAT-T01-Q01 | Integers and Fractions | Difficulty: Advanced

Calculate $15^2 + 2$.

A. 224

B. 227

C. 237

D. 230

Answer: B. $15^2 = 225$, then $225 + 2 = 227$.

STEM-G7-MAT-T01-Q02 | Integers and Fractions | Difficulty: Advanced

If a = 17 and b = 10, calculate 2a + 3b.

A. 64

B. 74

C. 67

D. 61

Answer: A. 2(17) + 3(10) = 34 + 30 = 64.

STEM-G7-MAT-T01-Q03 | Integers and Fractions | Difficulty: Advanced

A rectangle is 3 cm long and 9 cm wide. What is its perimeter?

A. 34

B. 27

C. 21

D. 24

Answer: D. Perimeter = 2(length + width) = 2(3 + 9) = 24 cm.

STEM-G7-MAT-T01-Q04 | Integers and Fractions | Difficulty: Advanced

A rectangular area measures 7 cm by 2 cm. What is its area?

A. 14

B. 17

C. 11

D. 24

Answer: A. Area = 7 × 2 = 14 cm^2.

STEM-G7-MAT-T01-Q05 | Integers and Fractions | Difficulty: Advanced

If x = 12 and y = 4, what is x × y?

A. 58

B. 45

C. 48

D. 51

Answer: C. Substitute the values: 12 × 4 = 48.

STEM-G7-MAT-T01-Q06 | Integers and Fractions | Difficulty: Advanced

Calculate $11^2 + 6$.

A. 127

B. 124

C. 137

D. 130

Answer: A. $11^2 = 121$, then $121 + 6 = 127$.

STEM-G7-MAT-T01-Q07 | Integers and Fractions | Difficulty: Advanced

If a = 17 and b = 6, calculate 2a + 3b.

A. 49

B. 62

C. 55

D. 52

Answer: D. 2(17) + 3(6) = 34 + 18 = 52.

STEM-G7-MAT-T01-Q08 | Integers and Fractions | Difficulty: Advanced

A rectangle is 4 cm long and 5 cm wide. What is its perimeter?

A. 28

B. 18

C. 21

D. 15

Answer: B. Perimeter = 2(length + width) = 2(4 + 5) = 18 cm.

STEM-G7-MAT-T01-Q09 | Integers and Fractions | Difficulty: Advanced

A rectangular area measures 3 cm by 12 cm. What is its area?

A. 39

B. 33

C. 46

D. 36

Answer: D. Area = 3 × 12 = 36 cm².

STEM-G7-MAT-T01-Q10 | Integers and Fractions | Difficulty: Advanced

If x = 17 and y = 4, what is x × y?

A. 68

B. 71

C. 78

D. 65

Answer: A. Substitute the values: 17 × 4 = 68.

Theme 2: Algebraic Expressions — 10 questions

STEM-G7-MAT-T02-Q01 | Algebraic Expressions | Difficulty: Advanced

Calculate $12^2 + 3$.

A. 157

B. 144

C. 150

D. 147

Answer: D. $12^2 = 144$, then 144 + 3 = 147.

STEM-G7-MAT-T02-Q02 | Algebraic Expressions | Difficulty: Advanced

If a = 8 and b = 11, calculate 2a + 3b.

A. 52

B. 49

C. 59

D. 46

Answer: B. 2(8) + 3(11) = 16 + 33 = 49.

STEM-G7-MAT-T02-Q03 | Algebraic Expressions | Difficulty: Advanced

A rectangle is 13 cm long and 11 cm wide. What is its perimeter?

A. 51

B. 58

C. 45

D. 48

Answer: D. Perimeter = 2(length + width) = 2(13 + 11) = 48 cm.

STEM-G7-MAT-T02-Q04 | Algebraic Expressions | Difficulty: Advanced

A rectangular area measures 4 cm by 8 cm. What is its area?

A. 29

B. 32

C. 35

D. 42

Answer: B. Area = 4 × 8 = 32 cm².

STEM-G7-MAT-T02-Q05 | Algebraic Expressions | Difficulty: Advanced

If x = 6 and y = 4, what is x × y?

A. 21

B. 24

C. 27

D. 34

Answer: B. Substitute the values: 6 × 4 = 24.

STEM-G7-MAT-T02-Q06 | Algebraic Expressions | Difficulty: Advanced

Calculate $9^2 + 4$.

A. 85

B. 95

C. 88

D. 82

Answer: A. $9^2 = 81$, then 81 + 4 = 85.

STEM-G7-MAT-T02-Q07 | Algebraic Expressions | Difficulty: Advanced

If a = 11 and b = 2, calculate 2a + 3b.

A. 38

B. 31

C. 28

D. 25

Answer: C. 2(11) + 3(2) = 22 + 6 = 28.

STEM-G7-MAT-T02-Q08 | Algebraic Expressions | Difficulty: Advanced

A rectangle is 10 cm long and 3 cm wide. What is its perimeter?

A. 36

B. 29

C. 23

D. 26

Answer: D. Perimeter = 2(length + width) = 2(10 + 3) = 26 cm.

STEM-G7-MAT-T02-Q09 | Algebraic Expressions | Difficulty: Advanced

A rectangular area measures 14 cm by 8 cm. What is its area?

A. 122

B. 112

C. 115

D. 109

Answer: B. Area = 14 × 8 = 112 cm².

STEM-G7-MAT-T02-Q10 | Algebraic Expressions | Difficulty: Advanced

If x = 7 and y = 12, what is x × y?

A. 84

B. 94

C. 87

D. 81

Answer: A. Substitute the values: 7 × 12 = 84.

Theme 3: Ratios and Financial Arithmetic — 10 questions

STEM-G7-MAT-T03-Q01 | Ratios and Financial Arithmetic | Difficulty: Advanced

Calculate $17^2 + 10$.

A. 302

B. 299

C. 309

D. 296

Answer: B. $17^2 = 289$, then $289 + 10 = 299$.

STEM-G7-MAT-T03-Q02 | Ratios and Financial Arithmetic | Difficulty: Advanced

If a = 10 and b = 5, calculate 2a + 3b.

A. 45

B. 35

C. 32

D. 38

Answer: B. 2(10) + 3(5) = 20 + 15 = 35.

STEM-G7-MAT-T03-Q03 | Ratios and Financial Arithmetic | Difficulty: Advanced

A rectangle is 13 cm long and 3 cm wide. What is its perimeter?

A. 32

B. 42

C. 35

D. 29

Answer: A. Perimeter = 2(length + width) = 2(13 + 3) = 32 cm.

STEM-G7-MAT-T03-Q04 | Ratios and Financial Arithmetic | Difficulty: Advanced

A rectangular area measures 9 cm by 7 cm. What is its area?

A. 60

B. 66

C. 63

D. 73

Answer: C. Area = 9 × 7 = 63 cm^2.

STEM-G7-MAT-T03-Q05 | Ratios and Financial Arithmetic | Difficulty: Advanced

If x = 15 and y = 5, what is x × y?

A. 75

B. 78

C. 85

D. 72

Answer: A. Substitute the values: 15 × 5 = 75.

STEM-G7-MAT-T03-Q06 | Ratios and Financial Arithmetic | Difficulty: Advanced

Calculate $15^2 + 3$.

A. 228

B. 225

C. 238

D. 231

Answer: A. $15^2 = 225$, then $225 + 3 = 228$.

STEM-G7-MAT-T03-Q07 | Ratios and Financial Arithmetic | Difficulty: Advanced

If a = 13 and b = 5, calculate 2a + 3b.

A. 51

B. 44

C. 38

D. 41

Answer: D. 2(13) + 3(5) = 26 + 15 = 41.

STEM-G7-MAT-T03-Q08 | Ratios and Financial Arithmetic | Difficulty: Advanced

A rectangle is 17 cm long and 10 cm wide. What is its perimeter?

A. 54

B. 57

C. 51

D. 64

Answer: A. Perimeter = 2(length + width) = 2(17 + 10) = 54 cm.

STEM-G7-MAT-T03-Q09 | Ratios and Financial Arithmetic | Difficulty: Advanced

A rectangular area measures 10 cm by 10 cm. What is its area?

A. 100

B. 110

C. 103

D. 97

Answer: A. Area = 10 × 10 = 100 cm².

STEM-G7-MAT-T03-Q10 | Ratios and Financial Arithmetic | Difficulty: Advanced

If x = 14 and y = 7, what is x × y?

A. 98

B. 108

C. 95

D. 101

Answer: A. Substitute the values: 14 × 7 = 98.

Theme 4: Angles and Plane Shapes — 10 questions

STEM-G7-MAT-T04-Q01 | Angles and Plane Shapes | Difficulty: Advanced

Calculate $16^2 + 9$.

A. 265

B. 275

C. 262

D. 268

Answer: A. $16^2 = 256$, then 256 + 9 = 265.

STEM-G7-MAT-T04-Q02 | Angles and Plane Shapes | Difficulty: Advanced

If a = 17 and b = 2, calculate 2a + 3b.

A. 43

B. 50

C. 37

D. 40

Answer: D. 2(17) + 3(2) = 34 + 6 = 40.

STEM-G7-MAT-T04-Q03 | Angles and Plane Shapes | Difficulty: Advanced

A rectangle is 12 cm long and 9 cm wide. What is its perimeter?

A. 45

B. 52

C. 42

D. 39

Answer: C. Perimeter = 2(length + width) = 2(12 + 9) = 42 cm.

STEM-G7-MAT-T04-Q04 | Angles and Plane Shapes | Difficulty: Advanced

A rectangular area measures 5 cm by 12 cm. What is its area?

A. 63

B. 70

C. 57

D. 60

Answer: D. Area = 5 × 12 = 60 cm².

STEM-G7-MAT-T04-Q05 | Angles and Plane Shapes | Difficulty: Advanced

If x = 7 and y = 8, what is x × y?

A. 59

B. 53

C. 66

D. 56

Answer: D. Substitute the values: 7 × 8 = 56.

STEM-G7-MAT-T04-Q06 | Angles and Plane Shapes | Difficulty: Advanced

Calculate $13^2 + 4$.

A. 170

B. 173

C. 183

D. 176

Answer: B. $13^2 = 169$, then 169 + 4 = 173.

STEM-G7-MAT-T04-Q07 | Angles and Plane Shapes | Difficulty: Advanced

If a = 17 and b = 7, calculate 2a + 3b.

A. 55

B. 58

C. 65

D. 52

Answer: A. 2(17) + 3(7) = 34 + 21 = 55.

STEM-G7-MAT-T04-Q08 | Angles and Plane Shapes | Difficulty: Advanced

A rectangle is 10 cm long and 5 cm wide. What is its perimeter?

A. 33

B. 27

C. 30

D. 40

Answer: C. Perimeter = 2(length + width) = 2(10 + 5) = 30 cm.

STEM-G7-MAT-T04-Q09 | Angles and Plane Shapes | Difficulty: Advanced

A rectangular area measures 5 cm by 7 cm. What is its area?

A. 45

B. 38

C. 32

D. 35

Answer: D. Area = 5 × 7 = 35 cm^2.

STEM-G7-MAT-T04-Q10 | Angles and Plane Shapes | Difficulty: Advanced

If x = 11 and y = 3, what is x × y?

A. 33

B. 43

C. 30

D. 36

Answer: A. Substitute the values: 11 × 3 = 33.

Science

Theme 1: Classification of Living Things — 10 questions

STEM-G7-SCI-T01-Q01 | Classification of Living Things | Difficulty: Advanced

An organism that makes its own food through photosynthesis is a ...

A. Obligate parasite

B. Decomposer only

C. Producer

D. Top consumer

Answer: C. Producers form the energy base of many food chains.

STEM-G7-SCI-T01-Q02 | Classification of Living Things | Difficulty: Advanced

An organism that makes its own food through photosynthesis is a ...

A. Decomposer only

B. Producer

C. Obligate parasite

D. Top consumer

Answer: B. Producers form the energy base of many food chains.

STEM-G7-SCI-T01-Q03 | Classification of Living Things | Difficulty: Advanced

When investigating classification of living things, which approach is most scientifically sound?

A. Collect evidence, test explanations, and draw conclusions

B. Change results to match a prediction

C. Choose an answer by guessing

D. Ignore data that disagrees with an idea

Answer: A. Scientific inquiry uses testable evidence and conclusions supported by data.

STEM-G7-SCI-T01-Q04 | Classification of Living Things | Difficulty: Advanced

An organism that makes its own food through photosynthesis is a ...

A. Top consumer

B. Obligate parasite

C. Producer

D. Decomposer only

Answer: C. Producers form the energy base of many food chains.

STEM-G7-SCI-T01-Q05 | Classification of Living Things | Difficulty: Advanced

An organism that makes its own food through photosynthesis is a ...

A. Top consumer

B. Decomposer only

C. Obligate parasite

D. Producer

Answer: D. Producers form the energy base of many food chains.

STEM-G7-SCI-T01-Q06 | Classification of Living Things | Difficulty: Advanced

When investigating classification of living things, which approach is most scientifically sound?

A. Collect evidence, test explanations, and draw conclusions

B. Ignore data that disagrees with an idea

C. Change results to match a prediction

D. Choose an answer by guessing

Answer: A. Scientific inquiry uses testable evidence and conclusions supported by data.

STEM-G7-SCI-T01-Q07 | Classification of Living Things | Difficulty: Advanced

An organism that makes its own food through photosynthesis is a ...

A. Decomposer only

B. Top consumer

C. Obligate parasite

D. Producer

Answer: D. Producers form the energy base of many food chains.

STEM-G7-SCI-T01-Q08 | Classification of Living Things | Difficulty: Advanced

An organism that makes its own food through photosynthesis is a ...

A. Top consumer

B. Obligate parasite

C. Decomposer only

D. Producer

Answer: D. Producers form the energy base of many food chains.

STEM-G7-SCI-T01-Q09 | Classification of Living Things | Difficulty: Advanced

When investigating classification of living things, which approach is most scientifically sound?

A. Change results to match a prediction

B. Ignore data that disagrees with an idea

C. Choose an answer by guessing

D. Collect evidence, test explanations, and draw conclusions

Answer: D. Scientific inquiry uses testable evidence and conclusions supported by data.

STEM-G7-SCI-T01-Q10 | Classification of Living Things | Difficulty: Advanced

An organism that makes its own food through photosynthesis is a ...

A. Producer

B. Decomposer only

C. Obligate parasite

D. Top consumer

Answer: A. Producers form the energy base of many food chains.

Theme 2: Temperature, Heat, and Expansion — 10 questions

STEM-G7-SCI-T02-Q01 | Temperature, Heat, and Expansion | Difficulty: Advanced

Heat naturally flows from an object at ...

A. Lower mass to higher mass only

B. Darker color to lighter color always

C. Higher temperature to lower temperature

D. Lower temperature to higher temperature

Answer: C. Heat transfers spontaneously from higher to lower temperature.

STEM-G7-SCI-T02-Q02 | Temperature, Heat, and Expansion | Difficulty: Advanced

Heat naturally flows from an object at ...

A. Lower mass to higher mass only

B. Lower temperature to higher temperature

C. Higher temperature to lower temperature

D. Darker color to lighter color always

Answer: C. Heat transfers spontaneously from higher to lower temperature.

STEM-G7-SCI-T02-Q03 | Temperature, Heat, and Expansion | Difficulty: Advanced

When investigating temperature, heat, and expansion, which approach is most scientifically sound?

A. Change results to match a prediction

B. Choose an answer by guessing

C. Collect evidence, test explanations, and draw conclusions

D. Ignore data that disagrees with an idea

Answer: C. Scientific inquiry uses testable evidence and conclusions supported by data.

STEM-G7-SCI-T02-Q04 | Temperature, Heat, and Expansion | Difficulty: Advanced

Heat naturally flows from an object at ...

A. Lower temperature to higher temperature

B. Lower mass to higher mass only

C. Darker color to lighter color always

D. Higher temperature to lower temperature

Answer: D. Heat transfers spontaneously from higher to lower temperature.

STEM-G7-SCI-T02-Q05 | Temperature, Heat, and Expansion | Difficulty: Advanced

Heat naturally flows from an object at ...

A. Higher temperature to lower temperature

B. Darker color to lighter color always

C. Lower temperature to higher temperature

D. Lower mass to higher mass only

Answer: A. Heat transfers spontaneously from higher to lower temperature.

STEM-G7-SCI-T02-Q06 | Temperature, Heat, and Expansion | Difficulty: Advanced

When investigating temperature, heat, and expansion, which approach is most scientifically sound?

A. Ignore data that disagrees with an idea

B. Choose an answer by guessing

C. Change results to match a prediction

D. Collect evidence, test explanations, and draw conclusions

Answer: D. Scientific inquiry uses testable evidence and conclusions supported by data.

STEM-G7-SCI-T02-Q07 | Temperature, Heat, and Expansion | Difficulty: Advanced

Heat naturally flows from an object at ...

A. Lower temperature to higher temperature

B. Darker color to lighter color always

C. Lower mass to higher mass only

D. Higher temperature to lower temperature

Answer: D. Heat transfers spontaneously from higher to lower temperature.

STEM-G7-SCI-T02-Q08 | Temperature, Heat, and Expansion | Difficulty: Advanced

Heat naturally flows from an object at ...

A. Lower mass to higher mass only

B. Darker color to lighter color always

C. Lower temperature to higher temperature

D. Higher temperature to lower temperature

Answer: D. Heat transfers spontaneously from higher to lower temperature.

STEM-G7-SCI-T02-Q09 | Temperature, Heat, and Expansion | Difficulty: Advanced

When investigating temperature, heat, and expansion, which approach is most scientifically sound?

A. Collect evidence, test explanations, and draw conclusions

B. Ignore data that disagrees with an idea

C. Change results to match a prediction

D. Choose an answer by guessing

Answer: A. Scientific inquiry uses testable evidence and conclusions supported by data.

STEM-G7-SCI-T02-Q10 | Temperature, Heat, and Expansion | Difficulty: Advanced

Heat naturally flows from an object at ...

A. Darker color to lighter color always

B. Higher temperature to lower temperature

C. Lower mass to higher mass only

D. Lower temperature to higher temperature

Answer: B. Heat transfers spontaneously from higher to lower temperature.

Theme 3: Organization of Life — 10 questions

STEM-G7-SCI-T03-Q01 | Organization of Life | Difficulty: Advanced

What is the smallest structural and functional unit of a living organism?

A. A cell

B. A tissue

C. An organ

D. An organ system

Answer: A. The cell is the basic unit of life.

STEM-G7-SCI-T03-Q02 | Organization of Life | Difficulty: Advanced

What is the smallest structural and functional unit of a living organism?

A. An organ system

B. A tissue

C. A cell

D. An organ

Answer: C. The cell is the basic unit of life.

STEM-G7-SCI-T03-Q03 | Organization of Life | Difficulty: Advanced

When investigating organization of life, which approach is most scientifically sound?

A. Collect evidence, test explanations, and draw conclusions

B. Ignore data that disagrees with an idea

C. Choose an answer by guessing

D. Change results to match a prediction

Answer: A. Scientific inquiry uses testable evidence and conclusions supported by data.

STEM-G7-SCI-T03-Q04 | Organization of Life | Difficulty: Advanced

What is the smallest structural and functional unit of a living organism?

A. A tissue

B. A cell

C. An organ system

D. An organ

Answer: B. The cell is the basic unit of life.

STEM-G7-SCI-T03-Q05 | Organization of Life | Difficulty: Advanced

What is the smallest structural and functional unit of a living organism?

A. An organ system

B. An organ

C. A tissue

D. A cell

Answer: D. The cell is the basic unit of life.

STEM-G7-SCI-T03-Q06 | Organization of Life | Difficulty: Advanced

When investigating organization of life, which approach is most scientifically sound?

A. Choose an answer by guessing

B. Ignore data that disagrees with an idea

C. Collect evidence, test explanations, and draw conclusions

D. Change results to match a prediction

Answer: C. Scientific inquiry uses testable evidence and conclusions supported by data.

STEM-G7-SCI-T03-Q07 | Organization of Life | Difficulty: Advanced

What is the smallest structural and functional unit of a living organism?

A. A tissue

B. An organ system

C. An organ

D. A cell

Answer: D. The cell is the basic unit of life.

STEM-G7-SCI-T03-Q08 | Organization of Life | Difficulty: Advanced

What is the smallest structural and functional unit of a living organism?

A. A tissue

B. An organ system

C. An organ

D. A cell

Answer: D. The cell is the basic unit of life.

STEM-G7-SCI-T03-Q09 | Organization of Life | Difficulty: Advanced

When investigating organization of life, which approach is most scientifically sound?

A. Change results to match a prediction

B. Ignore data that disagrees with an idea

C. Collect evidence, test explanations, and draw conclusions

D. Choose an answer by guessing

Answer: C. Scientific inquiry uses testable evidence and conclusions supported by data.

STEM-G7-SCI-T03-Q10 | Organization of Life | Difficulty: Advanced

What is the smallest structural and functional unit of a living organism?

A. An organ

B. An organ system

C. A tissue

D. A cell

Answer: D. The cell is the basic unit of life.

Theme 4: Motion and Forces — 10 questions

STEM-G7-SCI-T04-Q01 | Motion and Forces | Difficulty: Advanced

What is the SI unit of force?

A. Newton

B. Watt

C. Pascal

D. Joule

Answer: A. Force is measured in newtons.

STEM-G7-SCI-T04-Q02 | Motion and Forces | Difficulty: Advanced

What is the SI unit of force?

A. Joule

B. Newton

C. Pascal

D. Watt

Answer: B. Force is measured in newtons.

STEM-G7-SCI-T04-Q03 | Motion and Forces | Difficulty: Advanced

When investigating motion and forces, which approach is most scientifically sound?

A. Collect evidence, test explanations, and draw conclusions

B. Ignore data that disagrees with an idea

C. Choose an answer by guessing

D. Change results to match a prediction

Answer: A. Scientific inquiry uses testable evidence and conclusions supported by data.

STEM-G7-SCI-T04-Q04 | Motion and Forces | Difficulty: Advanced

What is the SI unit of force?

A. Watt

B. Pascal

C. Newton

D. Joule

Answer: C. Force is measured in newtons.

STEM-G7-SCI-T04-Q05 | Motion and Forces | Difficulty: Advanced

What is the SI unit of force?

A. Watt

B. Pascal

C. Newton

D. Joule

Answer: C. Force is measured in newtons.

STEM-G7-SCI-T04-Q06 | Motion and Forces | Difficulty: Advanced

When investigating motion and forces, which approach is most scientifically sound?

A. Ignore data that disagrees with an idea

B. Choose an answer by guessing

C. Change results to match a prediction

D. Collect evidence, test explanations, and draw conclusions

Answer: D. Scientific inquiry uses testable evidence and conclusions supported by data.

STEM-G7-SCI-T04-Q07 | Motion and Forces | Difficulty: Advanced

What is the SI unit of force?

A. Newton

B. Watt

C. Pascal

D. Joule

Answer: A. Force is measured in newtons.

STEM-G7-SCI-T04-Q08 | Motion and Forces | Difficulty: Advanced

What is the SI unit of force?

A. Pascal

B. Watt

C. Newton

D. Joule

Answer: C. Force is measured in newtons.

STEM-G7-SCI-T04-Q09 | Motion and Forces | Difficulty: Advanced

When investigating motion and forces, which approach is most scientifically sound?

A. Choose an answer by guessing

B. Change results to match a prediction

C. Collect evidence, test explanations, and draw conclusions

D. Ignore data that disagrees with an idea

Answer: C. Scientific inquiry uses testable evidence and conclusions supported by data.

STEM-G7-SCI-T04-Q10 | Motion and Forces | Difficulty: Advanced

What is the SI unit of force?

A. Joule

B. Pascal

C. Newton

D. Watt

Answer: C. Force is measured in newtons.

Engineering

Theme 1: Forces, Pressure, and Structures — 10 questions

STEM-G7-ENG-T01-Q01 | Forces, Pressure, and Structures | Difficulty: Advanced

Why should an engineering prototype be tested before use?

A. To avoid making improvements

B. To find out whether the design works and is safe

C. To make it look more expensive

D. To make it larger every time

Answer: B. Testing reveals weaknesses and helps improve a design.

STEM-G7-ENG-T01-Q02 | Forces, Pressure, and Structures | Difficulty: Advanced

If a model structure tips over easily, what is a sensible first improvement?

A. Add weight to the top

B. Remove all supports

C. Remove the base

D. Widen the base or strengthen the joints

Answer: D. A wider base and strong joints can improve stability.

STEM-G7-ENG-T01-Q03 | Forces, Pressure, and Structures | Difficulty: Advanced

What is the purpose of making a sketch before building an object?

A. To plan its shape, dimensions, and parts

B. To make the object build itself

C. To avoid measuring

D. To replace all testing

Answer: A. A sketch communicates a design before construction.

STEM-G7-ENG-T01-Q04 | Forces, Pressure, and Structures | Difficulty: Advanced

For a fair test, a team should ...

A. Choose only the preferred result

B. Change every factor at once

C. Change one factor while keeping other factors the same

D. Avoid recording results

Answer: C. Controlling variables helps identify what caused a result.

STEM-G7-ENG-T01-Q05 | Forces, Pressure, and Structures | Difficulty: Advanced

When selecting a material for a load-bearing part, what should be considered?

A. The designer's name

B. The packaging

C. Only its favorite color

D. Its strength and suitability for the task

Answer: D. Material choice depends on the load and operating conditions.

STEM-G7-ENG-T01-Q06 | Forces, Pressure, and Structures | Difficulty: Advanced

Why should joints in a model be inspected?

A. Joints remove gravity

B. Joints transfer forces between parts

C. Joints are only decoration

D. Joints make mass zero

Answer: B. Weak joints can become failure points in a structure.

STEM-G7-ENG-T01-Q07 | Forces, Pressure, and Structures | Difficulty: Advanced

After a test reveals a weakness, what should engineers do next?

A. Ignore safety

B. Hide the result

C. Improve the design and test it again

D. Delete the notes

Answer: C. Engineering is iterative and uses test results to guide improvements.

STEM-G7-ENG-T01-Q08 | Forces, Pressure, and Structures | Difficulty: Advanced

Which measurement is useful when comparing two bridge designs?

A. The number of stickers

B. The model's name

C. The maximum load each bridge can support

D. The bridge color

Answer: C. Load capacity is a relevant measure of bridge performance.

STEM-G7-ENG-T01-Q09 | Forces, Pressure, and Structures | Difficulty: Advanced

Why should an engineering prototype be tested before use?

A. To find out whether the design works and is safe

B. To make it look more expensive

C. To avoid making improvements

D. To make it larger every time

Answer: A. Testing reveals weaknesses and helps improve a design.

STEM-G7-ENG-T01-Q10 | Forces, Pressure, and Structures | Difficulty: Advanced

If a model structure tips over easily, what is a sensible first improvement?

A. Remove all supports

B. Add weight to the top

C. Widen the base or strengthen the joints

D. Remove the base

Answer: C. A wider base and strong joints can improve stability.

Theme 2: Mechanisms and Motion Systems — 10 questions

STEM-G7-ENG-T02-Q01 | Mechanisms and Motion Systems | Difficulty: Advanced

Why should an engineering prototype be tested before use?

A. To avoid making improvements

B. To make it look more expensive

C. To make it larger every time

D. To find out whether the design works and is safe

Answer: D. Testing reveals weaknesses and helps improve a design.

STEM-G7-ENG-T02-Q02 | Mechanisms and Motion Systems | Difficulty: Advanced

If a model structure tips over easily, what is a sensible first improvement?

A. Widen the base or strengthen the joints

B. Remove all supports

C. Remove the base

D. Add weight to the top

Answer: A. A wider base and strong joints can improve stability.

STEM-G7-ENG-T02-Q03 | Mechanisms and Motion Systems | Difficulty: Advanced

What is the purpose of making a sketch before building an object?

A. To replace all testing

B. To make the object build itself

C. To plan its shape, dimensions, and parts

D. To avoid measuring

Answer: C. A sketch communicates a design before construction.

STEM-G7-ENG-T02-Q04 | Mechanisms and Motion Systems | Difficulty: Advanced

For a fair test, a team should ...

A. Choose only the preferred result

B. Avoid recording results

C. Change one factor while keeping other factors the same

D. Change every factor at once

Answer: C. Controlling variables helps identify what caused a result.

STEM-G7-ENG-T02-Q05 | Mechanisms and Motion Systems | Difficulty: Advanced

When selecting a material for a load-bearing part, what should be considered?

A. The packaging

B. The designer's name

C. Only its favorite color

D. Its strength and suitability for the task

Answer: D. Material choice depends on the load and operating conditions.

STEM-G7-ENG-T02-Q06 | Mechanisms and Motion Systems | Difficulty: Advanced

Why should joints in a model be inspected?

A. Joints transfer forces between parts

B. Joints remove gravity

C. Joints make mass zero

D. Joints are only decoration

Answer: A. Weak joints can become failure points in a structure.

STEM-G7-ENG-T02-Q07 | Mechanisms and Motion Systems | Difficulty: Advanced

After a test reveals a weakness, what should engineers do next?

A. Hide the result

B. Improve the design and test it again

C. Ignore safety

D. Delete the notes

Answer: B. Engineering is iterative and uses test results to guide improvements.

STEM-G7-ENG-T02-Q08 | Mechanisms and Motion Systems | Difficulty: Advanced

Which measurement is useful when comparing two bridge designs?

A. The model's name

B. The number of stickers

C. The bridge color

D. The maximum load each bridge can support

Answer: D. Load capacity is a relevant measure of bridge performance.

STEM-G7-ENG-T02-Q09 | Mechanisms and Motion Systems | Difficulty: Advanced

Why should an engineering prototype be tested before use?

A. To make it larger every time

B. To make it look more expensive

C. To find out whether the design works and is safe

D. To avoid making improvements

Answer: C. Testing reveals weaknesses and helps improve a design.

STEM-G7-ENG-T02-Q10 | Mechanisms and Motion Systems | Difficulty: Advanced

If a model structure tips over easily, what is a sensible first improvement?

A. Remove the base

B. Remove all supports

C. Add weight to the top

D. Widen the base or strengthen the joints

Answer: D. A wider base and strong joints can improve stability.

Theme 3: Electrical Circuits and Components — 10 questions

STEM-G7-ENG-T03-Q01 | Electrical Circuits and Components | Difficulty: Advanced

Why should an engineering prototype be tested before use?

A. To make it larger every time

B. To avoid making improvements

C. To find out whether the design works and is safe

D. To make it look more expensive

Answer: C. Testing reveals weaknesses and helps improve a design.

STEM-G7-ENG-T03-Q02 | Electrical Circuits and Components | Difficulty: Advanced

If a model structure tips over easily, what is a sensible first improvement?

A. Remove all supports

B. Remove the base

C. Add weight to the top

D. Widen the base or strengthen the joints

Answer: D. A wider base and strong joints can improve stability.

STEM-G7-ENG-T03-Q03 | Electrical Circuits and Components | Difficulty: Advanced

What is the purpose of making a sketch before building an object?

A. To replace all testing

B. To plan its shape, dimensions, and parts

C. To avoid measuring

D. To make the object build itself

Answer: B. A sketch communicates a design before construction.

STEM-G7-ENG-T03-Q04 | Electrical Circuits and Components | Difficulty: Advanced

For a fair test, a team should ...

A. Change every factor at once

B. Avoid recording results

C. Change one factor while keeping other factors the same

D. Choose only the preferred result

Answer: C. Controlling variables helps identify what caused a result.

STEM-G7-ENG-T03-Q05 | Electrical Circuits and Components | Difficulty: Advanced

When selecting a material for a load-bearing part, what should be considered?

A. The designer's name

B. Its strength and suitability for the task

C. Only its favorite color

D. The packaging

Answer: B. Material choice depends on the load and operating conditions.

STEM-G7-ENG-T03-Q06 | Electrical Circuits and Components | Difficulty: Advanced

Why should joints in a model be inspected?

A. Joints transfer forces between parts

B. Joints remove gravity

C. Joints are only decoration

D. Joints make mass zero

Answer: A. Weak joints can become failure points in a structure.

STEM-G7-ENG-T03-Q07 | Electrical Circuits and Components | Difficulty: Advanced

After a test reveals a weakness, what should engineers do next?

A. Improve the design and test it again

B. Hide the result

C. Ignore safety

D. Delete the notes

Answer: A. Engineering is iterative and uses test results to guide improvements.

STEM-G7-ENG-T03-Q08 | Electrical Circuits and Components | Difficulty: Advanced

Which measurement is useful when comparing two bridge designs?

A. The number of stickers

B. The bridge color

C. The model's name

D. The maximum load each bridge can support

Answer: D. Load capacity is a relevant measure of bridge performance.

STEM-G7-ENG-T03-Q09 | Electrical Circuits and Components | Difficulty: Advanced

Why should an engineering prototype be tested before use?

A. To avoid making improvements

B. To find out whether the design works and is safe

C. To make it larger every time

D. To make it look more expensive

Answer: B. Testing reveals weaknesses and helps improve a design.

STEM-G7-ENG-T03-Q10 | Electrical Circuits and Components | Difficulty: Advanced

If a model structure tips over easily, what is a sensible first improvement?

A. Remove all supports

B. Remove the base

C. Add weight to the top

D. Widen the base or strengthen the joints

Answer: D. A wider base and strong joints can improve stability.

Theme 4: The Engineering Design Cycle — 10 questions

STEM-G7-ENG-T04-Q01 | The Engineering Design Cycle | Difficulty: Advanced

Why should an engineering prototype be tested before use?

A. To make it look more expensive

B. To find out whether the design works and is safe

C. To avoid making improvements

D. To make it larger every time

Answer: B. Testing reveals weaknesses and helps improve a design.

STEM-G7-ENG-T04-Q02 | The Engineering Design Cycle | Difficulty: Advanced

If a model structure tips over easily, what is a sensible first improvement?

A. Add weight to the top

B. Remove all supports

C. Widen the base or strengthen the joints

D. Remove the base

Answer: C. A wider base and strong joints can improve stability.

STEM-G7-ENG-T04-Q03 | The Engineering Design Cycle | Difficulty: Advanced

What is the purpose of making a sketch before building an object?

A. To make the object build itself

B. To plan its shape, dimensions, and parts

C. To avoid measuring

D. To replace all testing

Answer: B. A sketch communicates a design before construction.

STEM-G7-ENG-T04-Q04 | The Engineering Design Cycle | Difficulty: Advanced

For a fair test, a team should ...

A. Change one factor while keeping other factors the same

B. Choose only the preferred result

C. Avoid recording results

D. Change every factor at once

Answer: A. Controlling variables helps identify what caused a result.

STEM-G7-ENG-T04-Q05 | The Engineering Design Cycle | Difficulty: Advanced

When selecting a material for a load-bearing part, what should be considered?

A. Its strength and suitability for the task

B. Only its favorite color

C. The designer's name

D. The packaging

Answer: A. Material choice depends on the load and operating conditions.

STEM-G7-ENG-T04-Q06 | The Engineering Design Cycle | Difficulty: Advanced

Why should joints in a model be inspected?

A. Joints transfer forces between parts

B. Joints make mass zero

C. Joints remove gravity

D. Joints are only decoration

Answer: A. Weak joints can become failure points in a structure.

STEM-G7-ENG-T04-Q07 | The Engineering Design Cycle | Difficulty: Advanced

After a test reveals a weakness, what should engineers do next?

A. Delete the notes

B. Improve the design and test it again

C. Hide the result

D. Ignore safety

Answer: B. Engineering is iterative and uses test results to guide improvements.

STEM-G7-ENG-T04-Q08 | The Engineering Design Cycle | Difficulty: Advanced

Which measurement is useful when comparing two bridge designs?

A. The model's name

B. The maximum load each bridge can support

C. The bridge color

D. The number of stickers

Answer: B. Load capacity is a relevant measure of bridge performance.

STEM-G7-ENG-T04-Q09 | The Engineering Design Cycle | Difficulty: Advanced

Why should an engineering prototype be tested before use?

A. To find out whether the design works and is safe

B. To make it look more expensive

C. To make it larger every time

D. To avoid making improvements

Answer: A. Testing reveals weaknesses and helps improve a design.

STEM-G7-ENG-T04-Q10 | The Engineering Design Cycle | Difficulty: Advanced

If a model structure tips over easily, what is a sensible first improvement?

A. Remove all supports

B. Widen the base or strengthen the joints

C. Remove the base

D. Add weight to the top

Answer: B. A wider base and strong joints can improve stability.

ICT (Technology)

Theme 1: Computer Components and Operating Systems — 10 questions

STEM-G7-ICT-T01-Q01 | Computer Components and Operating Systems | Difficulty: Advanced

Finding and fixing errors in a program is called ...

A. Formatting

B. Debugging

C. Browsing

D. Streaming

Answer: B. Debugging is the process of locating and correcting errors.

STEM-G7-ICT-T01-Q02 | Computer Components and Operating Systems | Difficulty: Advanced

Which device is commonly used to enter text into a computer?

A. Keyboard

B. Monitor

C. Speaker

D. Projector

Answer: A. A keyboard is an input device used for typing.

STEM-G7-ICT-T01-Q03 | Computer Components and Operating Systems | Difficulty: Advanced

In the context of computer components and operating systems, which practice is most appropriate?

A. Assume every source is always correct

B. Share account credentials

C. Ignore validation

D. Use data and permissions responsibly, and verify results

Answer: D. Responsible technology use requires validation, security, and digital responsibility.

STEM-G7-ICT-T01-Q04 | Computer Components and Operating Systems | Difficulty: Advanced

What should you do if an unknown person sends you a suspicious link?

A. Download every attachment

B. Do not open it; check with a trusted adult or person

C. Forward it to every contact

D. Enter your password

Answer: B. Suspicious links may lead to scams or harmful software.

STEM-G7-ICT-T01-Q05 | Computer Components and Operating Systems | Difficulty: Advanced

Why should you keep your password private?

A. To extend battery life

B. To protect your account and personal information

C. To make the internet faster

D. To make the screen brighter

Answer: B. A password provides access to an account and should be protected.

STEM-G7-ICT-T01-Q06 | Computer Components and Operating Systems | Difficulty: Advanced

In the context of computer components and operating systems, which practice is most appropriate?

A. Use data and permissions responsibly, and verify results

B. Ignore validation

C. Share account credentials

D. Assume every source is always correct

Answer: A. Responsible technology use requires validation, security, and digital responsibility.

STEM-G7-ICT-T01-Q07 | Computer Components and Operating Systems | Difficulty: Advanced

In a spreadsheet, a cell is ...

A. An application icon

B. A type of cable

C. The intersection of a row and a column

D. A webpage

Answer: C. A cell is a box at the intersection of a row and a column.

STEM-G7-ICT-T01-Q08 | Computer Components and Operating Systems | Difficulty: Advanced

How should information found online be checked?

A. Count the emojis

B. Compare sources and supporting evidence

C. Choose the most dramatic headline

D. Share it before reading

Answer: B. Checking sources helps assess accuracy and credibility.

STEM-G7-ICT-T01-Q09 | Computer Components and Operating Systems | Difficulty: Advanced

In the context of computer components and operating systems, which practice is most appropriate?

A. Share account credentials

B. Assume every source is always correct

C. Ignore validation

D. Use data and permissions responsibly, and verify results

Answer: D. Responsible technology use requires validation, security, and digital responsibility.

STEM-G7-ICT-T01-Q10 | Computer Components and Operating Systems | Difficulty: Advanced

Which device is commonly used to enter text into a computer?

A. Keyboard

B. Projector

C. Monitor

D. Speaker

Answer: A. A keyboard is an input device used for typing.

Theme 2: Computer Networks and the Internet — 10 questions

STEM-G7-ICT-T02-Q01 | Computer Networks and the Internet | Difficulty: Advanced

Finding and fixing errors in a program is called ...

A. Browsing

B. Formatting

C. Debugging

D. Streaming

Answer: C. Debugging is the process of locating and correcting errors.

STEM-G7-ICT-T02-Q02 | Computer Networks and the Internet | Difficulty: Advanced

Which device is commonly used to enter text into a computer?

A. Monitor

B. Projector

C. Keyboard

D. Speaker

Answer: C. A keyboard is an input device used for typing.

STEM-G7-ICT-T02-Q03 | Computer Networks and the Internet | Difficulty: Advanced

In the context of computer networks and the internet, which practice is most appropriate?

A. Use data and permissions responsibly, and verify results

B. Ignore validation

C. Share account credentials

D. Assume every source is always correct

Answer: A. Responsible technology use requires validation, security, and digital responsibility.

STEM-G7-ICT-T02-Q04 | Computer Networks and the Internet | Difficulty: Advanced

What should you do if an unknown person sends you a suspicious link?

A. Do not open it; check with a trusted adult or person

B. Forward it to every contact

C. Enter your password

D. Download every attachment

Answer: A. Suspicious links may lead to scams or harmful software.

STEM-G7-ICT-T02-Q05 | Computer Networks and the Internet | Difficulty: Advanced

Why should you keep your password private?

A. To extend battery life

B. To make the internet faster

C. To make the screen brighter

D. To protect your account and personal information

Answer: D. A password provides access to an account and should be protected.

STEM-G7-ICT-T02-Q06 | Computer Networks and the Internet | Difficulty: Advanced

In the context of computer networks and the internet, which practice is most appropriate?

A. Use data and permissions responsibly, and verify results

B. Share account credentials

C. Assume every source is always correct

D. Ignore validation

Answer: A. Responsible technology use requires validation, security, and digital responsibility.

STEM-G7-ICT-T02-Q07 | Computer Networks and the Internet | Difficulty: Advanced

In a spreadsheet, a cell is ...

A. A type of cable

B. A webpage

C. An application icon

D. The intersection of a row and a column

Answer: D. A cell is a box at the intersection of a row and a column.

STEM-G7-ICT-T02-Q08 | Computer Networks and the Internet | Difficulty: Advanced

How should information found online be checked?

A. Compare sources and supporting evidence

B. Share it before reading

C. Count the emojis

D. Choose the most dramatic headline

Answer: A. Checking sources helps assess accuracy and credibility.

STEM-G7-ICT-T02-Q09 | Computer Networks and the Internet | Difficulty: Advanced

In the context of computer networks and the internet, which practice is most appropriate?

A. Assume every source is always correct

B. Use data and permissions responsibly, and verify results

C. Ignore validation

D. Share account credentials

Answer: B. Responsible technology use requires validation, security, and digital responsibility.

STEM-G7-ICT-T02-Q10 | Computer Networks and the Internet | Difficulty: Advanced

Which device is commonly used to enter text into a computer?

A. Monitor

B. Projector

C. Speaker

D. Keyboard

Answer: D. A keyboard is an input device used for typing.

Theme 3: Digital Data Representation — 10 questions

STEM-G7-ICT-T03-Q01 | Digital Data Representation | Difficulty: Advanced

Finding and fixing errors in a program is called ...

A. Debugging

B. Streaming

C. Formatting

D. Browsing

Answer: A. Debugging is the process of locating and correcting errors.

STEM-G7-ICT-T03-Q02 | Digital Data Representation | Difficulty: Advanced

Which device is commonly used to enter text into a computer?

A. Speaker

B. Projector

C. Monitor

D. Keyboard

Answer: D. A keyboard is an input device used for typing.

STEM-G7-ICT-T03-Q03 | Digital Data Representation | Difficulty: Advanced

In the context of digital data representation, which practice is most appropriate?

A. Share account credentials

B. Use data and permissions responsibly, and verify results

C. Assume every source is always correct

D. Ignore validation

Answer: B. Responsible technology use requires validation, security, and digital responsibility.

STEM-G7-ICT-T03-Q04 | Digital Data Representation | Difficulty: Advanced

What should you do if an unknown person sends you a suspicious link?

A. Download every attachment

B. Forward it to every contact

C. Do not open it; check with a trusted adult or person

D. Enter your password

Answer: C. Suspicious links may lead to scams or harmful software.

STEM-G7-ICT-T03-Q05 | Digital Data Representation | Difficulty: Advanced

Why should you keep your password private?

A. To protect your account and personal information

B. To make the screen brighter

C. To make the internet faster

D. To extend battery life

Answer: A. A password provides access to an account and should be protected.

STEM-G7-ICT-T03-Q06 | Digital Data Representation | Difficulty: Advanced

In the context of digital data representation, which practice is most appropriate?

A. Share account credentials

B. Assume every source is always correct

C. Use data and permissions responsibly, and verify results

D. Ignore validation

Answer: C. Responsible technology use requires validation, security, and digital responsibility.

STEM-G7-ICT-T03-Q07 | Digital Data Representation | Difficulty: Advanced

In a spreadsheet, a cell is ...

A. A type of cable

B. The intersection of a row and a column

C. An application icon

D. A webpage

Answer: B. A cell is a box at the intersection of a row and a column.

STEM-G7-ICT-T03-Q08 | Digital Data Representation | Difficulty: Advanced

How should information found online be checked?

A. Share it before reading

B. Compare sources and supporting evidence

C. Count the emojis

D. Choose the most dramatic headline

Answer: B. Checking sources helps assess accuracy and credibility.

STEM-G7-ICT-T03-Q09 | Digital Data Representation | Difficulty: Advanced

In the context of digital data representation, which practice is most appropriate?

A. Share account credentials

B. Ignore validation

C. Assume every source is always correct

D. Use data and permissions responsibly, and verify results

Answer: D. Responsible technology use requires validation, security, and digital responsibility.

STEM-G7-ICT-T03-Q10 | Digital Data Representation | Difficulty: Advanced

Which device is commonly used to enter text into a computer?

A. Projector

B. Speaker

C. Monitor

D. Keyboard

Answer: D. A keyboard is an input device used for typing.

Theme 4: Algorithms and Introductory Programming — 10 questions

STEM-G7-ICT-T04-Q01 | Algorithms and Introductory Programming | Difficulty: Advanced

Finding and fixing errors in a program is called ...

A. Browsing

B. Debugging

C. Streaming

D. Formatting

Answer: B. Debugging is the process of locating and correcting errors.

STEM-G7-ICT-T04-Q02 | Algorithms and Introductory Programming | Difficulty: Advanced

Which device is commonly used to enter text into a computer?

A. Speaker

B. Monitor

C. Keyboard

D. Projector

Answer: C. A keyboard is an input device used for typing.

STEM-G7-ICT-T04-Q03 | Algorithms and Introductory Programming | Difficulty: Advanced

In the context of algorithms and introductory programming, which practice is most appropriate?

A. Ignore validation

B. Assume every source is always correct

C. Use data and permissions responsibly, and verify results

D. Share account credentials

Answer: C. Responsible technology use requires validation, security, and digital responsibility.

STEM-G7-ICT-T04-Q04 | Algorithms and Introductory Programming | Difficulty: Advanced

What should you do if an unknown person sends you a suspicious link?

A. Do not open it; check with a trusted adult or person

B. Forward it to every contact

C. Enter your password

D. Download every attachment

Answer: A. Suspicious links may lead to scams or harmful software.

STEM-G7-ICT-T04-Q05 | Algorithms and Introductory Programming | Difficulty: Advanced

Why should you keep your password private?

A. To make the screen brighter

B. To make the internet faster

C. To protect your account and personal information

D. To extend battery life

Answer: C. A password provides access to an account and should be protected.

STEM-G7-ICT-T04-Q06 | Algorithms and Introductory Programming | Difficulty: Advanced

In the context of algorithms and introductory programming, which practice is most appropriate?

A. Assume every source is always correct

B. Share account credentials

C. Use data and permissions responsibly, and verify results

D. Ignore validation

Answer: C. Responsible technology use requires validation, security, and digital responsibility.

STEM-G7-ICT-T04-Q07 | Algorithms and Introductory Programming | Difficulty: Advanced

In a spreadsheet, a cell is ...

A. A webpage

B. An application icon

C. A type of cable

D. The intersection of a row and a column

Answer: D. A cell is a box at the intersection of a row and a column.

STEM-G7-ICT-T04-Q08 | Algorithms and Introductory Programming | Difficulty: Advanced

How should information found online be checked?

A. Share it before reading

B. Choose the most dramatic headline

C. Compare sources and supporting evidence

D. Count the emojis

Answer: C. Checking sources helps assess accuracy and credibility.

STEM-G7-ICT-T04-Q09 | Algorithms and Introductory Programming | Difficulty: Advanced

In the context of algorithms and introductory programming, which practice is most appropriate?

A. Assume every source is always correct

B. Use data and permissions responsibly, and verify results

C. Share account credentials

D. Ignore validation

Answer: B. Responsible technology use requires validation, security, and digital responsibility.

STEM-G7-ICT-T04-Q10 | Algorithms and Introductory Programming | Difficulty: Advanced

Which device is commonly used to enter text into a computer?

A. Projector

B. Speaker

C. Keyboard

D. Monitor

Answer: C. A keyboard is an input device used for typing.

Junior High School Grade 8

160 questions • 4 subjects • 4 themes per subject • 10 questions per theme

Mathematics

Theme 1: Number Patterns and Sequences — 10 questions

STEM-G8-MAT-T01-Q01 | Number Patterns and Sequences | Difficulty: Advanced

Calculate $14^2 + 3$.

A. 209

B. 199

C. 202

D. 196

Answer: B. $14^2 = 196$, then $196 + 3 = 199$.

STEM-G8-MAT-T01-Q02 | Number Patterns and Sequences | Difficulty: Advanced

If a = 17 and b = 5, calculate 2a + 3b.

A. 46

B. 52

C. 59

D. 49

Answer: D. 2(17) + 3(5) = 34 + 15 = 49.

STEM-G8-MAT-T01-Q03 | Number Patterns and Sequences | Difficulty: Advanced

A rectangle is 18 cm long and 10 cm wide. What is its perimeter?

A. 66

B. 56

C. 59

D. 53

Answer: B. Perimeter = 2(length + width) = 2(18 + 10) = 56 cm.

STEM-G8-MAT-T01-Q04 | Number Patterns and Sequences | Difficulty: Advanced

A rectangular area measures 8 cm by 6 cm. What is its area?

A. 45

B. 48

C. 58

D. 51

Answer: B. Area = 8 × 6 = 48 cm^2.

STEM-G8-MAT-T01-Q05 | Number Patterns and Sequences | Difficulty: Advanced

If x = 17 and y = 4, what is x × y?

A. 78

B. 71

C. 68

D. 65

Answer: C. Substitute the values: 17 × 4 = 68.

STEM-G8-MAT-T01-Q06 | Number Patterns and Sequences | Difficulty: Advanced

Calculate $8^2 + 11$.

A. 85

B. 78

C. 72

D. 75

Answer: D. $8^2 = 64$, then $64 + 11 = 75$.

STEM-G8-MAT-T01-Q07 | Number Patterns and Sequences | Difficulty: Advanced

If a = 16 and b = 6, calculate 2a + 3b.

A. 50

B. 60

C. 47

D. 53

Answer: A. 2(16) + 3(6) = 32 + 18 = 50.

STEM-G8-MAT-T01-Q08 | Number Patterns and Sequences | Difficulty: Advanced

A rectangle is 15 cm long and 4 cm wide. What is its perimeter?

A. 38

B. 48

C. 35

D. 41

Answer: A. Perimeter = 2(length + width) = 2(15 + 4) = 38 cm.

STEM-G8-MAT-T01-Q09 | Number Patterns and Sequences | Difficulty: Advanced

A rectangular area measures 18 cm by 4 cm. What is its area?

A. 82

B. 72

C. 69

D. 75

Answer: B. Area = 18 × 4 = 72 cm².

STEM-G8-MAT-T01-Q10 | Number Patterns and Sequences | Difficulty: Advanced

If x = 18 and y = 11, what is x × y?

A. 195

B. 208

C. 198

D. 201

Answer: C. Substitute the values: 18 × 11 = 198.

Theme 2: Linear Equations and Graphs — 10 questions

STEM-G8-MAT-T02-Q01 | Linear Equations and Graphs | Difficulty: Advanced

Calculate $7^2 + 2$.

A. 54

B. 51

C. 61

D. 48

Answer: B. $7^2 = 49$, then 49 + 2 = 51.

STEM-G8-MAT-T02-Q02 | Linear Equations and Graphs | Difficulty: Advanced

If a = 8 and b = 11, calculate 2a + 3b.

A. 59

B. 52

C. 49

D. 46

Answer: C. 2(8) + 3(11) = 16 + 33 = 49.

STEM-G8-MAT-T02-Q03 | Linear Equations and Graphs | Difficulty: Advanced

A rectangle is 11 cm long and 8 cm wide. What is its perimeter?

A. 38

B. 35

C. 41

D. 48

Answer: A. Perimeter = 2(length + width) = 2(11 + 8) = 38 cm.

STEM-G8-MAT-T02-Q04 | Linear Equations and Graphs | Difficulty: Advanced

A rectangular area measures 6 cm by 10 cm. What is its area?

A. 60

B. 57

C. 63

D. 70

Answer: A. Area = 6 × 10 = 60 cm².

STEM-G8-MAT-T02-Q05 | Linear Equations and Graphs | Difficulty: Advanced

If x = 12 and y = 5, what is x × y?

A. 63

B. 60

C. 57

D. 70

Answer: B. Substitute the values: 12 × 5 = 60.

STEM-G8-MAT-T02-Q06 | Linear Equations and Graphs | Difficulty: Advanced

Calculate $18^2 + 5$.

A. 329

B. 332

C. 339

D. 326

Answer: A. $18^2 = 324$, then 324 + 5 = 329.

STEM-G8-MAT-T02-Q07 | Linear Equations and Graphs | Difficulty: Advanced

If a = 12 and b = 2, calculate 2a + 3b.

A. 40

B. 33

C. 30

D. 27

Answer: C. 2(12) + 3(2) = 24 + 6 = 30.

STEM-G8-MAT-T02-Q08 | Linear Equations and Graphs | Difficulty: Advanced

A rectangle is 3 cm long and 9 cm wide. What is its perimeter?

A. 27

B. 34

C. 21

D. 24

Answer: D. Perimeter = 2(length + width) = 2(3 + 9) = 24 cm.

STEM-G8-MAT-T02-Q09 | Linear Equations and Graphs | Difficulty: Advanced

A rectangular area measures 3 cm by 3 cm. What is its area?

A. 12

B. 6

C. 9

D. 19

Answer: C. Area = 3 × 3 = 9 cm².

STEM-G8-MAT-T02-Q10 | Linear Equations and Graphs | Difficulty: Advanced

If x = 13 and y = 5, what is x × y?

A. 65

B. 68

C. 75

D. 62

Answer: A. Substitute the values: 13 × 5 = 65.

Theme 3: Pythagorean Theorem — 10 questions

STEM-G8-MAT-T03-Q01 | Pythagorean Theorem | Difficulty: Advanced

Calculate $10^2 + 4$.

A. 101

B. 107

C. 104

D. 114

Answer: C. $10^2 = 100$, then 100 + 4 = 104.

STEM-G8-MAT-T03-Q02 | Pythagorean Theorem | Difficulty: Advanced

If a = 10 and b = 2, calculate 2a + 3b.

A. 29

B. 26

C. 36

D. 23

Answer: B. 2(10) + 3(2) = 20 + 6 = 26.

STEM-G8-MAT-T03-Q03 | Pythagorean Theorem | Difficulty: Advanced

A rectangle is 14 cm long and 3 cm wide. What is its perimeter?

A. 37

B. 31

C. 34

D. 44

Answer: C. Perimeter = 2(length + width) = 2(14 + 3) = 34 cm.

STEM-G8-MAT-T03-Q04 | Pythagorean Theorem | Difficulty: Advanced

A rectangular area measures 9 cm by 6 cm. What is its area?

A. 57

B. 51

C. 64

D. 54

Answer: D. Area = 9 × 6 = 54 cm^2.

STEM-G8-MAT-T03-Q05 | Pythagorean Theorem | Difficulty: Advanced

If x = 17 and y = 6, what is x × y?

A. 99

B. 105

C. 102

D. 112

Answer: C. Substitute the values: 17 × 6 = 102.

STEM-G8-MAT-T03-Q06 | Pythagorean Theorem | Difficulty: Advanced

Calculate $4^2 + 12$.

A. 28

B. 31

C. 25

D. 38

Answer: A. $4^2 = 16$, then 16 + 12 = 28.

STEM-G8-MAT-T03-Q07 | Pythagorean Theorem | Difficulty: Advanced

If a = 16 and b = 12, calculate 2a + 3b.

A. 71

B. 65

C. 78

D. 68

Answer: D. 2(16) + 3(12) = 32 + 36 = 68.

STEM-G8-MAT-T03-Q08 | Pythagorean Theorem | Difficulty: Advanced

A rectangle is 7 cm long and 6 cm wide. What is its perimeter?

A. 29

B. 26

C. 23

D. 36

Answer: B. Perimeter = 2(length + width) = 2(7 + 6) = 26 cm.

STEM-G8-MAT-T03-Q09 | Pythagorean Theorem | Difficulty: Advanced

A rectangular area measures 17 cm by 6 cm. What is its area?

A. 99

B. 102

C. 112

D. 105

Answer: B. Area = 17 × 6 = 102 cm².

STEM-G8-MAT-T03-Q10 | Pythagorean Theorem | Difficulty: Advanced

If x = 12 and y = 8, what is x × y?

A. 93

B. 106

C. 99

D. 96

Answer: D. Substitute the values: 12 × 8 = 96.

Theme 4: Relations, Functions, and Statistics — 10 questions

STEM-G8-MAT-T04-Q01 | Relations, Functions, and Statistics | Difficulty: Advanced

Calculate $16^2 + 6$.

A. 265

B. 259

C. 272

D. 262

Answer: D. $16^2 = 256$, then 256 + 6 = 262.

STEM-G8-MAT-T04-Q02 | Relations, Functions, and Statistics | Difficulty: Advanced

If a = 5 and b = 5, calculate 2a + 3b.

A. 35

B. 28

C. 22

D. 25

Answer: D. 2(5) + 3(5) = 10 + 15 = 25.

STEM-G8-MAT-T04-Q03 | Relations, Functions, and Statistics | Difficulty: Advanced

A rectangle is 4 cm long and 2 cm wide. What is its perimeter?

A. 12

B. 15

C. 22

D. 9

Answer: A. Perimeter = 2(length + width) = 2(4 + 2) = 12 cm.

STEM-G8-MAT-T04-Q04 | Relations, Functions, and Statistics | Difficulty: Advanced

A rectangular area measures 16 cm by 3 cm. What is its area?

A. 48

B. 45

C. 51

D. 58

Answer: A. Area = 16 × 3 = 48 cm².

STEM-G8-MAT-T04-Q05 | Relations, Functions, and Statistics | Difficulty: Advanced

If x = 7 and y = 12, what is x × y?

A. 94

B. 84

C. 81

D. 87

Answer: B. Substitute the values: 7 × 12 = 84.

STEM-G8-MAT-T04-Q06 | Relations, Functions, and Statistics | Difficulty: Advanced

Calculate $16^2 + 2$.

A. 258

B. 255

C. 268

D. 261

Answer: A. $16^2 = 256$, then 256 + 2 = 258.

STEM-G8-MAT-T04-Q07 | Relations, Functions, and Statistics | Difficulty: Advanced

If a = 3 and b = 9, calculate 2a + 3b.

A. 43

B. 33

C. 30

D. 36

Answer: B. 2(3) + 3(9) = 6 + 27 = 33.

STEM-G8-MAT-T04-Q08 | Relations, Functions, and Statistics | Difficulty: Advanced

A rectangle is 11 cm long and 12 cm wide. What is its perimeter?

A. 43

B. 46

C. 56

D. 49

Answer: B. Perimeter = 2(length + width) = 2(11 + 12) = 46 cm.

STEM-G8-MAT-T04-Q09 | Relations, Functions, and Statistics | Difficulty: Advanced

A rectangular area measures 14 cm by 4 cm. What is its area?

A. 53

B. 59

C. 66

D. 56

Answer: D. Area = 14 × 4 = 56 cm^2.

STEM-G8-MAT-T04-Q10 | Relations, Functions, and Statistics | Difficulty: Advanced

If x = 16 and y = 9, what is x × y?

A. 147

B. 154

C. 144

D. 141

Answer: C. Substitute the values: 16 × 9 = 144.

Science

Theme 1: Digestive and Respiratory Systems — 10 questions

STEM-G8-SCI-T01-Q01 | Digestive and Respiratory Systems | Difficulty: Advanced

Where does most nutrient absorption take place?

A. The small intestine

B. The nasal cavity

C. The trachea

D. The esophagus

Answer: A. The small intestine has a large surface area for nutrient absorption.

STEM-G8-SCI-T01-Q02 | Digestive and Respiratory Systems | Difficulty: Advanced

Where does most nutrient absorption take place?

A. The small intestine

B. The nasal cavity

C. The esophagus

D. The trachea

Answer: A. The small intestine has a large surface area for nutrient absorption.

STEM-G8-SCI-T01-Q03 | Digestive and Respiratory Systems | Difficulty: Advanced

When investigating digestive and respiratory systems, which approach is most scientifically sound?

A. Collect evidence, test explanations, and draw conclusions

B. Change results to match a prediction

C. Choose an answer by guessing

D. Ignore data that disagrees with an idea

Answer: A. Scientific inquiry uses testable evidence and conclusions supported by data.

STEM-G8-SCI-T01-Q04 | Digestive and Respiratory Systems | Difficulty: Advanced

Where does most nutrient absorption take place?

A. The small intestine

B. The trachea

C. The nasal cavity

D. The esophagus

Answer: A. The small intestine has a large surface area for nutrient absorption.

STEM-G8-SCI-T01-Q05 | Digestive and Respiratory Systems | Difficulty: Advanced

Where does most nutrient absorption take place?

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B. The esophagus

C. The trachea

D. The nasal cavity

Answer: A. The small intestine has a large surface area for nutrient absorption.

STEM-G8-SCI-T01-Q06 | Digestive and Respiratory Systems | Difficulty: Advanced

When investigating digestive and respiratory systems, which approach is most scientifically sound?

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B. Change results to match a prediction

C. Choose an answer by guessing

D. Ignore data that disagrees with an idea

Answer: A. Scientific inquiry uses testable evidence and conclusions supported by data.

STEM-G8-SCI-T01-Q07 | Digestive and Respiratory Systems | Difficulty: Advanced

Where does most nutrient absorption take place?

A. The esophagus

B. The nasal cavity

C. The trachea

D. The small intestine

Answer: D. The small intestine has a large surface area for nutrient absorption.

STEM-G8-SCI-T01-Q08 | Digestive and Respiratory Systems | Difficulty: Advanced

Where does most nutrient absorption take place?

A. The small intestine

B. The trachea

C. The esophagus

D. The nasal cavity

Answer: A. The small intestine has a large surface area for nutrient absorption.

STEM-G8-SCI-T01-Q09 | Digestive and Respiratory Systems | Difficulty: Advanced

When investigating digestive and respiratory systems, which approach is most scientifically sound?

A. Choose an answer by guessing

B. Collect evidence, test explanations, and draw conclusions

C. Ignore data that disagrees with an idea

D. Change results to match a prediction

Answer: B. Scientific inquiry uses testable evidence and conclusions supported by data.

STEM-G8-SCI-T01-Q10 | Digestive and Respiratory Systems | Difficulty: Advanced

Where does most nutrient absorption take place?

A. The small intestine

B. The trachea

C. The nasal cavity

D. The esophagus

Answer: A. The small intestine has a large surface area for nutrient absorption.

Theme 2: Pressure in Solids, Liquids, and Gases — 10 questions

STEM-G8-SCI-T02-Q01 | Pressure in Solids, Liquids, and Gases | Difficulty: Advanced

Pressure is defined as ...

A. Energy per unit time

B. Mass multiplied by volume

C. Force per unit area

D. Distance per unit time

Answer: C. Pressure equals force divided by area.

STEM-G8-SCI-T02-Q02 | Pressure in Solids, Liquids, and Gases | Difficulty: Advanced

Pressure is defined as ...

A. Energy per unit time

B. Force per unit area

C. Mass multiplied by volume

D. Distance per unit time

Answer: B. Pressure equals force divided by area.

STEM-G8-SCI-T02-Q03 | Pressure in Solids, Liquids, and Gases | Difficulty: Advanced

When investigating pressure in solids, liquids, and gases, which approach is most scientifically sound?

A. Choose an answer by guessing

B. Change results to match a prediction

C. Collect evidence, test explanations, and draw conclusions

D. Ignore data that disagrees with an idea

Answer: C. Scientific inquiry uses testable evidence and conclusions supported by data.

STEM-G8-SCI-T02-Q04 | Pressure in Solids, Liquids, and Gases | Difficulty: Advanced

Pressure is defined as ...

A. Mass multiplied by volume

B. Energy per unit time

C. Distance per unit time

D. Force per unit area

Answer: D. Pressure equals force divided by area.

STEM-G8-SCI-T02-Q05 | Pressure in Solids, Liquids, and Gases | Difficulty: Advanced

Pressure is defined as ...

A. Mass multiplied by volume

B. Force per unit area

C. Energy per unit time

D. Distance per unit time

Answer: B. Pressure equals force divided by area.

STEM-G8-SCI-T02-Q06 | Pressure in Solids, Liquids, and Gases | Difficulty: Advanced

When investigating pressure in solids, liquids, and gases, which approach is most scientifically sound?

A. Collect evidence, test explanations, and draw conclusions

B. Ignore data that disagrees with an idea

C. Choose an answer by guessing

D. Change results to match a prediction

Answer: A. Scientific inquiry uses testable evidence and conclusions supported by data.

STEM-G8-SCI-T02-Q07 | Pressure in Solids, Liquids, and Gases | Difficulty: Advanced

Pressure is defined as ...

A. Energy per unit time

B. Mass multiplied by volume

C. Distance per unit time

D. Force per unit area

Answer: D. Pressure equals force divided by area.

STEM-G8-SCI-T02-Q08 | Pressure in Solids, Liquids, and Gases | Difficulty: Advanced

Pressure is defined as ...

A. Mass multiplied by volume

B. Distance per unit time

C. Energy per unit time

D. Force per unit area

Answer: D. Pressure equals force divided by area.

STEM-G8-SCI-T02-Q09 | Pressure in Solids, Liquids, and Gases | Difficulty: Advanced

When investigating pressure in solids, liquids, and gases, which approach is most scientifically sound?

A. Choose an answer by guessing

B. Ignore data that disagrees with an idea

C. Collect evidence, test explanations, and draw conclusions

D. Change results to match a prediction

Answer: C. Scientific inquiry uses testable evidence and conclusions supported by data.

STEM-G8-SCI-T02-Q10 | Pressure in Solids, Liquids, and Gases | Difficulty: Advanced

Pressure is defined as ...

A. Force per unit area

B. Distance per unit time

C. Mass multiplied by volume

D. Energy per unit time

Answer: A. Pressure equals force divided by area.

Theme 3: Vibrations, Waves, and Sound — 10 questions

STEM-G8-SCI-T03-Q01 | Vibrations, Waves, and Sound | Difficulty: Advanced

Sound cannot travel through ...

A. Iron

B. A vacuum

C. Air

D. Water

Answer: B. Sound needs a material medium to travel.

STEM-G8-SCI-T03-Q02 | Vibrations, Waves, and Sound | Difficulty: Advanced

Sound cannot travel through ...

A. A vacuum

B. Water

C. Air

D. Iron

Answer: A. Sound needs a material medium to travel.

STEM-G8-SCI-T03-Q03 | Vibrations, Waves, and Sound | Difficulty: Advanced

When investigating vibrations, waves, and sound, which approach is most scientifically sound?

A. Change results to match a prediction

B. Collect evidence, test explanations, and draw conclusions

C. Ignore data that disagrees with an idea

D. Choose an answer by guessing

Answer: B. Scientific inquiry uses testable evidence and conclusions supported by data.

STEM-G8-SCI-T03-Q04 | Vibrations, Waves, and Sound | Difficulty: Advanced

Sound cannot travel through ...

A. Iron

B. Air

C. A vacuum

D. Water

Answer: C. Sound needs a material medium to travel.

STEM-G8-SCI-T03-Q05 | Vibrations, Waves, and Sound | Difficulty: Advanced

Sound cannot travel through ...

A. A vacuum

B. Iron

C. Air

D. Water

Answer: A. Sound needs a material medium to travel.

STEM-G8-SCI-T03-Q06 | Vibrations, Waves, and Sound | Difficulty: Advanced

When investigating vibrations, waves, and sound, which approach is most scientifically sound?

A. Collect evidence, test explanations, and draw conclusions

B. Ignore data that disagrees with an idea

C. Choose an answer by guessing

D. Change results to match a prediction

Answer: A. Scientific inquiry uses testable evidence and conclusions supported by data.

STEM-G8-SCI-T03-Q07 | Vibrations, Waves, and Sound | Difficulty: Advanced

Sound cannot travel through ...

A. A vacuum

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Answer: A. Sound needs a material medium to travel.

STEM-G8-SCI-T03-Q08 | Vibrations, Waves, and Sound | Difficulty: Advanced

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STEM-G8-SCI-T03-Q09 | Vibrations, Waves, and Sound | Difficulty: Advanced

When investigating vibrations, waves, and sound, which approach is most scientifically sound?

A. Choose an answer by guessing

B. Change results to match a prediction

C. Collect evidence, test explanations, and draw conclusions

D. Ignore data that disagrees with an idea

Answer: C. Scientific inquiry uses testable evidence and conclusions supported by data.

STEM-G8-SCI-T03-Q10 | Vibrations, Waves, and Sound | Difficulty: Advanced

Sound cannot travel through ...

A. A vacuum

B. Iron

C. Water

D. Air

Answer: A. Sound needs a material medium to travel.

Theme 4: Plant Structure and Function — 10 questions

STEM-G8-SCI-T04-Q01 | Plant Structure and Function | Difficulty: Advanced

What does xylem mainly transport?

A. Blood to the heart

B. Water and minerals from the roots

C. Only sugar from leaves

D. Oxygen to the alveoli

Answer: B. Xylem transports water and minerals through a plant.

STEM-G8-SCI-T04-Q02 | Plant Structure and Function | Difficulty: Advanced

What does xylem mainly transport?

A. Only sugar from leaves

B. Blood to the heart

C. Water and minerals from the roots

D. Oxygen to the alveoli

Answer: C. Xylem transports water and minerals through a plant.

STEM-G8-SCI-T04-Q03 | Plant Structure and Function | Difficulty: Advanced

When investigating plant structure and function, which approach is most scientifically sound?

A. Collect evidence, test explanations, and draw conclusions

B. Change results to match a prediction

C. Ignore data that disagrees with an idea

D. Choose an answer by guessing

Answer: A. Scientific inquiry uses testable evidence and conclusions supported by data.

STEM-G8-SCI-T04-Q04 | Plant Structure and Function | Difficulty: Advanced

What does xylem mainly transport?

A. Oxygen to the alveoli

B. Only sugar from leaves

C. Water and minerals from the roots

D. Blood to the heart

Answer: C. Xylem transports water and minerals through a plant.

STEM-G8-SCI-T04-Q05 | Plant Structure and Function | Difficulty: Advanced

What does xylem mainly transport?

A. Blood to the heart

B. Water and minerals from the roots

C. Only sugar from leaves

D. Oxygen to the alveoli

Answer: B. Xylem transports water and minerals through a plant.

STEM-G8-SCI-T04-Q06 | Plant Structure and Function | Difficulty: Advanced

When investigating plant structure and function, which approach is most scientifically sound?

A. Ignore data that disagrees with an idea

B. Change results to match a prediction

C. Choose an answer by guessing

D. Collect evidence, test explanations, and draw conclusions

Answer: D. Scientific inquiry uses testable evidence and conclusions supported by data.

STEM-G8-SCI-T04-Q07 | Plant Structure and Function | Difficulty: Advanced

What does xylem mainly transport?

A. Water and minerals from the roots

B. Only sugar from leaves

C. Blood to the heart

D. Oxygen to the alveoli

Answer: A. Xylem transports water and minerals through a plant.

STEM-G8-SCI-T04-Q08 | Plant Structure and Function | Difficulty: Advanced

What does xylem mainly transport?

A. Water and minerals from the roots

B. Blood to the heart

C. Only sugar from leaves

D. Oxygen to the alveoli

Answer: A. Xylem transports water and minerals through a plant.

STEM-G8-SCI-T04-Q09 | Plant Structure and Function | Difficulty: Advanced

When investigating plant structure and function, which approach is most scientifically sound?

A. Ignore data that disagrees with an idea

B. Collect evidence, test explanations, and draw conclusions

C. Choose an answer by guessing

D. Change results to match a prediction

Answer: B. Scientific inquiry uses testable evidence and conclusions supported by data.

STEM-G8-SCI-T04-Q10 | Plant Structure and Function | Difficulty: Advanced

What does xylem mainly transport?

A. Only sugar from leaves

B. Oxygen to the alveoli

C. Blood to the heart

D. Water and minerals from the roots

Answer: D. Xylem transports water and minerals through a plant.

Engineering

Theme 1: Simple Machines and Efficiency — 10 questions

STEM-G8-ENG-T01-Q01 | Simple Machines and Efficiency | Difficulty: Advanced

If a model structure tips over easily, what is a sensible first improvement?

A. Add weight to the top

B. Remove the base

C. Widen the base or strengthen the joints

D. Remove all supports

Answer: C. A wider base and strong joints can improve stability.

STEM-G8-ENG-T01-Q02 | Simple Machines and Efficiency | Difficulty: Advanced

For a project about simple machines and efficiency, which design decision is best supported by engineering practice?

A. Avoid testing

B. Choose based on requirements, constraints, test data, and safety

C. Ignore cost and materials

D. Choose only by appearance

Answer: B. Engineering decisions consider performance, constraints, evidence, and safety.

STEM-G8-ENG-T01-Q03 | Simple Machines and Efficiency | Difficulty: Advanced

For a fair test, a team should ...

A. Avoid recording results

B. Choose only the preferred result

C. Change one factor while keeping other factors the same

D. Change every factor at once

Answer: C. Controlling variables helps identify what caused a result.

STEM-G8-ENG-T01-Q04 | Simple Machines and Efficiency | Difficulty: Advanced

For a project about simple machines and efficiency, which design decision is best supported by engineering practice?

A. Ignore cost and materials

B. Avoid testing

C. Choose only by appearance

D. Choose based on requirements, constraints, test data, and safety

Answer: D. Engineering decisions consider performance, constraints, evidence, and safety.

STEM-G8-ENG-T01-Q05 | Simple Machines and Efficiency | Difficulty: Advanced

Why should joints in a model be inspected?

A. Joints transfer forces between parts

B. Joints are only decoration

C. Joints remove gravity

D. Joints make mass zero

Answer: A. Weak joints can become failure points in a structure.

STEM-G8-ENG-T01-Q06 | Simple Machines and Efficiency | Difficulty: Advanced

For a project about simple machines and efficiency, which design decision is best supported by engineering practice?

A. Avoid testing

B. Choose only by appearance

C. Ignore cost and materials

D. Choose based on requirements, constraints, test data, and safety

Answer: D. Engineering decisions consider performance, constraints, evidence, and safety.

STEM-G8-ENG-T01-Q07 | Simple Machines and Efficiency | Difficulty: Advanced

Which measurement is useful when comparing two bridge designs?

A. The maximum load each bridge can support

B. The number of stickers

C. The bridge color

D. The model's name

Answer: A. Load capacity is a relevant measure of bridge performance.

STEM-G8-ENG-T01-Q08 | Simple Machines and Efficiency | Difficulty: Advanced

For a project about simple machines and efficiency, which design decision is best supported by engineering practice?

A. Choose only by appearance

B. Choose based on requirements, constraints, test data, and safety

C. Avoid testing

D. Ignore cost and materials

Answer: B. Engineering decisions consider performance, constraints, evidence, and safety.

STEM-G8-ENG-T01-Q09 | Simple Machines and Efficiency | Difficulty: Advanced

If a model structure tips over easily, what is a sensible first improvement?

A. Remove the base

B. Add weight to the top

C. Remove all supports

D. Widen the base or strengthen the joints

Answer: D. A wider base and strong joints can improve stability.

STEM-G8-ENG-T01-Q10 | Simple Machines and Efficiency | Difficulty: Advanced

For a project about simple machines and efficiency, which design decision is best supported by engineering practice?

A. Ignore cost and materials

B. Choose only by appearance

C. Choose based on requirements, constraints, test data, and safety

D. Avoid testing

Answer: C. Engineering decisions consider performance, constraints, evidence, and safety.

Theme 2: Structures, Stress, and Loads — 10 questions

STEM-G8-ENG-T02-Q01 | Structures, Stress, and Loads | Difficulty: Advanced

If a model structure tips over easily, what is a sensible first improvement?

A. Add weight to the top

B. Remove all supports

C. Remove the base

D. Widen the base or strengthen the joints

Answer: D. A wider base and strong joints can improve stability.

STEM-G8-ENG-T02-Q02 | Structures, Stress, and Loads | Difficulty: Advanced

For a project about structures, stress, and loads, which design decision is best supported by engineering practice?

A. Choose based on requirements, constraints, test data, and safety

B. Choose only by appearance

C. Avoid testing

D. Ignore cost and materials

Answer: A. Engineering decisions consider performance, constraints, evidence, and safety.

STEM-G8-ENG-T02-Q03 | Structures, Stress, and Loads | Difficulty: Advanced

For a fair test, a team should ...

A. Avoid recording results

B. Change every factor at once

C. Choose only the preferred result

D. Change one factor while keeping other factors the same

Answer: D. Controlling variables helps identify what caused a result.

STEM-G8-ENG-T02-Q04 | Structures, Stress, and Loads | Difficulty: Advanced

For a project about structures, stress, and loads, which design decision is best supported by engineering practice?

A. Choose based on requirements, constraints, test data, and safety

B. Choose only by appearance

C. Ignore cost and materials

D. Avoid testing

Answer: A. Engineering decisions consider performance, constraints, evidence, and safety.

STEM-G8-ENG-T02-Q05 | Structures, Stress, and Loads | Difficulty: Advanced

Why should joints in a model be inspected?

A. Joints make mass zero

B. Joints remove gravity

C. Joints are only decoration

D. Joints transfer forces between parts

Answer: D. Weak joints can become failure points in a structure.

STEM-G8-ENG-T02-Q06 | Structures, Stress, and Loads | Difficulty: Advanced

For a project about structures, stress, and loads, which design decision is best supported by engineering practice?

A. Choose only by appearance

B. Choose based on requirements, constraints, test data, and safety

C. Avoid testing

D. Ignore cost and materials

Answer: B. Engineering decisions consider performance, constraints, evidence, and safety.

STEM-G8-ENG-T02-Q07 | Structures, Stress, and Loads | Difficulty: Advanced

Which measurement is useful when comparing two bridge designs?

A. The maximum load each bridge can support

B. The number of stickers

C. The model's name

D. The bridge color

Answer: A. Load capacity is a relevant measure of bridge performance.

STEM-G8-ENG-T02-Q08 | Structures, Stress, and Loads | Difficulty: Advanced

For a project about structures, stress, and loads, which design decision is best supported by engineering practice?

A. Choose only by appearance

B. Ignore cost and materials

C. Choose based on requirements, constraints, test data, and safety

D. Avoid testing

Answer: C. Engineering decisions consider performance, constraints, evidence, and safety.

STEM-G8-ENG-T02-Q09 | Structures, Stress, and Loads | Difficulty: Advanced

If a model structure tips over easily, what is a sensible first improvement?

A. Widen the base or strengthen the joints

B. Add weight to the top

C. Remove the base

D. Remove all supports

Answer: A. A wider base and strong joints can improve stability.

STEM-G8-ENG-T02-Q10 | Structures, Stress, and Loads | Difficulty: Advanced

For a project about structures, stress, and loads, which design decision is best supported by engineering practice?

A. Choose only by appearance

B. Ignore cost and materials

C. Choose based on requirements, constraints, test data, and safety

D. Avoid testing

Answer: C. Engineering decisions consider performance, constraints, evidence, and safety.

Theme 3: Energy Systems and Conversion — 10 questions

STEM-G8-ENG-T03-Q01 | Energy Systems and Conversion | Difficulty: Advanced

If a model structure tips over easily, what is a sensible first improvement?

A. Add weight to the top

B. Remove all supports

C. Remove the base

D. Widen the base or strengthen the joints

Answer: D. A wider base and strong joints can improve stability.

STEM-G8-ENG-T03-Q02 | Energy Systems and Conversion | Difficulty: Advanced

For a project about energy systems and conversion, which design decision is best supported by engineering practice?

A. Choose based on requirements, constraints, test data, and safety

B. Avoid testing

C. Choose only by appearance

D. Ignore cost and materials

Answer: A. Engineering decisions consider performance, constraints, evidence, and safety.

STEM-G8-ENG-T03-Q03 | Energy Systems and Conversion | Difficulty: Advanced

For a fair test, a team should ...

A. Change one factor while keeping other factors the same

B. Choose only the preferred result

C. Avoid recording results

D. Change every factor at once

Answer: A. Controlling variables helps identify what caused a result.

STEM-G8-ENG-T03-Q04 | Energy Systems and Conversion | Difficulty: Advanced

For a project about energy systems and conversion, which design decision is best supported by engineering practice?

A. Avoid testing

B. Choose based on requirements, constraints, test data, and safety

C. Ignore cost and materials

D. Choose only by appearance

Answer: B. Engineering decisions consider performance, constraints, evidence, and safety.

STEM-G8-ENG-T03-Q05 | Energy Systems and Conversion | Difficulty: Advanced

Why should joints in a model be inspected?

A. Joints are only decoration

B. Joints transfer forces between parts

C. Joints remove gravity

D. Joints make mass zero

Answer: B. Weak joints can become failure points in a structure.

STEM-G8-ENG-T03-Q06 | Energy Systems and Conversion | Difficulty: Advanced

For a project about energy systems and conversion, which design decision is best supported by engineering practice?

A. Avoid testing

B. Choose only by appearance

C. Choose based on requirements, constraints, test data, and safety

D. Ignore cost and materials

Answer: C. Engineering decisions consider performance, constraints, evidence, and safety.

STEM-G8-ENG-T03-Q07 | Energy Systems and Conversion | Difficulty: Advanced

Which measurement is useful when comparing two bridge designs?

A. The model's name

B. The number of stickers

C. The maximum load each bridge can support

D. The bridge color

Answer: C. Load capacity is a relevant measure of bridge performance.

STEM-G8-ENG-T03-Q08 | Energy Systems and Conversion | Difficulty: Advanced

For a project about energy systems and conversion, which design decision is best supported by engineering practice?

A. Choose based on requirements, constraints, test data, and safety

B. Avoid testing

C. Choose only by appearance

D. Ignore cost and materials

Answer: A. Engineering decisions consider performance, constraints, evidence, and safety.

STEM-G8-ENG-T03-Q09 | Energy Systems and Conversion | Difficulty: Advanced

If a model structure tips over easily, what is a sensible first improvement?

A. Widen the base or strengthen the joints

B. Add weight to the top

C. Remove all supports

D. Remove the base

Answer: A. A wider base and strong joints can improve stability.

STEM-G8-ENG-T03-Q10 | Energy Systems and Conversion | Difficulty: Advanced

For a project about energy systems and conversion, which design decision is best supported by engineering practice?

A. Avoid testing

B. Choose only by appearance

C. Choose based on requirements, constraints, test data, and safety

D. Ignore cost and materials

Answer: C. Engineering decisions consider performance, constraints, evidence, and safety.

Theme 4: Prototypes, Variables, and Testing — 10 questions

STEM-G8-ENG-T04-Q01 | Prototypes, Variables, and Testing | Difficulty: Advanced

If a model structure tips over easily, what is a sensible first improvement?

A. Widen the base or strengthen the joints

B. Remove all supports

C. Remove the base

D. Add weight to the top

Answer: A. A wider base and strong joints can improve stability.

STEM-G8-ENG-T04-Q02 | Prototypes, Variables, and Testing | Difficulty: Advanced

For a project about prototypes, variables, and testing, which design decision is best supported by engineering practice?

A. Choose based on requirements, constraints, test data, and safety

B. Avoid testing

C. Ignore cost and materials

D. Choose only by appearance

Answer: A. Engineering decisions consider performance, constraints, evidence, and safety.

STEM-G8-ENG-T04-Q03 | Prototypes, Variables, and Testing | Difficulty: Advanced

For a fair test, a team should ...

A. Choose only the preferred result

B. Change every factor at once

C. Avoid recording results

D. Change one factor while keeping other factors the same

Answer: D. Controlling variables helps identify what caused a result.

STEM-G8-ENG-T04-Q04 | Prototypes, Variables, and Testing | Difficulty: Advanced

For a project about prototypes, variables, and testing, which design decision is best supported by engineering practice?

A. Avoid testing

B. Choose only by appearance

C. Choose based on requirements, constraints, test data, and safety

D. Ignore cost and materials

Answer: C. Engineering decisions consider performance, constraints, evidence, and safety.

STEM-G8-ENG-T04-Q05 | Prototypes, Variables, and Testing | Difficulty: Advanced

Why should joints in a model be inspected?

A. Joints remove gravity

B. Joints make mass zero

C. Joints transfer forces between parts

D. Joints are only decoration

Answer: C. Weak joints can become failure points in a structure.

STEM-G8-ENG-T04-Q06 | Prototypes, Variables, and Testing | Difficulty: Advanced

For a project about prototypes, variables, and testing, which design decision is best supported by engineering practice?

A. Ignore cost and materials

B. Choose based on requirements, constraints, test data, and safety

C. Choose only by appearance

D. Avoid testing

Answer: B. Engineering decisions consider performance, constraints, evidence, and safety.

STEM-G8-ENG-T04-Q07 | Prototypes, Variables, and Testing | Difficulty: Advanced

Which measurement is useful when comparing two bridge designs?

A. The model's name

B. The maximum load each bridge can support

C. The bridge color

D. The number of stickers

Answer: B. Load capacity is a relevant measure of bridge performance.

STEM-G8-ENG-T04-Q08 | Prototypes, Variables, and Testing | Difficulty: Advanced

For a project about prototypes, variables, and testing, which design decision is best supported by engineering practice?

A. Choose only by appearance

B. Avoid testing

C. Choose based on requirements, constraints, test data, and safety

D. Ignore cost and materials

Answer: C. Engineering decisions consider performance, constraints, evidence, and safety.

STEM-G8-ENG-T04-Q09 | Prototypes, Variables, and Testing | Difficulty: Advanced

If a model structure tips over easily, what is a sensible first improvement?

A. Remove the base

B. Remove all supports

C. Widen the base or strengthen the joints

D. Add weight to the top

Answer: C. A wider base and strong joints can improve stability.

STEM-G8-ENG-T04-Q10 | Prototypes, Variables, and Testing | Difficulty: Advanced

For a project about prototypes, variables, and testing, which design decision is best supported by engineering practice?

A. Ignore cost and materials

B. Choose only by appearance

C. Avoid testing

D. Choose based on requirements, constraints, test data, and safety

Answer: D. Engineering decisions consider performance, constraints, evidence, and safety.

ICT (Technology)

Theme 1: Spreadsheets and Data Analysis — 10 questions

STEM-G8-ICT-T01-Q01 | Spreadsheets and Data Analysis | Difficulty: Advanced

Software designed to help a user perform a task is commonly called an ...

A. Application

B. Power cable

C. Fan

D. Desk

Answer: A. An application is software that helps users perform specific tasks.

STEM-G8-ICT-T01-Q02 | Spreadsheets and Data Analysis | Difficulty: Advanced

What should you do if an unknown person sends you a suspicious link?

A. Forward it to every contact

B. Do not open it; check with a trusted adult or person

C. Enter your password

D. Download every attachment

Answer: B. Suspicious links may lead to scams or harmful software.

STEM-G8-ICT-T01-Q03 | Spreadsheets and Data Analysis | Difficulty: Advanced

In the context of spreadsheets and data analysis, which practice is most appropriate?

A. Share account credentials

B. Use data and permissions responsibly, and verify results

C. Assume every source is always correct

D. Ignore validation

Answer: B. Responsible technology use requires validation, security, and digital responsibility.

STEM-G8-ICT-T01-Q04 | Spreadsheets and Data Analysis | Difficulty: Advanced

A clear, ordered set of steps for solving a problem is called an ...

A. Wallpaper

B. Speaker

C. Resolution

D. Algorithm

Answer: D. An algorithm is an ordered set of instructions.

STEM-G8-ICT-T01-Q05 | Spreadsheets and Data Analysis | Difficulty: Advanced

In a spreadsheet, a cell is ...

A. A type of cable

B. A webpage

C. An application icon

D. The intersection of a row and a column

Answer: D. A cell is a box at the intersection of a row and a column.

STEM-G8-ICT-T01-Q06 | Spreadsheets and Data Analysis | Difficulty: Advanced

In the context of spreadsheets and data analysis, which practice is most appropriate?

A. Use data and permissions responsibly, and verify results

B. Share account credentials

C. Ignore validation

D. Assume every source is always correct

Answer: A. Responsible technology use requires validation, security, and digital responsibility.

STEM-G8-ICT-T01-Q07 | Spreadsheets and Data Analysis | Difficulty: Advanced

Finding and fixing errors in a program is called ...

A. Browsing

B. Debugging

C. Formatting

D. Streaming

Answer: B. Debugging is the process of locating and correcting errors.

STEM-G8-ICT-T01-Q08 | Spreadsheets and Data Analysis | Difficulty: Advanced

Which device is commonly used to enter text into a computer?

A. Speaker

B. Projector

C. Monitor

D. Keyboard

Answer: D. A keyboard is an input device used for typing.

STEM-G8-ICT-T01-Q09 | Spreadsheets and Data Analysis | Difficulty: Advanced

In the context of spreadsheets and data analysis, which practice is most appropriate?

A. Ignore validation

B. Share account credentials

C. Assume every source is always correct

D. Use data and permissions responsibly, and verify results

Answer: D. Responsible technology use requires validation, security, and digital responsibility.

STEM-G8-ICT-T01-Q10 | Spreadsheets and Data Analysis | Difficulty: Advanced

What should you do if an unknown person sends you a suspicious link?

A. Do not open it; check with a trusted adult or person

B. Enter your password

C. Download every attachment

D. Forward it to every contact

Answer: A. Suspicious links may lead to scams or harmful software.

Theme 2: Cybersecurity and Privacy — 10 questions

STEM-G8-ICT-T02-Q01 | Cybersecurity and Privacy | Difficulty: Advanced

Software designed to help a user perform a task is commonly called an ...

A. Power cable

B. Fan

C. Application

D. Desk

Answer: C. An application is software that helps users perform specific tasks.

STEM-G8-ICT-T02-Q02 | Cybersecurity and Privacy | Difficulty: Advanced

What should you do if an unknown person sends you a suspicious link?

A. Download every attachment

B. Do not open it; check with a trusted adult or person

C. Forward it to every contact

D. Enter your password

Answer: B. Suspicious links may lead to scams or harmful software.

STEM-G8-ICT-T02-Q03 | Cybersecurity and Privacy | Difficulty: Advanced

In the context of cybersecurity and privacy, which practice is most appropriate?

A. Ignore validation

B. Share account credentials

C. Use data and permissions responsibly, and verify results

D. Assume every source is always correct

Answer: C. Responsible technology use requires validation, security, and digital responsibility.

STEM-G8-ICT-T02-Q04 | Cybersecurity and Privacy | Difficulty: Advanced

A clear, ordered set of steps for solving a problem is called an ...

A. Resolution

B. Speaker

C. Wallpaper

D. Algorithm

Answer: D. An algorithm is an ordered set of instructions.

STEM-G8-ICT-T02-Q05 | Cybersecurity and Privacy | Difficulty: Advanced

In a spreadsheet, a cell is ...

A. A type of cable

B. A webpage

C. The intersection of a row and a column

D. An application icon

Answer: C. A cell is a box at the intersection of a row and a column.

STEM-G8-ICT-T02-Q06 | Cybersecurity and Privacy | Difficulty: Advanced

In the context of cybersecurity and privacy, which practice is most appropriate?

A. Ignore validation

B. Use data and permissions responsibly, and verify results

C. Assume every source is always correct

D. Share account credentials

Answer: B. Responsible technology use requires validation, security, and digital responsibility.

STEM-G8-ICT-T02-Q07 | Cybersecurity and Privacy | Difficulty: Advanced

Finding and fixing errors in a program is called ...

A. Formatting

B. Debugging

C. Browsing

D. Streaming

Answer: B. Debugging is the process of locating and correcting errors.

STEM-G8-ICT-T02-Q08 | Cybersecurity and Privacy | Difficulty: Advanced

Which device is commonly used to enter text into a computer?

A. Monitor

B. Keyboard

C. Projector

D. Speaker

Answer: B. A keyboard is an input device used for typing.

STEM-G8-ICT-T02-Q09 | Cybersecurity and Privacy | Difficulty: Advanced

In the context of cybersecurity and privacy, which practice is most appropriate?

A. Use data and permissions responsibly, and verify results

B. Ignore validation

C. Share account credentials

D. Assume every source is always correct

Answer: A. Responsible technology use requires validation, security, and digital responsibility.

STEM-G8-ICT-T02-Q10 | Cybersecurity and Privacy | Difficulty: Advanced

What should you do if an unknown person sends you a suspicious link?

A. Download every attachment

B. Enter your password

C. Do not open it; check with a trusted adult or person

D. Forward it to every contact

Answer: C. Suspicious links may lead to scams or harmful software.

Theme 3: HTML and Basic Web Structure — 10 questions

STEM-G8-ICT-T03-Q01 | HTML and Basic Web Structure | Difficulty: Advanced

Software designed to help a user perform a task is commonly called an ...

A. Application

B. Fan

C. Desk

D. Power cable

Answer: A. An application is software that helps users perform specific tasks.

STEM-G8-ICT-T03-Q02 | HTML and Basic Web Structure | Difficulty: Advanced

What should you do if an unknown person sends you a suspicious link?

A. Forward it to every contact

B. Enter your password

C. Download every attachment

D. Do not open it; check with a trusted adult or person

Answer: D. Suspicious links may lead to scams or harmful software.

STEM-G8-ICT-T03-Q03 | HTML and Basic Web Structure | Difficulty: Advanced

In the context of html and basic web structure, which practice is most appropriate?

A. Share account credentials

B. Ignore validation

C. Use data and permissions responsibly, and verify results

D. Assume every source is always correct

Answer: C. Responsible technology use requires validation, security, and digital responsibility.

STEM-G8-ICT-T03-Q04 | HTML and Basic Web Structure | Difficulty: Advanced

A clear, ordered set of steps for solving a problem is called an ...

A. Wallpaper

B. Algorithm

C. Resolution

D. Speaker

Answer: B. An algorithm is an ordered set of instructions.

STEM-G8-ICT-T03-Q05 | HTML and Basic Web Structure | Difficulty: Advanced

In a spreadsheet, a cell is ...

A. The intersection of a row and a column

B. A type of cable

C. An application icon

D. A webpage

Answer: A. A cell is a box at the intersection of a row and a column.

STEM-G8-ICT-T03-Q06 | HTML and Basic Web Structure | Difficulty: Advanced

In the context of html and basic web structure, which practice is most appropriate?

A. Use data and permissions responsibly, and verify results

B. Ignore validation

C. Share account credentials

D. Assume every source is always correct

Answer: A. Responsible technology use requires validation, security, and digital responsibility.

STEM-G8-ICT-T03-Q07 | HTML and Basic Web Structure | Difficulty: Advanced

Finding and fixing errors in a program is called ...

A. Browsing

B. Formatting

C. Debugging

D. Streaming

Answer: C. Debugging is the process of locating and correcting errors.

STEM-G8-ICT-T03-Q08 | HTML and Basic Web Structure | Difficulty: Advanced

Which device is commonly used to enter text into a computer?

A. Speaker

B. Monitor

C. Keyboard

D. Projector

Answer: C. A keyboard is an input device used for typing.

STEM-G8-ICT-T03-Q09 | HTML and Basic Web Structure | Difficulty: Advanced

In the context of html and basic web structure, which practice is most appropriate?

A. Share account credentials

B. Ignore validation

C. Use data and permissions responsibly, and verify results

D. Assume every source is always correct

Answer: C. Responsible technology use requires validation, security, and digital responsibility.

STEM-G8-ICT-T03-Q10 | HTML and Basic Web Structure | Difficulty: Advanced

What should you do if an unknown person sends you a suspicious link?

A. Download every attachment

B. Do not open it; check with a trusted adult or person

C. Enter your password

D. Forward it to every contact

Answer: B. Suspicious links may lead to scams or harmful software.

Theme 4: Conditionals, Loops, and Functions — 10 questions

STEM-G8-ICT-T04-Q01 | Conditionals, Loops, and Functions | Difficulty: Advanced

Software designed to help a user perform a task is commonly called an ...

A. Fan

B. Desk

C. Power cable

D. Application

Answer: D. An application is software that helps users perform specific tasks.

STEM-G8-ICT-T04-Q02 | Conditionals, Loops, and Functions | Difficulty: Advanced

What should you do if an unknown person sends you a suspicious link?

A. Enter your password

B. Do not open it; check with a trusted adult or person

C. Forward it to every contact

D. Download every attachment

Answer: B. Suspicious links may lead to scams or harmful software.

STEM-G8-ICT-T04-Q03 | Conditionals, Loops, and Functions | Difficulty: Advanced

In the context of conditionals, loops, and functions, which practice is most appropriate?

A. Ignore validation

B. Assume every source is always correct

C. Use data and permissions responsibly, and verify results

D. Share account credentials

Answer: C. Responsible technology use requires validation, security, and digital responsibility.

STEM-G8-ICT-T04-Q04 | Conditionals, Loops, and Functions | Difficulty: Advanced

A clear, ordered set of steps for solving a problem is called an ...

A. Resolution

B. Speaker

C. Wallpaper

D. Algorithm

Answer: D. An algorithm is an ordered set of instructions.

STEM-G8-ICT-T04-Q05 | Conditionals, Loops, and Functions | Difficulty: Advanced

In a spreadsheet, a cell is ...

A. A type of cable

B. The intersection of a row and a column

C. An application icon

D. A webpage

Answer: B. A cell is a box at the intersection of a row and a column.

STEM-G8-ICT-T04-Q06 | Conditionals, Loops, and Functions | Difficulty: Advanced

In the context of conditionals, loops, and functions, which practice is most appropriate?

A. Ignore validation

B. Use data and permissions responsibly, and verify results

C. Assume every source is always correct

D. Share account credentials

Answer: B. Responsible technology use requires validation, security, and digital responsibility.

STEM-G8-ICT-T04-Q07 | Conditionals, Loops, and Functions | Difficulty: Advanced

Finding and fixing errors in a program is called ...

A. Debugging

B. Formatting

C. Streaming

D. Browsing

Answer: A. Debugging is the process of locating and correcting errors.

STEM-G8-ICT-T04-Q08 | Conditionals, Loops, and Functions | Difficulty: Advanced

Which device is commonly used to enter text into a computer?

A. Keyboard

B. Monitor

C. Projector

D. Speaker

Answer: A. A keyboard is an input device used for typing.

STEM-G8-ICT-T04-Q09 | Conditionals, Loops, and Functions | Difficulty: Advanced

In the context of conditionals, loops, and functions, which practice is most appropriate?

A. Ignore validation

B. Assume every source is always correct

C. Use data and permissions responsibly, and verify results

D. Share account credentials

Answer: C. Responsible technology use requires validation, security, and digital responsibility.

STEM-G8-ICT-T04-Q10 | Conditionals, Loops, and Functions | Difficulty: Advanced

What should you do if an unknown person sends you a suspicious link?

A. Do not open it; check with a trusted adult or person

B. Enter your password

C. Download every attachment

D. Forward it to every contact

Answer: A. Suspicious links may lead to scams or harmful software.

Junior High School Grade 9

160 questions • 4 subjects • 4 themes per subject • 10 questions per theme

Mathematics

Theme 1: Exponents and Radicals — 10 questions

STEM-G9-MAT-T01-Q01 | Exponents and Radicals | Difficulty: Advanced

Calculate $3^2 + 10$.

A. 16

B. 29

C. 22

D. 19

Answer: D. $3^2 = 9$, then 9 + 10 = 19.

STEM-G9-MAT-T01-Q02 | Exponents and Radicals | Difficulty: Advanced

If a = 6 and b = 10, calculate 2a + 3b.

A. 42

B. 52

C. 39

D. 45

Answer: A. 2(6) + 3(10) = 12 + 30 = 42.

STEM-G9-MAT-T01-Q03 | Exponents and Radicals | Difficulty: Advanced

A rectangle is 5 cm long and 4 cm wide. What is its perimeter?

A. 21

B. 28

C. 15

D. 18

Answer: D. Perimeter = 2(length + width) = 2(5 + 4) = 18 cm.

STEM-G9-MAT-T01-Q04 | Exponents and Radicals | Difficulty: Advanced

A rectangular area measures 6 cm by 9 cm. What is its area?

A. 54

B. 57

C. 64

D. 51

Answer: A. Area = 6 × 9 = 54 cm^2.

STEM-G9-MAT-T01-Q05 | Exponents and Radicals | Difficulty: Advanced

If x = 5 and y = 3, what is x × y?

A. 18

B. 25

C. 15

D. 12

Answer: C. Substitute the values: 5 × 3 = 15.

STEM-G9-MAT-T01-Q06 | Exponents and Radicals | Difficulty: Advanced

Calculate $8^2 + 6$.

A. 70

B. 73

C. 80

D. 67

Answer: A. $8^2 = 64$, then 64 + 6 = 70.

STEM-G9-MAT-T01-Q07 | Exponents and Radicals | Difficulty: Advanced

If a = 6 and b = 9, calculate 2a + 3b.

A. 42

B. 39

C. 49

D. 36

Answer: B. 2(6) + 3(9) = 12 + 27 = 39.

STEM-G9-MAT-T01-Q08 | Exponents and Radicals | Difficulty: Advanced

A rectangle is 18 cm long and 10 cm wide. What is its perimeter?

A. 53

B. 66

C. 59

D. 56

Answer: D. Perimeter = 2(length + width) = 2(18 + 10) = 56 cm.

STEM-G9-MAT-T01-Q09 | Exponents and Radicals | Difficulty: Advanced

A rectangular area measures 13 cm by 9 cm. What is its area?

A. 127

B. 120

C. 117

D. 114

Answer: C. Area = 13 × 9 = 117 cm².

STEM-G9-MAT-T01-Q10 | Exponents and Radicals | Difficulty: Advanced

If x = 16 and y = 11, what is x × y?

A. 176

B. 179

C. 173

D. 186

Answer: A. Substitute the values: 16 × 11 = 176.

Theme 2: Quadratic Equations and Functions — 10 questions

STEM-G9-MAT-T02-Q01 | Quadratic Equations and Functions | Difficulty: Advanced

Calculate $4^2 + 3$.

A. 16

B. 29

C. 19

D. 22

Answer: C. $4^2 = 16$, then 16 + 3 = 19.

STEM-G9-MAT-T02-Q02 | Quadratic Equations and Functions | Difficulty: Advanced

If a = 7 and b = 10, calculate 2a + 3b.

A. 44

B. 47

C. 41

D. 54

Answer: A. 2(7) + 3(10) = 14 + 30 = 44.

STEM-G9-MAT-T02-Q03 | Quadratic Equations and Functions | Difficulty: Advanced

A rectangle is 17 cm long and 7 cm wide. What is its perimeter?

A. 45

B. 58

C. 51

D. 48

Answer: D. Perimeter = 2(length + width) = 2(17 + 7) = 48 cm.

STEM-G9-MAT-T02-Q04 | Quadratic Equations and Functions | Difficulty: Advanced

A rectangular area measures 15 cm by 4 cm. What is its area?

A. 63

B. 70

C. 60

D. 57

Answer: C. Area = 15 × 4 = 60 cm².

STEM-G9-MAT-T02-Q05 | Quadratic Equations and Functions | Difficulty: Advanced

If x = 6 and y = 9, what is x × y?

A. 54

B. 64

C. 51

D. 57

Answer: A. Substitute the values: 6 × 9 = 54.

STEM-G9-MAT-T02-Q06 | Quadratic Equations and Functions | Difficulty: Advanced

Calculate $9^2 + 4$.

A. 88

B. 82

C. 95

D. 85

Answer: D. $9^2 = 81$, then 81 + 4 = 85.

STEM-G9-MAT-T02-Q07 | Quadratic Equations and Functions | Difficulty: Advanced

If a = 6 and b = 6, calculate 2a + 3b.

A. 33

B. 30

C. 27

D. 40

Answer: B. 2(6) + 3(6) = 12 + 18 = 30.

STEM-G9-MAT-T02-Q08 | Quadratic Equations and Functions | Difficulty: Advanced

A rectangle is 17 cm long and 7 cm wide. What is its perimeter?

A. 48

B. 45

C. 58

D. 51

Answer: A. Perimeter = 2(length + width) = 2(17 + 7) = 48 cm.

STEM-G9-MAT-T02-Q09 | Quadratic Equations and Functions | Difficulty: Advanced

A rectangular area measures 13 cm by 9 cm. What is its area?

A. 114

B. 127

C. 117

D. 120

Answer: C. Area = 13 × 9 = 117 cm².

STEM-G9-MAT-T02-Q10 | Quadratic Equations and Functions | Difficulty: Advanced

If x = 17 and y = 3, what is x × y?

A. 54

B. 51

C. 48

D. 61

Answer: B. Substitute the values: 17 × 3 = 51.

Theme 3: Similarity and Solid Geometry — 10 questions

STEM-G9-MAT-T03-Q01 | Similarity and Solid Geometry | Difficulty: Advanced

Calculate $12^2 + 5$.

A. 152

B. 159

C. 146

D. 149

Answer: D. $12^2 = 144$, then $144 + 5 = 149$.

STEM-G9-MAT-T03-Q02 | Similarity and Solid Geometry | Difficulty: Advanced

If a = 11 and b = 5, calculate 2a + 3b.

A. 40

B. 37

C. 47

D. 34

Answer: B. 2(11) + 3(5) = 22 + 15 = 37.

STEM-G9-MAT-T03-Q03 | Similarity and Solid Geometry | Difficulty: Advanced

A rectangle is 5 cm long and 12 cm wide. What is its perimeter?

A. 31

B. 44

C. 34

D. 37

Answer: C. Perimeter = 2(length + width) = 2(5 + 12) = 34 cm.

STEM-G9-MAT-T03-Q04 | Similarity and Solid Geometry | Difficulty: Advanced

A rectangular area measures 14 cm by 3 cm. What is its area?

A. 42

B. 45

C. 39

D. 52

Answer: A. Area = 14 × 3 = 42 cm².

STEM-G9-MAT-T03-Q05 | Similarity and Solid Geometry | Difficulty: Advanced

If x = 11 and y = 11, what is x × y?

A. 121

B. 124

C. 131

D. 118

Answer: A. Substitute the values: 11 × 11 = 121.

STEM-G9-MAT-T03-Q06 | Similarity and Solid Geometry | Difficulty: Advanced

Calculate $8^2 + 10$.

A. 84

B. 71

C. 77

D. 74

Answer: D. $8^2 = 64$, then $64 + 10 = 74$.

STEM-G9-MAT-T03-Q07 | Similarity and Solid Geometry | Difficulty: Advanced

If a = 8 and b = 4, calculate 2a + 3b.

A. 31

B. 28

C. 25

D. 38

Answer: B. 2(8) + 3(4) = 16 + 12 = 28.

STEM-G9-MAT-T03-Q08 | Similarity and Solid Geometry | Difficulty: Advanced

A rectangle is 4 cm long and 7 cm wide. What is its perimeter?

A. 19

B. 25

C. 32

D. 22

Answer: D. Perimeter = 2(length + width) = 2(4 + 7) = 22 cm.

STEM-G9-MAT-T03-Q09 | Similarity and Solid Geometry | Difficulty: Advanced

A rectangular area measures 17 cm by 10 cm. What is its area?

A. 170

B. 180

C. 173

D. 167

Answer: A. Area = 17 × 10 = 170 cm².

STEM-G9-MAT-T03-Q10 | Similarity and Solid Geometry | Difficulty: Advanced

If x = 6 and y = 2, what is x × y?

A. 12

B. 9

C. 15

D. 22

Answer: A. Substitute the values: 6 × 2 = 12.

Theme 4: Statistics and Probability — 10 questions

STEM-G9-MAT-T04-Q01 | Statistics and Probability | Difficulty: Advanced

Calculate $7^2 + 4$.

A. 50

B. 56

C. 53

D. 63

Answer: C. $7^2 = 49$, then 49 + 4 = 53.

STEM-G9-MAT-T04-Q02 | Statistics and Probability | Difficulty: Advanced

If a = 11 and b = 10, calculate 2a + 3b.

A. 49

B. 52

C. 62

D. 55

Answer: B. 2(11) + 3(10) = 22 + 30 = 52.

STEM-G9-MAT-T04-Q03 | Statistics and Probability | Difficulty: Advanced

A rectangle is 17 cm long and 7 cm wide. What is its perimeter?

A. 51

B. 48

C. 58

D. 45

Answer: B. Perimeter = 2(length + width) = 2(17 + 7) = 48 cm.

STEM-G9-MAT-T04-Q04 | Statistics and Probability | Difficulty: Advanced

A rectangular area measures 16 cm by 6 cm. What is its area?

A. 93

B. 106

C. 99

D. 96

Answer: D. Area = 16 × 6 = 96 cm².

STEM-G9-MAT-T04-Q05 | Statistics and Probability | Difficulty: Advanced

If x = 12 and y = 4, what is x × y?

A. 45

B. 58

C. 48

D. 51

Answer: C. Substitute the values: 12 × 4 = 48.

STEM-G9-MAT-T04-Q06 | Statistics and Probability | Difficulty: Advanced

Calculate $17^2 + 4$.

A. 293

B. 303

C. 290

D. 296

Answer: A. $17^2 = 289$, then 289 + 4 = 293.

STEM-G9-MAT-T04-Q07 | Statistics and Probability | Difficulty: Advanced

If a = 12 and b = 4, calculate 2a + 3b.

A. 46

B. 33

C. 36

D. 39

Answer: C. 2(12) + 3(4) = 24 + 12 = 36.

STEM-G9-MAT-T04-Q08 | Statistics and Probability | Difficulty: Advanced

A rectangle is 8 cm long and 3 cm wide. What is its perimeter?

A. 22

B. 25

C. 19

D. 32

Answer: A. Perimeter = 2(length + width) = 2(8 + 3) = 22 cm.

STEM-G9-MAT-T04-Q09 | Statistics and Probability | Difficulty: Advanced

A rectangular area measures 8 cm by 5 cm. What is its area?

A. 50

B. 37

C. 43

D. 40

Answer: D. Area = 8 × 5 = 40 cm^2.

STEM-G9-MAT-T04-Q10 | Statistics and Probability | Difficulty: Advanced

If x = 17 and y = 2, what is x × y?

A. 34

B. 37

C. 44

D. 31

Answer: A. Substitute the values: 17 × 2 = 34.

Science

Theme 1: Reproduction and Inheritance — 10 questions

STEM-G9-SCI-T01-Q01 | Reproduction and Inheritance | Difficulty: Advanced

A segment of DNA that carries hereditary information is called a ...

A. Gene

B. Chlorophyll

C. Alveolus

D. Neuron

Answer: A. A gene is a segment of DNA that contributes to inherited traits.

STEM-G9-SCI-T01-Q02 | Reproduction and Inheritance | Difficulty: Advanced

A segment of DNA that carries hereditary information is called a ...

A. Alveolus

B. Neuron

C. Gene

D. Chlorophyll

Answer: C. A gene is a segment of DNA that contributes to inherited traits.

STEM-G9-SCI-T01-Q03 | Reproduction and Inheritance | Difficulty: Advanced

When investigating reproduction and inheritance, which approach is most scientifically sound?

A. Change results to match a prediction

B. Collect evidence, test explanations, and draw conclusions

C. Ignore data that disagrees with an idea

D. Choose an answer by guessing

Answer: B. Scientific inquiry uses testable evidence and conclusions supported by data.

STEM-G9-SCI-T01-Q04 | Reproduction and Inheritance | Difficulty: Advanced

A segment of DNA that carries hereditary information is called a ...

A. Neuron

B. Alveolus

C. Gene

D. Chlorophyll

Answer: C. A gene is a segment of DNA that contributes to inherited traits.

STEM-G9-SCI-T01-Q05 | Reproduction and Inheritance | Difficulty: Advanced

A segment of DNA that carries hereditary information is called a ...

A. Chlorophyll

B. Alveolus

C. Gene

D. Neuron

Answer: C. A gene is a segment of DNA that contributes to inherited traits.

STEM-G9-SCI-T01-Q06 | Reproduction and Inheritance | Difficulty: Advanced

When investigating reproduction and inheritance, which approach is most scientifically sound?

A. Collect evidence, test explanations, and draw conclusions

B. Choose an answer by guessing

C. Change results to match a prediction

D. Ignore data that disagrees with an idea

Answer: A. Scientific inquiry uses testable evidence and conclusions supported by data.

STEM-G9-SCI-T01-Q07 | Reproduction and Inheritance | Difficulty: Advanced

A segment of DNA that carries hereditary information is called a ...

A. Alveolus

B. Chlorophyll

C. Neuron

D. Gene

Answer: D. A gene is a segment of DNA that contributes to inherited traits.

STEM-G9-SCI-T01-Q08 | Reproduction and Inheritance | Difficulty: Advanced

A segment of DNA that carries hereditary information is called a ...

A. Neuron

B. Gene

C. Chlorophyll

D. Alveolus

Answer: B. A gene is a segment of DNA that contributes to inherited traits.

STEM-G9-SCI-T01-Q09 | Reproduction and Inheritance | Difficulty: Advanced

When investigating reproduction and inheritance, which approach is most scientifically sound?

A. Collect evidence, test explanations, and draw conclusions

B. Change results to match a prediction

C. Choose an answer by guessing

D. Ignore data that disagrees with an idea

Answer: A. Scientific inquiry uses testable evidence and conclusions supported by data.

STEM-G9-SCI-T01-Q10 | Reproduction and Inheritance | Difficulty: Advanced

A segment of DNA that carries hereditary information is called a ...

A. Alveolus

B. Chlorophyll

C. Gene

D. Neuron

Answer: C. A gene is a segment of DNA that contributes to inherited traits.

Theme 2: Electric Circuits and Magnetism — 10 questions

STEM-G9-SCI-T02-Q01 | Electric Circuits and Magnetism | Difficulty: Advanced

According to Ohm's law, electric current equals ...

A. Power multiplied by mass

B. Voltage divided by resistance

C. Resistance divided by time

D. Voltage multiplied by distance

Answer: B. Ohm's law is I = V/R.

STEM-G9-SCI-T02-Q02 | Electric Circuits and Magnetism | Difficulty: Advanced

According to Ohm's law, electric current equals ...

A. Voltage multiplied by distance

B. Power multiplied by mass

C. Voltage divided by resistance

D. Resistance divided by time

Answer: C. Ohm's law is I = V/R.

STEM-G9-SCI-T02-Q03 | Electric Circuits and Magnetism | Difficulty: Advanced

When investigating electric circuits and magnetism, which approach is most scientifically sound?

A. Collect evidence, test explanations, and draw conclusions

B. Change results to match a prediction

C. Ignore data that disagrees with an idea

D. Choose an answer by guessing

Answer: A. Scientific inquiry uses testable evidence and conclusions supported by data.

STEM-G9-SCI-T02-Q04 | Electric Circuits and Magnetism | Difficulty: Advanced

According to Ohm's law, electric current equals ...

A. Voltage multiplied by distance

B. Voltage divided by resistance

C. Power multiplied by mass

D. Resistance divided by time

Answer: B. Ohm's law is I = V/R.

STEM-G9-SCI-T02-Q05 | Electric Circuits and Magnetism | Difficulty: Advanced

According to Ohm's law, electric current equals ...

A. Resistance divided by time

B. Voltage multiplied by distance

C. Power multiplied by mass

D. Voltage divided by resistance

Answer: D. Ohm's law is I = V/R.

STEM-G9-SCI-T02-Q06 | Electric Circuits and Magnetism | Difficulty: Advanced

When investigating electric circuits and magnetism, which approach is most scientifically sound?

A. Ignore data that disagrees with an idea

B. Collect evidence, test explanations, and draw conclusions

C. Choose an answer by guessing

D. Change results to match a prediction

Answer: B. Scientific inquiry uses testable evidence and conclusions supported by data.

STEM-G9-SCI-T02-Q07 | Electric Circuits and Magnetism | Difficulty: Advanced

According to Ohm's law, electric current equals ...

A. Resistance divided by time

B. Voltage divided by resistance

C. Voltage multiplied by distance

D. Power multiplied by mass

Answer: B. Ohm's law is I = V/R.

STEM-G9-SCI-T02-Q08 | Electric Circuits and Magnetism | Difficulty: Advanced

According to Ohm's law, electric current equals ...

A. Voltage multiplied by distance

B. Resistance divided by time

C. Voltage divided by resistance

D. Power multiplied by mass

Answer: C. Ohm's law is I = V/R.

STEM-G9-SCI-T02-Q09 | Electric Circuits and Magnetism | Difficulty: Advanced

When investigating electric circuits and magnetism, which approach is most scientifically sound?

A. Choose an answer by guessing

B. Collect evidence, test explanations, and draw conclusions

C. Change results to match a prediction

D. Ignore data that disagrees with an idea

Answer: B. Scientific inquiry uses testable evidence and conclusions supported by data.

STEM-G9-SCI-T02-Q10 | Electric Circuits and Magnetism | Difficulty: Advanced

According to Ohm's law, electric current equals ...

A. Resistance divided by time

B. Voltage divided by resistance

C. Voltage multiplied by distance

D. Power multiplied by mass

Answer: B. Ohm's law is I = V/R.

Theme 3: Chemical Reactions and Changes in Matter — 10 questions

STEM-G9-SCI-T03-Q01 | Chemical Reactions and Changes in Matter | Difficulty: Advanced

Which observation can indicate that a chemical reaction has occurred?

A. Paper is folded

B. Ice melts

C. An object is only cut into pieces

D. A new gas or precipitate forms

Answer: D. Chemical reactions form new substances; gas or precipitate formation can be evidence.

STEM-G9-SCI-T03-Q02 | Chemical Reactions and Changes in Matter | Difficulty: Advanced

Which observation can indicate that a chemical reaction has occurred?

A. Paper is folded

B. A new gas or precipitate forms

C. An object is only cut into pieces

D. Ice melts

Answer: B. Chemical reactions form new substances; gas or precipitate formation can be evidence.

STEM-G9-SCI-T03-Q03 | Chemical Reactions and Changes in Matter | Difficulty: Advanced

When investigating chemical reactions and changes in matter, which approach is most scientifically sound?

A. Choose an answer by guessing

B. Ignore data that disagrees with an idea

C. Collect evidence, test explanations, and draw conclusions

D. Change results to match a prediction

Answer: C. Scientific inquiry uses testable evidence and conclusions supported by data.

STEM-G9-SCI-T03-Q04 | Chemical Reactions and Changes in Matter | Difficulty: Advanced

Which observation can indicate that a chemical reaction has occurred?

A. A new gas or precipitate forms

B. Ice melts

C. An object is only cut into pieces

D. Paper is folded

Answer: A. Chemical reactions form new substances; gas or precipitate formation can be evidence.

STEM-G9-SCI-T03-Q05 | Chemical Reactions and Changes in Matter | Difficulty: Advanced

Which observation can indicate that a chemical reaction has occurred?

A. A new gas or precipitate forms

B. Paper is folded

C. An object is only cut into pieces

D. Ice melts

Answer: A. Chemical reactions form new substances; gas or precipitate formation can be evidence.

STEM-G9-SCI-T03-Q06 | Chemical Reactions and Changes in Matter | Difficulty: Advanced

When investigating chemical reactions and changes in matter, which approach is most scientifically sound?

A. Ignore data that disagrees with an idea

B. Collect evidence, test explanations, and draw conclusions

C. Change results to match a prediction

D. Choose an answer by guessing

Answer: B. Scientific inquiry uses testable evidence and conclusions supported by data.

STEM-G9-SCI-T03-Q07 | Chemical Reactions and Changes in Matter | Difficulty: Advanced

Which observation can indicate that a chemical reaction has occurred?

A. An object is only cut into pieces

B. Paper is folded

C. Ice melts

D. A new gas or precipitate forms

Answer: D. Chemical reactions form new substances; gas or precipitate formation can be evidence.

STEM-G9-SCI-T03-Q08 | Chemical Reactions and Changes in Matter | Difficulty: Advanced

Which observation can indicate that a chemical reaction has occurred?

A. An object is only cut into pieces

B. A new gas or precipitate forms

C. Paper is folded

D. Ice melts

Answer: B. Chemical reactions form new substances; gas or precipitate formation can be evidence.

STEM-G9-SCI-T03-Q09 | Chemical Reactions and Changes in Matter | Difficulty: Advanced

When investigating chemical reactions and changes in matter, which approach is most scientifically sound?

A. Collect evidence, test explanations, and draw conclusions

B. Change results to match a prediction

C. Choose an answer by guessing

D. Ignore data that disagrees with an idea

Answer: A. Scientific inquiry uses testable evidence and conclusions supported by data.

STEM-G9-SCI-T03-Q10 | Chemical Reactions and Changes in Matter | Difficulty: Advanced

Which observation can indicate that a chemical reaction has occurred?

A. Paper is folded

B. A new gas or precipitate forms

C. An object is only cut into pieces

D. Ice melts

Answer: B. Chemical reactions form new substances; gas or precipitate formation can be evidence.

Theme 4: The Solar System, Gravity, and the Universe — 10 questions

STEM-G9-SCI-T04-Q01 | The Solar System, Gravity, and the Universe | Difficulty: Advanced

Which force keeps planets gravitationally bound to the Sun?

A. Gravity

B. Air resistance

C. Earth's magnetic force alone

D. Buoyancy

Answer: A. The Sun's gravity attracts planets and helps maintain their orbits.

STEM-G9-SCI-T04-Q02 | The Solar System, Gravity, and the Universe | Difficulty: Advanced

Which force keeps planets gravitationally bound to the Sun?

A. Earth's magnetic force alone

B. Buoyancy

C. Air resistance

D. Gravity

Answer: D. The Sun's gravity attracts planets and helps maintain their orbits.

STEM-G9-SCI-T04-Q03 | The Solar System, Gravity, and the Universe | Difficulty: Advanced

When investigating the solar system, gravity, and the universe, which approach is most scientifically sound?

A. Collect evidence, test explanations, and draw conclusions

B. Choose an answer by guessing

C. Ignore data that disagrees with an idea

D. Change results to match a prediction

Answer: A. Scientific inquiry uses testable evidence and conclusions supported by data.

STEM-G9-SCI-T04-Q04 | The Solar System, Gravity, and the Universe | Difficulty: Advanced

Which force keeps planets gravitationally bound to the Sun?

A. Earth's magnetic force alone

B. Gravity

C. Buoyancy

D. Air resistance

Answer: B. The Sun's gravity attracts planets and helps maintain their orbits.

STEM-G9-SCI-T04-Q05 | The Solar System, Gravity, and the Universe | Difficulty: Advanced

Which force keeps planets gravitationally bound to the Sun?

A. Gravity

B. Earth's magnetic force alone

C. Buoyancy

D. Air resistance

Answer: A. The Sun's gravity attracts planets and helps maintain their orbits.

STEM-G9-SCI-T04-Q06 | The Solar System, Gravity, and the Universe | Difficulty: Advanced

When investigating the solar system, gravity, and the universe, which approach is most scientifically sound?

A. Ignore data that disagrees with an idea

B. Collect evidence, test explanations, and draw conclusions

C. Change results to match a prediction

D. Choose an answer by guessing

Answer: B. Scientific inquiry uses testable evidence and conclusions supported by data.

STEM-G9-SCI-T04-Q07 | The Solar System, Gravity, and the Universe | Difficulty: Advanced

Which force keeps planets gravitationally bound to the Sun?

A. Air resistance

B. Gravity

C. Buoyancy

D. Earth's magnetic force alone

Answer: B. The Sun's gravity attracts planets and helps maintain their orbits.

STEM-G9-SCI-T04-Q08 | The Solar System, Gravity, and the Universe | Difficulty: Advanced

Which force keeps planets gravitationally bound to the Sun?

A. Earth's magnetic force alone

B. Gravity

C. Air resistance

D. Buoyancy

Answer: B. The Sun's gravity attracts planets and helps maintain their orbits.

STEM-G9-SCI-T04-Q09 | The Solar System, Gravity, and the Universe | Difficulty: Advanced

When investigating the solar system, gravity, and the universe, which approach is most scientifically sound?

A. Ignore data that disagrees with an idea

B. Collect evidence, test explanations, and draw conclusions

C. Change results to match a prediction

D. Choose an answer by guessing

Answer: B. Scientific inquiry uses testable evidence and conclusions supported by data.

STEM-G9-SCI-T04-Q10 | The Solar System, Gravity, and the Universe | Difficulty: Advanced

Which force keeps planets gravitationally bound to the Sun?

A. Buoyancy

B. Air resistance

C. Earth's magnetic force alone

D. Gravity

Answer: D. The Sun's gravity attracts planets and helps maintain their orbits.

Engineering

Theme 1: Mechanical Systems Analysis — 10 questions

STEM-G9-ENG-T01-Q01 | Mechanical Systems Analysis | Difficulty: Advanced

What is the purpose of making a sketch before building an object?

A. To plan its shape, dimensions, and parts

B. To replace all testing

C. To avoid measuring

D. To make the object build itself

Answer: A. A sketch communicates a design before construction.

STEM-G9-ENG-T01-Q02 | Mechanical Systems Analysis | Difficulty: Advanced

For a project about mechanical systems analysis, which design decision is best supported by engineering practice?

A. Choose based on requirements, constraints, test data, and safety

B. Avoid testing

C. Ignore cost and materials

D. Choose only by appearance

Answer: A. Engineering decisions consider performance, constraints, evidence, and safety.

STEM-G9-ENG-T01-Q03 | Mechanical Systems Analysis | Difficulty: Advanced

When selecting a material for a load-bearing part, what should be considered?

A. Only its favorite color

B. The designer's name

C. The packaging

D. Its strength and suitability for the task

Answer: D. Material choice depends on the load and operating conditions.

STEM-G9-ENG-T01-Q04 | Mechanical Systems Analysis | Difficulty: Advanced

For a project about mechanical systems analysis, which design decision is best supported by engineering practice?

A. Choose only by appearance

B. Ignore cost and materials

C. Avoid testing

D. Choose based on requirements, constraints, test data, and safety

Answer: D. Engineering decisions consider performance, constraints, evidence, and safety.

STEM-G9-ENG-T01-Q05 | Mechanical Systems Analysis | Difficulty: Advanced

After a test reveals a weakness, what should engineers do next?

A. Delete the notes

B. Ignore safety

C. Improve the design and test it again

D. Hide the result

Answer: C. Engineering is iterative and uses test results to guide improvements.

STEM-G9-ENG-T01-Q06 | Mechanical Systems Analysis | Difficulty: Advanced

For a project about mechanical systems analysis, which design decision is best supported by engineering practice?

A. Choose only by appearance

B. Choose based on requirements, constraints, test data, and safety

C. Ignore cost and materials

D. Avoid testing

Answer: B. Engineering decisions consider performance, constraints, evidence, and safety.

STEM-G9-ENG-T01-Q07 | Mechanical Systems Analysis | Difficulty: Advanced

Why should an engineering prototype be tested before use?

A. To make it larger every time

B. To find out whether the design works and is safe

C. To avoid making improvements

D. To make it look more expensive

Answer: B. Testing reveals weaknesses and helps improve a design.

STEM-G9-ENG-T01-Q08 | Mechanical Systems Analysis | Difficulty: Advanced

For a project about mechanical systems analysis, which design decision is best supported by engineering practice?

A. Choose only by appearance

B. Choose based on requirements, constraints, test data, and safety

C. Ignore cost and materials

D. Avoid testing

Answer: B. Engineering decisions consider performance, constraints, evidence, and safety.

STEM-G9-ENG-T01-Q09 | Mechanical Systems Analysis | Difficulty: Advanced

What is the purpose of making a sketch before building an object?

A. To replace all testing

B. To plan its shape, dimensions, and parts

C. To make the object build itself

D. To avoid measuring

Answer: B. A sketch communicates a design before construction.

STEM-G9-ENG-T01-Q10 | Mechanical Systems Analysis | Difficulty: Advanced

For a project about mechanical systems analysis, which design decision is best supported by engineering practice?

A. Choose based on requirements, constraints, test data, and safety

B. Choose only by appearance

C. Avoid testing

D. Ignore cost and materials

Answer: A. Engineering decisions consider performance, constraints, evidence, and safety.

Theme 2: Electricity in Design — 10 questions

STEM-G9-ENG-T02-Q01 | Electricity in Design | Difficulty: Advanced

What is the purpose of making a sketch before building an object?

A. To replace all testing

B. To avoid measuring

C. To plan its shape, dimensions, and parts

D. To make the object build itself

Answer: C. A sketch communicates a design before construction.

STEM-G9-ENG-T02-Q02 | Electricity in Design | Difficulty: Advanced

For a project about electricity in design, which design decision is best supported by engineering practice?

A. Ignore cost and materials

B. Avoid testing

C. Choose only by appearance

D. Choose based on requirements, constraints, test data, and safety

Answer: D. Engineering decisions consider performance, constraints, evidence, and safety.

STEM-G9-ENG-T02-Q03 | Electricity in Design | Difficulty: Advanced

When selecting a material for a load-bearing part, what should be considered?

A. Its strength and suitability for the task

B. The packaging

C. Only its favorite color

D. The designer's name

Answer: A. Material choice depends on the load and operating conditions.

STEM-G9-ENG-T02-Q04 | Electricity in Design | Difficulty: Advanced

For a project about electricity in design, which design decision is best supported by engineering practice?

A. Choose only by appearance

B. Avoid testing

C. Choose based on requirements, constraints, test data, and safety

D. Ignore cost and materials

Answer: C. Engineering decisions consider performance, constraints, evidence, and safety.

STEM-G9-ENG-T02-Q05 | Electricity in Design | Difficulty: Advanced

After a test reveals a weakness, what should engineers do next?

A. Delete the notes

B. Hide the result

C. Ignore safety

D. Improve the design and test it again

Answer: D. Engineering is iterative and uses test results to guide improvements.

STEM-G9-ENG-T02-Q06 | Electricity in Design | Difficulty: Advanced

For a project about electricity in design, which design decision is best supported by engineering practice?

A. Ignore cost and materials

B. Avoid testing

C. Choose only by appearance

D. Choose based on requirements, constraints, test data, and safety

Answer: D. Engineering decisions consider performance, constraints, evidence, and safety.

STEM-G9-ENG-T02-Q07 | Electricity in Design | Difficulty: Advanced

Why should an engineering prototype be tested before use?

A. To avoid making improvements

B. To make it larger every time

C. To make it look more expensive

D. To find out whether the design works and is safe

Answer: D. Testing reveals weaknesses and helps improve a design.

STEM-G9-ENG-T02-Q08 | Electricity in Design | Difficulty: Advanced

For a project about electricity in design, which design decision is best supported by engineering practice?

A. Ignore cost and materials

B. Avoid testing

C. Choose based on requirements, constraints, test data, and safety

D. Choose only by appearance

Answer: C. Engineering decisions consider performance, constraints, evidence, and safety.

STEM-G9-ENG-T02-Q09 | Electricity in Design | Difficulty: Advanced

What is the purpose of making a sketch before building an object?

A. To avoid measuring

B. To make the object build itself

C. To replace all testing

D. To plan its shape, dimensions, and parts

Answer: D. A sketch communicates a design before construction.

STEM-G9-ENG-T02-Q10 | Electricity in Design | Difficulty: Advanced

For a project about electricity in design, which design decision is best supported by engineering practice?

A. Avoid testing

B. Choose only by appearance

C. Choose based on requirements, constraints, test data, and safety

D. Ignore cost and materials

Answer: C. Engineering decisions consider performance, constraints, evidence, and safety.

Theme 3: Energy, Power, and Efficiency — 10 questions

STEM-G9-ENG-T03-Q01 | Energy, Power, and Efficiency | Difficulty: Advanced

What is the purpose of making a sketch before building an object?

A. To avoid measuring

B. To replace all testing

C. To make the object build itself

D. To plan its shape, dimensions, and parts

Answer: D. A sketch communicates a design before construction.

STEM-G9-ENG-T03-Q02 | Energy, Power, and Efficiency | Difficulty: Advanced

For a project about energy, power, and efficiency, which design decision is best supported by engineering practice?

A. Ignore cost and materials

B. Choose based on requirements, constraints, test data, and safety

C. Choose only by appearance

D. Avoid testing

Answer: B. Engineering decisions consider performance, constraints, evidence, and safety.

STEM-G9-ENG-T03-Q03 | Energy, Power, and Efficiency | Difficulty: Advanced

When selecting a material for a load-bearing part, what should be considered?

A. The packaging

B. The designer's name

C. Its strength and suitability for the task

D. Only its favorite color

Answer: C. Material choice depends on the load and operating conditions.

STEM-G9-ENG-T03-Q04 | Energy, Power, and Efficiency | Difficulty: Advanced

For a project about energy, power, and efficiency, which design decision is best supported by engineering practice?

A. Choose only by appearance

B. Avoid testing

C. Ignore cost and materials

D. Choose based on requirements, constraints, test data, and safety

Answer: D. Engineering decisions consider performance, constraints, evidence, and safety.

STEM-G9-ENG-T03-Q05 | Energy, Power, and Efficiency | Difficulty: Advanced

After a test reveals a weakness, what should engineers do next?

A. Ignore safety

B. Improve the design and test it again

C. Hide the result

D. Delete the notes

Answer: B. Engineering is iterative and uses test results to guide improvements.

STEM-G9-ENG-T03-Q06 | Energy, Power, and Efficiency | Difficulty: Advanced

For a project about energy, power, and efficiency, which design decision is best supported by engineering practice?

A. Choose based on requirements, constraints, test data, and safety

B. Ignore cost and materials

C. Avoid testing

D. Choose only by appearance

Answer: A. Engineering decisions consider performance, constraints, evidence, and safety.

STEM-G9-ENG-T03-Q07 | Energy, Power, and Efficiency | Difficulty: Advanced

Why should an engineering prototype be tested before use?

A. To avoid making improvements

B. To make it larger every time

C. To find out whether the design works and is safe

D. To make it look more expensive

Answer: C. Testing reveals weaknesses and helps improve a design.

STEM-G9-ENG-T03-Q08 | Energy, Power, and Efficiency | Difficulty: Advanced

For a project about energy, power, and efficiency, which design decision is best supported by engineering practice?

A. Avoid testing

B. Choose only by appearance

C. Choose based on requirements, constraints, test data, and safety

D. Ignore cost and materials

Answer: C. Engineering decisions consider performance, constraints, evidence, and safety.

STEM-G9-ENG-T03-Q09 | Energy, Power, and Efficiency | Difficulty: Advanced

What is the purpose of making a sketch before building an object?

A. To replace all testing

B. To avoid measuring

C. To make the object build itself

D. To plan its shape, dimensions, and parts

Answer: D. A sketch communicates a design before construction.

STEM-G9-ENG-T03-Q10 | Energy, Power, and Efficiency | Difficulty: Advanced

For a project about energy, power, and efficiency, which design decision is best supported by engineering practice?

A. Ignore cost and materials

B. Avoid testing

C. Choose based on requirements, constraints, test data, and safety

D. Choose only by appearance

Answer: C. Engineering decisions consider performance, constraints, evidence, and safety.

Theme 4: Data-Driven Design Optimization — 10 questions

STEM-G9-ENG-T04-Q01 | Data-Driven Design Optimization | Difficulty: Advanced

What is the purpose of making a sketch before building an object?

A. To replace all testing

B. To plan its shape, dimensions, and parts

C. To avoid measuring

D. To make the object build itself

Answer: B. A sketch communicates a design before construction.

STEM-G9-ENG-T04-Q02 | Data-Driven Design Optimization | Difficulty: Advanced

For a project about data-driven design optimization, which design decision is best supported by engineering practice?

A. Choose only by appearance

B. Choose based on requirements, constraints, test data, and safety

C. Avoid testing

D. Ignore cost and materials

Answer: B. Engineering decisions consider performance, constraints, evidence, and safety.

STEM-G9-ENG-T04-Q03 | Data-Driven Design Optimization | Difficulty: Advanced

When selecting a material for a load-bearing part, what should be considered?

A. Its strength and suitability for the task

B. The designer's name

C. The packaging

D. Only its favorite color

Answer: A. Material choice depends on the load and operating conditions.

STEM-G9-ENG-T04-Q04 | Data-Driven Design Optimization | Difficulty: Advanced

For a project about data-driven design optimization, which design decision is best supported by engineering practice?

A. Choose only by appearance

B. Avoid testing

C. Ignore cost and materials

D. Choose based on requirements, constraints, test data, and safety

Answer: D. Engineering decisions consider performance, constraints, evidence, and safety.

STEM-G9-ENG-T04-Q05 | Data-Driven Design Optimization | Difficulty: Advanced

After a test reveals a weakness, what should engineers do next?

A. Improve the design and test it again

B. Delete the notes

C. Ignore safety

D. Hide the result

Answer: A. Engineering is iterative and uses test results to guide improvements.

STEM-G9-ENG-T04-Q06 | Data-Driven Design Optimization | Difficulty: Advanced

For a project about data-driven design optimization, which design decision is best supported by engineering practice?

A. Avoid testing

B. Ignore cost and materials

C. Choose based on requirements, constraints, test data, and safety

D. Choose only by appearance

Answer: C. Engineering decisions consider performance, constraints, evidence, and safety.

STEM-G9-ENG-T04-Q07 | Data-Driven Design Optimization | Difficulty: Advanced

Why should an engineering prototype be tested before use?

A. To make it look more expensive

B. To make it larger every time

C. To find out whether the design works and is safe

D. To avoid making improvements

Answer: C. Testing reveals weaknesses and helps improve a design.

STEM-G9-ENG-T04-Q08 | Data-Driven Design Optimization | Difficulty: Advanced

For a project about data-driven design optimization, which design decision is best supported by engineering practice?

A. Choose only by appearance

B. Ignore cost and materials

C. Choose based on requirements, constraints, test data, and safety

D. Avoid testing

Answer: C. Engineering decisions consider performance, constraints, evidence, and safety.

STEM-G9-ENG-T04-Q09 | Data-Driven Design Optimization | Difficulty: Advanced

What is the purpose of making a sketch before building an object?

A. To replace all testing

B. To make the object build itself

C. To plan its shape, dimensions, and parts

D. To avoid measuring

Answer: C. A sketch communicates a design before construction.

STEM-G9-ENG-T04-Q10 | Data-Driven Design Optimization | Difficulty: Advanced

For a project about data-driven design optimization, which design decision is best supported by engineering practice?

A. Ignore cost and materials

B. Choose only by appearance

C. Avoid testing

D. Choose based on requirements, constraints, test data, and safety

Answer: D. Engineering decisions consider performance, constraints, evidence, and safety.

ICT (Technology)

Theme 1: Databases and Information Management — 10 questions

STEM-G9-ICT-T01-Q01 | Databases and Information Management | Difficulty: Advanced

Why should you keep your password private?

A. To make the screen brighter

B. To protect your account and personal information

C. To make the internet faster

D. To extend battery life

Answer: B. A password provides access to an account and should be protected.

STEM-G9-ICT-T01-Q02 | Databases and Information Management | Difficulty: Advanced

A clear, ordered set of steps for solving a problem is called an ...

A. Resolution

B. Algorithm

C. Speaker

D. Wallpaper

Answer: B. An algorithm is an ordered set of instructions.

STEM-G9-ICT-T01-Q03 | Databases and Information Management | Difficulty: Advanced

In the context of databases and information management, which practice is most appropriate?

A. Share account credentials

B. Assume every source is always correct

C. Use data and permissions responsibly, and verify results

D. Ignore validation

Answer: C. Responsible technology use requires validation, security, and digital responsibility.

STEM-G9-ICT-T01-Q04 | Databases and Information Management | Difficulty: Advanced

How should information found online be checked?

A. Compare sources and supporting evidence

B. Share it before reading

C. Count the emojis

D. Choose the most dramatic headline

Answer: A. Checking sources helps assess accuracy and credibility.

STEM-G9-ICT-T01-Q05 | Databases and Information Management | Difficulty: Advanced

Finding and fixing errors in a program is called ...

A. Browsing

B. Streaming

C. Debugging

D. Formatting

Answer: C. Debugging is the process of locating and correcting errors.

STEM-G9-ICT-T01-Q06 | Databases and Information Management | Difficulty: Advanced

In the context of databases and information management, which practice is most appropriate?

A. Ignore validation

B. Share account credentials

C. Use data and permissions responsibly, and verify results

D. Assume every source is always correct

Answer: C. Responsible technology use requires validation, security, and digital responsibility.

STEM-G9-ICT-T01-Q07 | Databases and Information Management | Difficulty: Advanced

Software designed to help a user perform a task is commonly called an ...

A. Fan

B. Power cable

C. Application

D. Desk

Answer: C. An application is software that helps users perform specific tasks.

STEM-G9-ICT-T01-Q08 | Databases and Information Management | Difficulty: Advanced

What should you do if an unknown person sends you a suspicious link?

A. Forward it to every contact

B. Enter your password

C. Download every attachment

D. Do not open it; check with a trusted adult or person

Answer: D. Suspicious links may lead to scams or harmful software.

STEM-G9-ICT-T01-Q09 | Databases and Information Management | Difficulty: Advanced

In the context of databases and information management, which practice is most appropriate?

A. Share account credentials

B. Assume every source is always correct

C. Use data and permissions responsibly, and verify results

D. Ignore validation

Answer: C. Responsible technology use requires validation, security, and digital responsibility.

STEM-G9-ICT-T01-Q10 | Databases and Information Management | Difficulty: Advanced

A clear, ordered set of steps for solving a problem is called an ...

A. Algorithm

B. Resolution

C. Wallpaper

D. Speaker

Answer: A. An algorithm is an ordered set of instructions.

Theme 2: Networks, Protocols, and Internet Services — 10 questions

STEM-G9-ICT-T02-Q01 | Networks, Protocols, and Internet Services | Difficulty: Advanced

Why should you keep your password private?

A. To make the internet faster

B. To protect your account and personal information

C. To make the screen brighter

D. To extend battery life

Answer: B. A password provides access to an account and should be protected.

STEM-G9-ICT-T02-Q02 | Networks, Protocols, and Internet Services | Difficulty: Advanced

A clear, ordered set of steps for solving a problem is called an ...

A. Resolution

B. Speaker

C. Algorithm

D. Wallpaper

Answer: C. An algorithm is an ordered set of instructions.

STEM-G9-ICT-T02-Q03 | Networks, Protocols, and Internet Services | Difficulty: Advanced

In the context of networks, protocols, and internet services, which practice is most appropriate?

A. Use data and permissions responsibly, and verify results

B. Share account credentials

C. Ignore validation

D. Assume every source is always correct

Answer: A. Responsible technology use requires validation, security, and digital responsibility.

STEM-G9-ICT-T02-Q04 | Networks, Protocols, and Internet Services | Difficulty: Advanced

How should information found online be checked?

A. Share it before reading

B. Choose the most dramatic headline

C. Compare sources and supporting evidence

D. Count the emojis

Answer: C. Checking sources helps assess accuracy and credibility.

STEM-G9-ICT-T02-Q05 | Networks, Protocols, and Internet Services | Difficulty: Advanced

Finding and fixing errors in a program is called ...

A. Streaming

B. Browsing

C. Formatting

D. Debugging

Answer: D. Debugging is the process of locating and correcting errors.

STEM-G9-ICT-T02-Q06 | Networks, Protocols, and Internet Services | Difficulty: Advanced

In the context of networks, protocols, and internet services, which practice is most appropriate?

A. Share account credentials

B. Ignore validation

C. Assume every source is always correct

D. Use data and permissions responsibly, and verify results

Answer: D. Responsible technology use requires validation, security, and digital responsibility.

STEM-G9-ICT-T02-Q07 | Networks, Protocols, and Internet Services | Difficulty: Advanced

Software designed to help a user perform a task is commonly called an ...

A. Fan

B. Desk

C. Power cable

D. Application

Answer: D. An application is software that helps users perform specific tasks.

STEM-G9-ICT-T02-Q08 | Networks, Protocols, and Internet Services | Difficulty: Advanced

What should you do if an unknown person sends you a suspicious link?

A. Forward it to every contact

B. Do not open it; check with a trusted adult or person

C. Enter your password

D. Download every attachment

Answer: B. Suspicious links may lead to scams or harmful software.

STEM-G9-ICT-T02-Q09 | Networks, Protocols, and Internet Services | Difficulty: Advanced

In the context of networks, protocols, and internet services, which practice is most appropriate?

A. Ignore validation

B. Share account credentials

C. Assume every source is always correct

D. Use data and permissions responsibly, and verify results

Answer: D. Responsible technology use requires validation, security, and digital responsibility.

STEM-G9-ICT-T02-Q10 | Networks, Protocols, and Internet Services | Difficulty: Advanced

A clear, ordered set of steps for solving a problem is called an ...

A. Resolution

B. Algorithm

C. Speaker

D. Wallpaper

Answer: B. An algorithm is an ordered set of instructions.

Theme 3: Programming and Problem Solving — 10 questions

STEM-G9-ICT-T03-Q01 | Programming and Problem Solving | Difficulty: Advanced

Why should you keep your password private?

A. To make the internet faster

B. To make the screen brighter

C. To protect your account and personal information

D. To extend battery life

Answer: C. A password provides access to an account and should be protected.

STEM-G9-ICT-T03-Q02 | Programming and Problem Solving | Difficulty: Advanced

A clear, ordered set of steps for solving a problem is called an ...

A. Algorithm

B. Resolution

C. Speaker

D. Wallpaper

Answer: A. An algorithm is an ordered set of instructions.

STEM-G9-ICT-T03-Q03 | Programming and Problem Solving | Difficulty: Advanced

In the context of programming and problem solving, which practice is most appropriate?

A. Use data and permissions responsibly, and verify results

B. Ignore validation

C. Assume every source is always correct

D. Share account credentials

Answer: A. Responsible technology use requires validation, security, and digital responsibility.

STEM-G9-ICT-T03-Q04 | Programming and Problem Solving | Difficulty: Advanced

How should information found online be checked?

A. Share it before reading

B. Compare sources and supporting evidence

C. Choose the most dramatic headline

D. Count the emojis

Answer: B. Checking sources helps assess accuracy and credibility.

STEM-G9-ICT-T03-Q05 | Programming and Problem Solving | Difficulty: Advanced

Finding and fixing errors in a program is called ...

A. Streaming

B. Browsing

C. Debugging

D. Formatting

Answer: C. Debugging is the process of locating and correcting errors.

STEM-G9-ICT-T03-Q06 | Programming and Problem Solving | Difficulty: Advanced

In the context of programming and problem solving, which practice is most appropriate?

A. Use data and permissions responsibly, and verify results

B. Ignore validation

C. Share account credentials

D. Assume every source is always correct

Answer: A. Responsible technology use requires validation, security, and digital responsibility.

STEM-G9-ICT-T03-Q07 | Programming and Problem Solving | Difficulty: Advanced

Software designed to help a user perform a task is commonly called an ...

A. Application

B. Desk

C. Power cable

D. Fan

Answer: A. An application is software that helps users perform specific tasks.

STEM-G9-ICT-T03-Q08 | Programming and Problem Solving | Difficulty: Advanced

What should you do if an unknown person sends you a suspicious link?

A. Download every attachment

B. Do not open it; check with a trusted adult or person

C. Enter your password

D. Forward it to every contact

Answer: B. Suspicious links may lead to scams or harmful software.

STEM-G9-ICT-T03-Q09 | Programming and Problem Solving | Difficulty: Advanced

In the context of programming and problem solving, which practice is most appropriate?

A. Ignore validation

B. Assume every source is always correct

C. Use data and permissions responsibly, and verify results

D. Share account credentials

Answer: C. Responsible technology use requires validation, security, and digital responsibility.

STEM-G9-ICT-T03-Q10 | Programming and Problem Solving | Difficulty: Advanced

A clear, ordered set of steps for solving a problem is called an ...

A. Resolution

B. Speaker

C. Algorithm

D. Wallpaper

Answer: C. An algorithm is an ordered set of instructions.

Theme 4: AI, Digital Ethics, and Information Evaluation — 10 questions

STEM-G9-ICT-T04-Q01 | AI, Digital Ethics, and Information Evaluation | Difficulty: Advanced

Why should you keep your password private?

A. To make the internet faster

B. To protect your account and personal information

C. To extend battery life

D. To make the screen brighter

Answer: B. A password provides access to an account and should be protected.

STEM-G9-ICT-T04-Q02 | AI, Digital Ethics, and Information Evaluation | Difficulty: Advanced

A clear, ordered set of steps for solving a problem is called an ...

A. Resolution

B. Algorithm

C. Speaker

D. Wallpaper

Answer: B. An algorithm is an ordered set of instructions.

STEM-G9-ICT-T04-Q03 | AI, Digital Ethics, and Information Evaluation | Difficulty: Advanced

In the context of ai, digital ethics, and information evaluation, which practice is most appropriate?

A. Share account credentials

B. Ignore validation

C. Use data and permissions responsibly, and verify results

D. Assume every source is always correct

Answer: C. Responsible technology use requires validation, security, and digital responsibility.

STEM-G9-ICT-T04-Q04 | AI, Digital Ethics, and Information Evaluation | Difficulty: Advanced

How should information found online be checked?

A. Count the emojis

B. Share it before reading

C. Compare sources and supporting evidence

D. Choose the most dramatic headline

Answer: C. Checking sources helps assess accuracy and credibility.

STEM-G9-ICT-T04-Q05 | AI, Digital Ethics, and Information Evaluation | Difficulty: Advanced

Finding and fixing errors in a program is called ...

A. Debugging

B. Streaming

C. Formatting

D. Browsing

Answer: A. Debugging is the process of locating and correcting errors.

STEM-G9-ICT-T04-Q06 | AI, Digital Ethics, and Information Evaluation | Difficulty: Advanced

In the context of ai, digital ethics, and information evaluation, which practice is most appropriate?

A. Assume every source is always correct

B. Share account credentials

C. Ignore validation

D. Use data and permissions responsibly, and verify results

Answer: D. Responsible technology use requires validation, security, and digital responsibility.

STEM-G9-ICT-T04-Q07 | AI, Digital Ethics, and Information Evaluation | Difficulty: Advanced

Software designed to help a user perform a task is commonly called an ...

A. Fan

B. Desk

C. Power cable

D. Application

Answer: D. An application is software that helps users perform specific tasks.

STEM-G9-ICT-T04-Q08 | AI, Digital Ethics, and Information Evaluation | Difficulty: Advanced

What should you do if an unknown person sends you a suspicious link?

A. Do not open it; check with a trusted adult or person

B. Forward it to every contact

C. Download every attachment

D. Enter your password

Answer: A. Suspicious links may lead to scams or harmful software.

STEM-G9-ICT-T04-Q09 | AI, Digital Ethics, and Information Evaluation | Difficulty: Advanced

In the context of ai, digital ethics, and information evaluation, which practice is most appropriate?

A. Ignore validation

B. Share account credentials

C. Assume every source is always correct

D. Use data and permissions responsibly, and verify results

Answer: D. Responsible technology use requires validation, security, and digital responsibility.

STEM-G9-ICT-T04-Q10 | AI, Digital Ethics, and Information Evaluation | Difficulty: Advanced

A clear, ordered set of steps for solving a problem is called an ...

A. Resolution

B. Speaker

C. Wallpaper

D. Algorithm

Answer: D. An algorithm is an ordered set of instructions.

READY-TO-USE PROMPT FOR GOOGLE AI STUDIO

I have uploaded the STEM Nexus English Master Question Bank PDF. Integrate all 1,440 questions into the existing application. Extract and store every question as structured application data; do not merely summarize the PDF or search its text during gameplay. Preserve each unique question ID, grade level, subject, theme, question text, four answer options, correct answer, and explanation. Implement mandatory filtering using the player's selected grade + subject + theme. Each quiz session must only retrieve questions matching all three selected metadata fields. Never mix topics, subjects, or grade levels, and never use cross-grade or cross-subject fallback questions. If a filtered question pool is empty, show a clear message or request additional questions that match the same filters. Ensure all existing game modes use the same question source and quiz session state. Preserve the current UI, scoring, progress, achievements, and features. Validate answer keys, prevent duplicate IDs, and report the imported question count by grade, subject, and theme. Run tests to verify that no question appears outside the selected filters.

Integration note: Store the extracted questions as structured JSON, TypeScript objects, or database records. Do not rely on free-text PDF searching every time a player starts a question.